Theory of Quantum Mechanics

From Mathematical Foundations to the Quantum Hall Effect

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Table of Contents

Preface

This book develops the mathematical and physical ideas needed to understand modern quantum mechanics and the Quantum Hall Effect. It begins with vectors, motion, and calculus, then advances through classical mechanics, electromagnetism, analytical mechanics, and quantum theory.

The presentation emphasizes geometric intuition, careful derivation, and explicit links between classical and quantum descriptions. Each chapter uses a consistent structure: learning objectives, conceptual development, physical interpretation, worked examples, common mistakes, a summary, and a preview of the next chapter.

Theory of Quantum Mechanics

Volume I

Theory of Quantum Mechanics

From Mathematical Foundations

to the Quantum Hall Effect

Volume I

Mathematical Foundations

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This manuscript is a working draft intended for educational and research purposes.

Future editions may contain additional chapters, figures, exercises and historical notes.

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Dedicated to everyone who wants to understand
Nature through mathematics.

Preface

Quantum mechanics is one of the greatest intellectual achievements in science. Rather than presenting the subject as a collection of formulas, this book builds the theory from first principles.

The journey begins with vectors, geometry and calculus before progressing to

- classical mechanics,
- electromagnetism,
- analytical mechanics,
- quantum mechanics,
- Landau levels,
- the Quantum Hall Effect,
- modern topological physics.

Every chapter follows the same educational structure:

1. Learning Objectives
2. Conceptual Motivation
3. Definitions
4. Theory
5. Worked Examples
6. Physical Interpretation
7. Historical Notes
8. Common Mistakes
9. Summary
10. Exercises
11. Looking Ahead

How to Use This Book

Different readers may use this text in different ways.

First reading: Focus on intuition and worked examples.

Second reading: Study the mathematical derivations.

Reference: Use the notation tables, summaries and index to locate concepts quickly.

Frequently Used Symbols

Symbol	Meaning	SI Unit
$\mathbf{r}$	Position vector	m
$\mathbf{v}$	Velocity	m/s
$\mathbf{a}$	Acceleration	m/s^2
$\mathbf{F}$	Force	N
$\mathbf{p}$	Momentum	kg m/s
∇	Nabla operator	–
ψ	Wavefunction	–

Fundamental Physical Constants

Constant	Symbol	Value
Speed of light	c	2.99792458×10^{8} m/s
Planck constant	h	$6.62607015 \times 10^{-34}$ J s
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$ J s
Elementary charge	e	$1.602176634 \times 10^{-19}$ C
Electron mass	m_e	$9.1093837015 \times 10^{-31}$ kg
Vacuum permittivity	ε_0	$8.8541878128 \times 10^{-12}$ F/m

Roadmap of the Book

Part I — Mathematical Foundations
Part II — Classical Mechanics
Part III — Electromagnetism
Part IV — Analytical Mechanics
Part V — Quantum Mechanics
Part VI — Quantum Hall Effect
Part VII — Topology and Modern Quantum Matter

Notation Table

Symbol	Meaning
$\mathbf{A}, \mathbf{B}$	Generic vectors
$\|\mathbf{A}\|$	Magnitude or norm of $\mathbf{A}$
$\mathbf{A} \cdot \mathbf{B}$	Dot product
$\mathbf{A} \times \mathbf{B}$	Cross product
$\operatorname{proj}_{\mathbf{B}} \mathbf{A}$	Projection of $\mathbf{A}$ onto $\mathbf{B}$
$\mathbf{e}_i$	Basis vector
$[\mathbf{v}]_{\mathcal{B}}$	Coordinate column of $\mathbf{v}$ in basis $\mathcal{B}$
$R(\theta)$	Rotation matrix through angle θ
T or A	Linear transformation or its matrix representation
λ	Eigenvalue
$\mathbf{0}$	Zero vector
$\hbar$	Reduced Planck constant
$f : D \to C$	Function from domain D to codomain C
$\lim_{x \to a} f(x)$	Limit of $f(x)$ as x approaches a
$f'(x), \frac{df}{dx}$	Ordinary derivative
$\partial_i f, \frac{\partial f}{\partial x^i}$	Partial derivative with respect to x^i
$df_{\mathbf{a}}, Df(\mathbf{a})$	Differential or derivative map of f at $\mathbf{a}$
$J_F(\mathbf{a})$	Jacobian matrix of F at $\mathbf{a}$
∇f	Gradient of a scalar field
$D_{\mathbf{u}} f$	Directional derivative in the unit direction $\mathbf{u}$
H_f	Hessian matrix of second partial derivatives
ψ	Quantum-mechanical wavefunction
$\dot{x}, \ddot{x}$	First and second derivatives with respect to time
$\mathbf{x}(t)$	State vector of a dynamical system
$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x})$	Canonical first-order dynamical system
$\mathbf{x}(t_0) = \mathbf{x}_0$	Initial condition at time t_0
e^{At}	Matrix exponential generating a linear flow
ω_0	Natural angular frequency
γ or ζ	Damping parameter or damping ratio, as locally defined
h	Numerical integration step size

Table of Contents

PART I

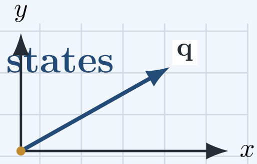
Mathematical Foundations

Chapter 1

Mathematical Language of Physics

From measurement to quantum states

Physics becomes predictive when observations are organized by mathematical structures. We begin with quantities and geometry, then advance toward the vector spaces and operators used in quantum mechanics.



"The book of Nature is written in the language of mathematics."

— Galileo Galilei

Chapter Roadmap

Physical quantities $\longrightarrow$ geometry and vectors $\longrightarrow$ coordinates and vector algebra $\longrightarrow$ products and projections $\longrightarrow$ bases and abstract vector spaces.

Learning Objectives

After completing this chapter, the reader should be able to:

- explain why mathematical modeling and experimental testing are inseparable in physics;
- distinguish a physical quantity from its numerical representation and unit;
- interpret measurement as comparison with a standard and recognize the role of uncertainty;
- distinguish scalars from vectors and interpret vectors geometrically;
- use coordinates, vector algebra, products, projections, bases, and dimension;
- recognize how geometric vectors lead toward abstract state vectors in quantum mechanics.

1.1 Mathematical Language of Physics

Physics seeks to answer some of the most fundamental questions about Nature. Although these questions arise from observation, their answers cannot be expressed with sufficient precision in ordinary language alone. The remarkable success of modern physics stems from the discovery that the laws governing the Universe can be formulated as mathematical relationships whose predictions can be tested experimentally.

Mathematics is therefore far more than a computational tool. It provides the conceptual language in which physical theories are formulated, analyzed, compared, and extended. In this chapter we begin developing that language, starting with measurable quantities and geometric vectors and then moving toward the abstract structures required by quantum mechanics.

1.1.1 Why Mathematics?

Long before modern science, humanity recognized regular patterns in Nature: the alternation of day and night, the changing seasons, the motion of celestial bodies, and the fall of terrestrial objects. Such observations made calendars, navigation, and the prediction of eclipses possible. Qualitative descriptions, however, could not by themselves reveal the precise relationships connecting these phenomena or generate reliable predictions in unfamiliar situations.

The emergence of theoretical physics required measurement, experiment, and mathematical reasoning to become inseparable. Galileo showed that motion could be studied quantitatively. Newton expressed terrestrial and celestial mechanics through the same dynamical principles. Maxwell united electricity, magnetism, and optics in a system of field equations. Einstein reformulated the concepts of space, time, and gravitation. Schrödinger and Heisenberg introduced new mathematical structures capable of describing microscopic phenomena for which classical trajectories were no longer adequate.

In each case, mathematics did more than summarize observations. Once a physical principle had been formulated mathematically, consequences could be derived that were not evident from the original experiments. A theory could therefore predict new phenomena, expose hidden relations among apparently different processes, and identify the limits within which an older description remained valid.

Mathematics is universal in a further sense. An equation does not depend on the spoken language of the scientist who writes it. Its symbols acquire meaning through definitions and operational rules, allowing the same physical law to be examined, reproduced, and criticized across cultures and generations.

Experiment nevertheless remains the final judge of a physical theory. Mathematics contains many internally consistent structures, but only a small subset describes the physical world. Theoretical physics therefore advances through a continuing dialogue:

$$\text{observation} \longrightarrow \text{model} \longrightarrow \text{prediction} \longrightarrow \text{experiment} \longrightarrow \text{revision.}$$

Mathematics and physical reality

A mathematical theory of physics is neither a collection of measurements nor a free construction detached from experiment. It is a structured model whose concepts are tied to possible observations and whose consequences must survive experimental tests.

The progression of this chapter reflects that viewpoint. We begin with the quantities obtained through measurement, pass through geometry and vectors, and gradually move toward vector spaces, matrices, eigenvalue problems, inner products, and operators. The final destination is the mathematical language in which a quantum state can be understood as a vector and a measurable quantity as an operator.

1.1.2 Physical Quantities

Every physical theory begins with quantities that can, at least in principle, be connected to measurement. Length, duration, mass, electric charge, temperature, momentum, and energy are familiar examples. To specify such a quantity completely, a numerical value is generally not enough; one must also state the unit and, for directional quantities, the relevant direction or components.

Thus the statement

$$L = 2.0\ \mathrm{m}$$

contains both a number and a unit. The number tells us how many copies of the chosen standard are present, while the unit identifies the standard against which the comparison has been made. Changing the unit changes the numerical value but not the underlying physical length:

$$2.0\ \mathrm{m} = 200\ \mathrm{cm}.$$

Physical quantities may be classified as *fundamental* or *derived*. A chosen system of units begins with a small set of base quantities. Other quantities are then defined through physical or geometric relations. Velocity, for example, is constructed from displacement and time, while force is related to mass and acceleration.

Physical quantity

A physical quantity is a property of a system or process that can be represented quantitatively through a specified measurement procedure, together with an appropriate unit and mathematical type.

The phrase *mathematical type* is important. Temperature and energy can be represented by scalars, whereas displacement and momentum require vectors. Later, quantum states will require a still more general kind of vector, and observables will be represented by operators. The mathematical object chosen to represent a quantity must preserve the structure observed in the physical phenomenon.

1.1.3 Measurements and Units

Measurement is a comparison between a physical system and an agreed standard. No real measurement produces an infinitely precise number. Every result is limited by the instrument, the procedure, the surrounding conditions, and the statistical variability of repeated observations. A measured quantity should therefore be understood not merely as a number, but as a reported value together with information about its uncertainty.

A simple measurement may be written schematically as

$$q = q_{\mathrm{meas}} \pm \Delta q,$$

where q_{meas} is the reported value and Δq characterizes the measurement uncertainty. The interpretation of Δq depends on the experimental and statistical context; a careful theory of uncertainty will be developed later in the book.

Units make comparisons reproducible. The International System of Units, or SI, uses seven base units, including the metre, kilogram, second, ampere, kelvin, mole, and candela. Derived units abbreviate recurring combinations of these base units. For example,

$$1\ \mathrm{N} = 1\ \mathrm{kg\,m\,s^{-2}},$$

so the newton is a convenient name for the SI unit of force.

Units also provide an immediate consistency test. An equation intended to represent a physical law must be dimensionally homogeneous: quantities added to one another must have the same physical dimensions, and the dimensions on both sides of an equality must agree.

A number without its unit

Except for dimensionless quantities, a numerical value written without its unit is incomplete. The statements $L = 2$, $t = 5$, and $E = 10$ do not yet specify physical quantities.

The next section introduces the first major distinction among mathematical representations of physical quantities: the distinction between scalars and vectors.

1.2 Scalars and Vectors

1.2.1 Why Direction Matters

A numerical value does not always provide a complete physical description. The statement

$$v = 20\,\mathrm{m\,s^{-1}}$$

specifies a speed, but it does not say whether the object moves east, west, upward, or in some other direction. By contrast,

$$\mathbf{v} = 20\,\mathrm{m\,s^{-1}}\,\mathbf{e}_x$$

specifies both the magnitude and the direction of the motion. This distinction separates *scalar* quantities from *vector* quantities.

Two kinds of physical information

A scalar answers "how much?" A vector answers both "how much?" and "in which direction?"

1.2.2 Scalars

A scalar is completely characterized by a numerical value and a unit. Typical examples are mass, temperature, electric potential, pressure, energy, and probability density. For example,

$$T = 300\,\mathrm{K}, \qquad m_e = 9.11 \times 10^{-31}\,\mathrm{kg}.$$

A scalar may vary from point to point or with time, but it does not acquire a spatial direction. Temperature in a room, for instance, can be represented by a scalar field $T(\mathbf{r}, t)$.

1.2.3 Vectors

A vector possesses a magnitude and a direction. Displacement, velocity, acceleration, force, momentum, electric field, magnetic field, and probability current are vector quantities. We write vectors in boldface,

$$\mathbf{A},$$

and reserve ordinary italic symbols such as A for scalars or for the magnitude $A = \|\mathbf{A}\|$ when the meaning is clear.

In a Cartesian basis, a vector is represented by components:

$$\mathbf{A} = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z.$$

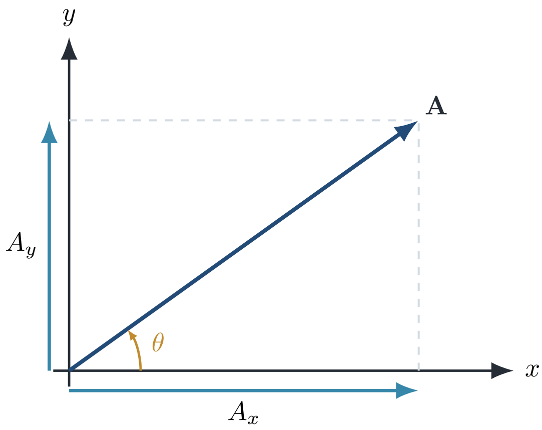


Figure 1.1: A vector resolved into Cartesian components. The diagonal arrow is the geometric vector; the horizontal and vertical arrows are its components in the chosen basis.

The components depend on the selected coordinate system, whereas the geometric vector does not. Rotating the axes changes the numbers A_x, A_y, A_z but does not change the physical displacement, velocity, or force represented by $\mathbf{A}$.

How to read the figure

The components are not separate vectors hidden inside $\mathbf{A}$. They are the projections of the same vector onto the selected basis directions.

1.2.4 Transformation Test

A useful way to distinguish scalars from vectors is to ask how the quantity changes when the coordinate axes are rotated. A true scalar is unchanged. A vector keeps its magnitude and geometric direction, but its components transform together according to the rotation. This transformation law, rather than the presence of an arrow in a diagram, is the precise mathematical criterion for vector character.

Common Mistake

A quantity is not a vector merely because it has several numerical entries. Those entries must transform as vector components under a change of basis.

1.3 Coordinate Systems

A physical point or vector exists independently of the labels used to describe it. A coordinate system is therefore not part of nature itself; it is a mathematical convention that assigns numbers to geometric objects. Good coordinates make a problem transparent, while poor coordinates can hide its symmetry.

1.3.1 Cartesian Coordinates

In three-dimensional Euclidean space, a Cartesian coordinate system consists of three mutually perpendicular axes with associated orthonormal unit vectors

$$\mathbf{e}_x, \qquad \mathbf{e}_y, \qquad \mathbf{e}_z.$$

They satisfy

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij},$$

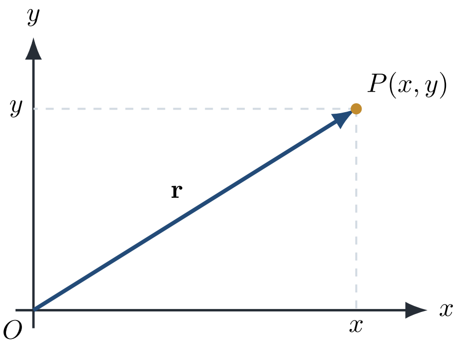


Figure 1.2: A point and its Cartesian position vector. The point is geometric; the coordinates are numerical labels relative to the selected axes.

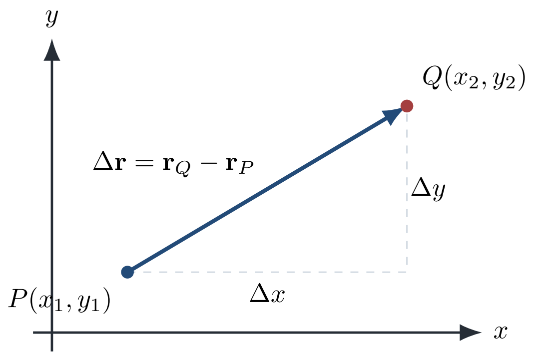


Figure 1.3: The displacement from P to Q is $\Delta\mathbf{r} = \mathbf{r}_Q - \mathbf{r}_P$. Its Cartesian components form a right triangle.

where δ_{ij} is the Kronecker delta. A point P is represented by the ordered triple (x, y, z), and its position vector is

$$\mathbf{r} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z.$$

Physical Interpretation

The point P is geometric; the numbers (x, y, z) are coordinate-dependent. Rotating the axes changes the components but does not move the point.

1.3.2 Coordinates and Components

For a general vector $\mathbf{A}$,

$$\mathbf{A} = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z.$$

The coefficients are obtained by projection:

$$A_x = \mathbf{A} \cdot \mathbf{e}_x, \qquad A_y = \mathbf{A} \cdot \mathbf{e}_y, \qquad A_z = \mathbf{A} \cdot \mathbf{e}_z.$$

Thus a basis converts an abstract vector into an ordered list of numbers.

Other Useful Coordinate Systems

Cartesian coordinates are not always optimal. Cylindrical coordinates (ρ, ϕ, z) are adapted to axial symmetry, while spherical coordinates (r, θ, ϕ) are adapted to central symmetry. Their relations to Cartesian coordinates are

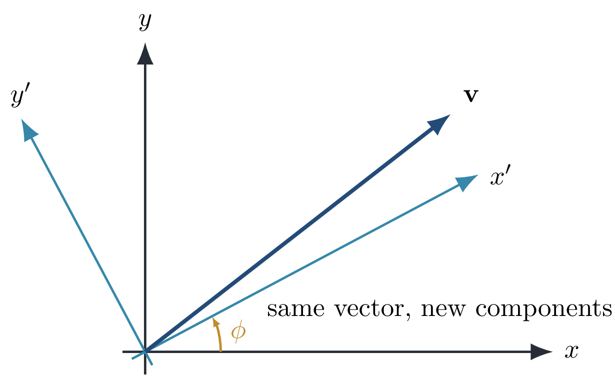


Figure 1.4: Rotating the coordinate axes changes the numerical components, but not the geometric vector itself.

$$x = \rho \cos \phi, \qquad y = \rho \sin \phi,$$

and

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$$

Unlike the Cartesian basis, curvilinear basis vectors generally vary from point to point.

Choosing coordinates

The gravitational field of an isolated spherical body is most naturally expressed in spherical coordinates because the field depends primarily on the radial distance r. A long straight wire suggests cylindrical coordinates because rotations around the wire leave the physical situation unchanged.

Common Mistake

Coordinates are not vectors. The triple (x, y, z) lists components relative to a chosen basis; the geometric position vector is $\mathbf{r} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z$.

1.4 Vector Algebra

Vector algebra formalizes the geometric operations of combining displacements, changing scale, and comparing directions.

1.4.1 Addition and Subtraction

If

$$\mathbf{A} = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z, \qquad \mathbf{B} = B_x\mathbf{e}_x + B_y\mathbf{e}_y + B_z\mathbf{e}_z,$$

then

$$\mathbf{A} + \mathbf{B} = (A_x + B_x)\mathbf{e}_x + (A_y + B_y)\mathbf{e}_y + (A_z + B_z)\mathbf{e}_z.$$

Geometrically, vector addition is represented by the head-to-tail rule or, equivalently, by the diagonal of a parallelogram. Subtraction is addition of the opposite vector:

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}).$$

1.4.2 Scalar Multiplication

Multiplying a vector by a scalar λ changes its magnitude and possibly its direction:

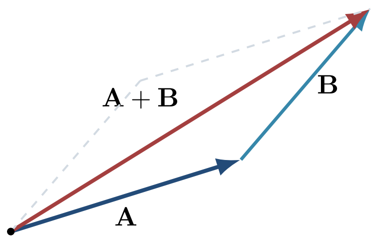


Figure 1.5: Head-to-tail vector addition. Translating a vector without rotating or rescaling it does not change the vector.

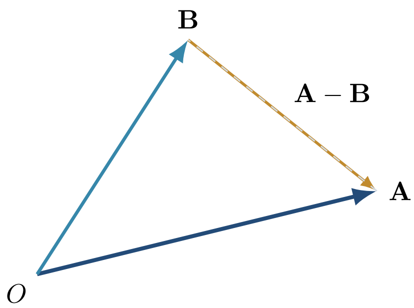


Figure 1.6: For vectors with a common origin, $\mathbf{A} - \mathbf{B}$ points from the tip of $\mathbf{B}$ to the tip of $\mathbf{A}$.

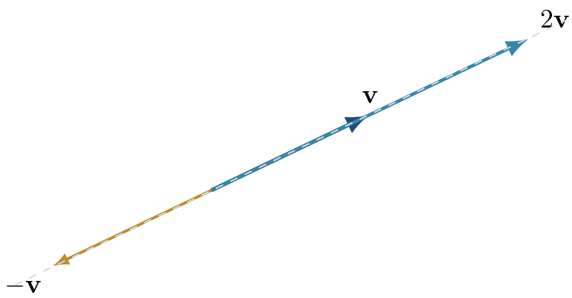


Figure 1.7: Multiplication by a scalar changes length; a negative scalar also reverses direction.

$$\lambda\mathbf{A} = (\lambda A_x)\mathbf{e}_x + (\lambda A_y)\mathbf{e}_y + (\lambda A_z)\mathbf{e}_z.$$

For $\lambda > 0$ the direction is unchanged; for $\lambda < 0$ it is reversed; and for $\lambda = 0$ the result is the zero vector.

Algebraic laws

For vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and scalars λ, μ,

$$\begin{aligned}\mathbf{A} + \mathbf{B} &= \mathbf{B} + \mathbf{A},\\ (\mathbf{A} + \mathbf{B}) + \mathbf{C} &= \mathbf{A} + (\mathbf{B} + \mathbf{C}),\\ \lambda(\mathbf{A} + \mathbf{B}) &= \lambda\mathbf{A} + \lambda\mathbf{B},\\ (\lambda + \mu)\mathbf{A} &= \lambda\mathbf{A} + \mu\mathbf{A}.\end{aligned}$$

Displacement Between Two Points

If points P and Q have position vectors $\mathbf{r}_P$ and $\mathbf{r}_Q$, the displacement from P to Q is

$$\Delta\mathbf{r} = \mathbf{r}_Q - \mathbf{r}_P.$$

This difference is independent of the arbitrary choice of coordinate origin.

Example 1.1

Let $P = (1, -2, 3)$ and $Q = (4, 2, -1)$. Then

$$\Delta\mathbf{r} = (4 - 1,\ 2 - (-2),\ -1 - 3) = (3, 4, -4).$$

1.5 Magnitude and Unit Vectors

The magnitude, or Euclidean norm, measures the length of a vector. For

$$\mathbf{A} = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z,$$

its magnitude is

$$\boxed{\|\mathbf{A}\| = \sqrt{A_x^2 + A_y^2 + A_z^2}}.$$

This is the three-dimensional Pythagorean theorem and may also be written as

$$\|\mathbf{A}\| = \sqrt{\mathbf{A}\cdot\mathbf{A}}.$$

1.5.1 Unit Vectors and Direction

A unit vector has magnitude one. Every nonzero vector can be separated into magnitude and direction:

$$\mathbf{A} = \|\mathbf{A}\|\,\widehat{\mathbf{A}}, \qquad \widehat{\mathbf{A}} = \frac{\mathbf{A}}{\|\mathbf{A}\|}.$$

Normalization removes the scale of a vector while preserving its direction.

Example 1.2

For $\mathbf{A} = 2\mathbf{e}_x - 3\mathbf{e}_y + 6\mathbf{e}_z$,

$$\|\mathbf{A}\| = \sqrt{4 + 9 + 36} = 7,$$

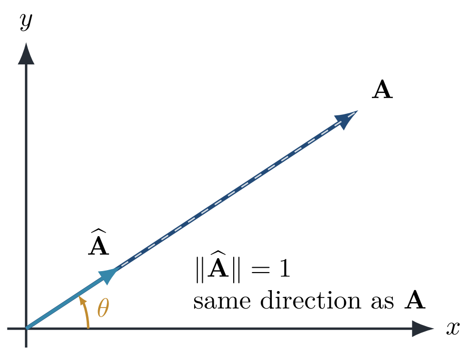


Figure 1.8: Normalization preserves direction and changes the magnitude to one.

so

$$\widehat{\mathbf{A}} = \frac{2}{7}\mathbf{e}_x - \frac{3}{7}\mathbf{e}_y + \frac{6}{7}\mathbf{e}_z.$$

A direct calculation confirms that $\|\widehat{\mathbf{A}}\| = 1$.

Triangle inequality

For any vectors $\mathbf{A}$ and $\mathbf{B}$,

$$\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|.$$

Geometrically, the direct path between two points cannot be longer than a broken path through an intermediate point.

Common Mistake

The expression $\mathbf{A}/\|\mathbf{A}\|$ is undefined for the zero vector. The zero vector has no unique direction.

1.6 Dot Product

Until now we have learned how to represent vectors and how to combine them through addition and scalar multiplication. These operations allow us to construct new vectors, but they do not answer an equally important question:

How can we compare two vectors?

For example, suppose two forces act on an object. Are they parallel? Are they perpendicular? Do they point in nearly the same direction, or in opposite directions?

To answer these questions we introduce one of the most important operations in mathematics and physics: the *dot product*, also called the *scalar product* or, in more general settings, an *inner product*.

Unlike vector addition, the dot product combines two vectors and produces a scalar rather than another vector. This scalar measures how strongly the two vectors are aligned with one another.

1.6.1 Motivation

Imagine pushing a heavy box across a floor. Suppose you apply a force of 100 N. If the force is directed exactly along the direction of motion, all of your effort contributes to moving the box.

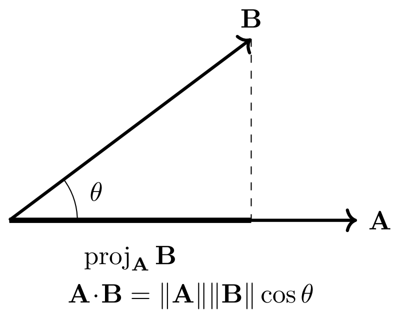


Figure 1.9: The dot product as an angular comparison and a projection. The sign is determined by whether the angle is acute, right, or obtuse.

Now imagine pushing at an angle of 60°. Although the force still has the same magnitude, only part of it acts in the direction of motion. The remainder presses the box downward against the floor and does not help it move forward.

Clearly, the effectiveness of a force depends not only on its magnitude but also on the angle between the force and the displacement. The mathematical operation that captures this dependence is the dot product.

1.6.2 Geometric Definition

Consider two nonzero vectors **A** and **B** separated by an angle θ, where $0 \leq \theta \leq \pi$. Their dot product is defined by

$$\boxed{\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\|\,\|\mathbf{B}\|\cos\theta}$$

This equation tells us that the dot product depends on three quantities:

- the magnitude of the first vector;
- the magnitude of the second vector;
- the cosine of the angle between them.

Figure 1.10 illustrates the geometric meaning of the dot product.

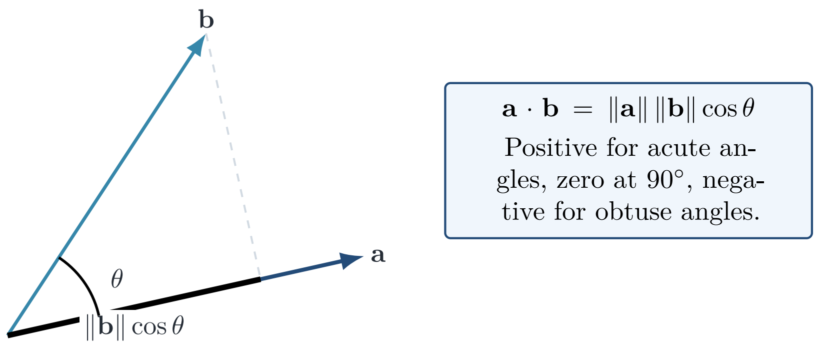


Figure 1.10: Two vectors separated by an angle θ. The dot product measures the component of one vector along the direction of the other.

1.6.3 Why the Cosine Appears

The cosine appears because it extracts the component of one vector along the direction of the other. Suppose we project **B** onto the direction of **A**. The scalar length of this projection is

$$\|\mathbf{B}\|\cos\theta.$$

Multiplying this projected length by the magnitude of $\mathbf{A}$ gives

$$\|\mathbf{A}\|\,\|\mathbf{B}\|\cos\theta,$$

which is precisely the definition of the dot product. Thus, the dot product measures how much of one vector lies along the direction of the other. This interpretation is far more important than simply memorizing the formula.

If $\widehat{\mathbf{A}} = \mathbf{A}/\|\mathbf{A}\|$ is the unit vector in the direction of $\mathbf{A}$, then

$$\mathbf{B}\cdot\widehat{\mathbf{A}} = \|\mathbf{B}\|\cos\theta.$$

is the scalar component of $\mathbf{B}$ along $\mathbf{A}$.

1.6.4 Component Formula

Although the geometric definition provides intuition, practical calculations are usually performed using vector components. Let

$$\mathbf{A} = A_x\mathbf{e}_x + A_y\mathbf{e}_y + A_z\mathbf{e}_z,$$

and

$$\mathbf{B} = B_x\mathbf{e}_x + B_y\mathbf{e}_y + B_z\mathbf{e}_z.$$

The Cartesian basis is orthonormal, so

$$\boxed{\mathbf{e}_i\cdot\mathbf{e}_j = \delta_{ij}}$$

where δ_{ij} is the Kronecker delta,

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Using distributivity and orthonormality,

$$\begin{aligned} \mathbf{A}\cdot\mathbf{B} &= (A_i\mathbf{e}_i)\cdot(B_j\mathbf{e}_j) \\ &= A_iB_j(\mathbf{e}_i\cdot\mathbf{e}_j) \\ &= A_iB_j\delta_{ij} \\ &= A_iB_i. \end{aligned}$$

In explicit Cartesian form,

$$\boxed{\mathbf{A}\cdot\mathbf{B} = A_xB_x + A_yB_y + A_zB_z}$$

where repeated indices are summed. This remarkably simple expression is one of the most frequently used formulas in all branches of physics.

1.6.5 Special Cases

The dot product immediately reveals the geometric relationship between two vectors.

Parallel Vectors

If $\theta = 0°$, then $\cos 0° = 1$, and therefore

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \, \|\mathbf{B}\|.$$

The dot product is positive and has its maximum possible value for fixed vector magnitudes.

Perpendicular Vectors

If $\theta = 90°$, then $\cos 90° = 0$, so

$$\boxed{\mathbf{A} \cdot \mathbf{B} = 0}$$

For nonzero vectors, this gives one of the most useful tests in mathematics and physics:

Two vectors are orthogonal if and only if their dot product is zero.

This criterion appears repeatedly in mechanics, electromagnetism, Fourier analysis, and quantum mechanics.

Opposite Directions

If $\theta = 180°$, then $\cos 180° = -1$, and

$$\mathbf{A} \cdot \mathbf{B} = -\|\mathbf{A}\| \, \|\mathbf{B}\|.$$

The dot product is negative, indicating that the vectors point in opposite directions.

1.6.6 Physical Interpretation

The sign of the dot product carries important physical information:

- positive: the vectors point generally in the same direction;
- zero: the vectors are perpendicular;
- negative: the vectors point generally in opposite directions.

Thus, a single scalar quantity contains information about both magnitude and relative orientation.

The angle can also be recovered from the dot product:

$$\boxed{\cos\theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \, \|\mathbf{B}\|}}$$

for nonzero vectors $\mathbf{A}$ and $\mathbf{B}$.

1.6.7 Example: Mechanical Work

One of the first major applications of the dot product appears in mechanics. Suppose a constant force $\mathbf{F}$ moves an object through a displacement $\mathbf{s}$. The mechanical work performed is defined as

$$\boxed{W = \mathbf{F} \cdot \mathbf{s}}$$

Substituting the geometric definition gives

$$W = Fs\cos\theta.$$

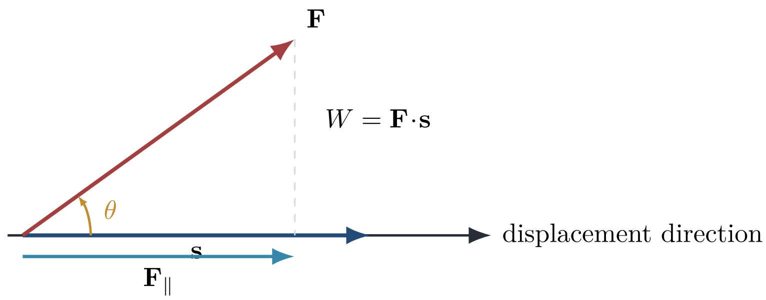


Figure 1.11: Only the force component parallel to the displacement contributes to mechanical work.

This equation explains several familiar situations:

- If the force acts along the displacement ($\theta = 0°$), the work is maximal.
- If the force is perpendicular to the displacement ($\theta = 90°$), the work is zero.
- If the force opposes the motion ($\theta = 180°$), the work is negative.

Thus, the dot product naturally quantifies energy transfer.

1.6.8 Worked Example 1.7

Consider the vectors

$$\mathbf{A} = (2, 3, 1), \qquad \mathbf{B} = (4, -1, 5).$$

Compute their dot product.

Solution. Using the component formula,

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= 2 \cdot 4 + 3(-1) + 1 \cdot 5 \\ &= 8 - 3 + 5 \\ &= 10. \end{aligned}$$

Therefore,

$$\boxed{\mathbf{A} \cdot \mathbf{B} = 10}$$

Since the result is positive, the angle between the vectors is less than 90°.

1.6.9 Coordinate Invariance

The component formula may appear to depend on the choice of Cartesian axes, but the dot product itself is a geometric scalar and is therefore independent of how the coordinate system is rotated.

Let an orthogonal transformation be represented by a matrix R, so that

$$\mathbf{A}' = R\mathbf{A}, \qquad \mathbf{B}' = R\mathbf{B},$$

with

$$R^{\mathsf{T}} R = I.$$

Then

$$\begin{aligned}
\mathbf{A}' \cdot \mathbf{B}' &= (R\mathbf{A})^{\mathsf{T}}(R\mathbf{B}) \\
&= \mathbf{A}^{\mathsf{T}} R^{\mathsf{T}} R\mathbf{B} \\
&= \mathbf{A}^{\mathsf{T}}\mathbf{B} \\
&= \mathbf{A} \cdot \mathbf{B}.
\end{aligned}$$

Thus, observers using rotated Cartesian axes calculate different components but the same dot product. This is why quantities such as length, angle, and work are coordinate-independent physical scalars.

1.6.10 Toward Metric Tensors and Inner Products

In Cartesian Euclidean space, the dot product can be written

$$\mathbf{A} \cdot \mathbf{B} = \delta_{ij} A^i B^j.$$

The Kronecker delta acts as the Euclidean metric. In more general coordinates or curved spaces, it is replaced by a metric tensor g_{ij}:

$$\boxed{\langle \mathbf{A}, \mathbf{B} \rangle = g_{ij} A^i B^j}$$

This is the natural extension of the dot product to differential geometry and general relativity.

Quantum mechanics generalizes the same idea further. Quantum states belong to a complex Hilbert space, and their inner product is usually written

$$\langle \phi \mid \psi \rangle.$$

Unlike the real Euclidean dot product, the Hilbert-space inner product involves complex conjugation. Nevertheless, it serves the same structural purpose: it defines lengths, angles, orthogonality, projections, and probabilities.

1.6.11 Historical Note

The scalar product emerged during the nineteenth century through the work of Hermann Grassmann and Josiah Willard Gibbs. Their development of vector analysis transformed mechanics and electromagnetism by providing a compact language for expressing physical laws. In the twentieth century, the idea was generalized to abstract inner products in Hilbert spaces, where it became one of the mathematical foundations of quantum mechanics. There, the inner product determines probabilities, expectation values, and the orthogonality of quantum states.

1.6.12 Common Mistakes

Several mistakes occur repeatedly when first learning the dot product:

- Confusing the dot product with ordinary multiplication of vectors. The result of a dot product is a scalar, not a vector.
- Forgetting that the dot product depends on the angle between the vectors. Two vectors with the same magnitudes can have very different dot products if they point in different directions.
- Assuming that a zero dot product means one of the vectors is zero. For nonzero vectors, it means that the vectors are perpendicular.

- Forgetting to divide by both magnitudes when using the dot product to calculate an angle.
- Treating $\mathbf{A} \cdot \mathbf{B}$ and $\mathbf{A} \times \mathbf{B}$ as interchangeable operations. They produce different types of objects and describe different geometry.

1.6.13 Summary

The dot product combines two vectors to produce a scalar. Geometrically,

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \, \|\mathbf{B}\| \cos\theta,$$

while in an orthonormal Cartesian basis,

$$\mathbf{A} \cdot \mathbf{B} = A_i B_i = A_x B_x + A_y B_y + A_z B_z.$$

It measures alignment, determines angles and orthogonality, extracts vector components, and expresses physical quantities such as work. Its invariance under rotations makes it a geometric scalar, while its generalizations lead to metric tensors and Hilbert-space inner products.

1.6.14 Looking Ahead

The dot product measures similarity and alignment between vectors. There is another fundamental operation that measures something entirely different: the area spanned by two vectors and the direction perpendicular to the plane containing them.

This operation is the cross product, which will be developed in the next section. Unlike the dot product, the cross product produces a new vector and plays a central role in rotational motion, angular momentum, torque, magnetic forces, and Maxwell's equations.

1.7 Cross Product

The dot product measures alignment. The cross product instead constructs a vector perpendicular to the plane of two vectors and therefore encodes orientation, area, and rotational effect.

1.7.1 Geometric Definition

For vectors $\mathbf{A}$ and $\mathbf{B}$ separated by an angle θ,

$$\boxed{\mathbf{A} \times \mathbf{B} = \|\mathbf{A}\|\|\mathbf{B}\| \sin\theta \, \hat{\mathbf{n}}},$$

where $\hat{\mathbf{n}}$ is perpendicular to both vectors. Its magnitude is the area of the parallelogram spanned by $\mathbf{A}$ and $\mathbf{B}$.

Right-hand rule

Curl the fingers of the right hand from $\mathbf{A}$ toward $\mathbf{B}$ through the smaller angle. The thumb points in the direction of $\mathbf{A} \times \mathbf{B}$.

1.7.2 Algebraic Form

For an orthonormal right-handed Cartesian basis,

$$\mathbf{e}_x \times \mathbf{e}_y = \mathbf{e}_z, \qquad \mathbf{e}_y \times \mathbf{e}_z = \mathbf{e}_x, \qquad \mathbf{e}_z \times \mathbf{e}_x = \mathbf{e}_y.$$

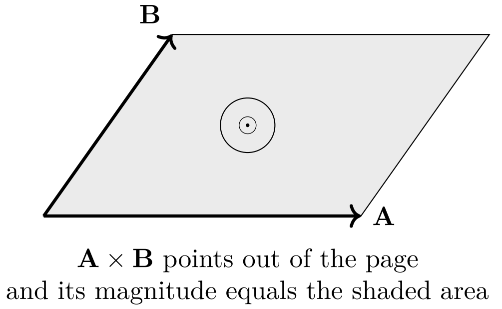


Figure 1.12: Area and orientation encoded by the cross product. Reversing the order of the vectors reverses the normal.

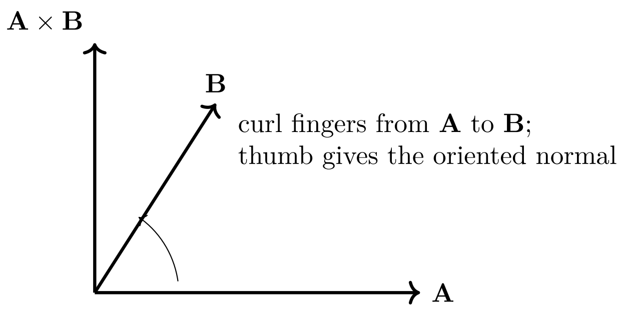


Figure 1.13: Operational version of the right-hand rule for an ordered pair of vectors.

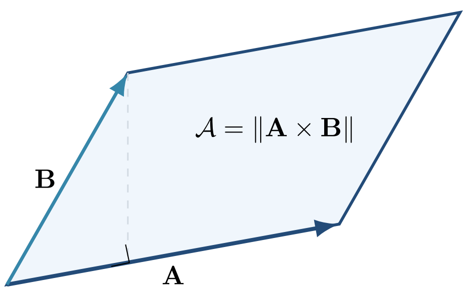


Figure 1.14: The cross-product magnitude equals the area of the parallelogram spanned by the two vectors.

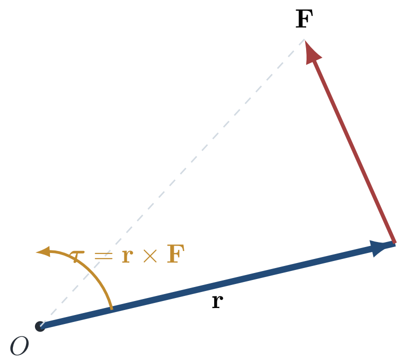


Figure 1.15: Torque measures the rotational action of a force about an origin.

Reversing the order reverses the sign:

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}.$$

For components,

$$\boxed{\mathbf{A} \times \mathbf{B} = \begin{pmatrix} A_y B_z - A_z B_y \\ A_z B_x - A_x B_z \\ A_x B_y - A_y B_x \end{pmatrix}}$$

or, as a mnemonic,

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.$$

Orthogonality

The cross product is orthogonal to both original vectors:

$$(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{A} = 0, \qquad (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{B} = 0.$$

It vanishes when the vectors are parallel or when either vector is zero.

1.7.3 Physical Applications

Cross products appear whenever orientation or rotation matters:

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \qquad \mathbf{L} = \mathbf{r} \times \mathbf{p}, \qquad \mathbf{F} = q\,\mathbf{v} \times \mathbf{B}, \qquad \mathbf{S} = \mathbf{E} \times \mathbf{H}.$$

These represent torque, angular momentum, magnetic force, and electromagnetic energy flow.

Example 1.3

For $\mathbf{A} = (2, 1, 0)$ and $\mathbf{B} = (1, 3, 4)$,

$$\mathbf{A} \times \mathbf{B} = (4, -8, 5).$$

Indeed,

$$(4, -8, 5) \cdot (2, 1, 0) = 0, \qquad (4, -8, 5) \cdot (1, 3, 4) = 0.$$

Its magnitude $\sqrt{105}$ is the area of the parallelogram generated by the two vectors.

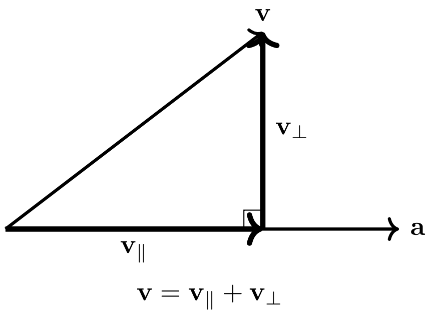


Figure 1.16: Orthogonal decomposition into components parallel and perpendicular to a chosen direction.

Common Mistake

The cross product is not commutative. Interchanging the vectors changes the sign and therefore reverses the direction.

1.8 Projection and Decomposition

One of the most powerful techniques in physics is to decompose a complicated quantity into simpler parts. Instead of analysing an entire vector at once, we often separate it into components that have clear physical meaning.

Consider, for example, a block resting on an inclined plane. The gravitational force acts vertically downward, yet only part of this force pulls the block down the slope. Another part presses the block against the surface. By resolving the gravitational force into two perpendicular components, the problem becomes much easier to analyse.

This idea appears throughout science and engineering. In mechanics we resolve forces and velocities; in electromagnetism we resolve electric and magnetic fields; and in quantum mechanics we project state vectors onto basis states. All of these applications rely on the mathematical concept of projection.

Motivation

Suppose that a force $\mathbf{F}$ acts on an object, but that we are interested only in the part of the force acting along a direction represented by a non-zero vector $\mathbf{a}$. The relevant question is not "What is the total force?" but rather "How much of this force acts in the chosen direction?"

The answer is provided by the projection of one vector onto another.

1.8.1 Scalar Projection

Let $\mathbf{A}$ and $\mathbf{B}$ be non-zero vectors separated by an angle θ. The *scalar projection*, also called the scalar component, of $\mathbf{A}$ onto $\mathbf{B}$ is

$$\boxed{\operatorname{comp}_{\mathbf{B}} \mathbf{A} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{B}\|} = \|\mathbf{A}\| \cos\theta}$$

The result is a scalar quantity with the same physical units as the original vector. Geometrically, it represents the signed length of the shadow cast by $\mathbf{A}$ onto the direction of $\mathbf{B}$. The sign distinguishes alignment from anti-alignment.

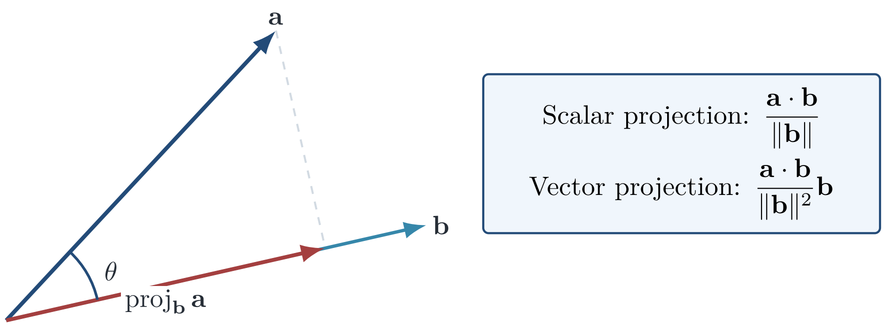


Figure 1.17: Scalar and vector projection of $\mathbf{A}$ onto the direction of $\mathbf{B}$. The dashed segment is perpendicular to $\mathbf{B}$.

Vector Projection

Frequently we require not only the length of the projection but also its direction. The *vector projection* of $\mathbf{A}$ onto $\mathbf{B}$ is

$$\boxed{\operatorname{proj}_{\mathbf{B}} \mathbf{A} = \left(\frac{\mathbf{A}\cdot\mathbf{B}}{\mathbf{B}\cdot\mathbf{B}}\right)\mathbf{B}}$$

Unlike the scalar projection, this expression is itself a vector. It is parallel to $\mathbf{B}$, and its signed magnitude is determined by the scalar projection.

1.8.2 Derivation of the Projection Formula

The projection formula follows directly from the dot product. First determine the signed length of the projection:

$$\|\mathbf{A}\|\cos\theta = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{B}\|}.$$

The unit vector in the direction of $\mathbf{B}$ is

$$\widehat{\mathbf{B}} = \frac{\mathbf{B}}{\|\mathbf{B}\|}.$$

Multiplying the projected length by this unit vector gives

$$\operatorname{proj}_{\mathbf{B}} \mathbf{A} = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{B}\|}\frac{\mathbf{B}}{\|\mathbf{B}\|}.$$

Since $\|\mathbf{B}\|^2 = \mathbf{B}\cdot\mathbf{B}$, this becomes

$$\boxed{\operatorname{proj}_{\mathbf{B}} \mathbf{A} = \frac{\mathbf{A}\cdot\mathbf{B}}{\mathbf{B}\cdot\mathbf{B}}\mathbf{B}.}$$

Thus, projection is not an unrelated formula to be memorized; it is a direct consequence of the geometric meaning of the dot product.

1.8.3 Orthogonal Decomposition

Every vector $\mathbf{A}$ may be decomposed uniquely into two parts relative to a chosen non-zero direction $\mathbf{B}$:

$$\boxed{\mathbf{A} = \mathbf{A}_{\parallel} + \mathbf{A}_{\perp}.}$$

The component parallel to $\mathbf{B}$ is

$$\boxed{\mathbf{A}_{\parallel} = \operatorname{proj}_{\mathbf{B}} \mathbf{A},}$$

while the perpendicular component is obtained by subtraction:

$$\boxed{\mathbf{A}_{\perp} = \mathbf{A} - \mathbf{A}_{\parallel}.}$$

By construction,

$$\mathbf{A}_{\perp} \cdot \mathbf{B} = 0.$$

This orthogonality can be verified directly:

$$\begin{aligned}
\mathbf{A}_{\perp} \cdot \mathbf{B} &= \left(\mathbf{A} - \frac{\mathbf{A} \cdot \mathbf{B}}{\mathbf{B} \cdot \mathbf{B}}\mathbf{B}\right) \cdot \mathbf{B} \\
&= \mathbf{A} \cdot \mathbf{B} - \frac{\mathbf{A} \cdot \mathbf{B}}{\mathbf{B} \cdot \mathbf{B}}(\mathbf{B} \cdot \mathbf{B}) \\
&= 0.
\end{aligned}$$

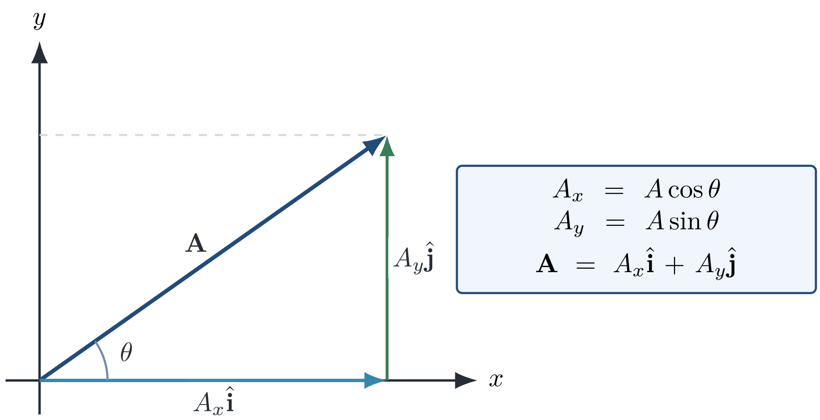


Figure 1.18: A vector decomposed into two mutually perpendicular components.

1.8.4 Application: Force on an Inclined Plane

Consider a block of mass m resting on a plane inclined at an angle α above the horizontal. The gravitational force is

$$\mathbf{F}_g = m\mathbf{g}.$$

Rather than analysing this force directly, we resolve it into components parallel and perpendicular to the plane. Their magnitudes are

$$\boxed{F_{\parallel} = mg\sin\alpha}$$

and

$$\boxed{F_{\perp} = mg\cos\alpha.}$$

The parallel component tends to accelerate the block down the incline. The perpendicular component determines the normal force exerted by the surface, provided that no other force has a component normal to the plane.

This decomposition converts a two-dimensional force problem into two coupled but individually one-dimensional equations.

1.8.5 Worked Example

Let

$$\mathbf{A} = (3, 4), \qquad \mathbf{B} = (5, 0).$$

Determine the projection of $\mathbf{A}$ onto $\mathbf{B}$ and the component of $\mathbf{A}$ perpendicular to $\mathbf{B}$.

First compute the dot products:

$$\mathbf{A} \cdot \mathbf{B} = 3(5) + 4(0) = 15,$$

and

$$\mathbf{B} \cdot \mathbf{B} = 5^2 + 0^2 = 25.$$

Therefore,

$$\operatorname{proj}_{\mathbf{B}} \mathbf{A} = \frac{15}{25}(5, 0) = (3, 0).$$

The perpendicular component is

$$\mathbf{A}_{\perp} = (3, 4) - (3, 0) = (0, 4).$$

Finally,

$$(3, 0) \cdot (0, 4) = 0,$$

which confirms that the two components are orthogonal.

1.8.6 Physical Interpretation

Projection isolates the part of a vector that is physically relevant to a chosen direction. Important examples include:

- the component of velocity directed toward an observer in the Doppler effect;
- the component of force along a displacement in mechanical work;
- the electric field normal to a conducting surface;
- the magnetic field perpendicular to the motion of a charged particle;
- the amplitude of a quantum state projected onto a measurement basis.

Although these examples belong to different branches of physics, they employ the same mathematical operation.

Historical Note

The systematic use of orthogonal decomposition developed during the nineteenth century through the work of Hermann Grassmann and later Josiah Willard Gibbs. In the twentieth century, David Hilbert generalized these ideas to infinite-dimensional inner-product spaces. In quantum mechanics, measurement is represented mathematically through projections onto subspaces associated with possible outcomes.

1.8.7 Common Mistakes

The scalar projection and vector projection are different quantities. The first is a signed number; the second is a vector. Projection also depends on the direction onto which the vector is projected, so in general

$$\operatorname{proj}_{\mathbf{B}} \mathbf{A} \neq \operatorname{proj}_{\mathbf{A}} \mathbf{B}.$$

Finally, the perpendicular component should be calculated by subtracting the parallel projection from the original vector:

$$\mathbf{A}_{\perp} = \mathbf{A} - \operatorname{proj}_{\mathbf{B}} \mathbf{A}.$$

1.8.8 Summary

The dot product measures alignment between vectors. A scalar projection extracts the signed magnitude of one vector along another, while a vector projection includes both magnitude and direction. Every vector can be decomposed uniquely into parallel and perpendicular parts relative to a non-zero reference direction. This decomposition simplifies physical problems by separating independent directions.

These ideas will be used repeatedly in Newtonian mechanics, electromagnetism, wave mechanics, and quantum theory.

Looking Ahead

The next stage develops coordinate transformations, including rotations, reflections, translations, and changes of basis. Projection and decomposition will provide the conceptual bridge between geometric vectors and their coordinate representations.

Projection Operators and Quantum Preview

Projection can be viewed as an operator P acting on vectors. A projection operator satisfies $P^2 = P$, meaning that applying the same projection twice has the same effect as applying it once. In linear algebra and quantum mechanics, projection operators play a central role in describing measurements.

Repeated projections can be used to construct orthogonal bases through the Gram–Schmidt orthogonalization procedure. In quantum mechanics, every measurement may be interpreted as projecting a state vector onto an orthonormal basis. The probability of observing a particular outcome is given by the squared magnitude of the corresponding projection, providing a direct bridge from elementary vector algebra to Hilbert spaces.

1.9 Linear Independence

A collection of vectors is useful as a coordinate framework only when none of its members is redundant.

1.9.1 Definition

Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent if

$$c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$$

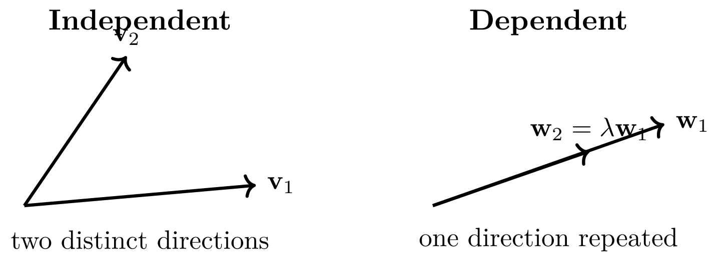


Figure 1.19: Independent vectors contribute genuinely different directions; dependent vectors repeat information already present.

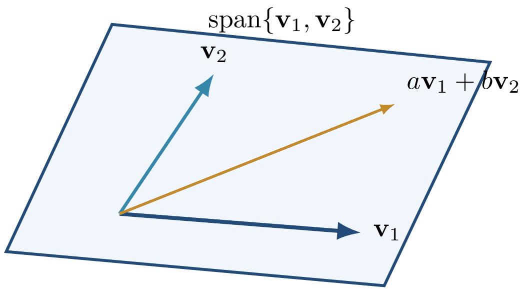


Figure 1.20: Two nonparallel vectors span a plane: every vector in it has the form $a\mathbf{v}_1 + b\mathbf{v}_2$.

implies

$$c_1 = \cdots = c_n = 0.$$

If a nontrivial choice of coefficients produces the zero vector, the vectors are linearly dependent.

Physical Interpretation

Linear dependence means that at least one proposed direction can be reconstructed from the others. It therefore contributes no new degree of freedom.

1.9.2 Geometric Interpretation

In two dimensions, two nonzero vectors are independent precisely when they are not parallel. In three dimensions, three vectors are independent precisely when they are not coplanar. For three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, the scalar triple product provides a test:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \neq 0.$$

Its absolute value is the volume of the parallelepiped spanned by the vectors.

Example 1.4

The vectors $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 1, 2)$ are dependent because

$$(1, 1, 2) = (1, 0, 1) + (0, 1, 1).$$

The third vector adds no new direction.

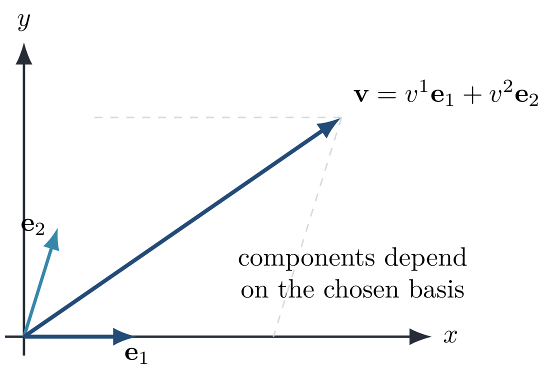


Figure 1.21: A vector is independent of its description, while its components are defined relative to a basis.

Matrix test

Place the candidate vectors as columns of a matrix M. For a square set, the vectors are linearly independent exactly when $\det M \neq 0$. More generally, independence means that the rank of M equals the number of columns.

1.10 Bases and Dimension

A basis is a minimal complete set of vectors from which every vector in a space can be constructed.

1.10.1 Basis and Span

A set $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is a basis when it is linearly independent and spans the space. Every vector then has a unique expansion

$$\mathbf{v} = \sum_{i=1}^{n} v^i \mathbf{e}_i.$$

The coefficients v^i are the components of $\mathbf{v}$ in that basis.

Definition 1.1

The dimension of a finite-dimensional vector space is the number of vectors in any basis of that space.

1.10.2 Change of Basis

The same geometric vector can have different components in different bases. If

$$\mathbf{e}'_i = \sum_j R_{ji} \mathbf{e}_j,$$

then the components transform so that the complete vector remains unchanged. For an orthonormal rotation, the matrix R satisfies

$$R^{\mathsf{T}} R = I.$$

Consequently lengths and angles are preserved.

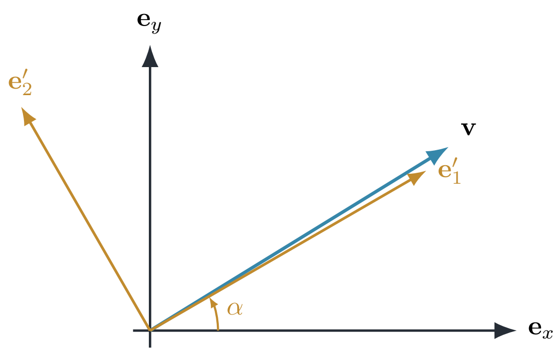


Figure 1.22: The same vector described in two rotated bases. The vector remains fixed while its numerical components change.

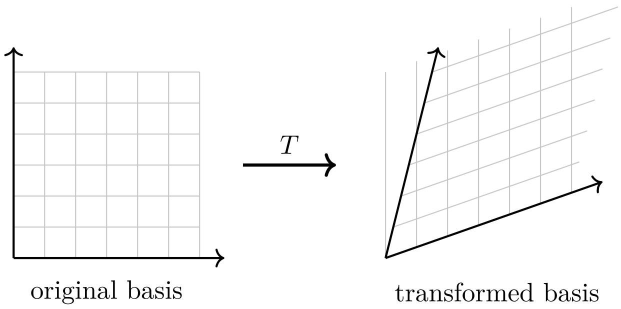


Figure 1.23: A linear transformation is fixed by its action on a basis. Grid lines remain straight and parallel under the transformation shown.

Physical Interpretation

A change of basis does not change the physical state. It changes only the numerical language used to describe that state. This principle later appears in rotations, normal modes, spin measurements, and transformations between position and momentum representations.

Example 1.5

In the plane, let

$$\mathbf{e}'_1 = \frac{\mathbf{e}_x + \mathbf{e}_y}{\sqrt{2}}, \qquad \mathbf{e}'_2 = \frac{-\mathbf{e}_x + \mathbf{e}_y}{\sqrt{2}}.$$

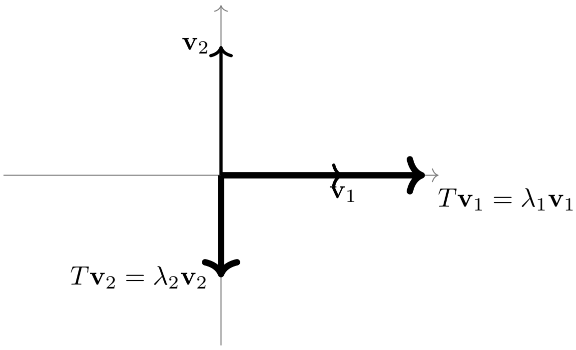
These vectors form an orthonormal basis rotated by 45°. The vector $\mathbf{v} = \mathbf{e}_x$ has components $(1, 0)$ in the original basis and $(1/\sqrt{2}, -1/\sqrt{2})$ in the rotated basis.

Eigenvectors: a Preview

For a linear transformation T, an eigenvector is a nonzero vector whose direction line is preserved:

$$T\mathbf{v} = \lambda\mathbf{v}.$$

The scalar λ is the eigenvalue. This simple relation later becomes central in normal-mode analysis and in quantum measurement.



eigenvectors keep their line under the transformation;
the eigenvalue sets scale and possible reversal

Figure 1.24: Two eigenvector directions under a linear transformation. An eigenvalue may stretch, shrink, or reverse a vector.

1.11 Vector Spaces

The arrows of Euclidean geometry are examples of a more general algebraic structure. This abstraction is essential because quantum states, functions, matrices, and solutions of differential equations can all behave like vectors.

1.11.1 Axiomatic Structure

A vector space V over a field of scalars $\mathbb{F}$ is a set equipped with vector addition and scalar multiplication. These operations satisfy closure, associativity, commutativity of addition, additive identity and inverses, and the distributive and compatibility laws for scalar multiplication.

Definition 1.2

A subset $W \subseteq V$ is a subspace if it contains the zero vector and is closed under vector addition and scalar multiplication.

1.11.2 Examples Beyond Arrows

Important vector spaces include

- ordered n-tuples $\mathbb{R}^n$ or $\mathbb{C}^n$;
- polynomials of degree at most n;
- matrices of fixed size;
- continuous functions on an interval;
- solutions of a homogeneous linear differential equation;
- quantum state vectors in a Hilbert space.

For functions, addition and scalar multiplication are defined pointwise:

$$(f+g)(x) = f(x) + g(x), \qquad (\lambda f)(x) = \lambda f(x).$$

Physical Interpretation

In quantum mechanics the word *vector* usually does not mean an arrow in ordinary space. It means an element of a complex vector space whose linear combinations describe superposition.

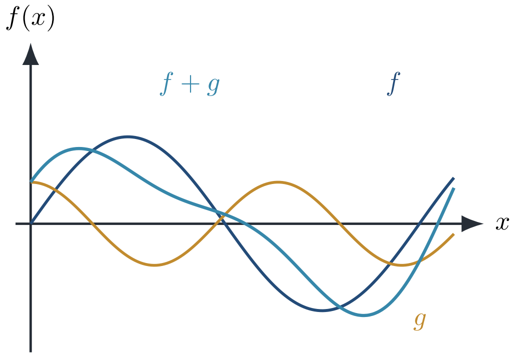


Figure 1.25: Functions form vector spaces because pointwise sums and scalar multiples remain functions in the space.

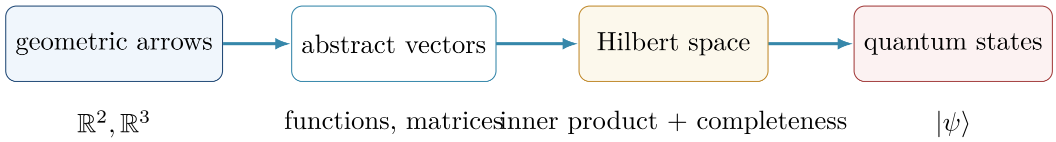


Figure 1.26: Geometric arrows generalize to abstract vectors, Hilbert-space vectors, and quantum states.

From Vector Spaces to Hilbert Spaces

A vector space alone provides addition and scalar multiplication. Adding an inner product introduces lengths and angles; completeness then leads to a Hilbert space. These ideas will later support operators, eigenstates, orthogonality, and quantum measurement.

Constructing an Orthonormal Basis

Given independent vectors, the Gram–Schmidt process subtracts from each new vector the components already accounted for. For two vectors,

$$\mathbf{u}_1 = \mathbf{v}_1, \qquad \mathbf{u}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1.$$

Normalizing $\mathbf{u}_1$ and $\mathbf{u}_2$ produces orthonormal basis vectors.

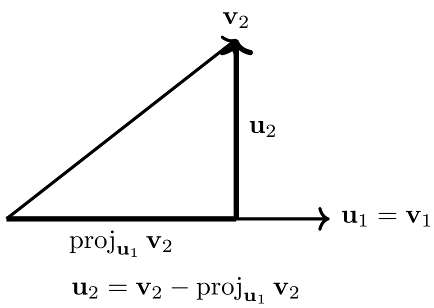


Figure 1.27: The Gram–Schmidt step removes the projection of $\mathbf{v}_2$ along $\mathbf{u}_1$, leaving an orthogonal direction.

1.12 Rigorous Foundations: Three Central Results

The geometric rules developed in this chapter rest on a small number of precise inequalities and decomposition theorems. Establishing them now clarifies why projections are well defined and why Euclidean length behaves as expected.

1.12.1 Cauchy–Schwarz Inequality

For vectors $\mathbf{u}$ and $\mathbf{v}$ in a real inner-product space,

$$\boxed{|\mathbf{u}\cdot\mathbf{v}| \le \|\mathbf{u}\|\,\|\mathbf{v}\|.}$$

If $\mathbf{v}=\mathbf{0}$ the statement is immediate. Otherwise consider, for real t, the nonnegative squared norm

$$0 \le \|\mathbf{u}-t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t(\mathbf{u}\cdot\mathbf{v}) + t^2\|\mathbf{v}\|^2.$$

Choose $t=(\mathbf{u}\cdot\mathbf{v})/\|\mathbf{v}\|^2$. Substitution gives

$$0 \le \|\mathbf{u}\|^2 - \frac{(\mathbf{u}\cdot\mathbf{v})^2}{\|\mathbf{v}\|^2},$$

and multiplication by $\|\mathbf{v}\|^2$ yields the result. Equality holds precisely when $\mathbf{u}=t\mathbf{v}$, so the vectors are linearly dependent only when the inequality is strict.

Geometric intuition

Cauchy–Schwarz states that the signed shadow of one vector on another can never be longer in magnitude than the original vector. It is the rigorous reason that $|\cos\theta|\le 1$ in the inner-product formula.

Common misconception: the inequality is usually strict

Equality is not evidence that two vectors are equal. It means that they are linearly dependent: they may point in the same direction or in opposite directions, and they may have different lengths.

1.12.2 Triangle Inequality

The length of a sum cannot exceed the sum of the lengths:

$$\boxed{\|\mathbf{u}+\mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|.}$$

Indeed,

$$\begin{aligned}
\|\mathbf{u}+\mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + 2\mathbf{u}\cdot\mathbf{v} + \|\mathbf{v}\|^2 \\
&\le \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \\
&= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2,
\end{aligned}$$

where Cauchy–Schwarz was used in the second line. Taking nonnegative square roots proves the claim.

Geometric intuition

Travelling directly from the initial point to the final point is never longer than travelling through an intermediate point. Equality occurs when the two displacements are aligned in the same direction.

1.12.3 Orthogonal Decomposition Theorem

Let $\mathbf{b} \neq \mathbf{0}$. Every vector $\mathbf{a}$ has a unique decomposition

$$\boxed{\mathbf{a} = \mathbf{a}_{\parallel} + \mathbf{a}_{\perp}}, \qquad \mathbf{a}_{\parallel} \parallel \mathbf{b}, \quad \mathbf{a}_{\perp} \perp \mathbf{b}.$$

Existence follows by defining

$$\mathbf{a}_{\parallel} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{b} \cdot \mathbf{b}}\mathbf{b}, \qquad \mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}.$$

The dot product $\mathbf{a}_{\perp} \cdot \mathbf{b}$ vanishes directly. For uniqueness, suppose $\mathbf{a} = \mathbf{p}_1 + \mathbf{q}_1 = \mathbf{p}_2 + \mathbf{q}_2$, where each $\mathbf{p}_i$ is parallel to $\mathbf{b}$ and each $\mathbf{q}_i$ is perpendicular to it. Then

$$\mathbf{p}_1 - \mathbf{p}_2 = -(\mathbf{q}_1 - \mathbf{q}_2).$$

The left side is parallel to $\mathbf{b}$ and the right side perpendicular to $\mathbf{b}$. The only vector with both properties is $\mathbf{0}$, hence $\mathbf{p}_1 = \mathbf{p}_2$ and $\mathbf{q}_1 = \mathbf{q}_2$.

Common misconception: projection requires a unit vector

The formula $\mathrm{proj}_{\mathbf{b}}\,\mathbf{a} = (\mathbf{a} \cdot \mathbf{b})\mathbf{b}$ is valid only when $\|\mathbf{b}\| = 1$. For a general nonzero $\mathbf{b}$, the denominator $\mathbf{b} \cdot \mathbf{b}$ is essential.

1.12.4 Why These Results Matter Later

Cauchy–Schwarz controls probabilities and expectation values in quantum mechanics. The triangle inequality underlies every metric and error estimate. Orthogonal decomposition becomes Fourier expansion, normal-mode analysis, least-squares approximation, and projection onto quantum measurement outcomes.

1.13 Historical and Physical Insight

The modern language of vectors emerged gradually. Geometry, coordinates, mechanics, vector analysis, and abstract spaces were developed in different centuries, then unified into the mathematical language now used throughout physics.

Historical Note 1.1: Euclid: geometry before coordinates

Euclid organized geometry around points, lines, angles, congruence, and proof. His framework was synthetic: geometric objects were studied directly, without assigning numerical coordinates to every point. This established the ideal of reasoning from explicit assumptions to necessary conclusions.

Connection to physics

Free-body diagrams, ray constructions, and spacetime diagrams retain this synthetic spirit. A diagram can expose invariant geometric relations before any coordinates are selected.

Historical Note 1.2: Descartes: geometry becomes algebra

René Descartes linked geometric curves with algebraic equations by introducing systematic coordinates. A point could now be represented by numbers, and a geometric problem could be translated into an algebraic calculation.

Connection to physics

Coordinates make trajectories, fields, and wavefunctions calculable. Yet the physical object must not be confused with its coordinate components: rotating the axes changes the numbers, not the vector itself.

Historical Note 1.3: Newton: directed quantities in mechanics

Newton formulated mechanics using geometric ideas of displacement, velocity, acceleration, force, and momentum. Modern vector notation came later, but the physical need to combine magnitudes with directions was already central to his dynamics.

Connection to physics

The equation $\mathbf{F} = m\mathbf{a}$ is meaningful only as a vector equation. It represents three coupled component equations in a chosen Cartesian frame and one coordinate-independent geometric law.

Historical Note 1.4: Gibbs: the practical vector calculus

In the nineteenth century, Josiah Willard Gibbs helped shape the compact dot- and cross-product notation now standard in physics and engineering. This language separated the scalar and vector products and made three-dimensional calculations considerably more efficient.

Connection to physics

Work uses $\mathbf{F} \cdot d\mathbf{r}$; torque and angular momentum use cross products; electromagnetism uses divergence and curl. The notation developed for classical physics becomes the working grammar of field theory.

Historical Note 1.5: Hilbert: vectors beyond arrows

David Hilbert's work helped establish infinite-dimensional inner-product spaces as mathematical objects in their own right. In such spaces, vectors may be functions rather than arrows, while length, orthogonality, bases, and projection remain meaningful.

Connection to physics

A quantum state $|\psi\rangle$ is a vector in a complex Hilbert space. A wavefunction $\psi(x) = \langle x|\psi\rangle$ is a coordinate representation of that state, just as (v_x, v_y, v_z) represents a geometric vector in a selected basis. This parallel is one of the central conceptual bridges of the book.

One language across physical theories

The progression from Euclidean geometry to Hilbert space does not discard the earlier ideas. It generalizes them. Direction becomes state, dot product becomes inner product, basis components become amplitudes, and projection becomes the mathematics of measurement.

1.14 Computational Companion

Analytical calculation remains the foundation of this chapter, but a short computational experiment can make geometry visible and can test algebra before it is used in a larger model. The examples below use NumPy for array operations and Matplotlib for visualization. Complete, runnable files are stored in `code/chapter01/python/`; Wolfram Language equivalents are collected in `code/chapter01/mathematica/chapter01_companion.wl`.

Computation as a check — not a substitute

A numerical result can verify a calculation for chosen inputs, reveal a sign error, or display a transformation. It does not replace a proof. For example, observing that two computed vectors are orthogonal to machine precision supports one calculation; the Gram–Schmidt derivation explains why orthogonality holds in general.

1.14.1 Projection and Orthogonal Decomposition

For vectors $\mathbf{a}$ and nonzero $\mathbf{b}$, the projection is

$$\mathrm{proj}_{\mathbf{b}}\,\mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{b}}{\mathbf{b}\cdot\mathbf{b}}\mathbf{b}.$$

The following program evaluates the formula and draws the parallel and perpendicular components.

Listing 1.1: Computing and visualizing an orthogonal projection.

```python
"""Visualize a vector and its orthogonal projection onto another vector."""
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUTPUT = Path(__file__).resolve().parents[3] / "figures/ch01/computational/
    vector_projection.pdf"
a = np.array([4.0, 3.0])
b = np.array([5.0, 1.0])
projection = (a @ b) / (b @ b) * b
perpendicular = a - projection

fig, ax = plt.subplots(figsize=(6.2, 4.5))
for vector, label in [(a, r"$\mathbf{a}$"), (b, r"$\mathbf{b}$"),
                      (projection, r"$\mathrm{proj}_{\mathbf{b}}\mathbf{a}$")]:
    ax.quiver(0, 0, *vector, angles="xy", scale_units="xy", scale=1, width=0.008)
    ax.text(*(vector + np.array([0.12, 0.12])), label)
ax.plot([projection[0], a[0]], [projection[1], a[1]], linestyle="--")
ax.set(xlim=(-0.5, 5.8), ylim=(-0.5, 4.2), xlabel="$x$", ylabel="$y$")
ax.set_aspect("equal")
ax.grid(True, alpha=0.3)
fig.tight_layout()
fig.savefig(OUTPUT)
```

```python
print(f"projection = {projection}; perpendicular = {perpendicular}")
```

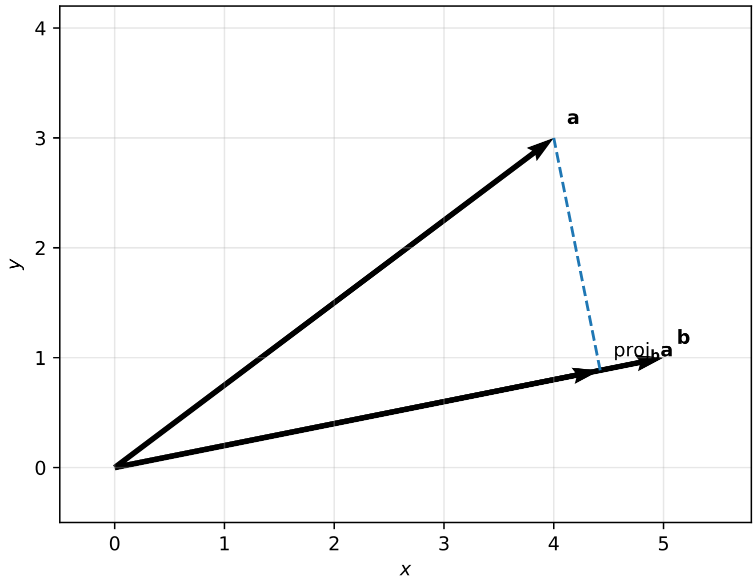


Figure 1.28: Numerical visualization of $\mathbf{a} = \mathrm{proj}_{\mathbf{b}}\,\mathbf{a} + \mathbf{a}_{\perp}$. The dashed segment is perpendicular to $\mathbf{b}$.

1.14.2 Rotations Preserve Lengths and Angles

A counterclockwise rotation through θ is represented by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Because $R^{\mathsf{T}}R = I$, one has $\|R\mathbf{v}\| = \|\mathbf{v}\|$. The script below evaluates several rotations of the same vector.

Listing 1.2: Applying a rotation matrix to a vector.

```python
"""Rotate a two-dimensional vector through several angles."""
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUTPUT = Path(__file__).resolve().parents[3] / "figures/ch01/computational/rotation.pdf"
v = np.array([3.0, 1.0])
angles = np.deg2rad([0, 30, 60, 90])
rotated = []
for theta in angles:
    R = np.array([[np.cos(theta), -np.sin(theta)],
                  [np.sin(theta),  np.cos(theta)]])
    rotated.append(R @ v)

fig, ax = plt.subplots(figsize=(5.8, 5.2))
for theta, vector in zip(np.rad2deg(angles).astype(int), rotated):
    ax.quiver(0, 0, *vector, angles="xy", scale_units="xy", scale=1, width=0.008)
    ax.text(*(vector * 1.08), rf"${theta}^\circ$")
ax.set(xlim=(-1.7, 3.8), ylim=(-0.5, 3.8), xlabel="$x$", ylabel="$y$")
```

```
ax.set_aspect("equal")
ax.grid(True, alpha=0.3)
fig.tight_layout()
fig.savefig(OUTPUT)
```

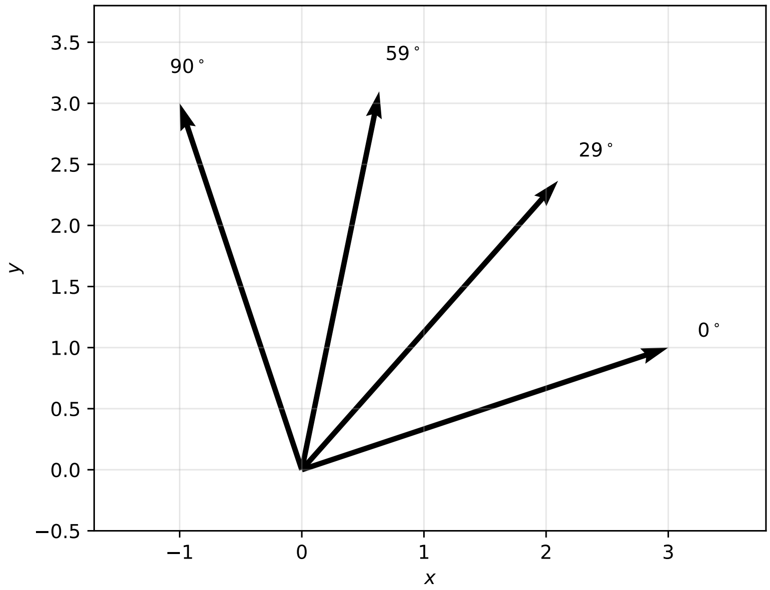


Figure 1.29: A vector rotated through several angles. Every arrow has the same length because an orthogonal rotation preserves the inner product.

1.14.3 Linear Transformations of Shapes and Bases

A matrix acts on every point of a geometric object and on the basis vectors used to describe it. Plotting both the original square and its image makes shear, stretch, and rotation immediately distinguishable.

Listing 1.3: Visualizing the action of a linear transformation.

```
"""Show how a matrix transforms a square and two basis vectors."""
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUTPUT = Path(__file__).resolve().parents[3] / "figures/ch01/computational/
    linear_transformation.pdf"
A = np.array([[1.5, 0.7], [0.3, 1.2]])
square = np.array([[0, 0], [1, 0], [1, 1], [0, 1], [0, 0]], dtype=float)
transformed = square @ A.T

fig, ax = plt.subplots(figsize=(6.0, 4.8))
ax.plot(square[:, 0], square[:, 1], linestyle="--", label="original square")
ax.plot(transformed[:, 0], transformed[:, 1], label="transformed square")
for vector, label in [(A @ np.array([1., 0.]), r"$A\mathbf{e}_1$"),
                      (A @ np.array([0., 1.]), r"$A\mathbf{e}_2$")]:
    ax.quiver(0, 0, *vector, angles="xy", scale_units="xy", scale=1, width=0.008)
```

```
    ax.text(*(vector * 1.08), label)
ax.set(xlim=(-0.25, 2.6), ylim=(-0.25, 1.8), xlabel="$x$", ylabel="$y$")
ax.set_aspect("equal")
ax.grid(True, alpha=0.3)
ax.legend(frameon=False)
fig.tight_layout()
fig.savefig(OUTPUT)
```

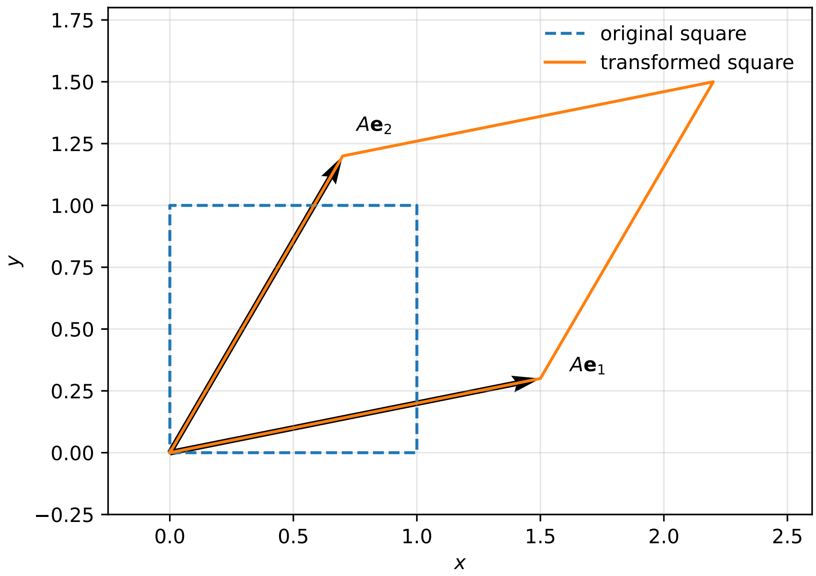


Figure 1.30: A unit square and its image under a linear transformation. The transformed basis vectors are the columns of the matrix.

1.14.4 Gram–Schmidt Orthogonalization

The Gram–Schmidt algorithm converts linearly independent vectors into an orthonormal basis. Numerically, the final dot product should be close to zero; small residual values arise from finite-precision arithmetic.

Listing 1.4: A two-dimensional Gram–Schmidt calculation.

```
"""Apply Gram-Schmidt orthogonalization and visualize the result."""
from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUTPUT = Path(__file__).resolve().parents[3] / "figures/ch01/computational/
    gram_schmidt.pdf"
v1 = np.array([3.0, 1.0])
v2 = np.array([2.0, 3.0])
e1 = v1 / np.linalg.norm(v1)
u2 = v2 - (v2 @ e1) * e1
e2 = u2 / np.linalg.norm(u2)

fig, ax = plt.subplots(figsize=(5.7, 4.7))
for vector, label in [(v1, r"$\mathbf{v}_1$"), (v2, r"$\mathbf{v}_2$"),
```

```
                      (2.2 * e1, r"$\mathbf{e}_1$"), (2.2 * e2, r"$\mathbf{e}_2$")]:
    ax.quiver(0, 0, *vector, angles="xy", scale_units="xy", scale=1, width=0.008)
    ax.text(*(vector * 1.08), label)
ax.set(xlim=(-1.2, 3.8), ylim=(-0.5, 3.8), xlabel="$x$", ylabel="$y$")
ax.set_aspect("equal")
ax.grid(True, alpha=0.3)
fig.tight_layout()
fig.savefig(OUTPUT)
print(f"e1 dot e2 = {e1 @ e2:.3e}")
```

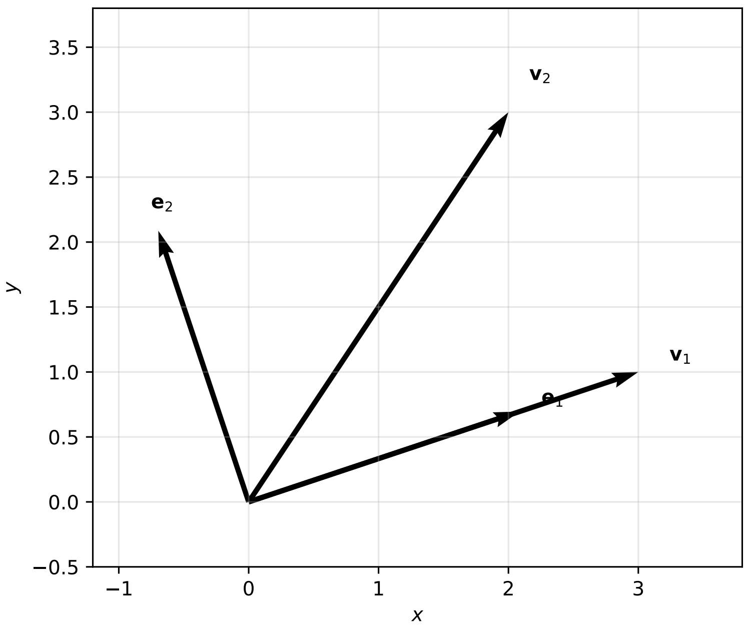


Figure 1.31: The original pair $\mathbf{v}_1, \mathbf{v}_2$ and the orthonormal pair $\mathbf{e}_1, \mathbf{e}_2$ produced by Gram–Schmidt.

1.14.5 Wolfram Language Equivalents

The same ideas can be explored symbolically. In Wolfram Language, `Orthogonalize` implements Gram–Schmidt, while matrix multiplication and graphics use nearly the same mathematical structure as the formulas in the text.

Listing 1.5: Projection, rotation, transformation, and orthogonalization in Wolfram Language.

```
(* Chapter 1 computational companion: Wolfram Language *)
a = {4, 3}; b = {5, 1};
projection = (a.b)/(b.b) b;
Graphics[{Arrow[{{0, 0}, a}], Arrow[{{0, 0}, b}],
  Dashed, Line[{projection, a}], Arrow[{{0, 0}, projection}]}, Axes -> True]

rotation[theta_] := {{Cos[theta], -Sin[theta]}, {Sin[theta], Cos[theta]}};
Table[rotation[theta].{3, 1}, {theta, 0, Pi/2, Pi/6}]

A = {{1.5, .7}, {.3, 1.2}};
A.# & /@ {{0, 0}, {1, 0}, {1, 1}, {0, 1}}

Orthogonalize[{{3, 1}, {2, 3}}, Method -> "GramSchmidt"]
```

Common computational mistake

Array multiplication is not universal across programming languages. In NumPy, `a @ b` denotes a matrix or dot product, while `a * b` multiplies components element by element. Always check the operation implemented by the language rather than inferring it from the printed symbols.

1.15 Integrated Worked Examples

The following examples combine several ideas from the chapter. Each solution separates calculation from physical interpretation.

Example 1.6

Displacement from components.
A particle moves 3 m east, 4 m north, and 12 m upward. Find the displacement vector, its magnitude, and its direction as a unit vector.

Solution

Using the Cartesian basis,

$$\Delta\mathbf{r} = 3\mathbf{e}_x + 4\mathbf{e}_y + 12\mathbf{e}_z \text{ m}.$$

Its magnitude is

$$\|\Delta\mathbf{r}\| = \sqrt{3^2 + 4^2 + 12^2}\,\text{m} = 13\,\text{m}.$$

Therefore

$$\widehat{\Delta\mathbf{r}} = \frac{3}{13}\mathbf{e}_x + \frac{4}{13}\mathbf{e}_y + \frac{12}{13}\mathbf{e}_z.$$

The unit vector contains only directional information; the factor 13 m contains the size of the displacement.

Example 1.7

Work done by a constant force.
A force $\mathbf{F} = (6, 2, 0)$ N acts through the displacement $\mathbf{s} = (3, -1, 0)$ m. Calculate the work.

Solution

Work is the scalar projection of the force along the displacement multiplied by the displacement magnitude, equivalently the dot product:

$$W = \mathbf{F} \cdot \mathbf{s} = (6)(3) + (2)(-1) = 16\,\text{J}.$$

The positive result means that the force has a net component in the direction of motion. A perpendicular component would contribute no work.

Example 1.8

Area and orientation from a cross product.
Let $\mathbf{a} = (2, 0, 0)$ m and $\mathbf{b} = (1, 3, 0)$ m. Find the oriented area vector of the parallelogram they span.

Solution

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ 2 & 0 & 0 \\ 1 & 3 & 0 \end{vmatrix} = 6\mathbf{e}_z \, \mathrm{m}^2.$$

Hence the area is $6\,\mathrm{m}^2$ and the orientation is $+\mathbf{e}_z$ for the ordering $\mathbf{a} \times \mathbf{b}$. Reversing the order reverses the orientation but not the area.

Example 1.9

Decomposing a force parallel and perpendicular to a line.
A force $\mathbf{F} = (5, 4, 0)\,\mathrm{N}$ acts near a rail directed along $\mathbf{d} = (3, 0, 0)$. Find the components parallel and perpendicular to the rail.

Solution

The rail unit vector is $\widehat{\mathbf{d}} = \mathbf{e}_x$. Thus

$$\mathbf{F}_{\parallel} = (\mathbf{F} \cdot \widehat{\mathbf{d}})\widehat{\mathbf{d}} = 5\mathbf{e}_x \, \mathrm{N},$$

and

$$\mathbf{F}_{\perp} = \mathbf{F} - \mathbf{F}_{\parallel} = 4\mathbf{e}_y \, \mathrm{N}.$$

The decomposition is verified by $\mathbf{F}_{\parallel} \cdot \mathbf{F}_{\perp} = 0$.

Example 1.10

Testing linear independence.
Determine whether $\mathbf{v}_1 = (1, 1, 0)$, $\mathbf{v}_2 = (0, 1, 1)$, and $\mathbf{v}_3 = (1, 2, 1)$ are linearly independent.

Solution

Direct inspection gives

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2.$$

Therefore

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0}$$

is a nontrivial linear relation. The three vectors are linearly dependent and span at most a two-dimensional subspace of $\mathbb{R}^3$.

Example 1.11

Angle between two vectors. For $\mathbf{a} = (1, 1, 0)$ and $\mathbf{b} = (1, 0, 1)$,

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} = \frac{1}{2},$$

so $\theta = 60^\circ$. The positive dot product signals an acute angle.

Example 1.12

Parallel and perpendicular components. Resolve $\mathbf{v} = (4, 2, 0)$ relative to $\mathbf{a} = (1, 1, 0)$. Since

$$\mathbf{v}_{\parallel} = \frac{\mathbf{v} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a} = 3(1, 1, 0) = (3, 3, 0),$$

we obtain $\mathbf{v}_{\perp} = (1, -1, 0)$. Directly, $\mathbf{v}_{\perp} \cdot \mathbf{a} = 0$.

Example 1.13

Torque from a force. A force $\mathbf{F} = (0, 12, 0)\,\mathrm{N}$ acts at $\mathbf{r} = (0.30, 0, 0)\,\mathrm{m}$. Then

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = (0, 0, 3.6)\,\mathrm{N\,m}.$$

The positive z direction corresponds to counterclockwise rotational tendency in the xy plane.

Example 1.14

Area of a triangle. Let $P = (0, 0, 0)$, $Q = (3, 0, 0)$, and $R = (1, 2, 0)$. The area is

$$\frac{1}{2}\|(Q - P) \times (R - P)\| = \frac{1}{2}\|(3, 0, 0) \times (1, 2, 0)\| = 3.$$

The corresponding parallelogram has area 6.

Example 1.15

Testing linear independence. For columns $\mathbf{v}_1 = (1, 0, 1)$, $\mathbf{v}_2 = (0, 1, 1)$, and $\mathbf{v}_3 = (1, 1, 0)$,

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} = -2 \neq 0.$$

The vectors are therefore independent and form a basis of $\mathbb{R}^3$.

Example 1.16

Coordinates in a rotated basis. Let $\mathbf{e}'_1 = (\mathbf{e}_x + \mathbf{e}_y)/\sqrt{2}$ and $\mathbf{e}'_2 = (-\mathbf{e}_x + \mathbf{e}_y)/\sqrt{2}$. For $\mathbf{v} = 3\mathbf{e}_x + \mathbf{e}_y$,

$$v'_1 = \mathbf{v} \cdot \mathbf{e}'_1 = 2\sqrt{2}, \qquad v'_2 = \mathbf{v} \cdot \mathbf{e}'_2 = -\sqrt{2}.$$

Both descriptions give $\|\mathbf{v}\| = \sqrt{10}$.

Example 1.17

One Gram–Schmidt step. Take $\mathbf{v}_1 = (1, 1, 0)$ and $\mathbf{v}_2 = (1, 0, 1)$. Set $\mathbf{u}_1 = \mathbf{v}_1$ and compute

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right).$$

After normalization, $\hat{\mathbf{u}}_1 = (1, 1, 0)/\sqrt{2}$ and $\hat{\mathbf{u}}_2 = (1, -1, 2)/\sqrt{6}$.

Example 1.18

Matrix acting on a vector. For $T = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{v} = (3, -1)$,

$$T\mathbf{v} = (5, -1).$$

The transformation stretches and shears the plane; it is not a rotation because it does not preserve all lengths and angles.

Example 1.19

Eigenvectors of a diagonal matrix. For $D = \mathrm{diag}(3, -2)$,

$$D\mathbf{e}_x = 3\mathbf{e}_x, \qquad D\mathbf{e}_y = -2\mathbf{e}_y.$$

Thus the coordinate axes are eigenvector directions. The second direction is reversed because its eigenvalue is negative.

Example 1.20

Functions as vectors. In the vector space of polynomials of degree at most two, write $p(x) = 2 - 3x + x^2$ in the basis $\{1, x, x^2\}$. Its coordinate vector is $(2, -3, 1)$. This example shows that a vector is an object admitting linear combination, not necessarily a geometric arrow.

1.16 Chapter Exercise Bank

Unless stated otherwise, use a right-handed Cartesian basis and give exact answers before decimal approximations.

Foundations

Exercise 1.1: Scalar or vector?

Difficulty: ⋆ ∘ ∘ ∘ ∘ Classify each quantity as scalar or vector and justify the classification: energy, angular momentum, electric potential, magnetic field, probability density, and probability current.

Exercise 1.2: Components and magnitude

Difficulty: ⋆ ∘ ∘ ∘ ∘ For $\mathbf{A} = 2\mathbf{e}_x - 3\mathbf{e}_y + 6\mathbf{e}_z$, calculate $\|\mathbf{A}\|$ and $\widehat{\mathbf{A}}$.

Exercise 1.3: Vector addition

Difficulty: ⋆ ∘ ∘ ∘ ∘ A hiker walks 5 km east and then 12 km north. Write the resultant displacement in component form and find its magnitude.

Exercise 1.4: Dot product and angle

Difficulty: ⋆ ∘ ∘ ∘ ∘ For $\mathbf{A} = (1, 2, 2)$ and $\mathbf{B} = (2, 1, -2)$, calculate $\mathbf{A} \cdot \mathbf{B}$ and the angle between the vectors.

Intermediate

Exercise 1.5: Projection

Difficulty: $\star\star\circ\circ\circ$ Resolve $\mathbf{A} = (4, 3, 0)$ into components parallel and perpendicular to $\mathbf{B} = (1, 1, 0)$.

Exercise 1.6: Torque

Difficulty: $\star\star\circ\circ\circ$ A force $\mathbf{F} = (0, 8, 0)\,\mathrm{N}$ is applied at $\mathbf{r} = (0.25, 0, 0)\,\mathrm{m}$. Calculate $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ and interpret its direction.

Exercise 1.7: Area of a triangle

Difficulty: $\star\star\circ\circ\circ$ The vertices of a triangle are $P = (0, 0, 0)$, $Q = (2, 1, 0)$, and $R = (1, 3, 0)$. Use a cross product to find its area.

Exercise 1.8: Change of basis

Difficulty: $\star\star\circ\circ\circ$ Let

$$\mathbf{e}_1' = \frac{\mathbf{e}_x + \mathbf{e}_y}{\sqrt{2}}, \qquad \mathbf{e}_2' = \frac{-\mathbf{e}_x + \mathbf{e}_y}{\sqrt{2}}.$$

Find the components of $\mathbf{v} = 3\mathbf{e}_x + \mathbf{e}_y$ in the primed basis and verify that its magnitude is unchanged.

Advanced

Exercise 1.9: Linear independence

Difficulty: $\star\star\star\circ\circ$ Determine all values of λ for which the vectors $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 1, \lambda)$ are linearly dependent.

Exercise 1.10: Orthonormalization

Difficulty: $\star\star\star\circ\circ$ Apply the Gram–Schmidt procedure to $\mathbf{v}_1 = (1, 1, 0)$ and $\mathbf{v}_2 = (1, 0, 1)$ to construct an orthonormal basis for their span.

Exercise 1.11: Coordinate-free proof

Difficulty: $\star\star\star\circ\circ$ Starting from the dot-product axioms, prove the Pythagorean theorem for vectors: if $\mathbf{A} \cdot \mathbf{B} = 0$, then $\|\mathbf{A} + \mathbf{B}\|^2 = \|\mathbf{A}\|^2 + \|\mathbf{B}\|^2$.

Exercise 1.12: From arrows to functions

Difficulty: $\star\star\star\circ\circ$ Show that the functions 1, x, and x^2 are linearly independent on any interval containing more than two points. Explain why this demonstrates that vectors need not be geometric arrows.

Self-check

A complete solution should state the basis, preserve units, distinguish a vector from its components, and interpret the final direction or sign whenever it has physical meaning.

Additional Foundations

Exercise 1.13: Units and components

Difficulty: ⋆ ∘ ∘ ∘ ∘ A velocity is written as $\mathbf{v} = (4, -3, 0)\,\mathrm{m\,s^{-1}}$. State the units of each component, the magnitude, and the corresponding unit vector.

Exercise 1.14: Displacement between points

Difficulty: ⋆ ∘ ∘ ∘ ∘ Find the displacement from $P = (-2, 1, 4)$ to $Q = (3, -1, 7)$ and its length.

Exercise 1.15: Opposite vectors

Difficulty: ⋆ ∘ ∘ ∘ ∘ For what scalar λ are $\mathbf{a} = (2, -4, 6)$ and $\mathbf{b} = (-1, 2, -3)$ related by $\mathbf{a} = \lambda\mathbf{b}$?

Exercise 1.16: Direction cosines

Difficulty: ⋆ ∘ ∘ ∘ ∘ Find the angles made by $\mathbf{a} = (2, 2, 1)$ with the positive coordinate axes.

Exercise 1.17: Triangle inequality

Difficulty: ⋆ ∘ ∘ ∘ ∘ Verify the triangle inequality for $\mathbf{a} = (1, 2)$ and $\mathbf{b} = (-2, 2)$. When does equality occur geometrically?

Exercise 1.18: Orthogonality test

Difficulty: ⋆ ∘ ∘ ∘ ∘ Determine k so that $(2, k, -1)$ is perpendicular to $(1, 3, 4)$.

Exercise 1.19: Work and sign

Difficulty: ⋆ ∘ ∘ ∘ ∘ A force of magnitude $20\,\mathrm{N}$ acts through $5\,\mathrm{m}$ at angles $0°$, $90°$, and $150°$. Compute the work in each case and interpret the sign.

Exercise 1.20: Standard basis products

Difficulty: ⋆ ∘ ∘ ∘ ∘ List all nine products $\mathbf{e}_i \cdot \mathbf{e}_j$ and all ordered cross products of distinct Cartesian basis vectors.

Exercise 1.21: Parallelogram diagonal

Difficulty: ⋆ ∘ ∘ ∘ ∘ Two sides of a parallelogram are $\mathbf{a} = (3, 1)$ and $\mathbf{b} = (1, 2)$. Find both diagonals.

Exercise 1.22: Unit direction

Difficulty: ⋆ ∘ ∘ ∘ ∘ Find the unit vector pointing from $A = (1, 2, -1)$ to $B = (5, 5, 1)$.

Exercise 1.23: Scalar projection

Difficulty: ⋆ ∘ ∘ ∘ ∘ Find $\mathrm{comp}_{\mathbf{b}}\,\mathbf{a}$ for $\mathbf{a} = (3, 4)$ and $\mathbf{b} = (1, 0)$.

Exercise 1.24: Vector projection

Difficulty: ⋆◦◦◦◦ Project $\mathbf{a} = (2, 5)$ onto $\mathbf{b} = (2, 0)$ and explain why scaling $\mathbf{b}$ does not alter the projected vector.

Additional Intermediate Problems

Exercise 1.25: Distance from a line

Difficulty: ⋆⋆◦◦◦ For $\mathbf{r} = (4, 3, 0)$ and line direction $\mathbf{a} = (1, 1, 0)$ through the origin, use orthogonal decomposition to find the shortest distance from the point to the line.

Exercise 1.26: Inclined-plane decomposition

Difficulty: ⋆⋆◦◦◦ Resolve a weight $m\mathbf{g}$ into components parallel and perpendicular to a plane inclined at angle α above the horizontal.

Exercise 1.27: Magnetic-force direction

Difficulty: ⋆⋆◦◦◦ For a positive charge moving along $+x$ in a magnetic field along $+z$, determine the direction of $q\mathbf{v} \times \mathbf{B}$. Repeat for an electron.

Exercise 1.28: Angular momentum

Difficulty: ⋆⋆◦◦◦ A particle has $\mathbf{r} = (2, 0, 0)\,\mathrm{m}$ and $\mathbf{p} = (0, 3, 4)\,\mathrm{kg\,m\,s^{-1}}$. Find $\mathbf{L} = \mathbf{r} \times \mathbf{p}$.

Exercise 1.29: Coplanarity

Difficulty: ⋆⋆◦◦◦ Use the scalar triple product to test whether $(1, 0, 1)$, $(0, 2, 1)$, and $(2, 2, 3)$ are coplanar.

Exercise 1.30: Volume

Difficulty: ⋆⋆◦◦◦ Find the volume of the parallelepiped spanned by $(1, 1, 0)$, $(0, 2, 1)$, and $(2, 0, 1)$.

Exercise 1.31: Basis coordinates

Difficulty: ⋆⋆◦◦◦ Express $\mathbf{v} = (5, 1)$ in the basis $\mathbf{b}_1 = (1, 1)$, $\mathbf{b}_2 = (1, -1)$.

Exercise 1.32: Basis verification

Difficulty: ⋆⋆◦◦◦ Show that $(1, 1, 0)$, $(0, 1, 1)$, and $(1, 0, 1)$ form a basis of $\mathbb{R}^3$.

Exercise 1.33: Matrix representation

Difficulty: ⋆⋆◦◦◦ Find the matrix of the transformation $T(x, y) = (2x - y, x + 3y)$ in the standard basis.

Exercise 1.34: Image of a basis

Difficulty: ⋆⋆◦◦◦ A linear map satisfies $T\mathbf{e}_1 = (1, 2)$ and $T\mathbf{e}_2 = (-1, 3)$. Find $T(4, -2)$.

Exercise 1.35: Eigenvector check

Difficulty: ⋆⋆∘∘∘ Determine whether $(1,1)$ and $(1,-1)$ are eigenvectors of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and find the corresponding eigenvalues.

Exercise 1.36: Polynomial coordinates

Difficulty: ⋆⋆∘∘∘ Write $p(x) = 1 + 4x - 2x^2$ in the basis $\{1, x, x^2\}$ and in the basis $\{1, 1+x, (1+x)^2\}$.

Additional Advanced Problems

Exercise 1.37: Cauchy–Schwarz

Difficulty: ⋆⋆⋆∘∘ Use $\|\mathbf{a} - t\mathbf{b}\|^2 \geq 0$ for all real t to prove $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\|\|\mathbf{b}\|$.

Exercise 1.38: Equality in Cauchy–Schwarz

Difficulty: ⋆⋆⋆∘∘ Determine the condition for equality in the Cauchy–Schwarz inequality and interpret it geometrically.

Exercise 1.39: Projection operator

Difficulty: ⋆⋆⋆∘∘ For a unit vector $\hat{\mathbf{n}}$, define $P = \hat{\mathbf{n}}\hat{\mathbf{n}}^\mathsf{T}$. Prove that $P^2 = P$ and interpret this property.

Exercise 1.40: Orthogonal complement

Difficulty: ⋆⋆⋆∘∘ Find a basis for all vectors in $\mathbb{R}^3$ perpendicular to $(1, 2, 3)$.

Exercise 1.41: Gram–Schmidt in three dimensions

Difficulty: ⋆⋆⋆∘∘ Apply Gram–Schmidt to $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$.

Exercise 1.42: Change-of-basis matrix

Difficulty: ⋆⋆⋆∘∘ Construct the matrix taking standard coordinates to coordinates in the basis $(1, 1)$, $(1, -1)$, and verify its inverse.

Exercise 1.43: Rotation preserves products

Difficulty: ⋆⋆⋆∘∘ Prove that if $R^\mathsf{T} R = I$, then $(R\mathbf{a}) \cdot (R\mathbf{b}) = \mathbf{a} \cdot \mathbf{b}$.

Exercise 1.44: Determinant and orientation

Difficulty: ⋆⋆⋆∘∘ Explain why an orthogonal matrix with determinant -1 preserves lengths but reverses orientation.

Exercise 1.45: Subspace test

Difficulty: ⋆⋆⋆∘∘ Determine which of the following are subspaces of $\mathbb{R}^3$: $x + y + z = 0$; $x + y + z = 1$; $xy = 0$.

Exercise 1.46: Function-space inner product

Difficulty: ⋆⋆⋆∘∘ Using $\langle f,g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$, determine whether 1, x, and $x^2-1/3$ are mutually orthogonal.

Exercise 1.47: Eigenbasis

Difficulty: ⋆⋆⋆∘∘ Diagonalize $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ using an orthonormal eigenbasis.

Exercise 1.48: Quantum-state preview

Difficulty: ⋆⋆⋆∘∘ Let $|\psi\rangle = a|0\rangle + b|1\rangle$ with complex coefficients. State the normalization condition and explain how it is the analogue of a unit vector.

Challenge Problems

Exercise 1.49: Lagrange identity

Difficulty: ⋆⋆⋆⋆∘ Prove $\|\mathbf{a}\times\mathbf{b}\|^2 + (\mathbf{a}\cdot\mathbf{b})^2 = \|\mathbf{a}\|^2\|\mathbf{b}\|^2$.

Exercise 1.50: Best approximation

Difficulty: ⋆⋆⋆⋆∘ Prove that the orthogonal projection of $\mathbf{v}$ onto a subspace W is the unique vector in W closest to $\mathbf{v}$.

Exercise 1.51: Dual viewpoint

Difficulty: ⋆⋆⋆⋆∘ Explain how the dot product with a fixed vector defines a linear functional. In finite-dimensional Euclidean space, show how every linear functional arises this way.

Exercise 1.52: Spectral preview

Difficulty: ⋆⋆⋆⋆∘ For a real symmetric 2×2 matrix, show directly that eigenvectors associated with distinct eigenvalues are orthogonal.

1.17 Summary and Looking Ahead

1.17.1 Chapter Synthesis

Physics separates an object from its representation. A point, displacement, force, or state is the physical or mathematical object; coordinates are the numbers assigned after a basis has been chosen. This distinction allows equations to remain meaningful when axes, units, or representations change.

The chapter developed a continuous chain of ideas:

$$\begin{aligned} &\text{measurement} \longrightarrow \text{quantities} \longrightarrow \text{vectors and coordinates} \\ &\qquad \longrightarrow \text{products and projections} \longrightarrow \text{bases and vector spaces.} \end{aligned}$$

The dot product measures alignment and creates lengths, angles, work, and orthogonality. The cross product encodes oriented area and rotational direction in three-dimensional Euclidean space. Projection separates a vector into physically meaningful components. Linear independence removes redundancy, while a basis provides a minimal and complete coordinate

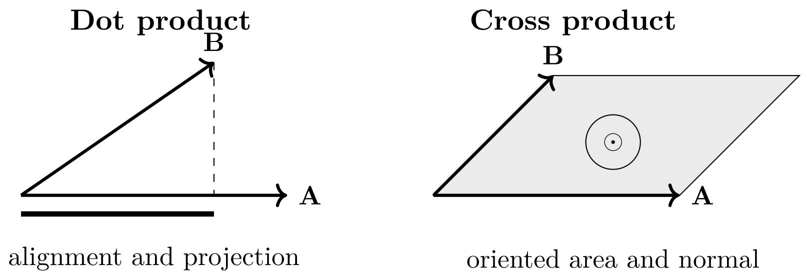


Figure 1.32: A compact visual comparison of the dot product and cross product.

framework. Abstract vector spaces then extend these ideas from arrows to functions, matrices, and quantum states.

Quick Formula Sheet

Quick Formula Sheet

Quantity or operation	Formula
Vector expansion	$\mathbf{v} = \sum_i v^i \mathbf{e}_i$
Magnitude	$\lVert\mathbf{v}\rVert = \sqrt{\mathbf{v}\cdot\mathbf{v}}$
Unit vector	$\hat{\mathbf{v}} = \mathbf{v}/\lVert\mathbf{v}\rVert$
Dot product	$\mathbf{a}\cdot\mathbf{b} = \lVert\mathbf{a}\rVert\lVert\mathbf{b}\rVert\cos\theta$
Cross product	$\lVert\mathbf{a}\times\mathbf{b}\rVert = \lVert\mathbf{a}\rVert\lVert\mathbf{b}\rVert\sin\theta$
Projection	$\mathrm{proj}_{\mathbf{a}}\,\mathbf{v} = (\mathbf{v}\cdot\mathbf{a}/\mathbf{a}\cdot\mathbf{a})\mathbf{a}$
Linear independence	$\sum_i c_i\mathbf{v}_i = \mathbf{0} \Rightarrow c_i = 0$ for all i
Eigenvector equation	$T\mathbf{v} = \lambda\mathbf{v}$

Notation and Units

Notation and Units

Symbol	Meaning	SI unit
$\mathbf{r}$	Position vector	m
$\mathbf{a}, \mathbf{b}, \mathbf{v}$	General vectors	context dependent
$\mathbf{e}_i$	Basis vector	usually dimensionless
$\hat{\mathbf{n}}$	Unit direction or unit normal	dimensionless
θ	Angle between vectors	rad
T	Linear transformation	context dependent
λ	Scalar or eigenvalue	context dependent

Common Mistake

Do not identify a vector with one particular column of components. Components change under a change of basis; the vector itself does not. Likewise, a nonzero magnitude does not determine a direction, and a zero vector cannot be normalized.

Learning Outcomes Check

Learning Outcomes Check

After studying this chapter, the reader should be able to:

- distinguish physical quantities from their coordinate representations;
- compute vector sums, magnitudes, dot products, cross products, and projections;
- interpret signs, directions, areas, and orthogonality physically;
- test linear independence and construct coordinates in a basis;
- recognize functions, matrices, and quantum states as vectors in suitable spaces;
- explain why a change of basis changes components but not the represented object.

Looking Ahead

Looking Ahead

Chapter 2 makes physical quantities vary continuously. Position becomes a function $\mathbf{r}(t)$; derivatives generate velocity and acceleration; and local linear approximation becomes the central tool connecting geometry with dynamics:

$$\mathbf{r}(t) \longrightarrow \frac{d\mathbf{r}}{dt} \longrightarrow \frac{d^2\mathbf{r}}{dt^2}.$$

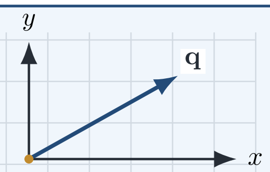
The computational experiments in Section 1.14 provide a numerical preview of this transition. The basis and projection ideas developed here will also reappear in gradients, fields, normal modes, and quantum measurement.

Chapter 2

Differential Calculus

Change becomes measurable

Limits and derivatives turn local change into a precise mathematical language.



Chapter Roadmap

Dependence (functions) $\longrightarrow$ **approach** (limits) $\longrightarrow$ **local change** (derivatives) $\longrightarrow$ **many-variable change** (partials and differentials) $\longrightarrow$ **geometric change** (gradients and directions).

Learning Objectives

After completing this chapter, the reader should be able to:

- distinguish domains, codomains, ranges, and scalar- or vector-valued functions;
- use limits to define continuity and instantaneous rates of change;
- derive and interpret ordinary derivatives, partial derivatives, and total differentials;
- apply the one-variable and multivariable chain rules;
- compute gradients and directional derivatives and interpret them geometrically;
- use Jacobian matrices to describe local linear transformations;
- connect differential calculus to motion, fields, optimization, and quantum mechanics.

How this chapter builds on Chapter 1

Chapter 1 established vectors, bases, inner products, and linear transformations. This chapter adds change. The derivative of a scalar function is a local slope; the derivative of a vector-valued function is a tangent vector; the Jacobian is a local linear map; and the gradient converts the differential of a scalar field into a vector through the Euclidean inner product.

2.1 Functions

Differential calculus begins with the idea of a *function*: a mathematical object that describes how one quantity depends upon another. Every concept introduced in this chapter—limits, derivatives, differentials, gradients, and differential equations—is ultimately formulated in terms of functions.

In physics, functions provide the language used to state quantitative laws. The position of a planet varies with time, the electric potential changes throughout space, the temperature inside a material depends upon position, and a quantum-mechanical wavefunction depends simultaneously upon spatial coordinates and time. Understanding functions is therefore the first essential step toward understanding the mathematical structure of physical theories.

2.1.1 Functions as Rules of Dependence

Function

Let A and B be non-empty sets. A function

$$f : A \longrightarrow B$$

assigns to every element $x \in A$ exactly one element

$$y = f(x) \in B.$$

The set A is called the *domain*, while B is called the *codomain*. The collection of values actually attained by the function is its *range*, or *image*, denoted by $f(A)$.

The notation used in this definition may be summarized as follows:

Symbol	Meaning
A	domain
B	codomain
x	independent variable or input
$y = f(x)$	dependent variable or output
$f : A \to B$	function or mapping
$f(A)$	range or image

The requirement of exactly one output for each allowed input is essential. Different inputs may produce the same output, but a single input may not be assigned two different outputs by the same function.

In elementary calculus the domain is often a subset of the real numbers,

$$A \subseteq \mathbb{R},$$

so that the function depends upon one independent variable. More generally, a function may depend upon several variables,

$$f = f(x, y, z),$$

or may take vector values. Later chapters will extend this idea still further to functions whose arguments are themselves functions.

2.1.2 Scalar- and Vector-Valued Functions

For a particle moving through three-dimensional space, the Cartesian coordinates may each depend on time:

$$x = x(t), \qquad y = y(t), \qquad z = z(t).$$

Together these coordinate functions define the complete position vector

$$\boxed{\mathbf{r}(t) = x(t)\hat{\mathbf{x}} + y(t)\hat{\mathbf{y}} + z(t)\hat{\mathbf{z}}.}$$

Thus, a moving particle traces a curve through space, and the function $\mathbf{r}(t)$ records where the particle is at each instant.

Scalar- and Vector-Valued Functions

The coordinate functions $x(t)$, $y(t)$, and $z(t)$ are scalar-valued because each returns a real number. The function $\mathbf{r}(t)$ is vector-valued because it returns a vector. Differential calculus applies to both kinds of function.

The position vector combines three coordinate functions into one geometric object. In later sections its first derivative will define the instantaneous velocity,

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt},$$

and its second derivative will define the instantaneous acceleration,

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}.$$

Many fundamental quantities in mechanics therefore arise directly from differentiating vector-valued functions.

A Temperature Distribution

Consider a thin metal plate whose temperature varies from point to point. Its temperature may be represented by

$$T = T(x, y),$$

where each point (x, y) in the plate is assigned one temperature. Unlike a particle trajectory, this function depends simultaneously upon two independent variables. Such multivariable functions motivate partial derivatives and gradients, which are introduced later in this chapter.

2.1.3 Why Calculus Is Needed

Imagine observing an electron moving along a straight line. At several moments we measure its position:

t (s)	x (m)
0	0
1	2
2	5
3	9

The particle is clearly moving. More importantly, the successive position changes are not equal, so its speed is changing. A finite table tells us what happened over several intervals, but it does not yet tell us the particle's velocity at one precise instant.

The natural question is therefore

How can continuous change be described mathematically?

Calculus answers this question by replacing finite changes with limiting processes. The central idea is the limit, which makes it possible to define an instantaneous rate of change rigorously.

Newton and Leibniz

Isaac Newton and Gottfried Wilhelm Leibniz developed differential calculus independently during the seventeenth century while studying motion, geometry, and continuously varying systems. Their notations emphasized different aspects of the subject, and both remain in widespread use. Newton's dot notation is especially natural for time derivatives in mechanics, while Leibniz's notation makes the variables involved in differentiation explicit.

Functions describe relationships between quantities, but they do not yet explain how rapidly those quantities change. To measure instantaneous change, we must understand what it means for one value of a function to be approached as its input approaches a chosen point. This leads naturally to the concept of a limit, the rigorous foundation of differential calculus.

2.2 Limits

A derivative describes change at a single point, but it is constructed from changes measured over a finite interval. The mathematical bridge between these two descriptions is the *limit*. A limit allows us to study the value approached by a function even when the function is not defined at the point being approached.

2.2.1 Approaching a Point

Consider the function

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

At $x = 1$, the denominator vanishes, so the expression is undefined. Nevertheless, its behavior near $x = 1$ is regular. Since

$$x^2 - 1 = (x - 1)(x + 1),$$

we may simplify the expression for every $x \neq 1$:

$$f(x) = x + 1, \qquad x \neq 1.$$

As x approaches 1, the values of $f(x)$ approach 2. We write

$$\boxed{\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.}$$

Informal Meaning of a Limit

The statement

$$\lim_{x \to a} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by choosing x sufficiently close to a, while allowing $x \neq a$.

The distinction between a function value and a limiting value is fundamental. In the example above, $f(1)$ is undefined, yet the limit exists and equals 2. A limit depends on the behavior of the function *near* the point, not necessarily on its value *at* the point.

A Punctured Neighborhood

When evaluating $\lim_{x \to a} f(x)$, one examines values of x in a small neighborhood around a while excluding the point $x = a$ itself. For this reason, limits can describe removable holes, discontinuities, and singular expressions without requiring the original formula to be defined at the limiting point.

2.2.2 The Precise Definition

The intuitive phrase "arbitrarily close" is made rigorous using two positive numbers. The number ε measures the permitted error in the function value, while δ measures how close the input must remain to the point a.

The ε–δ Definition

Let f be defined on a punctured neighborhood of a. We say that

$$\lim_{x\to a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x-a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The logical order is important. An arbitrary accuracy ε is specified first. We must then find a corresponding input tolerance δ that guarantees the required accuracy. The value of δ may depend on ε, but it may not depend on the particular value of x.

Proving a Linear Limit

Prove that

$$\lim_{x\to 2}(3x-1) = 5.$$

We begin with the desired condition

$$|(3x-1)-5| < \varepsilon.$$

Since

$$|(3x-1)-5| = |3x-6| = 3|x-2|,$$

it is sufficient to require

$$|x-2| < \frac{\varepsilon}{3}.$$

Therefore choose

$$\delta = \frac{\varepsilon}{3}.$$

Then

$$0 < |x-2| < \delta$$

implies

$$|(3x-1)-5| = 3|x-2| < 3\delta = \varepsilon.$$

Hence, by the definition of a limit,

$$\boxed{\lim_{x\to 2}(3x-1) = 5.}$$

2.2.3 One-Sided Limits

A function may approach different values from the left and from the right. These behaviors are described by one-sided limits:

$$\lim_{x\to a^-} f(x) \qquad \text{and} \qquad \lim_{x\to a^+} f(x).$$

The superscript $-$ indicates that $x < a$, while the superscript $+$ indicates that $x > a$.

Existence of a Two-Sided Limit

The two-sided limit

$$\lim_{x \to a} f(x) = L$$

exists if and only if both one-sided limits exist and are equal:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L.$$

A Jump Discontinuity

Let

$$f(x) = \begin{cases} -1, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then

$$\lim_{x \to 0^-} f(x) = -1, \qquad \lim_{x \to 0^+} f(x) = 1.$$

Because the one-sided limits are unequal, the two-sided limit $\lim_{x \to 0} f(x)$ does not exist.

2.2.4 Algebra of Limits

Once basic limits are known, more complicated limits may be evaluated using limit laws.

Limit Laws

Suppose

$$\lim_{x \to a} f(x) = L, \qquad \lim_{x \to a} g(x) = M,$$

and let c be a constant. Then

$$\lim_{x \to a} (f(x) + g(x)) = L + M,$$

$$\lim_{x \to a} (cf(x)) = cL,$$

$$\lim_{x \to a} (f(x)g(x)) = LM,$$

and, provided $M \neq 0$,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

For positive integers n,

$$\lim_{x \to a} [f(x)]^n = L^n.$$

These rules justify direct substitution for polynomials and for rational functions whose denominators do not vanish at the limiting point.

A Limit Evaluated by Algebra

Evaluate

$$\lim_{x \to 3} \frac{x^2 + 2x - 1}{x + 1}.$$

Because the denominator is nonzero at $x = 3$, the limit laws permit direct substitution:

$$\lim_{x \to 3} \frac{x^2 + 2x - 1}{x + 1} = \frac{3^2 + 2(3) - 1}{3 + 1} = \frac{14}{4} = \frac{7}{2}.$$

2.2.5 The Squeeze Principle

Some limits are difficult to evaluate directly but can be trapped between two simpler functions.

Squeeze Principle

Suppose that, for all x sufficiently close to a,

$$g(x) \leq f(x) \leq h(x),$$

and suppose that

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L.$$

Then

$$\lim_{x \to a} f(x) = L.$$

A central example is

$$\boxed{\lim_{x \to 0} \frac{\sin x}{x} = 1,}$$

where x is measured in radians. This result is one of the basic limits used to derive the derivatives of trigonometric functions.

2.2.6 Limits and Continuity

A function is continuous at a point when its limiting value agrees with its actual value.

Continuity at a Point

A function f is continuous at $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

Equivalently, all three of the following conditions must hold:

1. $f(a)$ is defined;
2. $\lim_{x \to a} f(x)$ exists;
3. the limit equals the function value.

Continuity expresses the absence of a break, jump, or hole at the point under consideration. Most elementary physical models assume that position, temperature, density, and field variables vary continuously over the scales at which the model is applied.

Continuity Is a Modeling Assumption

Continuity is not guaranteed by nature at every scale. A continuum model may treat matter as smoothly distributed even though matter is atomistic microscopically. The usefulness of continuity therefore depends on the physical scale and the approximation being made.

2.2.7 Physical Meaning of the Limiting Process

Suppose a particle moves along a line with position $x(t)$. Its average velocity over a finite interval Δt is

$$\bar{v} = \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

To obtain the velocity at the instant t, the interval must shrink toward zero:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t+\Delta t) - x(t)}{\Delta t}.$$

The interval never needs to be set equal to zero; doing so would produce division by zero. Instead, the ratio is examined for nonzero intervals that become arbitrarily small.

Why Limits Matter

An instantaneous rate of change cannot be measured directly from a zero-length interval. Limits retain the information contained in ratios over finite intervals while allowing those intervals to approach zero. This is the central mathematical idea behind differentiation.

With the limiting process now defined, we are prepared to construct the derivative and distinguish average change from instantaneous change.

2.3 Derivatives

A function describes how one quantity depends upon another, but in physics we are usually interested in more than the values of that function. We want to know how rapidly the quantity changes, whether it is increasing or decreasing, and how accurately it can be approximated near a chosen point. The derivative answers all of these questions.

The central idea is local. A finite interval gives an average rate of change; a limiting process then shrinks that interval toward zero and extracts the instantaneous rate of change. The same construction underlies velocity and acceleration in mechanics, current in electrodynamics, growth and decay laws, wave propagation, and the differential operators of quantum mechanics.

The Tangent and Motion Problems

The derivative emerged historically from two closely related questions. The geometric problem asked for the tangent to a curve at a specified point. The physical problem asked for the instantaneous velocity of a moving body. Newton emphasized changing quantities and motion, while Leibniz developed a powerful differential notation. Modern calculus unifies both viewpoints through the concept of a limit.

2.3.1 Average Rate of Change

Let $y = f(x)$. Suppose the independent variable changes from x to $x + \Delta x$. The corresponding change in the function is

$$\Delta f = f(x+\Delta x) - f(x).$$

The average rate of change of f over the interval is

$$\boxed{\frac{\Delta f}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}} \tag{2.1}$$

provided $\Delta x \neq 0$.

Geometrically, Eq. (2.1) is the slope of the secant line joining the two points

$$P = (x, f(x)), \qquad Q = (x+\Delta x, f(x+\Delta x)).$$

In mechanics, if $x(t)$ denotes position, then

$$\frac{x(t+\Delta t) - x(t)}{\Delta t}$$

is the average velocity during the time interval Δt.

Average Velocity

A particle moves along a line according to

$$x(t) = t^2 + 2t,$$

where x is measured in metres and t in seconds. Between $t = 1$ and $t = 1.5$,

$$\Delta x = x(1.5) - x(1) = 5.25 - 3 = 2.25 \text{ m},$$

and therefore

$$\overline{v} = \frac{\Delta x}{\Delta t} = \frac{2.25}{0.5} = 4.5 \text{ m s}^{-1}.$$

This number characterizes the motion over the entire interval. It does not yet give the velocity at one precise instant.

2.3.2 From Secant Line to Tangent Line

To obtain a local rate of change at x, the second point Q is moved toward P. Algebraically, this means allowing

$$\Delta x \to 0.$$

The secant line then approaches a limiting line: the tangent line at P.

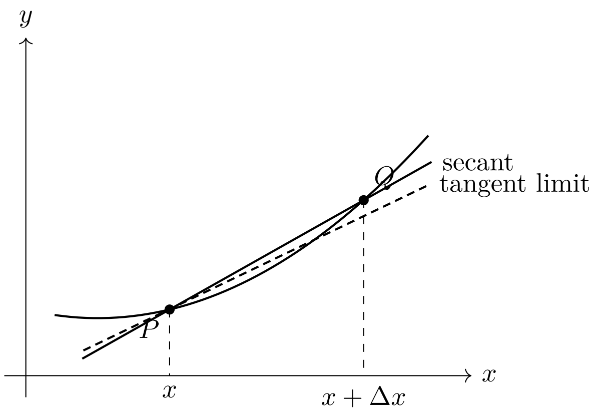


Figure 2.1: The derivative is obtained by allowing the secant line through P and Q to approach the tangent line at P.

Directly substituting $\Delta x = 0$ into the difference quotient would produce

$$\frac{f(x) - f(x)}{0} = \frac{0}{0}.$$

The expression $0/0$ is indeterminate: it is not a number and it does not mean that the derivative is zero. The point of the limiting process is to simplify or analyze the quotient for nonzero Δx and only then determine the value approached as Δx tends to zero.

Do Not Substitute Too Early

In a derivative calculation, Δx remains nonzero while algebraic cancellation is performed. Only after the quotient has been simplified do we take the limit $\Delta x \to 0$.

2.3.3 Definition of the Derivative

Derivative at a Point

Let f be defined on an open interval containing x. The derivative of f at x is

$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}} \tag{2.2}$$

provided the limit exists and is finite. In that case, f is said to be differentiable at x.

The increment is often denoted by h rather than Δx. Both notations express the same idea. In Eq. (2.2),

- $f(x+h) - f(x)$ is the change in the output;
- h is the change in the input;
- the quotient is an average rate of change over a finite interval;
- the limit extracts the local rate of change at x.

Equivalent notations are

$$f'(x), \qquad \frac{df}{dx}, \qquad D_x f, \qquad \left.\frac{df}{dx}\right|_x.$$

When the independent variable is time, Newton's dot notation is especially common:

$$\dot{x} = \frac{dx}{dt}, \qquad \ddot{x} = \frac{d^2x}{dt^2}.$$

Local Meaning of the Derivative

The derivative is simultaneously:

- an instantaneous rate of change;
- the slope of the tangent line;
- the coefficient of the best linear approximation near a point.

2.3.4 Differentiability and Continuity

Differentiability is stronger than continuity.

Differentiability Implies Continuity

If f is differentiable at $x = a$, then f is continuous at a.

To see why, write

$$f(a+h) - f(a) = h\frac{f(a+h) - f(a)}{h}.$$

If $f'(a)$ exists, then as $h \to 0$ the quotient approaches the finite number $f'(a)$, while $h \to 0$. Hence

$$\lim_{h \to 0}[f(a+h) - f(a)] = 0,$$

so

$$\lim_{h\to 0} f(a+h) = f(a).$$

The converse is false. A continuous function need not be differentiable.

A Corner

The function

$$f(x) = |x|$$

is continuous at $x = 0$, but its one-sided slopes are

$$\lim_{h\to 0^-} \frac{|h|}{h} = -1, \qquad \lim_{h\to 0^+} \frac{|h|}{h} = 1.$$

Because the one-sided limits disagree, $f'(0)$ does not exist. The graph has a sharp corner at the origin.

Other obstructions to differentiability include jumps, cusps, vertical tangents, and sufficiently irregular oscillation.

2.3.5 Derivatives from First Principles

The definition should be understood as a computational procedure, not merely memorized as a formula.

Example 1: The Derivative of x^2

Let $f(x) = x^2$. Then

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h\to 0} \frac{h(2x+h)}{h} \\ &= \lim_{h\to 0} (2x+h) \\ &= 2x. \end{aligned}$$

Therefore

$$\boxed{\frac{d}{dx} x^2 = 2x.}$$

The cancellation of h is valid because the difference quotient is evaluated for $h \neq 0$. Only afterward is the limit taken.

Example 2: The Derivative of x^3

For $f(x) = x^3$,

$$
\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{(x+h)^3 - x^3}{h} \\
&= \lim_{h\to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\
&= \lim_{h\to 0} \left(3x^2 + 3xh + h^2\right) \\
&= 3x^2.
\end{aligned}
$$

Thus

$$
\boxed{\frac{d}{dx} x^3 = 3x^2.}
$$

Example 3: The Derivative of $1/x$

For $x \neq 0$,

$$
\begin{aligned}
\frac{d}{dx}\left(\frac{1}{x}\right) &= \lim_{h\to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\
&= \lim_{h\to 0} \frac{x - (x+h)}{hx(x+h)} \\
&= \lim_{h\to 0} \frac{-h}{hx(x+h)} \\
&= \lim_{h\to 0} \left[-\frac{1}{x(x+h)}\right] \\
&= -\frac{1}{x^2}.
\end{aligned}
$$

Hence

$$
\boxed{\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}.}
$$

Example 4: The Derivative of $\sqrt{x}$

For $x > 0$,

$$
\begin{aligned}
\frac{d}{dx}\sqrt{x} &= \lim_{h\to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\
&= \lim_{h\to 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} \\
&= \lim_{h\to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\
&= \frac{1}{2\sqrt{x}}.
\end{aligned}
$$

Therefore

$$
\boxed{\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}.}
$$

2.3.6 Tangent Lines and Local Linearization

If f is differentiable at $x = a$, the tangent line has slope $f'(a)$ and passes through $(a, f(a))$. Its equation is

$$\boxed{y - f(a) = f'(a)(x - a)} \tag{2.3}$$

or equivalently

$$\boxed{L(x) = f(a) + f'(a)(x - a).} \tag{2.4}$$

The function $L(x)$ is the linearization of f at a. For x sufficiently close to a,

$$f(x) \approx f(a) + f'(a)(x - a).$$

This approximation is more than a convenient geometrical picture. Differentiability means precisely that the error is small compared with the displacement:

$$f(a + h) = f(a) + f'(a)h + r(h),$$

where

$$\lim_{h \to 0} \frac{r(h)}{h} = 0.$$

Approximating a Square Root

Linearize $f(x) = \sqrt{x}$ at $a = 4$. Since

$$f(4) = 2, \qquad f'(4) = \frac{1}{4},$$

we obtain

$$\sqrt{x} \approx 2 + \frac{1}{4}(x - 4).$$

For $x = 4.1$,

$$\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025.$$

The exact value is approximately 2.02485, so the local linear approximation is already accurate.

2.3.7 Differentials

If $y = f(x)$, the differential of y is defined by

$$\boxed{dy = f'(x)\,dx.} \tag{2.5}$$

Here dx represents a small change in the independent variable, while dy is the corresponding change predicted by the tangent-line approximation. The actual finite change is

$$\Delta y = f(x + \Delta x) - f(x),$$

whereas

$$dy = f'(x)\Delta x$$

is its leading linear approximation. Thus

$$\Delta y = dy + \text{higher-order error}.$$

Leibniz notation is especially useful because it displays the variables and transforms naturally under the chain rule. Nevertheless, dy/dx should be understood as a derivative operator or limiting ratio, not automatically as an ordinary fraction.

2.3.8 Physical Meaning of the Derivative

Suppose a particle moves along a line with position $x = x(t)$. Its average velocity over a finite time interval is

$$\overline{v} = \frac{x(t + \Delta t) - x(t)}{\Delta t}.$$

Taking the limit yields the instantaneous velocity:

$$\boxed{v(t) = \frac{dx}{dt}.} \tag{2.6}$$

Differentiating again gives the acceleration:

$$\boxed{a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2}.} \tag{2.7}$$

The same mathematical structure occurs throughout physics:

$$I = \frac{dQ}{dt}$$

for electric current,

$$P = \frac{dE}{dt}$$

for power, and

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}$$

for force in its momentum form.

A Derivative Requires Units

The derivative inherits units from the ratio of the dependent and independent variables. If x is measured in metres and t in seconds, then dx/dt has units of metres per second. Dimensional analysis therefore provides a quick consistency check on derivative formulas.

Position to Velocity and Acceleration

Let

$$x(t) = t^3 - 3t^2 + 2t.$$

Then

$$v(t) = \frac{dx}{dt} = 3t^2 - 6t + 2,$$

and

$$a(t) = \frac{dv}{dt} = 6t - 6.$$

At $t = 2$,

$$v(2) = 2, \qquad a(2) = 6.$$

The position function, its slope, and the rate of change of that slope encode three different levels of the motion.

2.3.9 Basic Differentiation Rules

The limit definition is fundamental, but repeated calculations are made efficient by general rules.

Constant and Sum Rules

If c is constant, then

$$\boxed{\frac{d}{dx}c = 0.}$$

If f and g are differentiable and α, β are constants, then

$$\boxed{\frac{d}{dx}[\alpha f(x) + \beta g(x)] = \alpha f'(x) + \beta g'(x).} \tag{2.8}$$

This is the linearity of differentiation.

Power Rule

For positive integers n, the binomial theorem gives

$$(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

Substituting into the difference quotient and dividing by h yields

$$nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}.$$

Taking $h \to 0$ leaves

$$\boxed{\frac{d}{dx}x^n = nx^{n-1}.} \tag{2.9}$$

The rule extends, under appropriate domain restrictions, to real powers.

Product Rule

Let $F(x) = f(x)g(x)$. Then

$$F'(x) = \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

Add and subtract $f(x+h)g(x)$ in the numerator:

$$F'(x) = \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}g(x) + f(x+h)\frac{g(x+h)-g(x)}{h}\right].$$

Because differentiability implies continuity, $f(x+h) \to f(x)$. Therefore

$$\boxed{\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).} \tag{2.10}$$

The Derivative of a Product

In general,

$$(fg)' \neq f'g'.$$

Both terms in Eq. (2.10) are required.

Quotient Rule

For $g(x) \neq 0$,

$$\boxed{\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}.} \tag{2.11}$$

One derivation writes $f/g = f\,g^{-1}$ and applies the product rule together with

$$\frac{d}{dx}g^{-1} = -g^{-2}g'.$$

Chain Rule

Suppose $y = f(u)$ and $u = g(x)$. A change in x first changes u, and the change in u then changes y. The derivative of the composition is

$$\boxed{\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).} \tag{2.12}$$

In Leibniz notation,

$$\boxed{\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.}$$

This notation resembles cancellation, but the chain rule is a theorem about limits and composition. The apparent cancellation is a reliable mnemonic when the differentiability assumptions are satisfied.

Using the Chain Rule

For

$$y = (3x^2+1)^5,$$

let $u = 3x^2 + 1$. Then

$$\frac{dy}{du} = 5u^4, \qquad \frac{du}{dx} = 6x.$$

Hence

$$\frac{dy}{dx} = 5(3x^2+1)^4(6x) = 30x(3x^2+1)^4.$$

2.3.10 Derivatives of Common Functions

The most frequently used elementary derivatives include

$$\frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x},$$

$$\frac{d}{dx}\sin x = \cos x, \qquad \frac{d}{dx}\cos x = -\sin x,$$

and

$$\frac{d}{dx}\tan x = \sec^2 x.$$

The trigonometric formulas assume that angles are measured in radians. This is essential: only in radian measure does the fundamental limit

$$\lim_{h\to 0}\frac{\sin h}{h} = 1$$

lead directly to $d(\sin x)/dx = \cos x$.

2.3.11 Implicit Differentiation

Some relations define y as a function of x without solving explicitly for y. Consider the circle

$$x^2 + y^2 = R^2.$$

Differentiate both sides with respect to x:

$$2x + 2y\frac{dy}{dx} = 0.$$

Therefore

$$\boxed{\frac{dy}{dx} = -\frac{x}{y}.} \tag{2.13}$$

The factor dy/dx appears because y depends on x and the chain rule must be applied to y^2.

Implicitly Defined Curve

Differentiate

$$x^3 + y^3 = 6xy.$$

We obtain

$$3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}.$$

Collecting the derivative terms gives

$$(3y^2 - 6x)\frac{dy}{dx} = 6y - 3x^2,$$

so

$$\boxed{\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}}.$$

2.3.12 Higher-Order Derivatives

Repeated differentiation produces

$$f''(x), \qquad f'''(x), \qquad f^{(n)}(x).$$

For motion in one dimension,

$$\dot{x} = v, \qquad \ddot{x} = a, \qquad \dddot{x} = \text{jerk}.$$

Higher derivatives describe increasingly detailed aspects of change. In differential equations, the highest derivative present often reflects the amount of initial data required to determine a solution. For example, a second-order equation for $x(t)$ generally requires both an initial position and an initial velocity.

The second derivative also describes curvature of a graph locally:

- $f''(x) > 0$: the graph is locally concave upward;
- $f''(x) < 0$: the graph is locally concave downward;
- $f''(x) = 0$: a possible inflection point, though further analysis is required.

2.3.13 Derivative Notation

Several notational systems coexist because each emphasizes a different aspect.

Notation	Example	Typical use
Lagrange	$f'(x), f''(x)$	functions and elementary analysis
Leibniz	$\frac{dy}{dx}$	variable dependence and chain rules
Newton	$\dot{x}, \ddot{x}$	time evolution in mechanics
Operator	$D_x f$	analysis and differential operators

No notation is universally superior. A mature calculation often moves between them.

2.3.14 Common Errors and Diagnostic Checks

Typical Differentiation Errors

Common mistakes include:

- substituting $h = 0$ before simplifying the difference quotient;
- writing $(fg)' = f'g'$ instead of using the product rule;
- forgetting the inner derivative in the chain rule;
- omitting dy/dx during implicit differentiation;
- ignoring the domain on which a derivative formula is valid;

- dropping physical units.

Three quick checks are especially useful:

1. *Units:* do the derivative units equal output units divided by input units?
2. *Scale:* does the answer have a plausible magnitude and sign?
3. *Structure:* were all product, quotient, and nested-function dependencies differentiated?

2.3.15 Connection to Quantum Mechanics

Derivatives become operators in quantum theory. The time-dependent Schrödinger equation contains a time derivative,

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi, \tag{2.14}$$

while the momentum operator in position space is

$$\boxed{\hat{\mathbf{p}} = -i\hbar\nabla.} \tag{2.15}$$

For a particle in one spatial dimension, the kinetic-energy operator contains a second derivative:

$$\hat{T} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}.$$

Thus the first derivative measures local variation of a wavefunction, while the second derivative measures its local curvature. These operations later determine momentum, kinetic energy, probability flow, and wave propagation.

From Number to Operator

In elementary calculus, differentiation maps one function to another. In quantum mechanics, the same mapping is promoted to a linear operator acting on state functions. The mathematics introduced here therefore becomes part of the observable structure of the theory.

2.3.16 Section Summary

Derivatives

- The average rate of change is the slope of a secant line.
- The derivative is the limit of the difference quotient as the interval shrinks to zero.
- Geometrically, the derivative is the tangent-line slope.
- Analytically, it gives the best local linear approximation.
- Differentiability implies continuity, but continuity alone does not guarantee differentiability.
- Product, quotient, and chain rules encode how derivatives behave under algebraic operations and composition.
- Higher derivatives describe higher-order change, including acceleration and curvature.
- In quantum mechanics, derivatives become operators such as momentum and kinetic energy.

Ordinary derivatives apply when a function depends on a single independent variable. Physical fields generally depend on several coordinates at once. The next section introduces partial derivatives, which isolate the rate of change with respect to one variable while the others are held fixed.

2.4 Partial Derivatives

Most physical systems depend on several independent variables. A temperature field may vary with three spatial coordinates and time, a thermodynamic state function may depend on pressure and volume, and a quantum wavefunction may depend on position, spin coordinates, and time. Partial differentiation extends the derivative to such multivariable functions by isolating the response to one variable while the remaining independent variables are held fixed.

The central idea is local. Near a point, a sufficiently smooth multivariable function is approximated by a plane or, in higher dimensions, by a linear map. Its partial derivatives are the coefficients of this approximation. They therefore provide the raw material for gradients, Jacobians, Hessians, differential equations, and the field theories used throughout physics.

From Curves to Fields

Ordinary differential calculus was developed for functions of one variable, but mechanics, thermodynamics, and field theory quickly required functions of several variables. Euler, Lagrange, Laplace, Fourier, and later Gibbs and Maxwell transformed partial differentiation into a basic language of mathematical physics. Modern quantum mechanics continues this tradition: the Schrödinger equation is a partial differential equation because the wavefunction depends simultaneously on space and time.

2.4.1 Functions of Several Variables

A scalar-valued function of two variables has the form

$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad (x, y) \mapsto f(x, y).$$

Its graph is the surface

$$z = f(x, y)$$

in three-dimensional space. A function of three spatial coordinates,

$$\phi(x, y, z),$$

is more naturally interpreted as a scalar field: each point of space is assigned a number such as temperature, pressure, density, or electric potential.

Temperature Field

Suppose

$$T(x, y, z) = T_0 - \alpha(x^2 + y^2) - \beta z,$$

where T_0, α, and β are constants. The temperature changes differently in different directions. Moving in the x-direction produces a change governed by $-2\alpha x$, moving in the y-direction produces $-2\alpha y$, and moving vertically produces the constant change $-\beta$.

2.4.2 Rigorous Definition from a Limit

Fix a point (a, b) in the domain of $f(x, y)$. To study change in the x-direction, hold $y = b$ fixed and vary only x. The resulting one-variable slice is

$$g(x) = f(x, b).$$

The ordinary derivative of this slice at $x = a$ defines the partial derivative with respect to x.

Partial Derivative with Respect to x

The partial derivative of f with respect to x at (a, b) is

$$\boxed{\frac{\partial f}{\partial x}(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h}} \tag{2.16}$$

provided the limit exists and is finite.

Similarly,

$$\boxed{\frac{\partial f}{\partial y}(a, b) = \lim_{k \to 0} \frac{f(a, b + k) - f(a, b)}{k}} \tag{2.17}$$

holds $x = a$ fixed and varies only y.
Common notations include

$$f_x, \qquad \partial_x f, \qquad \frac{\partial f}{\partial x}, \qquad D_x f.$$

What "Holding the Other Variables Fixed" Means

The phrase does not mean that the other quantities can never change physically. It means that the mathematical experiment defining one partial derivative compares nearby points that differ in only one coordinate. Other changes are included later through the total differential or total derivative.

2.4.3 Partial Derivatives from First Principles

Consider

$$f(x, y) = x^2 y + 3y^2.$$

Using the limit definition,

$$\begin{aligned}
\frac{\partial f}{\partial x}(x, y) &= \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h} \\
&= \lim_{h \to 0} \frac{(x + h)^2 y + 3y^2 - (x^2 y + 3y^2)}{h} \\
&= \lim_{h \to 0} \frac{2xyh + yh^2}{h} \\
&= \lim_{h \to 0} (2xy + yh) \\
&= 2xy.
\end{aligned}$$

Likewise,

$$\begin{aligned}\frac{\partial f}{\partial y}(x,y) &= \lim_{k\to 0}\frac{x^2(y+k)+3(y+k)^2-(x^2y+3y^2)}{k}\\ &= \lim_{k\to 0}\left(x^2+6y+3k\right)\\ &= x^2+6y.\end{aligned}$$

Therefore

$$\boxed{f_x = 2xy, \qquad f_y = x^2+6y.}$$

Exponential and Trigonometric Dependence

Let

$$f(x,y) = e^{xy}\sin(x+y).$$

Treating y as constant,

$$f_x = ye^{xy}\sin(x+y) + e^{xy}\cos(x+y).$$

Treating x as constant,

$$f_y = xe^{xy}\sin(x+y) + e^{xy}\cos(x+y).$$

The expressions differ because the product xy responds differently to changes in x and y.

Three Variables

For

$$f(x,y,z) = x^2e^{yz} + z\cos x,$$

one obtains

$$\begin{aligned}f_x &= 2xe^{yz} - z\sin x,\\ f_y &= x^2ze^{yz},\end{aligned}$$

and

$$f_z = x^2ye^{yz} + \cos x.$$

2.4.4 Geometric Interpretation

For the surface $z = f(x,y)$, fixing $y = b$ produces the curve

$$z = f(x,b).$$

The number $f_x(a,b)$ is the slope of the tangent line to this curve at $(a,b,f(a,b))$. Fixing $x = a$ gives a second coordinate curve whose tangent slope is $f_y(a,b)$.

The two partial derivatives describe slopes along the coordinate directions, but a surface has infinitely many possible directions. Directional derivatives will combine these coordinate slopes into a general directional rate of change.

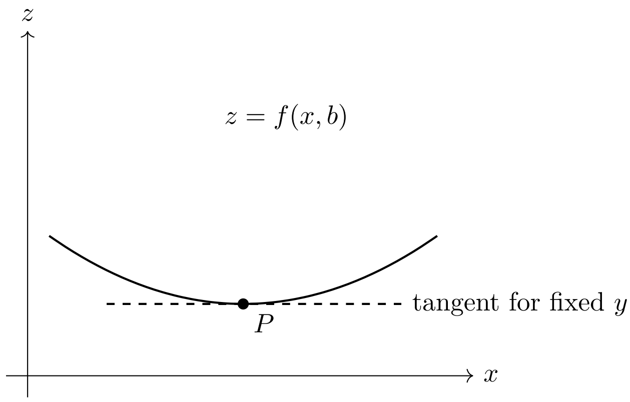


Figure 2.2: A partial derivative is an ordinary derivative along a coordinate slice of the surface.

2.4.5 Continuity, Partial Derivatives, and Differentiability

The existence of f_x and f_y at a point does not by itself guarantee that f is continuous or differentiable there. Multivariable differentiability is stronger: the function must admit one linear approximation that works for all sufficiently small directions simultaneously.

Differentiability in Two Variables

A function f is differentiable at (a, b) if there exists a linear expression

$$L(h, k) = Ah + Bk$$

such that

$$f(a + h, b + k) = f(a, b) + Ah + Bk + r(h, k), \tag{2.18}$$

where

$$\lim_{(h,k)\to(0,0)} \frac{r(h, k)}{\sqrt{h^2 + k^2}} = 0.$$

In that case $A = f_x(a, b)$ and $B = f_y(a, b)$.

A standard sufficient condition is especially useful.

Continuous Partials Imply Differentiability

If f_x and f_y exist in a neighbourhood of (a, b) and are continuous at (a, b), then f is differentiable at (a, b).

Existing Partials Are Not Enough

It is possible for both coordinate partial derivatives to exist at a point while the function behaves badly along diagonal or curved paths. Differentiability requires control in every direction, not only along the coordinate axes.

2.4.6 Tangent Plane and Linear Approximation

For a differentiable surface $z = f(x, y)$ near (a, b), Eq. (2.18) gives

$$f(a + h, b + k) \approx f(a, b) + f_x(a, b)h + f_y(a, b)k.$$

Since $h = x - a$ and $k = y - b$, the tangent plane is

$$\boxed{z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).} \tag{2.19}$$

Best Local Plane

The tangent plane is not merely a plane touching the surface. It is the unique affine function whose error becomes negligible compared with the distance from (a,b) as the point approaches (a,b).

Tangent Plane

Let

$$f(x,y) = x^2 + xy + y^2.$$

At $(1,2)$,

$$f(1,2) = 7, \qquad f_x = 2x + y, \qquad f_y = x + 2y,$$

so

$$f_x(1,2) = 4, \qquad f_y(1,2) = 5.$$

The tangent plane is therefore

$$\boxed{z = 7 + 4(x-1) + 5(y-2).}$$

At $(1.02, 1.97)$, the linear estimate gives

$$z \approx 7 + 4(0.02) + 5(-0.03) = 6.93.$$

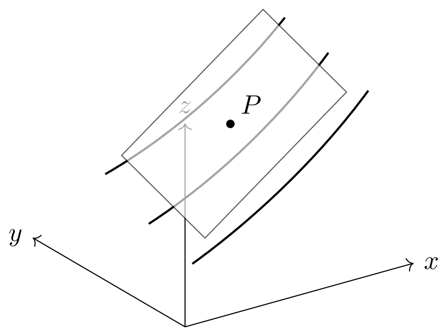


Figure 2.3: A differentiable surface is locally approximated by its tangent plane.

2.4.7 Total Differential

For a differentiable function $f(x,y)$, the first-order change is

$$\boxed{df = f_x\,dx + f_y\,dy.} \tag{2.20}$$

For $f(x,y,z)$,

$$\boxed{df = f_x\,dx + f_y\,dy + f_z\,dz.} \tag{2.21}$$

This follows directly from the linear approximation. If Δx, Δy are small, then

$$\Delta f = f_x \Delta x + f_y \Delta y + exthigher - ordererror,$$

and the total differential retains the first-order part.

Propagation of Measurement Uncertainty

For the area of a rectangle,

$$A(L, W) = LW,$$

the differential is

$$dA = W\,dL + L\,dW.$$

If $L = 2.00\,\mathrm{m}$, $W = 1.50\,\mathrm{m}$, $|dL| = 0.01\,\mathrm{m}$, and $|dW| = 0.02\,\mathrm{m}$, then a conservative first-order estimate is

$$|dA| \leq W|dL| + L|dW| = 1.50(0.01) + 2.00(0.02) = 0.055\,\mathrm{m}^2.$$

2.4.8 Total Derivative and the Multivariable Chain Rule

If $x = x(t)$ and $y = y(t)$, then

$$\boxed{\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.} \tag{2.22}$$

For $f(x, y, z, t)$ evaluated along a moving trajectory $\mathbf{r}(t)$,

$$\boxed{\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}\dot{x} + \frac{\partial f}{\partial y}\dot{y} + \frac{\partial f}{\partial z}\dot{z}.} \tag{2.23}$$

This distinction is fundamental in physics: $\partial f/\partial t$ measures local time change at a fixed point, whereas df/dt measures the change experienced by an observer moving through the field.

Temperature Experienced by a Moving Probe

Let

$$T(x, y, t) = 100 - 2x^2 - y^2 + 3t,$$

and let a probe move along

$$x = t, \qquad y = 2t.$$

Then

$$T_t = 3, \qquad T_x = -4x, \qquad T_y = -2y,$$

so

$$\frac{dT}{dt} = 3 - 4x\dot{x} - 2y\dot{y}.$$

Since $\dot{x} = 1$, $\dot{y} = 2$, $x = t$, and $y = 2t$,

$$\boxed{\frac{dT}{dt} = 3 - 12t.}$$

2.4.9 Directional Derivatives

Let $\mathbf{u} = (u_x, u_y)$ be a unit vector. The directional derivative of f at $\mathbf{a} = (a, b)$ in the direction $\mathbf{u}$ is

$$\boxed{D_{\mathbf{u}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{u}) - f(\mathbf{a})}{h}.} \tag{2.24}$$

For a differentiable function,

$$\boxed{D_{\mathbf{u}} f = f_x u_x + f_y u_y = \nabla f \cdot \mathbf{u}.} \tag{2.25}$$

Although the gradient is developed fully in the next section, Eq. (2.25) already shows why partial derivatives are its components.

Directional Rate of Change

Let

$$f(x, y) = x^2 + 3xy.$$

At $(1, 2)$,

$$f_x = 2x + 3y = 8, \qquad f_y = 3x = 3.$$

For the unit direction

$$\mathbf{u} = \frac{1}{5}(3, 4),$$

we find

$$D_{\mathbf{u}} f = 8\frac{3}{5} + 3\frac{4}{5} = \frac{36}{5}.$$

2.4.10 Second and Mixed Partial Derivatives

Repeated differentiation produces

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2},$$

and the mixed derivatives

$$f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right), \qquad f_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right).$$

Clairaut–Schwarz Theorem

If the mixed second partial derivatives are continuous in a neighbourhood of a point, then

$$\boxed{f_{xy} = f_{yx}.}$$

Mixed Partials

For

$$f(x, y) = x^3 y^2 + e^{xy},$$

one has

$$f_x = 3x^2 y^2 + y e^{xy},$$

so

$$f_{xy} = 6x^2 y + e^{xy} + xy e^{xy}.$$

Starting from

$$f_y = 2x^3 y + x e^{xy}$$

gives the same result for f_{yx}.

2.4.11 The Jacobian Matrix

For a vector-valued function

$$\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m, \qquad \mathbf{F} = (F_1, \dots, F_m),$$

the first-order partial derivatives form the Jacobian matrix

$$\boxed{J_{\mathbf{F}} = \begin{pmatrix} \dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_m}{\partial x_1} & \cdots & \dfrac{\partial F_m}{\partial x_n} \end{pmatrix}.} \tag{2.26}$$

The Jacobian is the best local linear map associated with $\mathbf{F}$. If $m = n$, its determinant measures the local signed change of volume.

Planar Mapping

Let

$$u = x^2 - y^2, \qquad v = 2xy.$$

Then

$$J = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}, \qquad \det J = 4(x^2 + y^2).$$

Away from the origin, the map locally expands area by the factor $4(x^2 + y^2)$.

2.4.12 Jacobian Determinants and Coordinate Transformations

For polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

the Jacobian is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r.$$

Thus the area element transforms as

$$\boxed{dx\, dy = r\, dr\, d\theta.} \tag{2.27}$$

For cylindrical coordinates,

$$dV = r\, dr\, d\theta\, dz,$$

and for spherical coordinates,

$$\boxed{dV = r^2 \sin\theta\, dr\, d\theta\, d\phi.} \tag{2.28}$$

The factors r and $r^2 \sin\theta$ are not arbitrary conventions. They are Jacobian determinants encoding how coordinate cells stretch in physical space.

2.4.13 The Hessian Matrix

For a twice differentiable scalar function $f : \mathbb{R}^n \to \mathbb{R}$, the Hessian is

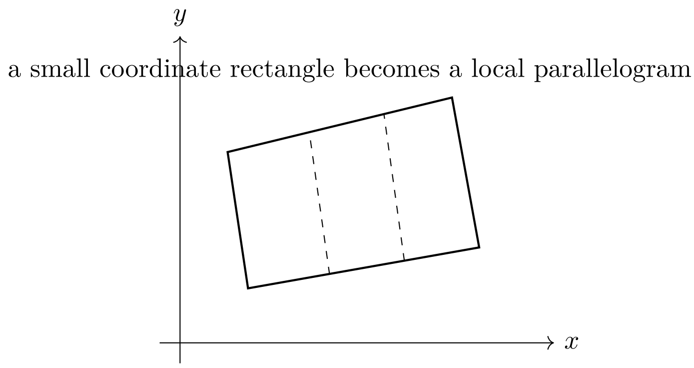


Figure 2.4: The Jacobian maps infinitesimal coordinate cells into locally distorted cells. Its determinant gives the signed area or volume scale factor.

$$\boxed{H_f = \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \cdots & \dfrac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_n^2} \end{pmatrix}.} \tag{2.29}$$

When mixed partials are continuous, the Hessian is symmetric. Its eigenvectors identify principal curvature directions, and its eigenvalues measure the corresponding second-order curvature.

For two variables,

$$H_f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

2.4.14 Local Extrema and the Second-Derivative Test

A critical point satisfies

$$f_x(a,b) = 0, \qquad f_y(a,b) = 0.$$

Define

$$D = f_{xx} f_{yy} - f_{xy}^2.$$

At a critical point:

- if $D > 0$ and $f_{xx} > 0$, f has a local minimum;
- if $D > 0$ and $f_{xx} < 0$, f has a local maximum;
- if $D < 0$, the point is a saddle point;
- if $D = 0$, the test is inconclusive.

Equivalently, in terms of Hessian eigenvalues:

- all positive eigenvalues imply a local minimum;
- all negative eigenvalues imply a local maximum;
- mixed signs imply a saddle point.

Minimum and Saddle

For

$$f(x,y) = x^2 + 4y^2,$$

the only critical point is $(0,0)$ and

$$H_f = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}.$$

Both eigenvalues are positive, so the origin is a strict local minimum.
For

$$g(x,y) = x^2 - y^2,$$

$$H_g = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix},$$

whose eigenvalues have opposite signs. The origin is therefore a saddle point.

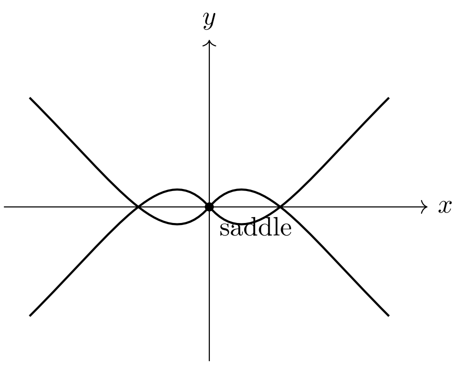


Figure 2.5: At a saddle point the function curves upward in one principal direction and downward in another.

2.4.15 Applications in Thermodynamics

Thermodynamic variables are often connected by state functions such as

$$U = U(S, V, N),$$

where U is internal energy, S entropy, V volume, and N particle number. Its differential is

$$\boxed{dU = \left(\frac{\partial U}{\partial S}\right)_{V,N} dS + \left(\frac{\partial U}{\partial V}\right)_{S,N} dV + \left(\frac{\partial U}{\partial N}\right)_{S,V} dN.} \tag{2.30}$$

Thermodynamics identifies

$$T = \left(\frac{\partial U}{\partial S}\right)_{V,N}, \qquad -p = \left(\frac{\partial U}{\partial V}\right)_{S,N}, \qquad \mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}.$$

Thus

$$\boxed{dU = T\,dS - p\,dV + \mu\,dN.}$$

The subscripts are essential: a thermodynamic partial derivative is defined by specifying which variables remain fixed.

2.4.16 Applications in Electromagnetism

An electrostatic potential $V(x, y, z)$ determines the electric field through

$$\mathbf{E} = -\nabla V.$$

Its Cartesian components are

$$E_x = -\frac{\partial V}{\partial x}, \qquad E_y = -\frac{\partial V}{\partial y}, \qquad E_z = -\frac{\partial V}{\partial z}.$$

The potential also satisfies Poisson's equation,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\varepsilon_0},$$

or Laplace's equation when $\rho = 0$.

2.4.17 Applications in Fluid Mechanics

A fluid velocity field is

$$\mathbf{v}(x, y, z, t) = (v_x, v_y, v_z).$$

The rate of change of a scalar quantity f experienced by a moving fluid element is

$$\boxed{\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f.} \tag{2.31}$$

This material derivative combines local time variation and advection through space. It is one of the central operators of continuum mechanics.

2.4.18 Applications in Quantum Mechanics

The non-relativistic wavefunction

$$\psi(x, y, z, t)$$

is a complex field over space and time. The time-dependent Schrödinger equation is

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z, t) \right] \psi.} \tag{2.32}$$

The first partial derivative in time controls evolution, while the second spatial partial derivatives represent kinetic energy through the Laplacian.

Separable Wavefunction

If

$$\psi(x, t) = X(x)T(t),$$

then

$$\frac{\partial \psi}{\partial t} = X(x)\frac{dT}{dt}, \qquad \frac{\partial^2 \psi}{\partial x^2} = T(t)\frac{d^2 X}{dx^2}.$$

Partial differentiation allows one factor to be treated as constant while differentiating the other, enabling separation of variables.

2.4.19 Common Mistakes

Frequent Errors

- Forgetting which variables are held fixed.
- Confusing $\partial f/\partial t$ with the total derivative df/dt.
- Assuming that existing partial derivatives automatically imply differentiability.
- Omitting the Jacobian determinant in a change of integration variables.
- Applying the Hessian test at a point that is not critical.
- Treating ∇ as an ordinary numerical vector before it acts on a field.

2.4.20 Exercises

1. Use the limit definition to calculate f_x and f_y for $f(x, y) = xy^2 + x^2$.
2. Find the tangent plane to $z = e^{xy}$ at $(1, 0)$.
3. Estimate the change in $f(x, y) = \sqrt{x^2 + y^2}$ when (x, y) changes from $(3, 4)$ to $(3.02, 3.99)$.
4. Compute the directional derivative of $f(x, y) = x^2 - y^2$ at $(2, 1)$ in the direction from $(2, 1)$ to $(5, 5)$.
5. Find the Jacobian determinant of the transformation $u = x + y$, $v = x - y$.
6. Derive the polar area element $dx\,dy = r\,dr\,d\theta$.
7. Classify the critical points of $f(x, y) = x^3 - 3x + y^2$.
8. For $T(x, y, t) = e^{-t}(x^2 + y^2)$ and a path $x = \cos t$, $y = \sin t$, compute dT/dt.
9. Show that the Hessian of $f(x, y) = \ln(x^2 + y^2)$ is symmetric away from the origin.
10. For a wavefunction $\psi(x, t) = Ae^{-\alpha x^2}e^{-i\omega t}$, compute $\partial\psi/\partial t$ and $\partial^2\psi/\partial x^2$.

Summary

Partial derivatives extend ordinary differentiation to functions of many variables. Each coordinate partial derivative is defined by an ordinary one-dimensional limit along a coordinate slice. When the function is differentiable, these partial derivatives combine into the best local linear approximation, the tangent plane, and the total differential.

Directional derivatives describe change in arbitrary directions. Jacobian matrices represent the first-order linearization of vector-valued mappings, while Jacobian determinants measure local changes of area and volume under coordinate transformations. Hessian matrices collect second derivatives and reveal curvature, stability, and the character of critical points.

In physics, these ideas appear in thermodynamic state relations, electric and gravitational potentials, fluid flow, continuum mechanics, and quantum wave equations. The next section develops the gradient, which packages the coordinate partial derivatives of a scalar field into a single geometric vector pointing in the direction of maximum increase.

2.5 The Gradient

A scalar field assigns one number to every point in a region of space. A mountain landscape assigns a height $h(x, y)$, a laboratory assigns a temperature $T(x, y, z)$, and electrostatics assigns a potential $V(x, y, z)$. A single partial derivative describes change only along one coordinate direction. Physical space, however, contains infinitely many directions. We therefore need one object that gathers all first-order directional information into a coordinate-independent geometric statement. That object is the *gradient*.

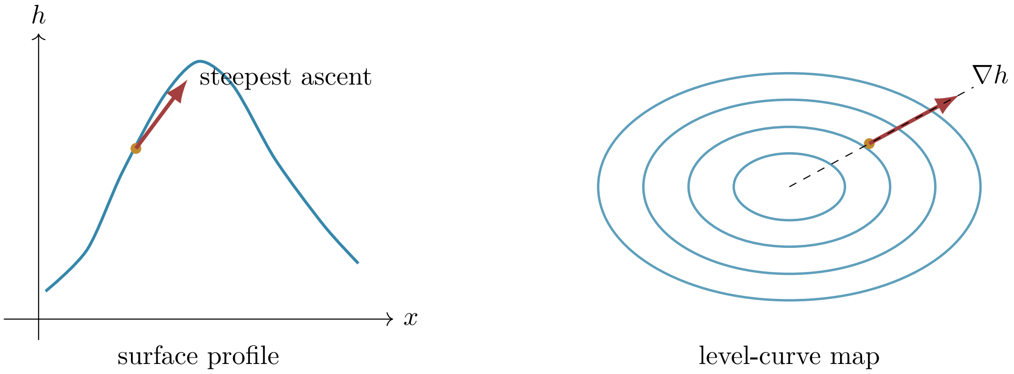


Figure 2.6: A scalar field may be represented by a surface or by level curves. The gradient points uphill and crosses each level curve orthogonally.

Motivation: Why One Partial Derivative Is Not Enough

Suppose that $h(x, y)$ gives the height of a mountain above a horizontal map. The derivative $\partial h/\partial x$ tells a hiker how rapidly the height changes while walking east, and $\partial h/\partial y$ gives the rate while walking north. Neither derivative alone answers the more natural question:

> In which direction should the hiker walk to climb as rapidly as possible?

The same question appears throughout physics. At a given point, in which direction does temperature rise fastest? In which direction does electric potential decrease fastest? Which way does pressure push a fluid element? The gradient answers all of these questions at once.

Definition of the Gradient

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be differentiable. In Cartesian coordinates the del operator is written

$$\nabla = \hat{\mathbf{x}}\frac{\partial}{\partial x} + \hat{\mathbf{y}}\frac{\partial}{\partial y} + \hat{\mathbf{z}}\frac{\partial}{\partial z}.$$

It is an operator: it instructs us to differentiate whatever scalar field follows it. Acting on f gives the gradient.

Gradient

For a differentiable scalar field $f(x, y, z)$,

$$\boxed{\nabla f = \frac{\partial f}{\partial x}\hat{\mathbf{x}} + \frac{\partial f}{\partial y}\hat{\mathbf{y}} + \frac{\partial f}{\partial z}\hat{\mathbf{z}}}$$

or, in component form,

$$\boxed{\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right).}$$

Each component measures the rate of change when only one coordinate varies. The three components together form a vector because first-order change depends linearly on displacement. For a small displacement

$$d\mathbf{r} = dx\,\hat{\mathbf{x}} + dy\,\hat{\mathbf{y}} + dz\,\hat{\mathbf{z}},$$

the total differential becomes

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz = \boxed{\nabla f \cdot d\mathbf{r}}.$$

Thus the gradient is the vector representation of the differential of f in Euclidean coordinates.

Units

If f has units $[f]$ and the spatial coordinates have units of length, then ∇f has units $[f]$/length. A temperature gradient is measured in kelvin per metre, a pressure gradient in pascals per metre, and an electric-potential gradient in volts per metre.

Geometric Interpretation

Let $\hat{\mathbf{u}}$ be a unit vector. A line through $\mathbf{r}_0$ in direction $\hat{\mathbf{u}}$ is

$$\mathbf{r}(s) = \mathbf{r}_0 + s\hat{\mathbf{u}}.$$

The rate at which f changes along this line is obtained with the chain rule:

$$\begin{aligned}\frac{d}{ds}f(\mathbf{r}(s)) &= \frac{\partial f}{\partial x}\frac{dx}{ds} + \frac{\partial f}{\partial y}\frac{dy}{ds} + \frac{\partial f}{\partial z}\frac{dz}{ds} \\ &= \nabla f \cdot \hat{\mathbf{u}}.\end{aligned}$$

This is the directional derivative.

Directional Derivative

For a differentiable scalar field f and a unit vector $\hat{\mathbf{u}}$,

$$\boxed{D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}}.}$$

The dot product appears because it extracts the component of ∇f along the chosen direction. If θ is the angle between ∇f and $\hat{\mathbf{u}}$, then

$$D_{\hat{\mathbf{u}}} f = \|\nabla f\| \cos\theta.$$

Because $-1 \leq \cos\theta \leq 1$,

$$-\|\nabla f\| \leq D_{\hat{\mathbf{u}}} f \leq \|\nabla f\|.$$

The largest value occurs for $\hat{\mathbf{u}} = \nabla f / \|\nabla f\|$.

Steepest-Ascent Theorem

At a point where $\nabla f \neq \mathbf{0}$:

- ∇f points in the direction of maximum increase of f;
- $\|\nabla f\|$ is the maximum directional derivative;
- $-\nabla f$ points in the direction of maximum decrease;
- every direction perpendicular to ∇f has zero first-order change.

Level Curves and Level Surfaces

A level surface of f is defined by

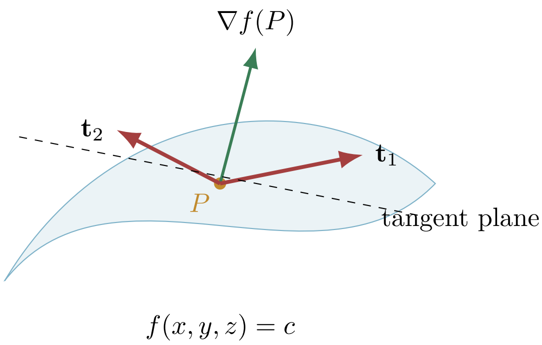


Figure 2.7: The gradient is normal to a level surface, while tangent vectors lie in the tangent plane.

$$f(x, y, z) = c,$$

where c is constant. Let $\mathbf{r}(s)$ be any curve lying within this surface. Since $f(\mathbf{r}(s)) = c$,

$$\frac{d}{ds} f(\mathbf{r}(s)) = 0.$$

Using the chain rule,

$$\nabla f \cdot \frac{d\mathbf{r}}{ds} = 0.$$

The tangent vector $d\mathbf{r}/ds$ may point in any tangent direction within the surface. Therefore ∇f is perpendicular to the whole tangent plane.

Orthogonality to Level Sets

At a regular point of the level surface $f = c$, the gradient ∇f is a normal vector to that surface. In two dimensions, it is normal to the level curve.

This gives the tangent-plane equation immediately. If $f(x_0, y_0, z_0) = c$, then

$$\boxed{\nabla f(x_0, y_0, z_0) \cdot [(x, y, z) - (x_0, y_0, z_0)] = 0.}$$

Worked Examples

Polynomial Surface

Let

$$f(x, y) = x^2 + 3xy - 2y^2.$$

Then

$$\nabla f = (2x + 3y,\ 3x - 4y).$$

At $(1, -1)$,

$$\nabla f(1, -1) = (-1, 7), \qquad \|\nabla f\| = \sqrt{50} = 5\sqrt{2}.$$

The unit direction of steepest ascent is

$$\hat{\mathbf{u}}_{\max} = \frac{1}{5\sqrt{2}}(-1, 7),$$

and the maximum rate of increase is $5\sqrt{2}$.

Directional Change Toward a Point

For $f(x, y, z) = x^2 + y^2 + z^2$, find the directional derivative at $P = (1, 2, 2)$ toward $Q = (3, 1, 6)$. First,

$$\nabla f = (2x, 2y, 2z), \qquad \nabla f(P) = (2, 4, 4).$$

The displacement is $Q - P = (2, -1, 4)$, so

$$\hat{\mathbf{u}} = \frac{(2, -1, 4)}{\sqrt{21}}.$$

Therefore

$$D_{\hat{\mathbf{u}}} f(P) = (2, 4, 4) \cdot \frac{(2, -1, 4)}{\sqrt{21}} = \frac{16}{\sqrt{21}}.$$

The normalization of $Q - P$ is essential; otherwise the result would depend on how the direction vector happened to be scaled.

Gaussian Field

Consider

$$f(x, y) = A \exp\left[-\frac{x^2 + y^2}{2\sigma^2}\right].$$

Differentiation gives

$$\nabla f = -\frac{f}{\sigma^2}(x, y).$$

For $A > 0$, the gradient points radially inward, toward the maximum at the origin. Its magnitude is

$$\|\nabla f\| = \frac{r}{\sigma^2} A e^{-r^2/(2\sigma^2)}, \qquad r = \sqrt{x^2 + y^2}.$$

The field is flat at the center, becomes steepest at $r = \sigma$, and then decays at large radius.

Tangent Plane to a Level Surface

The ellipsoid

$$F(x, y, z) = \frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1$$

contains the point $P = (1, 3/2, 1/\sqrt{2})$. Its normal vector is

$$\nabla F(P) = \left(\frac{1}{2}, \frac{1}{3}, \sqrt{2}\right).$$

Hence the tangent plane is

$$\frac{1}{2}(x - 1) + \frac{1}{3}\left(y - \frac{3}{2}\right) + \sqrt{2}\left(z - \frac{1}{\sqrt{2}}\right) = 0.$$

Temperature Fields and Heat Flow

Let $T(\mathbf{r}, t)$ be temperature. At fixed time, ∇T points toward increasing temperature. Heat flows in the opposite direction, from hot regions toward cold regions. Fourier's law is

$$\boxed{\mathbf{q} = -k\nabla T,}$$

where $\mathbf{q}$ is heat flux and $k > 0$ is thermal conductivity. The minus sign is physical: if temperature increases to the right, heat flows to the left.

A Three-Dimensional Temperature Field

Suppose

$$T(x, y, z) = T_0 + ax - by^2 - cz^2, \qquad a, b, c > 0.$$

Then

$$\nabla T = (a, -2by, -2cz), \qquad \mathbf{q} = (-ka, 2kby, 2kcz).$$

At points with $y, z > 0$, temperature increases in the negative y and negative z directions; accordingly, heat flux points toward positive y and positive z.

Combining Fourier's law with conservation of energy leads to the heat equation. If ρ is density and c_p is specific heat,

$$\rho c_p \frac{\partial T}{\partial t} = -\nabla \cdot \mathbf{q} = k\nabla^2 T.$$

Thus the gradient generates the heat flux, while the divergence measures the net flux leaving a small region.

Potential Energy and Conservative Forces

For a conservative system, the potential energy $U(\mathbf{r})$ determines the force:

$$\boxed{\mathbf{F} = -\nabla U.}$$

To understand the sign, displace the particle by $d\mathbf{r}$. The potential-energy change is

$$dU = \nabla U \cdot d\mathbf{r}.$$

The work done by the force is $dW = \mathbf{F} \cdot d\mathbf{r} = -dU$. Because this must hold for every displacement,

$$\mathbf{F} = -\nabla U.$$

Isotropic Harmonic Oscillator

For

$$U(x, y, z) = \frac{1}{2}k(x^2 + y^2 + z^2),$$

we obtain

$$\nabla U = k(x, y, z), \qquad \mathbf{F} = -k(x, y, z) = -k\mathbf{r}.$$

The potential rises radially outward, while the restoring force points back toward equilibrium.

Newtonian Gravitational Potential

Outside a spherical mass M,

$$\Phi(r) = -\frac{GM}{r}.$$

Since $\nabla r = \hat{\mathbf{r}}$ and $d(-GM/r)/dr = GM/r^2$,

$$\nabla\Phi = \frac{GM}{r^2}\hat{\mathbf{r}}.$$

The gravitational acceleration is

$$\mathbf{g} = -\nabla\Phi = -\frac{GM}{r^2}\hat{\mathbf{r}},$$

which points inward. The gradient itself points toward increasing potential, namely outward toward the less negative values of Φ.

Electric Potential and Electromagnetism

In electrostatics the electric field is the negative gradient of electric potential:

$$\boxed{\mathbf{E} = -\nabla V.}$$

Equipotential surfaces satisfy $V =$ constant. Since ∇V is normal to them, electric field lines cross equipotential surfaces orthogonally. A positive test charge accelerates toward decreasing potential energy qV.

Point-Charge Potential

For a point charge Q,

$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}.$$

Using $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$,

$$\nabla V = -\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}},$$

so

$$\mathbf{E} = -\nabla V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}}.$$

For $Q > 0$ the field points outward; for $Q < 0$ it points inward.

In time-dependent electromagnetism the electric field is more generally

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t},$$

where $\mathbf{A}$ is the vector potential. The gradient part remains the conservative, potential-derived contribution.

Pressure Gradients in Fluids

Pressure is a scalar field $p(\mathbf{r}, t)$. A small fluid element experiences greater force on the high-pressure side than on the low-pressure side. The net pressure force per unit volume is

$$\boxed{\mathbf{f}_p = -\nabla p.}$$

Accordingly, the pressure contribution to the acceleration is

$$\mathbf{a}_p = -\frac{1}{\rho}\nabla p.$$

This term appears in Euler's equation and in the Navier–Stokes equations.

Hydrostatic Equilibrium

Let z increase upward in a fluid at rest under gravity $\mathbf{g} = -g\hat{\mathbf{z}}$. Force balance requires

$$\mathbf{0} = -\nabla p + \rho \mathbf{g}.$$

Therefore

$$\frac{dp}{dz} = -\rho g.$$

Pressure decreases upward and increases with depth. The pressure gradient points upward, while $-\nabla p$ points downward and balances the gravitational body force in the chosen sign convention.

Curvilinear Coordinates

The geometric gradient is independent of coordinates, but its component formula changes because coordinate basis directions have different scale factors. In orthogonal coordinates (q_1, q_2, q_3) with scale factors h_1, h_2, h_3,

$$\boxed{\nabla f = \hat{\mathbf{e}}_1 \frac{1}{h_1}\frac{\partial f}{\partial q_1} + \hat{\mathbf{e}}_2 \frac{1}{h_2}\frac{\partial f}{\partial q_2} + \hat{\mathbf{e}}_3 \frac{1}{h_3}\frac{\partial f}{\partial q_3}.}$$

For cylindrical coordinates (ρ, ϕ, z),

$$\nabla f = \hat{\boldsymbol{\rho}} \frac{\partial f}{\partial \rho} + \hat{\boldsymbol{\phi}} \frac{1}{\rho}\frac{\partial f}{\partial \phi} + \hat{\mathbf{z}} \frac{\partial f}{\partial z}.$$

For spherical coordinates (r, θ, ϕ),

$$\nabla f = \hat{\mathbf{r}} \frac{\partial f}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}.$$

The factors $1/\rho$, $1/r$, and $1/(r\sin\theta)$ are not optional decorations; they compensate for the physical lengths associated with changes in angular coordinates.

Radial Scalar Field

If $f = f(r)$ depends only on distance from the origin, then all angular derivatives vanish and

$$\boxed{\nabla f = \frac{df}{dr}\hat{\mathbf{r}}.}$$

This identity underlies the gravitational and electrostatic examples above.

Critical Points and Optimization

A critical point satisfies

$$\nabla f = \mathbf{0}.$$

At such a point there is no first-order direction of increase or decrease. The point may be a local maximum, local minimum, saddle point, or a degenerate point. The Hessian matrix, introduced in the previous section, supplies the second-order information needed for classification.

Gradient ascent and gradient descent use the local linear model

$$f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot \Delta\mathbf{x}.$$

Choosing

$$\Delta\mathbf{x} = +\eta\nabla f$$

raises f locally, while

$$\boxed{\mathbf{x}_{n+1} = \mathbf{x}_n - \eta\nabla f(\mathbf{x}_n)}$$

implements gradient descent. The step size η must be chosen carefully: a very small value converges slowly, whereas a large value can overshoot or diverge.

The Del Operator and the Laplacian

The symbol ∇ is used to construct several differential operators. Acting on a scalar field, it produces the gradient. Taking the divergence of that gradient produces the Laplacian:

$$\boxed{\nabla^2 f \equiv \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.}$$

The gradient measures first-order spatial variation. The Laplacian compares a field value with its local surroundings and measures a form of net curvature or local spreading.

Do Not Treat ∇ as an Ordinary Vector

The notation resembles a vector, and many vector-algebra mnemonics are useful, but ∇ contains derivatives. Product rules, coordinate dependence, and the object on which it acts must always be respected.

Connection to Quantum Mechanics

A nonrelativistic wavefunction $\psi(\mathbf{r}, t)$ is a complex scalar field. Its gradient

$$\nabla\psi = \left(\frac{\partial\psi}{\partial x}, \frac{\partial\psi}{\partial y}, \frac{\partial\psi}{\partial z}\right)$$

encodes how both amplitude and phase vary in space. The momentum operator is

$$\boxed{\hat{\mathbf{p}} = -i\hbar\nabla.}$$

Applying momentum twice gives

$$\hat{\mathbf{p}}^2 = (-i\hbar)^2\nabla^2 = -\hbar^2\nabla^2.$$

Therefore the kinetic-energy operator is

$$\boxed{\hat{T} = \frac{\hat{\mathbf{p}}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2.}$$

The time-dependent Schrödinger equation is consequently

$$\boxed{i\hbar\frac{\partial\psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t)\right]\psi.}$$

Thus the gradient is not merely a geometric convenience. It is the first step toward the differential operators that generate momentum, kinetic energy, probability current, and wave propagation.

For a wavefunction written in polar form,

$$\psi = Re^{iS/\hbar},$$

its gradient is

$$\nabla\psi = e^{iS/\hbar}\left(\nabla R + \frac{i}{\hbar}R\nabla S\right).$$

The two terms distinguish amplitude variation from phase variation. In semiclassical mechanics, ∇S is identified with momentum, illustrating again that spatial phase gradients carry dynamical information.

Further Fully Solved Problems

Maximum Increase Under a Constraint on Direction

Let $f(x, y, z) = xy + yz + zx$. At $P = (1, 2, -1)$,

$$\nabla f = (y + z, z + x, x + y) = (1, 0, 3).$$

The maximum directional derivative is

$$\|\nabla f(P)\| = \sqrt{10},$$

and it occurs along

$$\hat{\mathbf{u}} = \frac{1}{\sqrt{10}}(1, 0, 3).$$

Any unit vector orthogonal to $(1, 0, 3)$ gives zero first-order change.

Direction of No Temperature Change

For $T(x, y) = x^2 + 4y^2$, at $(2, 1)$ we have

$$\nabla T = (4, 8).$$

A tangent direction to the level curve must satisfy

$$(4, 8) \cdot (u_x, u_y) = 0.$$

One choice is $(2, -1)$, so a unit direction is

$$\hat{\mathbf{u}} = \frac{1}{\sqrt{5}}(2, -1).$$

Indeed,

$$D_{\hat{\mathbf{u}}}T = (4, 8) \cdot \frac{1}{\sqrt{5}}(2, -1) = 0.$$

Electric Field from a Nonradial Potential

Let

$$V(x,y,z) = V_0\frac{z}{a} + \alpha xy.$$

Then

$$\nabla V = (\alpha y, \alpha x, V_0/a),$$

so

$$\mathbf{E} = (-\alpha y, -\alpha x, -V_0/a).$$

The constant z-component describes a uniform field, while the x and y components vary linearly across space.

Quantum Plane Wave

For

$$\psi(\mathbf{r},t) = Ae^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)},$$

we find

$$\nabla\psi = i\mathbf{k}\psi.$$

Therefore

$$\hat{\mathbf{p}}\psi = -i\hbar\nabla\psi = \hbar\mathbf{k}\psi.$$

The plane wave is an eigenfunction of momentum with eigenvalue $\hbar\mathbf{k}$.

Laplacian of a Gaussian

For the three-dimensional Gaussian

$$f(r) = e^{-\alpha r^2},$$

radial symmetry gives

$$\nabla f = -2\alpha\mathbf{r}\, e^{-\alpha r^2}.$$

Taking the divergence,

$$\nabla^2 f = \left(4\alpha^2 r^2 - 6\alpha\right) e^{-\alpha r^2}.$$

This expression is central in Gaussian trial wavefunctions and in the ground state of the harmonic oscillator.

Common Mistakes

1. **Forgetting to normalize a direction vector.** A directional derivative per unit distance requires a unit vector.
2. **Confusing ∇f with df.** The gradient is a vector; the differential is the scalar $df = \nabla f \cdot d\mathbf{r}$.
3. **Assuming the gradient points downhill.** It points toward increasing f; forces and fluxes often involve $-\nabla f$.
4. **Ignoring units.** The components of a gradient carry field units divided by coordinate units.
5. **Using Cartesian formulas in curvilinear coordinates.** Angular coordinates require scale factors.

6. **Concluding that $\nabla f = 0$ implies a minimum.** It identifies a critical point, not its type.
7. **Treating ∇ as an ordinary algebraic vector.** It is a differential operator and obeys product rules.

Exercises and Interpretation Questions

1. Compute ∇f for $f = x^3y - yz^2 + e^{xz}$ and evaluate it at $(1, 1, 0)$.
2. Find the direction and rate of greatest increase of $T = x^2 + 2y^2 + 3z^2$ at $(1, -1, 2)$.
3. Determine the tangent plane to $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.
4. A potential is $V = V_0 e^{-z/a} \cos(kx)$. Find $\mathbf{E} = -\nabla V$.
5. Show that if $f = f(r)$, then $\nabla f = f'(r)\hat{\mathbf{r}}$.
6. Explain geometrically why walking along a contour line gives zero directional derivative.
7. For $U = \frac{1}{2}(k_x x^2 + k_y y^2 + k_z z^2)$, find the force and identify the equilibrium point.
8. For $\psi = Ae^{i(kx-\omega t)}$, verify both $\hat{p}_x\psi = \hbar k\psi$ and $\hat{T}\psi = (\hbar^2k^2/2m)\psi$.

The Gradient

- The gradient packages all first-order spatial derivatives of a scalar field into one vector.
- ∇f points in the direction of steepest increase, and $\|\nabla f\|$ is the maximum rate of increase.
- The directional derivative is $D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}}$.
- Gradients are normal to level curves and level surfaces.
- Conservative forces, electric fields, heat fluxes, and pressure forces are generated by negative gradients.
- The del operator leads to the Laplacian ∇^2, the momentum operator $-i\hbar\nabla$, and the quantum kinetic-energy operator $-(\hbar^2/2m)\nabla^2$.

Looking Ahead

The gradient converts a scalar field into a vector field. The next chapter studies two complementary ways to differentiate vector fields. The divergence measures local source or sink strength, while the curl measures local rotational tendency. Together with the gradient and Laplacian, these operators form the language of Maxwell's equations, fluid mechanics, diffusion theory, general relativity in local coordinates, and quantum mechanics.

2.6 Directional Derivatives Along Paths

The gradient contains every first-order directional rate of change of a differentiable scalar field at one point. The directional derivative extracts one of those rates. This section emphasizes the distinction between choosing a direction at a point and following an actual path through space.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\mathbf{r}_0$, and let $\hat{\mathbf{u}}$ be a unit vector. The line through $\mathbf{r}_0$ in that direction is

$$\mathbf{r}(s) = \mathbf{r}_0 + s\hat{\mathbf{u}}.$$

Applying the chain rule gives

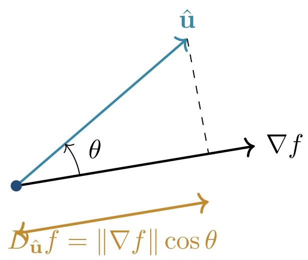


Figure 2.8: The directional derivative is the projection of the gradient onto the selected unit direction.

$$\left.\frac{d}{ds} f(\mathbf{r}_0 + s\hat{\mathbf{u}})\right|_{s=0} = \nabla f(\mathbf{r}_0) \cdot \hat{\mathbf{u}}.$$

Directional Derivative

For a differentiable scalar field f and a unit direction $\hat{\mathbf{u}}$,

$$\boxed{D_{\hat{\mathbf{u}}} f(\mathbf{r}_0) = \nabla f(\mathbf{r}_0) \cdot \hat{\mathbf{u}}.}$$

If θ is the angle between ∇f and $\hat{\mathbf{u}}$, then

$$D_{\hat{\mathbf{u}}} f = \|\nabla f\| \cos\theta.$$

This formula immediately explains the three important cases: maximal increase for $\theta = 0$, no first-order change for $\theta = \pi/2$, and maximal decrease for $\theta = \pi$.

Now let an observer follow a differentiable path $\mathbf{r}(t)$. The field value measured by the observer is the composite function $f(\mathbf{r}(t))$, whose derivative is

$$\boxed{\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt}.}$$

The velocity need not be a unit vector. Therefore this derivative includes both the direction of travel and the speed of travel. Writing $\mathbf{v} = v\hat{\mathbf{t}}$ gives

$$\frac{df}{dt} = v D_{\hat{\mathbf{t}}} f.$$

Physical Meaning

For a temperature field $T(\mathbf{r})$, the quantity $\nabla T \cdot \mathbf{v}$ is the rate at which a moving thermometer records temperature change. The field may be static, yet the reading changes because the observer moves through regions of different temperature.

Direction versus velocity

The definition $D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}}$ uses a unit vector. Replacing $\hat{\mathbf{u}}$ by a non-unit velocity vector computes change per unit time, not change per unit distance.

This distinction prepares the material derivative used in fluid mechanics and the convective derivative used throughout field theory.

2.7 A Visual Synthesis of Local Change

The principal ideas of differential calculus can be read as one sequence: a function assigns values, a limit describes controlled approach, a derivative supplies a local linear model, and a gradient or Jacobian packages that local model in several dimensions.

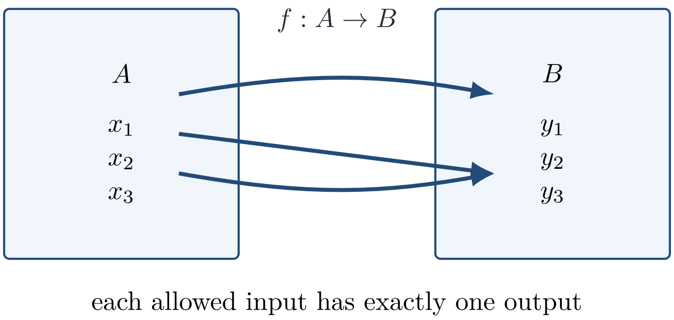


Figure 2.9: A function assigns exactly one output to every allowed input. Distinct inputs may share the same output.

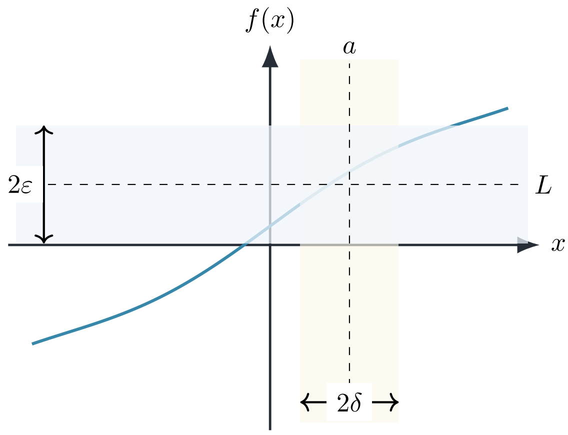


Figure 2.10: The ε–δ definition controls output error by restricting the input to a sufficiently small neighborhood.

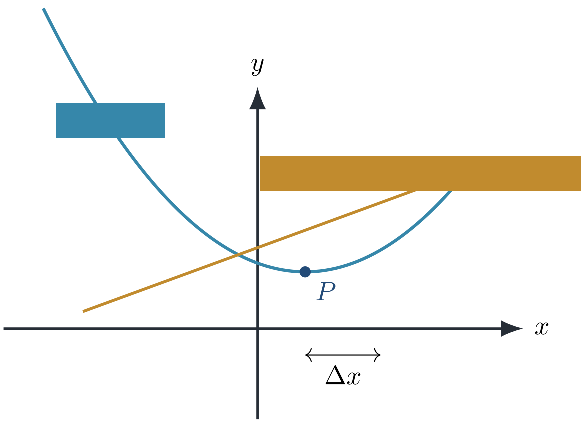


Figure 2.11: Near a differentiability point, a smooth curve is approximated by its tangent line. The approximation error is smaller than first order in Δx.

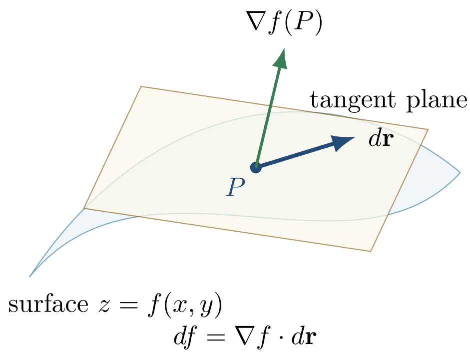


Figure 2.12: For a scalar field, the total differential is the linear response $df = \nabla f \cdot d\mathbf{r}$ represented on the tangent plane.

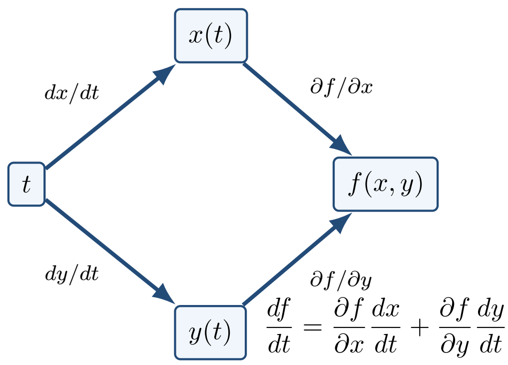


Figure 2.13: The multivariable chain rule sums the contributions from every dependency path.

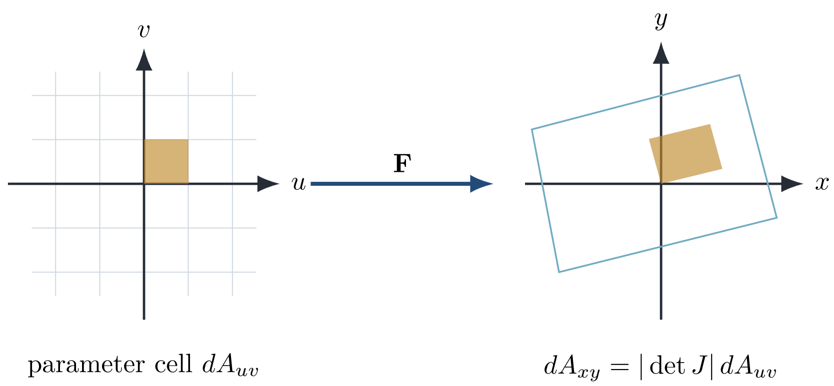
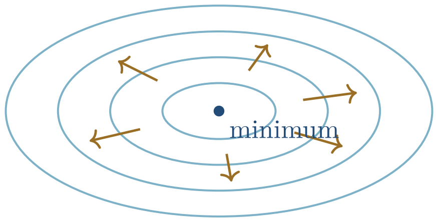


Figure 2.14: The Jacobian maps an infinitesimal parameter cell to a deformed cell. Its determinant gives the local oriented area-scaling factor.



gradient ascent crosses level curves orthogonally

Figure 2.15: Gradient trajectories cross level curves orthogonally and move toward increasing field values.

Together with Figures 2.1, 2.2, 2.3, 2.4, 2.6, and 2.7, this visual sequence supplies a geometric overview of the entire chapter.

2.8 Rigorous Synthesis: Differentiability as Local Linearity

The most economical way to unify one-variable and multivariable calculus is to define differentiability through approximation by a linear map.

Fréchet Differentiability in Euclidean Space

A function $F : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\mathbf{a}$ if there exists a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$F(\mathbf{a} + \mathbf{h}) = F(\mathbf{a}) + L\mathbf{h} + \mathbf{r}(\mathbf{h}),$$

where

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{\|\mathbf{r}(\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

The linear map L is the derivative $DF(\mathbf{a})$ and is represented in Cartesian coordinates by the Jacobian matrix.

This definition says more than the existence of coordinate partial derivatives. It says that one linear transformation predicts every sufficiently small displacement, regardless of the direction from which the point is approached.

Uniqueness of the Derivative

If a derivative $DF(\mathbf{a})$ exists, it is unique.

Proof. Suppose L_1 and L_2 both satisfy the definition. Then

$$(L_1 - L_2)\mathbf{h} = \mathbf{r}_2(\mathbf{h}) - \mathbf{r}_1(\mathbf{h}).$$

Choose $\mathbf{h} = t\mathbf{u}$ with $\|\mathbf{u}\| = 1$ and $t \neq 0$. Dividing by $|t|$ gives

$$\|(L_1 - L_2)\mathbf{u}\| \leq \frac{\|\mathbf{r}_1(t\mathbf{u})\|}{|t|} + \frac{\|\mathbf{r}_2(t\mathbf{u})\|}{|t|}.$$

The right-hand side tends to zero as $t \to 0$, so $(L_1 - L_2)\mathbf{u} = 0$ for every unit vector $\mathbf{u}$. Hence $L_1 = L_2$. ☐

Differentiability Implies Continuity

If F is differentiable at $\mathbf{a}$, then F is continuous at $\mathbf{a}$.

Proof. From the definition,

$$F(\mathbf{a}+\mathbf{h})-F(\mathbf{a})=DF(\mathbf{a})\mathbf{h}+\mathbf{r}(\mathbf{h}).$$

A linear map on finite-dimensional Euclidean space is bounded, so for some $C>0$,

$$\|DF(\mathbf{a})\mathbf{h}\|\le C\|\mathbf{h}\|.$$

Therefore

$$\|F(\mathbf{a}+\mathbf{h})-F(\mathbf{a})\|\le C\|\mathbf{h}\|+\|\mathbf{r}(\mathbf{h})\|\to 0.$$

Thus $F(\mathbf{a}+\mathbf{h})\to F(\mathbf{a})$. □

Multivariable Chain Rule

If $F:\mathbb{R}^n\to\mathbb{R}^m$ is differentiable at $\mathbf{a}$ and $G:\mathbb{R}^m\to\mathbb{R}^p$ is differentiable at $F(\mathbf{a})$, then $G\circ F$ is differentiable at $\mathbf{a}$ and

$$\boxed{D(G\circ F)(\mathbf{a})=DG(F(\mathbf{a}))\,DF(\mathbf{a}).}$$

displayed in coordinates as multiplication of Jacobian matrices.

Proof. Write

$$F(\mathbf{a}+\mathbf{h})=F(\mathbf{a})+A\mathbf{h}+\mathbf{r}_F(\mathbf{h})$$

and, with $\mathbf{k}=A\mathbf{h}+\mathbf{r}_F(\mathbf{h})$,

$$G(F(\mathbf{a})+\mathbf{k})=G(F(\mathbf{a}))+B\mathbf{k}+\mathbf{r}_G(\mathbf{k}).$$

Substitution yields

$$G(F(\mathbf{a}+\mathbf{h}))=G(F(\mathbf{a}))+BA\mathbf{h}+B\mathbf{r}_F(\mathbf{h})+\mathbf{r}_G(\mathbf{k}).$$

Both remaining terms are $o(\|\mathbf{h}\|)$ because $\mathbf{k}=O(\|\mathbf{h}\|)$. Therefore the derivative is BA. □

Steepest-Ascent Result

For a differentiable scalar field f at a point where $\nabla f\neq\mathbf{0}$,

$$\max_{\|\hat{\mathbf{u}}\|=1} D_{\hat{\mathbf{u}}}f=\|\nabla f\|,$$

attained uniquely at $\hat{\mathbf{u}}=\nabla f/\|\nabla f\|$.

Proof. By the Cauchy–Schwarz inequality from Chapter 1,

$$D_{\hat{\mathbf{u}}}f=\nabla f\cdot\hat{\mathbf{u}}\le\|\nabla f\|\|\hat{\mathbf{u}}\|=\|\nabla f\|.$$

Equality holds exactly when the two vectors are parallel with the same orientation. □

Partial derivatives are not the whole derivative

The existence of every first partial derivative at a point does not by itself guarantee differentiability there. Differentiability requires one linear approximation that controls the error in all directions simultaneously. Continuity of the first partial derivatives in a neighborhood is a widely used sufficient condition.

Why physicists should care

Local linearity is the hidden structure behind tangent lines, response coefficients, stability matrices, normal-mode analysis, coordinate transformations, error propagation, and perturbation theory. The Jacobian is not merely a table of partial derivatives; it is the best first-order linear model of a nonlinear map.

2.9 Computational Companion

Computation is most useful here when it makes local geometry visible or checks an analytic calculation. The source files for this section are stored in `examples/python/ch02/` and `examples/wolfram/ch02/`.

2.9.1 Python: Surfaces, Gradients, and Jacobians

The complete Python program uses NumPy for array calculations and Matplotlib for visualization. It generates the figures in this section and verifies a numerically approximated Jacobian against its exact analytic form.

Listing 2.1: Python computational companion for Chapter 2.

```python
"""Computational companion for Chapter 2: Differential Calculus."""
from __future__ import annotations

from pathlib import Path
import numpy as np
import matplotlib.pyplot as plt

OUT = Path(__file__).resolve().parents[3] / "generated" / "ch02"
OUT.mkdir(parents=True, exist_ok=True)


def save(fig: plt.Figure, name: str) -> None:
    fig.tight_layout()
    fig.savefig(OUT / name, bbox_inches="tight")
    plt.close(fig)


def surface_and_contours() -> None:
    x = np.linspace(-2.5, 2.5, 180)
    y = np.linspace(-2.5, 2.5, 180)
    X, Y = np.meshgrid(x, y)
    Z = X**2 + 0.5 * Y**2
    fig = plt.figure(figsize=(6.2, 4.7))
    ax = fig.add_subplot(111, projection="3d")
    ax.plot_surface(X, Y, Z, rstride=5, cstride=5, linewidth=0.15, alpha=0.85)
    ax.set_xlabel("x")
    ax.set_ylabel("y")
    ax.set_zlabel("f(x,y)")
    ax.set_title(r"Surface: $f(x,y)=x^2+\frac{1}{2} y^2$")
    save(fig, "surface_plot.pdf")


def gradient_field() -> None:
    x = np.linspace(-2.2, 2.2, 19)
    y = np.linspace(-2.2, 2.2, 19)
```

```
    X, Y = np.meshgrid(x, y)
    F = X**2 + Y**2
    U, V = 2 * X, 2 * Y
    mag = np.hypot(U, V)
    U = np.divide(U, mag, out=np.zeros_like(U), where=mag > 0)
    V = np.divide(V, mag, out=np.zeros_like(V), where=mag > 0)
    fig, ax = plt.subplots(figsize=(5.8, 5.0))
    ax.contour(X, Y, F, levels=9)
    ax.quiver(X, Y, U, V)
    ax.set_aspect("equal")
    ax.set_xlabel("x")
    ax.set_ylabel("y")
    ax.set_title(r"Normalized gradient field of $f=x^2+y^2$")
    save(fig, "gradient_field.pdf")


def jacobian_transformation() -> None:
    u = np.linspace(-1.5, 1.5, 17)
    v = np.linspace(-1.5, 1.5, 17)
    fig, ax = plt.subplots(figsize=(6.0, 5.0))
    for c in u:
        vv = np.linspace(-1.5, 1.5, 200)
        xx = c + 0.35 * vv
        yy = 0.25 * c + vv + 0.12 * c * vv
        ax.plot(xx, yy, linewidth=0.8)
    for c in v:
        uu = np.linspace(-1.5, 1.5, 200)
        xx = uu + 0.35 * c
        yy = 0.25 * uu + c + 0.12 * uu * c
        ax.plot(xx, yy, linewidth=0.8)
    ax.set_aspect("equal")
    ax.set_xlabel("x")
    ax.set_ylabel("y")
    ax.set_title("A nonlinear coordinate map and its local Jacobian")
    save(fig, "jacobian_grid.pdf")


def directional_derivative() -> None:
    theta = np.linspace(0, 2 * np.pi, 400)
    grad = np.array([3.0, 1.0])
    rates = grad[0] * np.cos(theta) + grad[1] * np.sin(theta)
    fig, ax = plt.subplots(figsize=(6.0, 4.2))
    ax.plot(theta, rates)
    ax.axhline(np.linalg.norm(grad), linestyle="--", linewidth=0.8)
    ax.axhline(-np.linalg.norm(grad), linestyle="--", linewidth=0.8)
    ax.set_xlabel(r"direction angle $\theta$")
    ax.set_ylabel(r"$D_{\hat{u}}f$")
    ax.set_title(r"Directional derivative for $\nabla f=(3,1)$")
    ax.set_xticks([0, np.pi / 2, np.pi, 3 * np.pi / 2, 2 * np.pi])
    ax.set_xticklabels(["0", r"$\pi/2$", r"$\pi$", r"$3\pi/2$", r"$2\pi$"])
    save(fig, "directional_derivative.pdf")


def numerical_jacobian(func, point: np.ndarray, step: float = 1e-6) -> np.ndarray:
```

```
    point = np.asarray(point, dtype=float)
    f0 = np.asarray(func(point), dtype=float)
    jac = np.empty((f0.size, point.size), dtype=float)
    for j in range(point.size):
        delta = np.zeros_like(point)
        delta[j] = step
        jac[:, j] = (np.asarray(func(point + delta)) - np.asarray(func(point - delta
    ))) / (2 * step)
    return jac


def verify_numerical_jacobian() -> None:
    func = lambda q: np.array([q[0] ** 2 * q[1], np.sin(q[0] + q[1])])
    p = np.array([1.2, -0.4])
    numeric = numerical_jacobian(func, p)
    exact = np.array([[2 * p[0] * p[1], p[0] ** 2], [np.cos(p.sum()), np.cos(p.sum()
    )]])
    if not np.allclose(numeric, exact, rtol=1e-5, atol=1e-7):
        raise RuntimeError("Numerical Jacobian verification failed")
    print("Numerical Jacobian verified at", p)
    print(numeric)


if __name__ == "__main__":
    surface_and_contours()
    gradient_field()
    jacobian_transformation()
    directional_derivative()
    verify_numerical_jacobian()
```

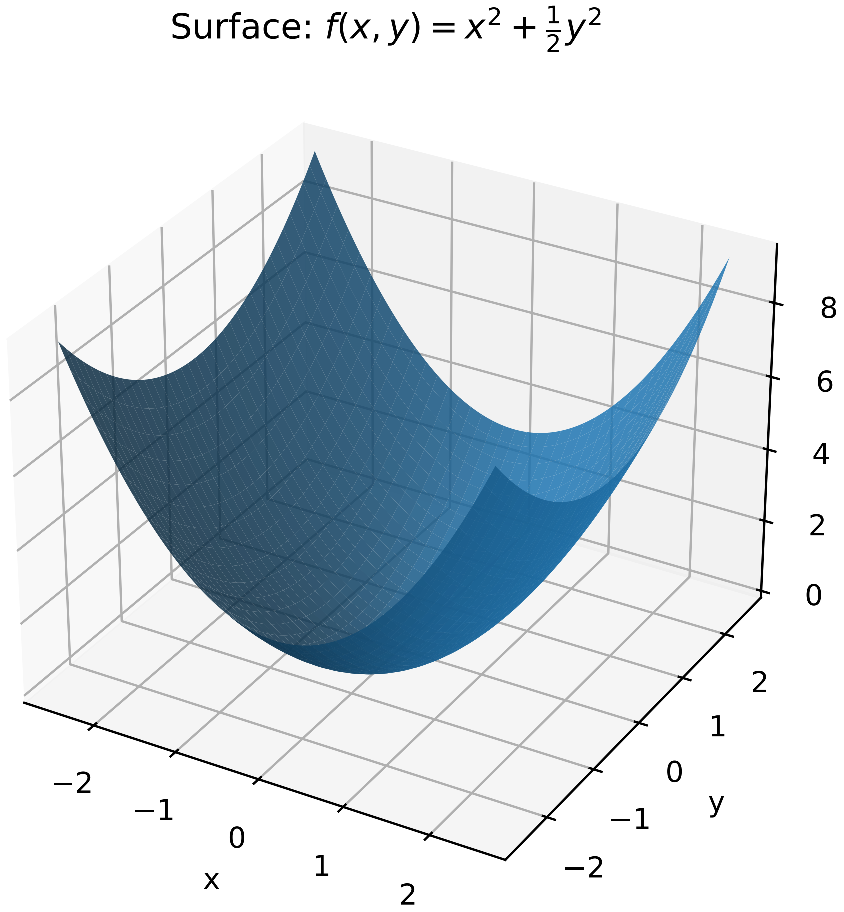


Figure 2.16: A surface plot makes the scalar field $f(x, y) = x^2 + \frac{1}{2}y^2$ visible as a geometric object.

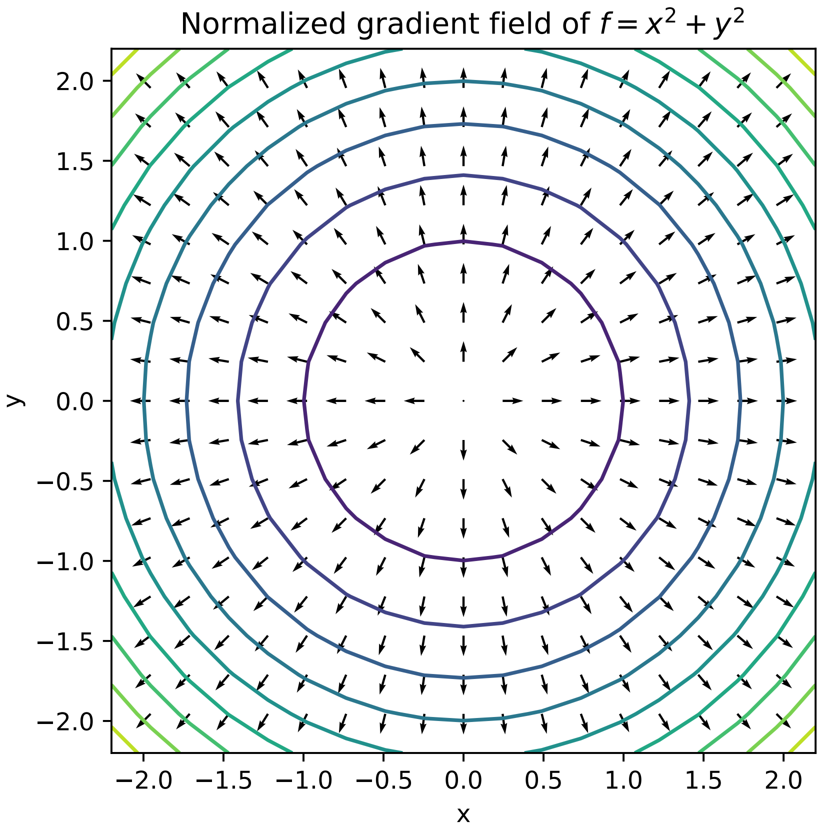


Figure 2.17: Normalized gradient vectors cross the level curves of $f = x^2 + y^2$ orthogonally.

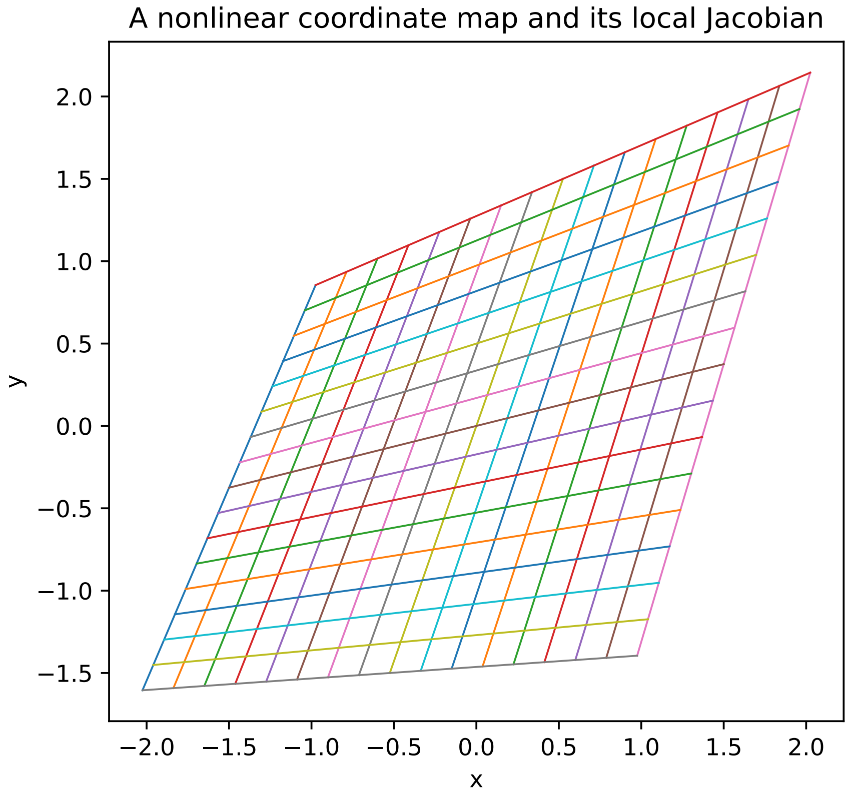


Figure 2.18: A nonlinear map deforms a rectangular parameter grid. The Jacobian at each point is the best local linear description of that deformation.

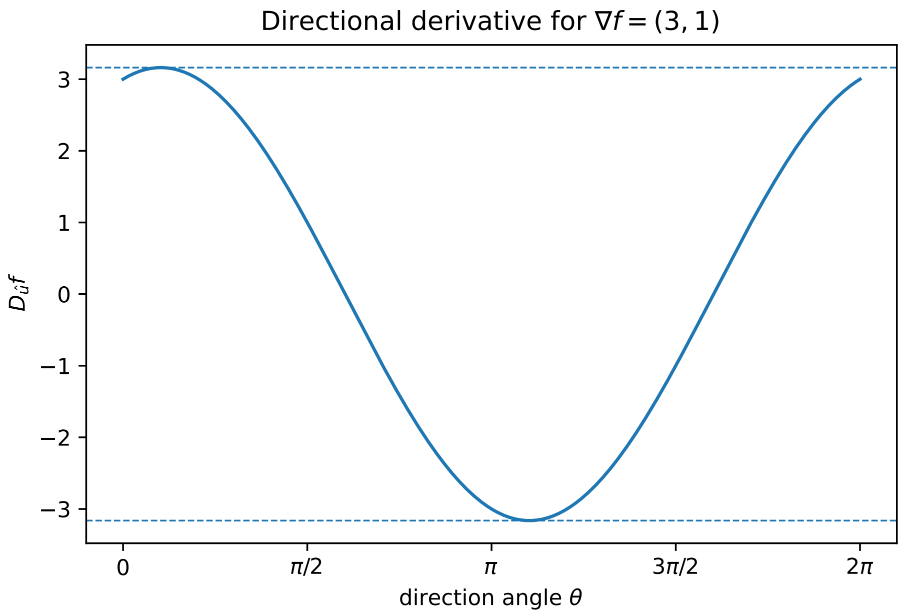


Figure 2.19: For a fixed gradient, the directional derivative varies sinusoidally with direction and is bounded by $\pm\|\nabla f\|$.

Numerical differentiation

A centered finite difference usually has smaller truncation error than a forward difference. Nevertheless, numerical derivatives depend on the step size: a step that is too large produces truncation error, while a step that is too small magnifies floating-point cancellation. Analytic derivatives remain the reference whenever they are available.

2.9.2 Wolfram Language: Symbolic and Visual Checks

The Wolfram Language companion differentiates symbolically, computes gradients and Hessians, plots scalar and vector fields, and evaluates a Jacobian determinant.

Listing 2.2: Wolfram Language companion for Chapter 2.

```
(* Chapter 2 computational companion: Wolfram Language *)
ClearAll[x, y, u, v, f, map];

f[x_, y_] := x^2 + y^2;

(* Gradient and Hessian *)
gradient = Grad[f[x, y], {x, y}]
hessian = D[f[x, y], {{x, y}, 2}]

(* Surface and level curves *)
Plot3D[f[x, y], {x, -2, 2}, {y, -2, 2},
 AxesLabel -> {"x", "y", "f"}, PlotTheme -> "Scientific"]
ContourPlot[f[x, y], {x, -2, 2}, {y, -2, 2},
 Contours -> 10, FrameLabel -> {"x", "y"}]

(* Gradient vector field *)
VectorPlot[Evaluate[gradient], {x, -2, 2}, {y, -2, 2},
 VectorScale -> {Small, Scaled[0.45], None}]

(* Jacobian of a nonlinear map *)
map[u_, v_] := {u + 0.35 v, 0.25 u + v + 0.12 u v};
jacobian = D[map[u, v], {{u, v}}]
Det[jacobian] // Simplify

(* Directional derivative at a point *)
point = {1, -1};
direction = Normalize[{2, 1}];
directionalDerivative = gradient.direction /. Thread[{x, y} -> point]
```

A plot is not a proof

A plot may suggest continuity, differentiability, or the location of an extremum, but it cannot establish these properties by itself. Sampling can miss singularities or narrow features. Use visualization to form and test intuition, then use definitions and analysis to justify the conclusion.

2.10 Historical Pathways into Differential Calculus

Differential calculus was not created in a single moment. Its modern language emerged from a sequence of attempts to describe motion, curvature, fields, and local approximation. The following notes emphasize the ideas that remain active in contemporary physics rather than treating history as a list of dates. Historical perspective in this section follows the broad development described by Boyer and Merzbach [1], while the mathematical language is aligned with modern analysis texts [2, 3].

2.10.1 Leibniz: differentials as a language of change

Historical Note

Gottfried Wilhelm Leibniz introduced the notation dx, dy, and $\int$, presenting differentiation and integration as mutually related operations. His notation made the chain rule, substitution, and higher differentials transparent enough to become a working language shared across mathematics and physics. Modern analysis gives limits and linear maps the logical role once assigned to infinitesimals, but the Leibniz notation remains powerful because it records which quantities vary and how their variations are composed.

Connection to physics: virtual displacements

In analytical mechanics, dq^i denotes an admissible infinitesimal change of a generalized coordinate, while dS, dA, and dV encode oriented elements of curves, surfaces, and volumes. Leibniz's notation therefore survives not as decoration but as a compact grammar for local physical variation.

2.10.2 Euler: functions become the basic objects

Historical Note

Leonhard Euler consolidated the function concept and developed systematic methods for differentiating exponential, logarithmic, and trigonometric expressions. He also connected differential equations to mechanics and fluid motion. The shift from isolated curves to functions of several variables was essential: a physical field is naturally a function assigning a value to every point of space and time.

Connection to physics: fields

Temperature $T(\mathbf{r}, t)$, pressure $p(\mathbf{r}, t)$, electric potential $\phi(\mathbf{r}, t)$, and a wavefunction $\psi(\mathbf{r}, t)$ are all Eulerian objects: functions whose derivatives express local physical response.

2.10.3 Lagrange: mechanics through derivatives of energy

Historical Note

Joseph-Louis Lagrange reformulated mechanics in generalized coordinates and made partial derivatives central to dynamics. Instead of resolving every constraint force geometrically, one differentiates a single scalar function, the Lagrangian $L(q^i, \dot{q}^i, t)$. This is an early and profound example of a physical theory being organized by the differential structure of a function.

Connection to physics: Euler–Lagrange equations

The equations

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0$$

combine ordinary, partial, and total derivatives. Every symbol in this formula depends on the calculus developed in this chapter.

2.10.4 Gauss: local sources and global flux

Historical Note

Carl Friedrich Gauss connected geometry, potential theory, and field measurement. His work helped establish the principle that a local source density can be related to the flux through a surrounding boundary. Although the divergence theorem belongs to the next chapter, its local ingredient is the derivative of a vector field.

Connection to physics: charge density

Gauss's law appears locally as $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ and globally as a flux law. Differential calculus supplies the local statement; integral calculus proves that all local contributions assemble into the boundary flux.

2.10.5 Green: potentials, boundaries, and field reconstruction

Historical Note

George Green developed identities linking derivatives inside a region to data on its boundary. His ideas introduced the Green function, which converts a differential equation with a localized source into an integral representation. This viewpoint now permeates electrostatics, wave theory, quantum mechanics, and quantum field theory.

Connection to physics: response to a point source

A Green function answers a universal question: what field is produced by a unit point source? Once that response is known, linearity and integration build the response to an arbitrary source distribution.

2.10.6 Stokes: circulation and local rotation

Historical Note

The theorem associated with George Gabriel Stokes relates circulation around a closed curve to the curl distributed across a spanning surface. Its modern form crystallizes a recurring pattern: differentiation measures local structure, while integration accumulates it into an observable boundary effect.

Connection to physics: induction

Faraday's law may be written locally as $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ and globally as an emf around a loop. The derivative tells how the field rotates locally; the line integral records the measurable circulation.

2.10.7 Hamilton: derivatives on phase space

Historical Note

William Rowan Hamilton recast mechanics in terms of coordinates q^i and conjugate momenta p_i. The Hamiltonian $H(q, p, t)$ generates time evolution through its partial derivatives. This formulation exposes the geometry of phase space and provides the closest classical precursor to the operator language of quantum mechanics.

Connection to physics: generators of motion

Hamilton's equations,

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i},$$

show that dynamics is encoded in the gradient of energy on phase space. Quantum evolution later replaces Poisson brackets by commutators.

2.10.8 Gibbs: vector calculus as the language of fields

Historical Note

Josiah Willard Gibbs helped standardize the vector notation now used for the gradient, divergence, and curl. This compressed quaternionic and component calculations into a practical language suited to Maxwell's electromagnetic theory and continuum physics. The symbols ∇f, $\nabla\cdot\mathbf{F}$, and $\nabla\times\mathbf{F}$ separate three distinct kinds of derivative: steepest scalar change, local source strength, and local rotation.

Connection to physics: one operator and several structures

The nabla operator acquires meaning from what follows it. Acting on a scalar it produces a vector; contracted with a vector it produces a scalar; crossed with a vector it produces a pseudovector. This dimensional and geometric bookkeeping is indispensable in electromagnetism and fluid mechanics.

Historical synthesis

Leibniz supplied a notation for infinitesimal change; Euler made functions the central objects; Lagrange and Hamilton encoded mechanics in derivatives of energy; Gauss, Green, and Stokes connected local derivatives to global measurements; Gibbs supplied the vector language in which modern field theory is written. The common thread is local linearization: near a point, a smooth physical law is controlled by its derivative.

2.11 Twenty-Five Integrated Worked Examples

These examples are deliberately cumulative. Each states the governing idea, performs the calculation, and closes with a physical or geometric reading of the result.

Example 2.1

Domain and level set.
Problem. For $f(x,y) = \sqrt{4 - x^2 - y^2}$, determine the domain and the level set $f = 1$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

Reality of the square root requires $x^2 + y^2 \leq 4$. Setting $f = 1$ gives $4 - x^2 - y^2 = 1$, hence $x^2 + y^2 = 3$, a circle of radius $\sqrt{3}$.
Interpretation. The graph is the upper hemisphere of radius 2; horizontal slicing produces circular level sets.

Example 2.2

A removable limit.
Problem. Evaluate $\lim_{x\to 2}(x^2 - 4)/(x - 2)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

Factor $x^2 - 4 = (x - 2)(x + 2)$. For $x \neq 2$ the quotient equals $x + 2$, so the limit is 4.
Interpretation. The value at the missing point is irrelevant to the limit; only nearby values matter.

Example 2.3

Two-path test.
Problem. Show that $f(x,y) = xy/(x^2 + y^2)$ has no limit at $(0,0)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

Along $y = x$, $f = 1/2$. Along $y = -x$, $f = -1/2$. Since two paths give different values, the limit does not exist.
Interpretation. Multivariable approach has infinitely many directions, so one-dimensional intuition must be strengthened.

Example 2.4

Derivative from first principles.
Problem. Use the definition to differentiate $f(x) = x^2$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$[f(x+h) - f(x)]/h = [2xh + h^2]/h = 2x + h$. Taking $h \to 0$ gives $f'(x) = 2x$.
Interpretation. The derivative is the coefficient of the best local linear approximation.

Example 2.5

Linearization error.
Problem. Approximate $\sqrt{4.1}$ by linearizing $f(x) = \sqrt{x}$ at $x = 4$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$f(4) = 2$ and $f'(4) = 1/4$. Thus $f(4.1) \approx 2 + (1/4)(0.1) = 2.025$.
Interpretation. A nonlinear function is replaced locally by its tangent line; the neglected error is second order.

Example 2.6

Velocity and acceleration.
Problem. For $x(t) = t^3 - 3t^2 + 2t$, find v and a at $t = 2$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$v = 3t^2 - 6t + 2$ and $a = 6t - 6$. Therefore $v(2) = 2$ and $a(2) = 6$.
Interpretation. Position, velocity, and acceleration are successive time derivatives.

Example 2.7

Product and chain rules.
Problem. Differentiate $y = x^2 e^{3x}$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$y' = 2xe^{3x} + 3x^2 e^{3x} = e^{3x}(2x + 3x^2)$.
Interpretation. The product rule combines independent changes; the chain rule propagates change through composition.

Example 2.8

Implicit differentiation.
Problem. For $x^2 + xy + y^2 = 7$, find dy/dx.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

Differentiating gives $2x + y + xy' + 2yy' = 0$, hence $y' = -(2x + y)/(x + 2y)$.
Interpretation. The curve can have a well-defined tangent even where solving globally for $y(x)$ is inconvenient.

Example 2.9

Partial derivatives.
Problem. For $T = x^2y + e^{yz}$, compute T_x, T_y, T_z.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$T_x = 2xy$, $T_y = x^2 + ze^{yz}$, and $T_z = ye^{yz}$.
Interpretation. A partial derivative varies one coordinate while holding the others fixed.

Example 2.10

Total differential.
Problem. Estimate the change in $f = x^2y$ from $(2, 3)$ under $(dx, dy) = (0.01, -0.02)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$df = 2xy\,dx + x^2\,dy$. At $(2, 3)$, $df = 12(0.01) + 4(-0.02) = 0.04$.
Interpretation. The differential combines all first-order coordinate effects into one predicted change.

Example 2.11

Tangent plane.
Problem. Find the tangent plane to $z = x^2 + xy$ at $(1, 2, 3)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$f_x = 2x + y = 4$ and $f_y = x = 1$. Thus $z - 3 = 4(x - 1) + (y - 2)$.
Interpretation. The plane is the graph of the derivative, the best local linear model of the surface.

Example 2.12

Gradient.
Problem. For $f = x^2 + 2y^2 + 3z^2$, find ∇f at $(1, -1, 2)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$\nabla f = (2x, 4y, 6z)$, hence $\nabla f(1, -1, 2) = (2, -4, 12)$.
Interpretation. The gradient points in the direction of greatest instantaneous increase.

Example 2.13

Directional derivative.
Problem. For $f = x^2 + y^2$, compute the derivative at $(1, 2)$ toward $(4, 6)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

The direction is $(3, 4)$, so $\hat{u} = (3/5, 4/5)$. Since $\nabla f = (2, 4)$, $D_{\hat{u}} f = 2(3/5)+4(4/5) = 22/5$.
Interpretation. Direction must be normalized; otherwise the result includes an arbitrary speed factor.

Example 2.14

Steepest descent.
Problem. For $V = x^2 + xy + 3y^2$, find the steepest-descent direction at $(1, 1)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$\nabla V = (2x + y, x + 6y) = (3, 7)$. The unit steepest-descent direction is $-(3, 7)/\sqrt{58}$.
Interpretation. For a potential energy, negative gradient gives the force direction.

Example 2.15

Jacobian matrix.
Problem. For $F(u, v) = (u^2 - v, uv)$, compute $J_F(1, 2)$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$J_F = \begin{pmatrix} 2u & -1 \\ v & u \end{pmatrix}$, hence $J_F(1, 2) = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix}$.
Interpretation. The Jacobian is the local linear transformation carrying input perturbations to output perturbations.

Example 2.16

Jacobian determinant.
Problem. For polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, find the area scale factor.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$$\det \partial(x,y)/\partial(r,\theta) = \det \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r.$$

Interpretation. A small polar rectangle $dr\, dheta$ maps to area $r\, dr\, dheta$.

Example 2.17

Multivariable chain rule.
Problem. Let $z = x^2 y$, $x = t^2$, $y = e^t$. Find dz/dt.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$z = t^4 e^t$, so $dz/dt = e^t(4t^3 + t^4)$. Equivalently, $z_x x' + z_y y'$ gives the same result.
Interpretation. The chain rule sums change transmitted through every dependency path.

Example 2.18

Critical point classification.
Problem. Classify the origin for $f = x^2 + 4xy + y^2$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

The Hessian is $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$ with determinant $-12 < 0$, so the origin is a saddle point.
Interpretation. A zero gradient is only a candidate; the Hessian determines the local quadratic geometry.

Example 2.19

Constrained optimization.
Problem. Maximize xy subject to $x^2 + y^2 = 2$ in the first quadrant.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$\nabla(xy) = \lambda\nabla(x^2 + y^2)$ gives $y = 2\lambda x$ and $x = 2\lambda y$, hence $x = y$. The constraint gives $x = y = 1$, so the maximum is 1.
Interpretation. At a constrained extremum, the objective gradient is normal to the constraint surface.

Example 2.20

Radial derivative.
Problem. For $f(\mathbf{r}) = r^{-1}$ in three dimensions, compute ∇f for $r \neq 0$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

Since $df/dr = -r^{-2}$ and $\nabla r = \mathbf{r}/r$, $\nabla(r^{-1}) = -\mathbf{r}/r^3$.
Interpretation. Inverse-square fields arise as gradients of inverse-distance potentials.

Example 2.21

Divergence.
Problem. For $\mathbf{F} = (x^2, xy, z^2)$, compute $\nabla \cdot \mathbf{F}$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$\nabla \cdot \mathbf{F} = 2x + x + 2z = 3x + 2z$.
Interpretation. Positive divergence represents local net outflow; negative divergence represents local convergence.

Example 2.22

Curl.
Problem. For $\mathbf{F} = (-y, x, 0)$, compute $\nabla \times \mathbf{F}$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$\nabla \times \mathbf{F} = (0, 0, \partial_x x - \partial_y(-y)) = (0, 0, 2)$.
Interpretation. The field has uniform local rotation about the z axis.

Example 2.23

Laplacian.
Problem. Compute $\nabla^2 e^{-\alpha(x^2+y^2)}$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

Differentiating twice gives $\nabla^2 f = [4\alpha^2(x^2 + y^2) - 4\alpha]e^{-\alpha(x^2+y^2)}$.
Interpretation. The Laplacian compares a field value with its local neighborhood and drives diffusion and wave equations.

Example 2.24

Complex phase gradient.
Problem. For $\psi = Ae^{i\mathbf{k}\cdot\mathbf{r}}$, compute $\nabla\psi$.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$\nabla\psi = i\mathbf{k}\psi$.
Interpretation. A plane wave is an eigenfunction of the gradient; its phase gradient is the wavevector.

Example 2.25

Error propagation.
Problem. If $R = V/I$, estimate dR for small independent changes dV, dI.
Theory. Identify the relevant derivative or local-linear principle before calculating.

Solution

$dR = (1/I)dV - (V/I^2)dI$. Thus $dR/R = dV/V - dI/I$ to first order.
Interpretation. Differentials propagate measurement errors through a derived quantity.

2.12 Graded Exercise Bank

The exercises are graded by mathematical maturity rather than by algebraic length. Basic problems establish fluency; intermediate problems combine ideas; advanced problems emphasize proof and structure; challenge problems connect the chapter to mechanics, field theory, and quantum mechanics.

2.12.1 Basic fluency

1. State the domain and range of $f(x) = \sqrt{9 - x^2}$.
2. Sketch the level sets of $f(x, y) = x + y$ for levels $-1, 0, 1$.
3. Evaluate $\lim_{x\to 3}(x^2 - 9)/(x - 3)$.
4. Evaluate $\lim_{h\to 0}[(2 + h)^3 - 8]/h$.
5. Determine whether $f(x) = |x|$ is differentiable at zero.
6. Differentiate $x^4 - 3x^2 + 7$.
7. Differentiate $\sin(x^2)$.
8. Differentiate xe^x.
9. Find the tangent line to $y = \ln x$ at $x = 1$.
10. For $x(t) = 5t^2 - 2t$, find velocity and acceleration.
11. Compute f_x and f_y for $f = x^3y^2$.
12. Compute all first partial derivatives of $f = e^{xyz}$.
13. Find df for $f = x^2 + y^2$.
14. Linearize $f(x, y) = \sqrt{x + y}$ at $(2, 2)$.
15. Find the tangent plane to $z = x^2 + y^2$ at $(1, 1, 2)$.
16. Compute ∇f for $f = xyz$.
17. Find $D_{\hat{u}}f$ for $f = x^2 + y^2$ at $(1, 0)$ with $\hat{u} = (0, 1)$.
18. Find the direction of steepest increase of $f = 3x - y$.
19. Compute the Jacobian of $F(x, y) = (x + y, x - y)$.
20. Compute the determinant of the Jacobian in Exercise 2.19.
21. Use the chain rule for $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$.
22. Find the critical points of $f = x^2 + y^2$.
23. Classify the origin for $f = x^2 - y^2$.
24. Compute $\nabla \cdot (x, y, z)$.
25. Compute $\nabla \times (-y, x, 0)$.
26. Compute $\nabla^2(x^2 + y^2 + z^2)$.
27. Differentiate $r = \sqrt{x^2 + y^2 + z^2}$ with respect to x.
28. Compute ∇r for $r \neq 0$.
29. Show that the directional derivative is linear in the direction vector.
30. Explain why a non-unit direction vector changes the numerical value of a directional derivative.

2.12.2 Intermediate synthesis

1. Use an ε–δ argument to prove $\lim_{x\to 2}(3x - 1) = 5$.
2. Investigate the limit of $x^2y/(x^4 + y^2)$ at the origin along several paths.

3. Show that $f(x, y) = xy/\sqrt{x^2 + y^2}$ is continuous at the origin when defined there as zero.
4. Derive the quotient rule from the product and chain rules.
5. Find d^2y/dx^2 for the implicit curve $x^2 + y^2 = 1$.
6. Estimate $1/1.98$ using linearization at $x = 2$.
7. Find the second-order Taylor polynomial of e^x at zero.
8. Compute the Hessian of $f = x^2y + xy^2$.
9. Find all tangent directions to the level curve $x^2 + xy + y^2 = 3$ at $(1, 1)$.
10. For $T = e^{-x^2-y^2}$, find the direction of fastest cooling at $(1, 1)$.
11. Find the normal line to $x^2 + y^2 + z^2 = 9$ at $(1, 2, 2)$.
12. Compute the Jacobian of spherical coordinates.
13. Show that the polar Jacobian vanishes at the coordinate singularity $r = 0$.
14. For $F(x, y) = (e^x \cos y, e^x \sin y)$, compute and interpret $\det J_F$.
15. Use the chain rule to differentiate $f(r(t))$ for $f = x^2y$ and $r(t) = (t, e^t)$.
16. Classify all critical points of $f = x^4 + y^4 - 4xy$.
17. Find extrema of $x + y$ on the circle $x^2 + y^2 = 1$.
18. Find the point on $z = x^2 + y^2$ nearest $(0, 0, 3)$.
19. Compute divergence and curl of $F = (yz, zx, xy)$.
20. Verify directly that $\nabla \times (\nabla f) = 0$ for $f = x^2yz$.
21. Verify directly that $\nabla \cdot (\nabla \times F) = 0$ for $F = (xy, yz, zx)$.
22. Compute $\nabla^2(r^2)$ in three dimensions.
23. Compute $\nabla^2(1/r)$ for $r \neq 0$.
24. Derive first-order relative-error propagation for $Q = ab^2/c$.
25. Estimate the change in kinetic energy $K = mv^2/2$ under small changes dm, dv.

2.12.3 Advanced analysis

1. Prove that differentiability at a point implies continuity there.
2. Give a function whose partial derivatives exist at the origin but which is not continuous there.
3. Prove the multivariable chain rule using remainder terms.
4. Show that the derivative, when it exists, is unique.
5. Derive the Hessian transformation rule under an orthogonal change of coordinates.
6. Prove that $\|\nabla f\|$ is the maximal unit directional derivative.
7. Derive the second-derivative test from the quadratic Taylor approximation.
8. Find and classify constrained extrema of xyz on $x^2 + y^2 + z^2 = 3$.
9. Derive the Jacobian for cylindrical coordinates and interpret each scale factor.
10. Show that a continuously differentiable map with nonzero Jacobian determinant is locally orientation preserving or reversing according to its sign.
11. For a radial field $F = f(r)\hat{r}$, derive $\nabla \cdot F$ for $r \neq 0$.
12. For a radial scalar $u(r)$, derive $\nabla^2 u = u'' + 2u'/r$ in three dimensions.
13. Show that $e^{i\mathbf{k}\cdot\mathbf{r}}$ is an eigenfunction of the Laplacian.
14. Derive the material derivative $D/Dt = \partial_t + \mathbf{v} \cdot \nabla$.

15. Use dimensional analysis to check every term in the diffusion equation $\partial_t T = \kappa \nabla^2 T$.

2.12.4 Challenge and physics bridge

1. Construct a function continuous at the origin whose directional derivatives all exist there but which is not differentiable there.
2. Prove the inverse-function theorem in one dimension and explain which step fails when $f'(a) = 0$.
3. Derive the Euler–Lagrange equation from a first variation of the action.
4. Show that Hamilton's equations imply conservation of H when $\partial H/\partial t = 0$.
5. Derive Liouville's theorem by computing the divergence of Hamiltonian phase-space flow.
6. Use the Hessian to derive the normal-mode approximation near a stable equilibrium.
7. Derive the probability current for a Schrödinger wavefunction and identify where gradients enter.
8. Show how gauge transformation $A \mapsto A + \nabla\chi$ leaves $B = \nabla \times A$ unchanged.
9. Explain why $\nabla^2(1/r) = 0$ away from the origin yet represents a point source distributionally.
10. Derive the local conservation equation from a balance law on an arbitrary small volume.

2.12.5 Selected hints and solutions

Basic 14. Use $df = f_x\,dx + f_y\,dy$ at $(2, 2)$; the denominator in both partial derivatives is $2\sqrt{x+y}$.

Basic 19. The matrix is $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$; its determinant is -2, so the map reverses orientation and doubles area locally.

Intermediate 2. Along $y = mx^2$ the expression becomes $m/(1 + m^2)$, which depends on m; therefore the limit does not exist.

Intermediate 10. $\nabla T = -2e^{-x^2-y^2}(x, y)$; fastest cooling is in the direction $-\nabla T$.

Intermediate 12. The absolute Jacobian determinant is $r^2 \sin\theta$.

Intermediate 17. Lagrange multipliers give $(1, 1)/\sqrt{2}$ for the maximum and its negative for the minimum.

Intermediate 23. For $r \neq 0$, direct differentiation yields zero; the origin must be treated separately.

Advanced 6. Apply Cauchy–Schwarz to $D_{\hat{u}} f = \nabla f \cdot \hat{u}$ and identify the equality condition.

Advanced 8. Symmetry and the multiplier equations imply $|x| = |y| = |z| = 1$ at nonzero extrema; inspect signs.

Advanced 11. Write $F_i = f(r)x_i/r$ and differentiate components before summing.

Advanced 13. Two differentiations give $\nabla^2 e^{i\mathbf{k}\cdot\mathbf{r}} = -|\mathbf{k}|^2 e^{i\mathbf{k}\cdot\mathbf{r}}$.

Challenge 3. Vary $q(t) \mapsto q(t) + \epsilon\eta(t)$ with η vanishing at both endpoints, differentiate the action with respect to ϵ, and integrate the $\dot{\eta}$ term by parts.

Challenge 5. Compute the phase-space divergence of $(\partial H/\partial p_i, -\partial H/\partial q^i)$; equality of mixed partials causes cancellation.

Challenge 8. Curl annihilates gradients wherever χ is sufficiently smooth.

Challenge 10. Apply the divergence theorem to the flux term, then use the arbitrariness of the volume.

2.13 Notation, Terminology, and Chapter Dependencies

This section consolidates the vocabulary and notation of the chapter so that later parts of the book can use them without redefining the underlying ideas. The central progression is

function $\longrightarrow$ limit $\longrightarrow$ derivative $\longrightarrow$ differential $\longrightarrow$ Jacobian matrix or gradient.

Canonical derivative notation

For a map $F : \mathbb{R}^n \to \mathbb{R}^m$, the coordinate-independent derivative at $\mathbf{a}$ is written $DF(\mathbf{a})$. Its matrix in the standard bases is $J_F(\mathbf{a})$. For a scalar field $f : \mathbb{R}^n \to \mathbb{R}$, the same linear map may be written

$$df_{\mathbf{a}}(\mathbf{h}) = Df(\mathbf{a})[\mathbf{h}] = \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

Thus $df_{\mathbf{a}}$ is a covector, while $\nabla f(\mathbf{a})$ is the vector that represents it through the Euclidean inner product.

Object	Canonical notation	Physical reading
ordinary derivative	df/dx or $f'(x)$	instantaneous response
partial derivative	$\partial f/\partial x^i$	coordinate response
total differential	df	first-order change
Jacobian	J_F	local linear transformation
gradient	∇f	steepest increase / field normal
directional derivative	$D_{\mathbf{u}} f$	response along $\mathbf{u}$
Hessian	H_f	local curvature / stability

Dependencies carried into later physics

Chapter 3 differentiates vector fields and integrates the resulting local quantities. Lagrangian and Hamiltonian mechanics use partial and total derivatives on configuration and phase space. Electromagnetism uses gradients of potentials and time derivatives of fields. Quantum mechanics uses derivatives as momentum and kinetic-energy operators. The notation fixed here is therefore a repository-wide contract, not a chapter-local convention.

Cross-reference map

Use Section 2.1 for mappings and domains, Section 2.2 for limiting arguments, Section 2.3 for one-variable differentiation, Section 2.4 for multivariable rates, Section 2.5 for scalar fields, and Section 2.6 for change along paths and directions. The rigorous linear-map formulation is collected in Section 2.8.

2.14 Summary and Looking Ahead

Functions describe dependence between quantities, while limits make controlled approach precise. A derivative is the best local linear approximation to a function. In one dimension it is represented by a slope; in several dimensions it is represented by a Jacobian matrix. Partial derivatives provide coordinate rates, but differentiability is the stronger statement that one linear map controls change in every direction.

For scalar fields, the total differential can be written as $df = \nabla f \cdot d\mathbf{r}$. The gradient points in the direction of steepest ascent, is normal to regular level sets, and determines every directional derivative. Along a path $\mathbf{r}(t)$, the chain rule gives $df/dt = \nabla f \cdot \dot{\mathbf{r}}$.

Chapter 2 in One Line

Differential calculus replaces a nonlinear function, near one point, by its unique best first-order linear model.

Bridge to Chapter

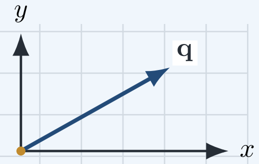
Chapter 3 applies the same local-derivative viewpoint to vector fields. Divergence measures local expansion, curl measures local rotation, and the Laplacian combines second derivatives. The integral theorems of Green, Gauss, and Stokes then connect these local quantities to global flux and circulation.

Chapter 3

Vector Calculus and Physical Fields

Local structure and global laws

Fields vary point by point, while integrals accumulate their effects over curves, surfaces, and volumes.



Chapter Roadmap

Scalar and vector fields $\longrightarrow$ gradient, divergence, curl, and Laplacian $\longrightarrow$ line, surface, and volume integrals $\longrightarrow$ Green, Stokes, and Gauss $\longrightarrow$ conservation laws and Maxwell's equations.

Learning Objectives

After completing this chapter, the reader should be able to:

- interpret scalar and vector fields geometrically and physically;
- calculate and explain gradient, divergence, curl, and Laplacians;
- construct line, surface, and volume integrals from parametrizations;
- distinguish circulation, flux, density, and accumulated quantity;
- connect differential and integral descriptions through Green's, Stokes', and Gauss' theorems;
- recognize the same structures in fluid dynamics, gravitation, electromagnetism, and quantum mechanics.

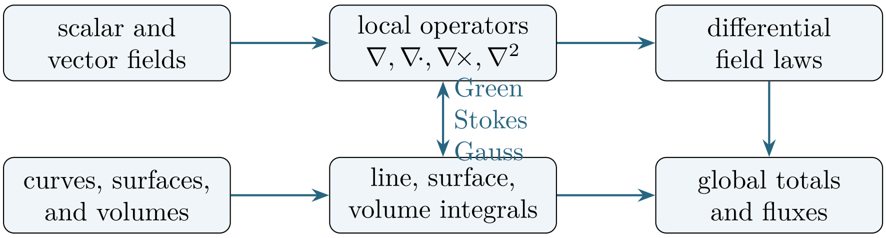


Figure 3.1: The architecture of vector calculus. Differential operators describe local field structure; integrals accumulate that structure; the integral theorems connect interiors to boundaries.

3.1 Fields, Geometry, and Global Laws

Chapter 2 established that a differentiable function is locally approximated by a linear map. Vector calculus develops this idea into a language for physical fields. A scalar field assigns one number to each point, while a vector field assigns a direction and magnitude. Temperature, pressure, mass density, and electric potential are scalar fields. Velocity, force, electric field, magnetic field, and probability current are vector fields.

The central theme is the relation between *local* and *global* descriptions. Local operators inspect an infinitesimal neighborhood:

$$\nabla f, \qquad \nabla \cdot \mathbf{F}, \qquad \nabla \times \mathbf{F}, \qquad \nabla^2 f.$$

Global operators accumulate information over extended geometric objects:

$$\int_C, \qquad \iint_S, \qquad \iiint_V.$$

The great integral theorems show that these are not separate languages. Under appropriate smoothness and orientation assumptions, an integral over a boundary equals an integral of a derivative over the interior.

One geometry: many physical theories

The same mathematical pattern appears repeatedly: a density fills a region, a current crosses its boundary, and a differential equation expresses local conservation. Fluid mass, electric charge, energy, and quantum probability differ physically, but their conservation laws share the same vector-calculus skeleton.

3.1.1 Oriented geometry

Line and surface integrals require orientation. A curve is oriented by increasing parameter. A surface is oriented by a chosen unit normal. For a closed surface, the conventional orientation is outward. Reversing orientation changes the sign of circulation or flux but not the underlying field.

3.1.2 Regularity and domains

Formulas involving derivatives require differentiability, while integral theorems require compatible smoothness of the field and the geometry. Singularities must be treated explicitly. The inverse-square field, for example, is smooth away from its source but undefined at the source itself. Whether that point lies inside the domain changes the validity and interpretation of an integral theorem.

Ignoring the domain

A symbolic derivative can be correct while a theorem is applied incorrectly. Always identify where the field is defined, whether the region contains singularities, and whether the boundary is sufficiently regular.

3.2 Vector Fields

Thus far, we have treated vectors as individual geometric objects. A displacement vector specifies the position of one point relative to another, a force vector acts on a particle, and a velocity vector describes the motion of an object. In each case, a single vector is associated with a particular location or event.

Many physical systems cannot be described by only one vector. The wind has a different speed and direction at different places, gravitational attraction changes with position, and the electric field surrounding a charge varies continuously throughout space. To describe such systems, we introduce the concept of a vector field.

Definition

Vector field A vector field assigns a vector to every point in a region of space.

3.2.1 Motivation

Imagine standing on a hill and observing the wind. At one location the wind blows gently from the west, a few metres away it is stronger, and farther uphill it changes direction. There is not one wind vector but a collection of vectors, one associated with every point. Together, these vectors form a wind field.

The same mathematical idea describes Earth's gravitational field, the electric field surrounding a charged particle, the magnetic field produced by a current, the velocity field of a flowing river, and the deformation field inside a solid. Although these systems are physically different, they use the same language.

3.2.2 Formal Definition

A vector field on a region $\Omega \subseteq \mathbb{R}^3$ is a function

$$\boxed{\mathbf{F} : \Omega \longrightarrow \mathbb{R}^3}$$

that assigns a vector $\mathbf{F}(\mathbf{r})$ to every position $\mathbf{r} = (x, y, z) \in \Omega$.

In Cartesian coordinates,

$$\boxed{\mathbf{F}(x, y, z) = F_x(x, y, z)\,\mathbf{e}_x + F_y(x, y, z)\,\mathbf{e}_y + F_z(x, y, z)\,\mathbf{e}_z.}$$

The quantities F_x, F_y, and F_z are no longer fixed numbers. They are functions of position. Consequently, both the magnitude and direction of the field may vary from point to point.

3.2.3 Visualizing a Vector Field

A continuous field contains infinitely many vectors, so a drawing can show only a representative sample. At each selected point, an arrow indicates the local value of the field. Its orientation shows direction and its length represents magnitude.

Increasing the sampling density may improve the visual approximation, but the arrows remain samples of an underlying continuous function.

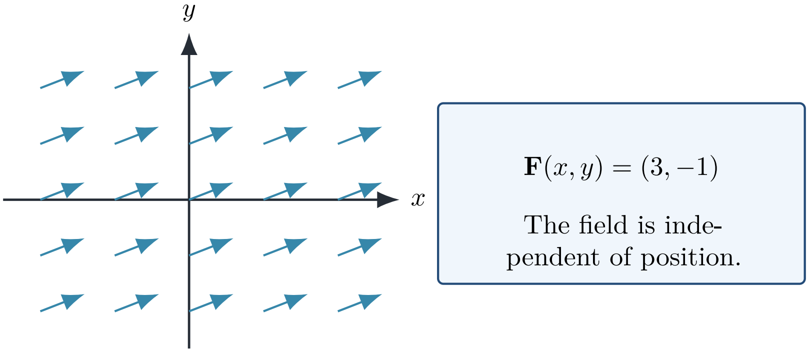


Figure 3.2: A uniform vector field. Every sampled arrow has the same direction and magnitude.

3.2.4 Uniform and Non-Uniform Fields

The simplest vector field is constant:

$$\mathbf{F}(x, y, z) = (3, -1, 2).$$

Every point in space is assigned exactly the same vector. Uniform fields are idealizations, but they are useful approximations over sufficiently small regions. Examples include the gravitational field near Earth's surface, the electric field between large parallel capacitor plates, and magnetic fields in certain laboratory regions.

Most physical fields are non-uniform. Consider

$$\boxed{\mathbf{F}(x, y) = (-y, x).}$$

Its direction changes continuously with position.

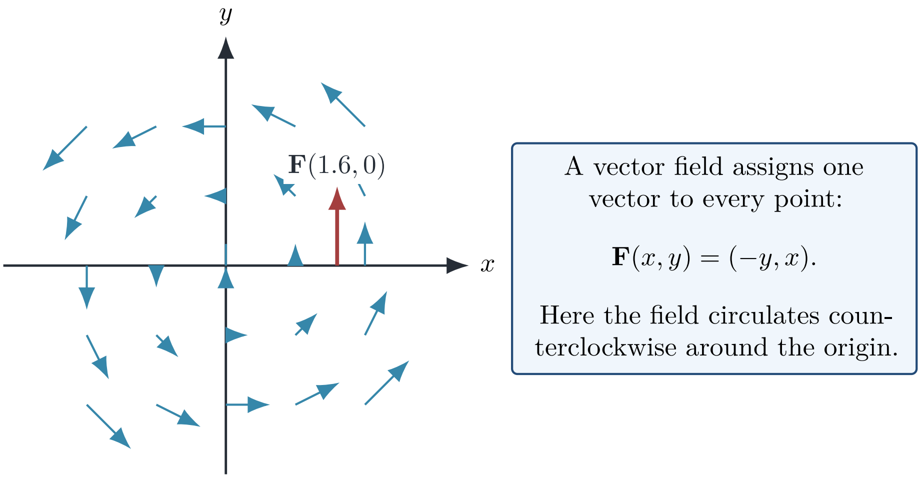


Figure 3.3: The rotational field $\mathbf{F}(x, y) = (-y, x)$. The vectors circulate counterclockwise around the origin.

Fields of this type arise naturally in rotational fluid flow and in simplified models of magnetic circulation.

3.2.5 Radial Fields

An especially important class of fields points directly toward or away from a central point. Let

$$\mathbf{r} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z, \qquad r = \|\mathbf{r}\|.$$

The radial unit-vector field is

$$\widehat{\mathbf{r}} = \frac{\mathbf{r}}{r}, \qquad r \neq 0.$$

Every vector points directly away from the origin. A radial inverse-square field has the form

$$\boxed{\mathbf{F}(\mathbf{r}) = \frac{C}{r^3}\mathbf{r} = \frac{C}{r^2}\widehat{\mathbf{r}}.}$$

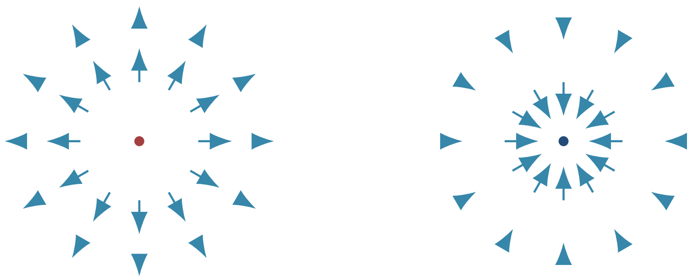


Figure 3.4: Outward and inward radial vector fields. The arrow lengths decrease with distance from the centre.

The factor $1/r^3$ multiplies the vector $\mathbf{r}$, whose magnitude is r; therefore the resulting field magnitude varies as $1/r^2$.

3.2.6 Example: Earth's Gravitational Field

Outside a spherically symmetric Earth of mass M, Newtonian gravity is described by

$$\boxed{\mathbf{g}(\mathbf{r}) = -\frac{GM}{r^3}\mathbf{r} = -\frac{GM}{r^2}\widehat{\mathbf{r}}.}$$

The minus sign indicates that the field points toward Earth's centre. Its magnitude decreases with distance, and its form is spherically symmetric.

This field provides the local acceleration that would be experienced by a small test mass placed at position $\mathbf{r}$:

$$\mathbf{F}_{\mathrm{g}} = m\mathbf{g}.$$

Example: Electric Field of a Point Charge

A stationary point charge q produces the electric field

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^3}\mathbf{r}.}$$

For $q > 0$, the field points outward; for $q < 0$, it points inward. Gravity and electrostatics describe different interactions, yet their central-field structures are mathematically similar.

3.2.7 Streamlines

A streamline is a curve whose tangent is everywhere parallel to the local field. If a curve is written as $\mathbf{r}(s)$, then a streamline satisfies

$$\frac{d\mathbf{r}}{ds} \parallel \mathbf{F}(\mathbf{r}(s)).$$

Equivalently,

$$\frac{d\mathbf{r}}{ds} = \lambda(s)\mathbf{F}(\mathbf{r}(s))$$

for some scalar function $\lambda(s)$.

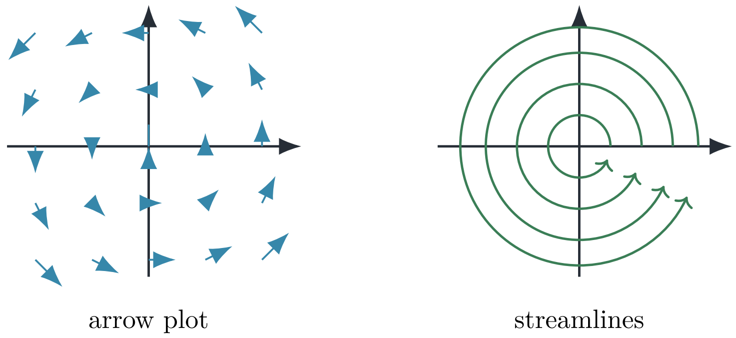


Figure 3.5: Arrow and streamline representations of the same rotational field.

Streamlines are visualization tools. They coincide with particle paths only under appropriate physical conditions, such as steady fluid flow.

3.2.8 Physical Interpretation

A vector field describes how a physical influence is distributed throughout space. Instead of asking only

> What force acts on this particular object?

we ask

> What force would act on a suitable test object placed at any point?

This shift from isolated vectors to continuous fields is one of the central conceptual developments of theoretical physics. It allows interactions to be described locally rather than solely through direct pairwise action at a distance.

Historical Note

The modern field concept developed during the nineteenth century. Michael Faraday introduced lines of force while studying electricity and magnetism. James Clerk Maxwell translated these physical pictures into differential equations for electromagnetic fields. In the twentieth century, Einstein described gravitation geometrically through spacetime, while quantum theory introduced wavefunctions and quantum fields defined over space and time.

3.2.9 Common Mistakes

A vector field is not one vector; it is a function assigning a vector to every point. The arrows in a field plot are samples, not separate physical objects. Their graphical length may also be rescaled for readability, so it need not represent the field magnitude literally.

A streamline should not automatically be interpreted as a particle trajectory. The distinction depends on whether the field is stationary and on the equations governing the motion.

Finally, the expression

$$\frac{\mathbf{r}}{r^3}$$

has magnitude $1/r^2$, not $1/r^3$.

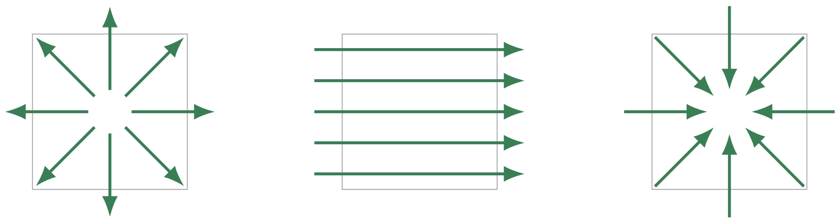


Figure 3.6: Positive divergence, zero divergence, and negative divergence visualized by the balance of arrows leaving and entering a small neighborhood.

3.2.10 Summary

A vector field is a function from positions to vectors. Its components are functions of position, so its magnitude and direction can vary throughout space. Uniform, rotational, and radial fields provide basic patterns that recur throughout mechanics, gravitation, electromagnetism, and fluid dynamics. Arrow plots and streamlines provide complementary visualizations of the same underlying field.

Looking Ahead

Vector fields provide the language of modern physics, but we still need tools that describe how they vary from point to point. Gradient, divergence, and curl measure directional change, local sources and sinks, and rotational structure. These operators underlie Maxwell's equations, fluid mechanics, heat flow, and the differential equations of quantum mechanics.

Building Toward Quantum Mechanics

The progression from vectors to vector fields prepares the reader for the wavefunction $\psi(\mathbf{r}, t)$, a field-like mathematical object defined over space and time. Differential operators act on that function to determine its spatial structure and evolution. Developing these ideas first in a geometric and physical setting makes the later quantum formalism much more transparent.

3.3 Divergence — Local Sources and Sinks

Divergence measures the net outward tendency of a vector field near a point. Imagine a tiny closed surface placed in a moving fluid. If more fluid leaves the surface than enters it, the point behaves locally like a source; if more enters than leaves, it behaves like a sink.

For

$$\mathbf{F} = F_x \mathbf{e}_x + F_y \mathbf{e}_y + F_z \mathbf{e}_z,$$

the divergence is the scalar field

$$\boxed{\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.}$$

Each derivative compares a component with displacement in its own coordinate direction. The sum records the first-order expansion or compression of an infinitesimal volume.

3.3.1 Flux-density definition

The coordinate formula is a consequence of a geometric limit. If V_ε is a small volume around $\mathbf{r}_0$ with outward area element $d\mathbf{A}$, then

$$\boxed{(\nabla \cdot \mathbf{F})(\mathbf{r}_0) = \lim_{\varepsilon \to 0} \frac{1}{\mathrm{Vol}(V_\varepsilon)} \iint_{\partial V_\varepsilon} \mathbf{F} \cdot d\mathbf{A}.}$$

Thus divergence is outward flux per unit volume in the infinitesimal limit.

3.3.2 Examples

For the radial expansion field $\mathbf{F} = (x, y, z)$,

$$\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3.$$

Every small volume expands uniformly. In contrast, for the rotational field $\mathbf{F} = (-y, x, 0)$,

$$\nabla \cdot \mathbf{F} = 0 + 0 + 0 = 0.$$

The field circulates but does not locally expand or compress area.

Example 3.1

Divergence of a non-uniform flow.
Let $\mathbf{v} = (x^2, -2xy, z^2)$. Then

$$\nabla \cdot \mathbf{v} = 2x - 2x + 2z = 2z.$$

The flow is locally expanding where $z > 0$, compressing where $z < 0$, and incompressible on the plane $z = 0$. Notice that divergence is a field, not a single number.

3.3.3 Continuity equations

If ρ is a conserved density and $\mathbf{J}$ its current density, local conservation is expressed by

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.}$$

A positive outward divergence of current must be accompanied by a decrease of density. This equation governs fluid mass, electric charge, and quantum probability.

Connection to electromagnetism

Gauss' law in differential form, $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$, states that electric charge is the local source of electric flux. The divergence theorem later converts this pointwise statement into a relation between enclosed charge and flux through a closed surface.

Divergence is not the direction of a field

Divergence is a scalar. A field can point strongly in one direction and still have zero divergence if neighboring arrows neither spread apart nor converge.

3.4 Curl — Measuring Rotation

We now arrive at the third and final fundamental differential operator of vector calculus. Together with the gradient and divergence, the curl completes the mathematical language needed to describe local change in vector fields. It is indispensable in fluid dynamics, electromagnetism, plasma physics, continuum mechanics, and many parts of modern theoretical physics.

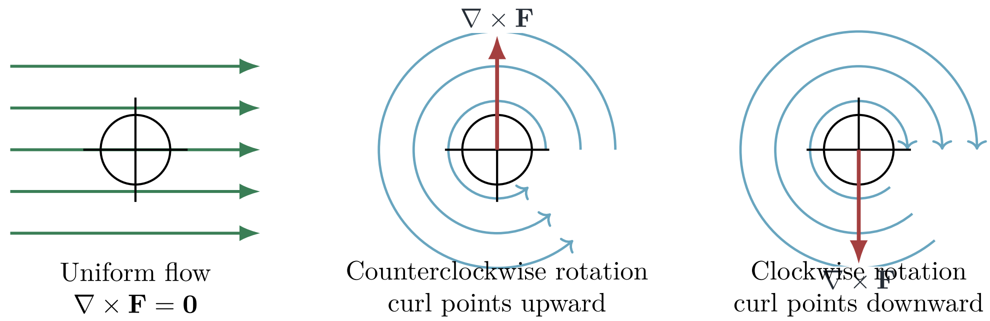


Figure 3.7: A small paddle wheel distinguishes uniform translation from local rotational flow. The direction of the curl follows the right-hand rule.

Suppose that a tiny paddle wheel is placed into a flowing stream. If the water moves uniformly in one direction, the wheel is carried downstream without spinning. If the water swirls around a whirlpool, however, the wheel begins to rotate. The amount of water entering and leaving a small region may remain balanced, yet the local flow can still possess rotation. The mathematical quantity that measures this rotational tendency is the *curl*.

Unlike divergence, which measures whether a field spreads outward from or converges inward toward a point, curl measures how strongly the field circulates around that point.

3.4.1 Motivation

Imagine stirring a cup of coffee. Near the spoon, the liquid rotates rapidly; farther away, the rotational motion becomes weaker. Nothing need be created or destroyed, so the divergence may be approximately zero. Nevertheless, each small fluid element can experience rotational motion.

This illustrates an important distinction. A vector field may possess

- divergence without rotation,
- rotation without divergence,
- both divergence and rotation,
- or neither.

To distinguish these possibilities, vector calculus introduces the curl operator.

3.4.2 Local Rotation and the Paddle-Wheel Test

Consider placing an infinitesimally small paddle wheel into a vector field $\mathbf{F}$.

- If the wheel does not rotate, the local curl vanishes:

$$\nabla \times \mathbf{F} = \mathbf{0}.$$

- If the wheel spins counterclockwise in the xy-plane, the curl points upward, in the positive z-direction, according to the right-hand rule.
- If the wheel spins clockwise, the curl points downward, in the negative z-direction.

3.4.3 Definition in Cartesian Coordinates

Let

$$\mathbf{F}(x, y, z) = F_x\, \mathbf{e}_x + F_y\, \mathbf{e}_y + F_z\, \mathbf{e}_z.$$

The curl of $\mathbf{F}$ is defined by

$$\boxed{\nabla \times \mathbf{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\mathbf{e}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\mathbf{e}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\mathbf{e}_z.}$$

Equivalently, in column-vector notation,

$$\boxed{\nabla \times \mathbf{F} = \begin{pmatrix} \partial_y F_z - \partial_z F_y \\ \partial_z F_x - \partial_x F_z \\ \partial_x F_y - \partial_y F_x \end{pmatrix}.}$$

Unlike the divergence, which maps a vector field to a scalar field, the curl maps a vector field to another vector field:

$$\boxed{\text{vector field} \longrightarrow \text{vector field.}}$$

Its direction specifies the local axis of rotation, while its magnitude measures the strength of that rotation.

3.4.4 Determinant Mnemonic

Many physicists remember the Cartesian formula using the symbolic determinant

$$\boxed{\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}.}$$

This expression is a mnemonic rather than an ordinary numerical determinant, because its first row contains basis vectors and its second row contains differential operators. Expanding it along the first row reproduces the component formula above.

3.4.5 Physical Meaning

The curl measures *local circulation.* A large magnitude $|\nabla \times \mathbf{F}|$ indicates strong local rotational motion. Zero curl means that the field is locally irrotational:

$$\nabla \times \mathbf{F} = \mathbf{0}.$$

This does not imply that the field itself vanishes. It means only that an infinitesimal paddle wheel placed at that point would not spin.

The word *local* is essential. A field can possess curved streamlines and yet have zero curl in a region. Conversely, a field may look nearly straight on a large scale while still having nonzero local curl.

3.4.6 Example 1: A Rotational Field

Consider

$$\mathbf{F}(x, y, z) = (-y, x, 0).$$

Then

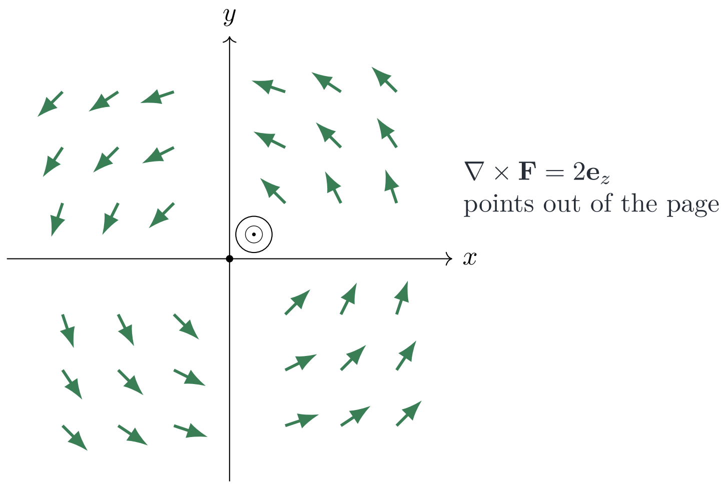


Figure 3.8: The rotational field $\mathbf{F} = (-y, x, 0)$ circulates around the origin. Its curl is uniform and points in the positive z-direction.

$$\nabla \times \mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ \dfrac{\partial x}{\partial x} - \dfrac{\partial(-y)}{\partial y} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = 2\mathbf{e}_z.$$

The positive z-component indicates counterclockwise rotation in the xy-plane. The vectors circulate around the origin while the curl points perpendicular to the plane of motion.

3.4.7 Example 2: A Radial Field

Consider the radial field

$$\mathbf{F}(x, y, z) = (x, y, z).$$

Its curl is

$$\nabla \times \mathbf{F} = \begin{pmatrix} \partial_y z - \partial_z y \\ \partial_z x - \partial_x z \\ \partial_x y - \partial_y x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Although the vectors point outward from the origin, there is no local rotation. This field possesses positive divergence,

$$\nabla \cdot \mathbf{F} = 3,$$

but zero curl.

3.4.8 Example 3: A Rotating Fluid

Suppose a fluid undergoes rigid rotation about the z-axis with angular velocity $\boldsymbol{\omega} = \omega \mathbf{e}_z$. Its velocity field is

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = (-\omega y, \omega x, 0).$$

Therefore,

$$\nabla \times \mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 2\omega \end{pmatrix} = 2\boldsymbol{\omega}.$$

Thus, for rigid-body rotation, the curl of the velocity field equals twice the local angular velocity. In fluid mechanics the quantity

$$\boldsymbol{\zeta} = \nabla \times \mathbf{v}$$

is called the *vorticity.* Meteorologists use vorticity to study hurricanes, tornadoes, cyclones, and atmospheric circulation.

3.4.9 Example 4: Electromagnetism

Two of Maxwell's equations are curl equations. Faraday's law states

$$\boxed{\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.}$$

A changing magnetic field therefore generates a circulating electric field. The Ampère–Maxwell law states

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}.}$$

Electric current density and changing electric fields generate circulating magnetic fields. These equations underlie the operation of

- electric generators,
- transformers,
- electric motors,
- radio antennas,
- and electromagnetic waves.

Without curl, Maxwell's equations could not be written in their compact and geometrically transparent differential form.

A further important example is the magnetic vector potential $\mathbf{A}$, for which

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.}$$

This representation automatically implies $\nabla \cdot \mathbf{B} = 0$, because the divergence of a curl vanishes wherever the fields are sufficiently smooth.

3.4.10 Integral Interpretation: Circulation Density

Let C be a small closed curve surrounding a point, and let S be the small surface bounded by C. The circulation of $\mathbf{F}$ around C is

$$\oint_C \mathbf{F} \cdot d\mathbf{l}.$$

For a chosen unit normal $\hat{\mathbf{n}}$, the normal component of the curl is obtained by shrinking the loop toward the point:

$$\boxed{(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = \lim_{A \to 0} \frac{1}{A} \oint_C \mathbf{F} \cdot d\mathbf{l}.}$$

Here A is the area of S, and the positive orientation of C is related to $\hat{\mathbf{n}}$ by the right-hand rule. This formula shows that curl is the local circulation per unit area.

The finite-surface version is Stokes' theorem:

$$\boxed{\oint_{\partial S} \mathbf{F} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A}.}$$

Thus the differential concept of curl and the integral concept of circulation are two descriptions of the same geometry.

3.4.11 Comparing Gradient, Divergence, and Curl

The three fundamental differential operators have complementary roles.

Operator	Acts on	Produces	Interpretation
∇f	scalar field	vector field	direction of greatest increase
$\nabla \cdot \mathbf{F}$	vector field	scalar field	local sources and sinks
$\nabla \times \mathbf{F}$	vector field	vector field	local circulation or rotation

Together, these operators form the foundation of classical field theory and provide the local differential language used throughout mathematical physics.

Curl Is Not the Same as a Curved Trajectory

A common misconception is that curl measures whether individual trajectories are curved. It does not. Curl measures the local rotation of the field itself. A particle may follow a curved path in an irrotational field, and a field with nonzero curl need not force every particle to move on a closed circular orbit. The paddle-wheel test concerns the relative motion of neighboring parts of the field, not merely the shape of one path.

3.4.12 Historical Note

The modern formulation of curl emerged during the nineteenth century through the work of William Rowan Hamilton, Peter Guthrie Tait, and Josiah Willard Gibbs. James Clerk Maxwell incorporated rotational derivatives into his theory of electricity and magnetism, revealing that electromagnetic phenomena are governed by an interplay between divergence and curl. Later developments in fluid mechanics, plasma physics, continuum mechanics, and quantum field theory established curl as one of the central tools of mathematical physics.

3.4.13 Common Mistakes

Several misconceptions should be avoided.

1. **Curl measures local rotation, not merely circular-looking trajectories.** A particle may move along a curved path even where the curl is zero.
2. **Zero divergence does not imply zero curl.** The field $\mathbf{F} = (-y, x, 0)$ has zero divergence but nonzero curl.
3. **Zero curl does not imply zero divergence.** The radial field $\mathbf{F} = (x, y, z)$ has nonzero divergence but zero curl.
4. **Curl is a vector.** Its direction is fixed by the right-hand rule and identifies the axis of local rotation.
5. **The determinant expression is only a mnemonic.** It must be expanded with the signs and component order used in the cross product.
6. **The circulation limit determines a component of the curl.** The expression must include a chosen surface normal; curl is not obtained by dividing a scalar circulation by area without specifying orientation.

3.4.14 Connection to Modern Physics

The operators developed in this chapter appear throughout theoretical physics. The Schrödinger equation contains the Laplacian, which can be written as the divergence of the gradient. Maxwell's equations are expressed almost entirely through divergence and curl. In quantum

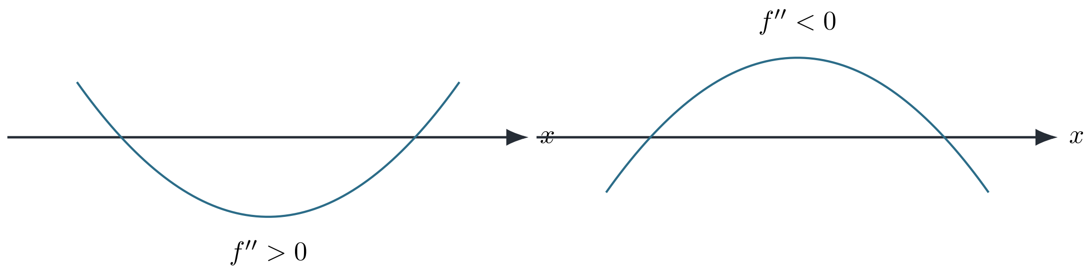


Figure 3.9: Positive and negative curvature in one dimension. In several dimensions the Laplacian sums curvature over mutually perpendicular directions.

mechanics, the magnetic field is the curl of the vector potential, and the vector potential enters the Hamiltonian through minimal coupling. In relativistic field theory, these ideas are reorganized into antisymmetric tensors and exterior derivatives. In general relativity they are extended to differentiable manifolds and curved spacetime.

By mastering gradient, divergence, and curl in ordinary three-dimensional space, the reader develops geometric intuition for the more abstract differential structures used later in advanced mechanics, electromagnetism, quantum theory, and geometry.

3.4.15 Looking Ahead

With gradient, divergence, and curl established, we possess the three fundamental first-order differential operators of vector calculus. The next section introduces the Laplacian, a second-order operator constructed from the divergence of the gradient:

$$\nabla^2 f = \nabla \cdot (\nabla f).$$

The Laplacian appears in diffusion, wave propagation, electrostatics, fluid mechanics, and, most importantly for this book, the kinetic-energy term of the Schrödinger equation. Later chapters will also express these operators in cylindrical and spherical coordinates, where they become especially useful for wires, planets, central-force systems, and atomic wavefunctions.

Chapter Summary

The curl $\nabla \times \mathbf{F}$ is a vector field that measures local circulation. Its direction gives the local axis of rotation by the right-hand rule, and its magnitude gives the strength of that rotation. In Cartesian coordinates it is built from antisymmetric combinations of first partial derivatives. Its integral interpretation is circulation per unit area, and its global form is Stokes' theorem. Curl is central to vorticity, Faraday induction, the Ampère–Maxwell law, and the relation $\mathbf{B} = \nabla \times \mathbf{A}$.

3.5 The Laplacian — Curvature, Diffusion, and Waves

Applying divergence to the gradient gives

$$\boxed{\nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.}$$

The Laplacian compares a value with nearby values in every spatial direction. A positive Laplacian means the point lies below its local average; a negative Laplacian means it lies above it.

3.5.1 Harmonic functions

A function satisfying $\nabla^2 f = 0$ is harmonic. Harmonic functions have no local maxima or minima inside a source-free domain unless they are constant. Electrostatic potential in a charge-free region and steady temperature without internal heat sources are harmonic.

3.5.2 Diffusion and waves

The diffusion equation is

$$\boxed{\frac{\partial u}{\partial t} = D\nabla^2 u.}$$

At a peak, $\nabla^2 u < 0$, so the peak falls; at a trough, $\nabla^2 u > 0$, so the trough rises. Diffusion smooths spatial variation. By contrast,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

is a wave equation: the same spatial curvature produces propagation because the time derivative is second order.

Example 3.2

Laplacian of a Gaussian.
For $f(x, y) = e^{-(x^2+y^2)}$ and $r^2 = x^2 + y^2$,

$$\nabla^2 f = 4(r^2 - 1)e^{-r^2}.$$

It is negative near the central peak and positive outside the unit circle.

Connection to quantum mechanics

The nonrelativistic kinetic-energy operator is

$$\widehat{T} = -\frac{\hbar^2}{2m}\nabla^2.$$

The Laplacian therefore measures spatial curvature of the wavefunction and converts that curvature into kinetic energy.

3.6 Line Integrals

Introduction—From Local Quantities to Global Physics

3.6.1 The transition from local to global descriptions

In the previous chapters we developed the mathematical language required to describe quantities at individual points in space. We introduced vectors, scalar and vector fields, coordinate systems, and the differential operators gradient, divergence, and curl. These concepts allow us to answer questions such as

- What is the electric field at this point?
- What is the temperature here?
- How rapidly is the pressure changing?
- Is the fluid rotating locally?

These are *local* questions. They concern the behavior of physical quantities in an infinitesimally small neighborhood of a single point.

Many of the most important physical questions, however, are not local. They concern the accumulated behavior of a field over an extended region of space. For example:

- How much work does a force perform while moving an object along a path?
- How much electric charge is enclosed inside a surface?
- How much fluid passes through a pipe every second?
- How much mass is contained inside a planet?
- What is the total magnetic flux through a superconducting loop?

These questions cannot be answered by differentiation alone. They require integration.

3.6.2 From infinitesimal pieces to complete systems

Integration is based on a remarkably simple idea. Suppose we wish to determine the length of a winding mountain road. Instead of measuring the entire road at once, we divide it into many extremely small segments. Each segment is nearly straight and therefore easy to measure. If the scalar length element is denoted by ds, then adding all infinitesimal pieces gives the total length

$$\boxed{L = \int_C ds.} \tag{3.1}$$

The symbol C denotes the curve followed by the road. It is important to distinguish the vector displacement $d\mathbf{r}$ from its scalar magnitude:

$$ds = |d\mathbf{r}|. \tag{3.2}$$

The same principle extends far beyond geometry. Instead of adding small lengths, we may add infinitesimal amounts of work, area, volume, electric charge, probability, or energy. Integration is therefore the mathematical operation that converts local information into global quantities.

3.6.3 Differential and integral descriptions

Throughout theoretical physics there exist two complementary descriptions of nature. The first describes what happens at each point. The second describes what happens over an entire region.

Local description	**Global description**
Differential equations	Integral equations
Pointwise behavior	Total accumulated behavior
Derivatives	Integrals
Local geometry	Global geometry
Infinitesimal change	Finite quantity

Neither description is universally more fundamental. Under suitable smoothness and geometric conditions, they are equivalent formulations of the same physical law. One of the deepest achievements of mathematical physics is the demonstration of this equivalence.

3.6.4 Work performed by a variable force

In elementary mechanics the expression

$$W = \mathbf{F} \cdot \mathbf{d} \tag{3.3}$$

gives the work done by a constant force during a fixed displacement $\mathbf{d}$. Suppose instead that the force varies continuously from one point to another, as it does in gravitational and electric fields, springs, and many forms of drag.

We divide the path into infinitesimal displacements $d\mathbf{r}$. Over each tiny segment the force is approximately constant, so the infinitesimal work is

$$dW = \mathbf{F} \cdot d\mathbf{r}. \tag{3.4}$$

Adding every contribution along the path C gives

$$\boxed{W = \int_C \mathbf{F} \cdot d\mathbf{r}.} \tag{3.5}$$

This expression is a *line integral.* The dot product selects the component of the force tangent to the motion. A force perpendicular to the instantaneous displacement contributes no work at that point.

3.6.5 Flux through a surface

Consider rain falling onto a roof. Instead of asking how strong the rain is at one point, we ask how much rain reaches the entire roof each second. The answer depends on the local rain field, the area of the roof, and the orientation of each piece of the surface.

For an infinitesimal surface element of area dA with unit normal $\hat{\mathbf{n}}$, define the oriented area vector

$$d\mathbf{A} = \hat{\mathbf{n}}\, dA. \tag{3.6}$$

The total flux of a vector field $\mathbf{F}$ through a surface S is

$$\boxed{\Phi = \int_S \mathbf{F} \cdot d\mathbf{A}.} \tag{3.7}$$

Flux appears throughout physics in electric and magnetic fields, fluid flow, heat transport, gravitational fields, and quantum probability current.

3.6.6 Quantities distributed throughout a volume

Some physical quantities occupy an entire three-dimensional region. Suppose a material has mass density $\rho(x, y, z)$. The density at a single point does not determine the total mass. We divide the object into infinitesimal volume elements dV. The mass of one element is

$$dm = \rho\, dV. \tag{3.8}$$

Adding all contributions yields

$$\boxed{M = \int_V \rho\, dV.} \tag{3.9}$$

The same construction applies to electric charge, particle number, internal energy, entropy, and quantum probability.

3.6.7 Why the integral theorems matter

At first glance, differentiation and integration appear to be different operations. The great integral theorems of vector calculus reveal that they are deeply connected. In schematic form,

$$\boxed{\text{boundary information} \quad \longleftrightarrow \quad \text{interior information.}} \tag{3.10}$$

More precisely, circulation around a closed boundary is related to the curl inside the bounded surface, while flux through a closed surface is related to the divergence inside the enclosed volume. These relations allow difficult volume calculations to be replaced by surface calculations, and conversely. They also provide the mathematical structure behind many conservation laws.

3.6.8 Historical perspective

The development of integral calculus began with Isaac Newton and Gottfried Wilhelm Leibniz in the late seventeenth century. Their independent formulations provided systematic methods for studying continuously varying quantities. During the eighteenth and nineteenth centuries, Leonhard Euler, Joseph-Louis Lagrange, Carl Friedrich Gauss, George Green, William Thomson, and George Gabriel Stokes extended these ideas into multidimensional geometry.

James Clerk Maxwell recognized that these mathematical results were ideally suited to electric and magnetic fields. In Maxwell's theory, integral theorems are not mathematical curiosities: they connect local field equations to measurable quantities such as enclosed charge, circulation, and flux.

3.6.9 Applications throughout physics

In classical mechanics, line integrals describe the work performed by variable forces. In electromagnetism, surface integrals define electric and magnetic flux, while volume integrals determine total charge. In fluid mechanics, flux integrals measure transport through boundaries, and divergence theorems express conservation of mass.

In general relativity, integration over curved spacetime regions is essential for defining invariant and conserved quantities. In quantum mechanics, volume integrals normalize the wavefunction:

$$\boxed{\int_{\mathbb{R}^3} |\psi(\mathbf{r}, t)|^2 \, dV = 1.} \tag{3.11}$$

The probability that a particle lies inside a region V is

$$P(V) = \int_V |\psi|^2 \, dV, \tag{3.12}$$

while probability flow through a surface is expressed by a flux integral of the probability-current density $\mathbf{j}$.

Chapter Summary

Differential operators describe the local behavior of fields. Integrals add local contributions over curves, surfaces, and volumes to produce global physical quantities. Line integrals describe work and circulation, surface integrals describe flux, and volume integrals convert densities into totals. The integral theorems connect these global quantities to local divergence and curl.

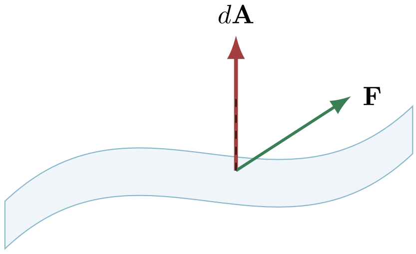


Figure 3.10: Flux through an oriented surface. Only the normal component of the field contributes to $\mathbf{F} \cdot d\mathbf{A}$.

Looking Ahead

The next section begins with the geometry of parameterized curves and arc length. This provides the rigorous foundation for line integrals, work, circulation, conservative forces, and potential energy.

Definition

For a curve C parametrized by $\mathbf{r}(t)$, the line integral of a vector field is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt.$$

It measures accumulated tangential action, such as work done by a force along a trajectory.

3.7 Surface Integrals and Flux

A surface integral accumulates a scalar field over a two-dimensional surface or measures the flux of a vector field through that surface. Let

$$\mathbf{r}(u, v)$$

parametrize a surface. The tangent vectors are $\mathbf{r}_u$ and $\mathbf{r}_v$, and their cross product determines the oriented area element,

$$\boxed{d\mathbf{A} = (\mathbf{r}_u \times \mathbf{r}_v)\, du\, dv.}$$

Its magnitude gives the scalar area element,

$$dA = \|\mathbf{r}_u \times \mathbf{r}_v\|\, du\, dv.$$

For a scalar field g,

$$\iint_S g\, dA$$

weights each small surface patch by the local value of g. For a vector field $\mathbf{F}$, the flux is

$$\boxed{\Phi_S = \iint_S \mathbf{F} \cdot d\mathbf{A}.}$$

The dot product selects the component normal to the surface.

Figure 3.11: Coordinate volume elements. The Jacobian accounts for the changing physical size of equal coordinate increments.

3.7.1 Orientation

A surface has two possible orientations. Reversing the normal changes $d\mathbf{A}$ to $-d\mathbf{A}$ and therefore reverses the sign of flux. For a closed surface the standard orientation is outward.

Example 3.3

Flux through a horizontal disk.
Let $\mathbf{F} = (0, 0, z)$ and let S be the disk $x^2 + y^2 \leq a^2$ in the plane $z = h$, oriented upward. Since $\hat{\mathbf{n}} = \mathbf{e}_z$,

$$\mathbf{F} \cdot \hat{\mathbf{n}} = h.$$

Therefore

$$\Phi_S = \iint_S h\, dA = h\pi a^2.$$

The result is field strength times projected area because the field is uniform over the disk.

Using dA when orientation matters

The scalar element dA has no direction. Flux requires the vector element $d\mathbf{A} = \hat{\mathbf{n}} dA$.

3.8 Volume Integrals and Densities

A volume integral converts a local density into a total quantity:

$$\boxed{Q = \iiint_V \rho(\mathbf{r})\, dV.}$$

Depending on the meaning of ρ, Q may represent mass, charge, particle number, energy, or probability.

3.8.1 Coordinate volume elements

A coordinate transformation changes the infinitesimal volume by a Jacobian. In Cartesian coordinates,

$$dV = dx\, dy\, dz.$$

In cylindrical and spherical coordinates,

$$\boxed{dV = s\, ds\, d\phi\, dz,} \qquad \boxed{dV = r^2 \sin\theta\, dr\, d\theta\, d\phi.}$$

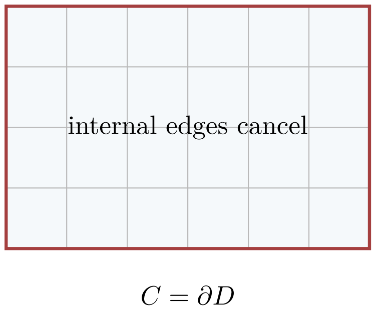


Figure 3.12: Green's theorem: internal edge contributions cancel when small cells are assembled, leaving only the outer boundary.

Example 3.4

Mass of a radially varying sphere.
Let a sphere of radius R have density $\rho(r) = \rho_0(1 - r/R)$. Then

$$M = \int_0^{2\pi}\int_0^{\pi}\int_0^{R} \rho_0 \left(1 - \frac{r}{R}\right) r^2 \sin\theta \, dr \, d\theta \, d\phi = \frac{\pi}{3}\rho_0 R^3.$$

The angular integrals contribute 4π, while the radial integral records the decreasing density.

Connection to quantum mechanics

A normalized wavefunction satisfies

$$\iiint_{\mathbb{R}^3} |\psi(\mathbf{r}, t)|^2 \, dV = 1.$$

The scalar field $|\psi|^2$ is a probability density; integration converts it into the probability of finding the particle in a finite region.

3.9 Green's Theorem — Boundary and Interior in the Plane

Let D be a planar region with positively oriented boundary $C = \partial D$. For a continuously differentiable field $\mathbf{F} = (P, Q)$,

$$\boxed{\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.}$$

The left side measures circulation around the boundary; the right side adds the local scalar curl throughout the interior.

The cancellation mechanism is central. Divide D into many small cells. Neighboring cells traverse their common edge in opposite directions, so those contributions cancel. Only the external boundary survives.

A second, flux form is

$$\boxed{\oint_C \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \iint_D \nabla \cdot \mathbf{F} \, dA.}$$

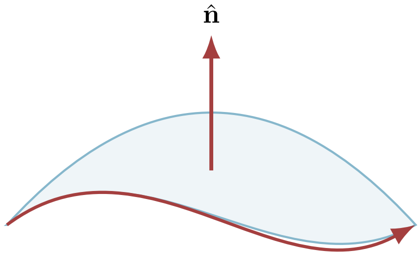


Figure 3.13: Stokes' theorem. The right-hand rule links the surface normal to the positive orientation of the boundary.

Example 3.5

Area from a boundary integral.
Choose $P = -y/2$ and $Q = x/2$. Then

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1.$$

Green's theorem gives

$$\text{Area}(D) = \frac{1}{2}\oint_C (x\,dy - y\,dx).$$

Thus a two-dimensional area can be recovered entirely from data on its one-dimensional boundary.

3.10 Stokes' Theorem — Curl Through a Surface

For an oriented surface S with compatible boundary orientation ∂S,

$$\boxed{\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A}.}$$

Stokes' theorem states that circulation around a boundary equals the accumulated normal component of curl through any spanning surface.

The theorem also explains why two surfaces sharing the same boundary give the same curl flux, provided the field is smooth in the region between them.

Example 3.6

Circulation without parametrizing the boundary.
Let $\mathbf{F} = (-y, x, 0)$ and let C be the circle $x^2 + y^2 = a^2$ in the plane $z = 0$, counterclockwise from above. Since

$$\nabla \times \mathbf{F} = (0, 0, 2),$$

Stokes' theorem gives

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S 2\,dA = 2\pi a^2.$$

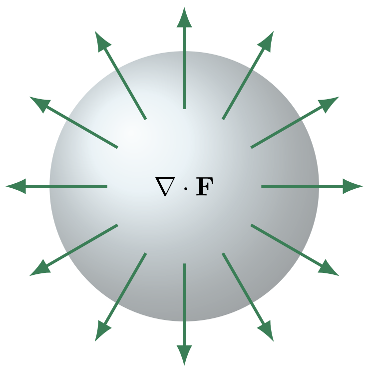


Figure 3.14: Gauss' theorem. Flux through internal faces cancels, while flux through the outer boundary remains.

Connection to electromagnetism

Faraday's law,

$$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{r} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot d\mathbf{A},$$

relates induced electric circulation to changing magnetic flux. Stokes' theorem converts it to the local equation $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$.

3.11 Gauss' Divergence Theorem — Sources and Closed Flux

Let V be a volume with outward-oriented closed boundary ∂V. Then

$$\boxed{\iint_{\partial V} \mathbf{F} \cdot d\mathbf{A} = \iiint_V \nabla \cdot \mathbf{F} \, dV.}$$

The net flux through the boundary equals the total source strength inside.

Example 3.7

Flux of a radial expansion field.
For $\mathbf{F} = (x, y, z)$ inside a sphere of radius R,

$$\nabla \cdot \mathbf{F} = 3.$$

Therefore

$$\iint_{\partial V} \mathbf{F} \cdot d\mathbf{A} = \iiint_V 3 \, dV = 3 \left(\frac{4\pi R^3}{3} \right) = 4\pi R^3.$$

On the sphere, $\mathbf{F} = R\hat{\mathbf{n}}$, so direct integration gives the same result.

3.11.1 Singular sources

The inverse-square field $\mathbf{F} = \mathbf{r}/r^3$ has zero divergence wherever $r \neq 0$, yet its flux through any sphere centered at the origin is 4π. The origin is a singularity and cannot be ignored. In distributional language,

$$\nabla \cdot \frac{\mathbf{r}}{r^3} = 4\pi \delta^{(3)}(\mathbf{r}).$$

Connection to electrostatics

For $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$, the divergence theorem yields

$$\iint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{1}{\varepsilon_0} \iiint_V \rho \, dV = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

The differential and integral forms of Gauss' law are therefore the same statement at different scales.

3.12 Integrated Worked Examples

The following examples form a continuous problem-solving sequence. Each calculation is paired with a geometric or physical interpretation so that the operators are not reduced to formal symbol manipulation.

3.12.1 Fields and differential operators

Example 3.8

Radial inverse-square field.
Problem. For $\mathbf{F}(\mathbf{r}) = k\mathbf{r}/r^3$ with $r \neq 0$, determine its magnitude and direction.

Solution

Since $|\mathbf{r}| = r$, one has $|\mathbf{F}| = |k|/r^2$. For $k > 0$ the field points radially outward; for $k < 0$ it points inward.
Interpretation. This is the geometric form shared by ideal point-source gravitational and electrostatic fields. The singular point must be excluded from ordinary differentiation.

Example 3.9

Gradient and equipotential curves.
Problem. Let $\phi(x, y) = x^2 + 2y^2$. Find $\nabla\phi$ and describe its relation to the level curve through $(1, 1)$.

Solution

$\nabla\phi = (2x, 4y)$, hence $\nabla\phi(1, 1) = (2, 4)$. The level curve is $x^2 + 2y^2 = 3$. Its tangent vectors are orthogonal to $(2, 4)$ at $(1, 1)$.
Interpretation. The gradient points across equipotential curves in the direction of greatest increase.

Example 3.10

Divergence of an expanding flow.
Problem. Compute $\nabla \cdot \mathbf{v}$ for $\mathbf{v} = (ax, ay, az)$.

Solution

$\nabla \cdot \mathbf{v} = a + a + a = 3a$.
Interpretation. Every small material volume expands at fractional rate $3a$ when $a > 0$, and contracts when $a < 0$.

Example 3.11

An incompressible shear.
Problem. For $\mathbf{v} = (\gamma y, 0, 0)$, calculate divergence and curl.

Solution

$\nabla \cdot \mathbf{v} = 0$, while

$$\nabla \times \mathbf{v} = (0, 0, -\gamma).$$

Interpretation. The flow preserves volume but possesses local rotational tendency. Divergence and curl measure independent aspects of a field.

Example 3.12

Rigid rotation.
Problem. Let $\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r}$ with constant $\boldsymbol{\Omega}$. Show that $\nabla \times \mathbf{v} = 2\boldsymbol{\Omega}$.

Solution

Choosing $\boldsymbol{\Omega} = \Omega\hat{\mathbf{z}}$ gives $\mathbf{v} = (-\Omega y, \Omega x, 0)$. Direct differentiation yields $\nabla \times \mathbf{v} = (0, 0, 2\Omega)$; rotational covariance gives the general result.
Interpretation. Fluid vorticity is twice the local angular-velocity vector for rigid rotation.

Example 3.13

Laplacian of a Gaussian.
Problem. Find $\nabla^2 e^{-\alpha r^2}$ in three dimensions.

Solution

For a radial function, $\nabla^2 f = r^{-2}\frac{d}{dr}(r^2 f')$. With $f' = -2\alpha r e^{-\alpha r^2}$,

$$\nabla^2 e^{-\alpha r^2} = (4\alpha^2 r^2 - 6\alpha)e^{-\alpha r^2}.$$

Interpretation. Near the peak the Laplacian is negative; farther away it changes sign, encoding local curvature relative to neighboring values.

3.12.2 Line integrals and potentials

Example 3.14

Work along a straight path.
Problem. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = (y, x)$ from $(0, 0)$ to $(1, 2)$ along the straight segment.

Solution

Use $\mathbf{r}(t) = (t, 2t)$, $0 \le t \le 1$. Then $\mathbf{F} = (2t, t)$ and $d\mathbf{r} = (1, 2)dt$, so

$$\int_0^1 (2t + 2t)dt = 2.$$

Interpretation. Since $\mathbf{F} = \nabla(xy)$, the result also equals $xy|_{(1,2)} - xy|_{(0,0)} = 2$.

Example 3.15

Path independence.
Problem. Find the work done by $\mathbf{F} = (2x, 2y, 2z)$ from $A = (1, 0, 0)$ to $B = (0, 2, 1)$.

Solution

$\mathbf{F} = \nabla(x^2 + y^2 + z^2)$. Therefore

$$W = (0^2 + 2^2 + 1^2) - (1^2 + 0 + 0) = 4.$$

Interpretation. A conservative field converts a potentially difficult path integral into an endpoint calculation.

Example 3.16

Circulation around a circle.
Problem. Compute $\oint_C (-y, x, 0) \cdot d\mathbf{r}$ around the circle $x^2 + y^2 = R^2$, counterclockwise.

Solution

With $\mathbf{r} = (R\cos t, R\sin t, 0)$, the dot product is $R^2 dt$. Hence

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} R^2 dt = 2\pi R^2.$$

Interpretation. The result is twice the enclosed area, consistent with curl $(0, 0, 2)$ and Stokes' theorem.

Example 3.17

A field that is locally but not globally conservative.
Problem. For $\mathbf{F} = (-y/(x^2+y^2), x/(x^2+y^2))$ on the punctured plane, evaluate circulation around the unit circle.

Solution

On $\mathbf{r} = (\cos t, \sin t)$, $\mathbf{F} = (-\sin t, \cos t)$ equals $\mathbf{r}'(t)$. Thus $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi$.
Interpretation. The curl vanishes away from the origin, but the domain is not simply connected. Topology prevents a single-valued global potential.

3.12.3 Surface and volume integrals

Example 3.18

Flux through a plane.
Problem. Find the flux of constant $\mathbf{F} = (1, 2, 3)$ through the rectangle $0 \le x \le 2$, $0 \le y \le 1$ in the plane $z = 0$, oriented upward.

Solution

$d\mathbf{S} = \hat{\mathbf{z}}\, dxdy$, so $\mathbf{F} \cdot d\mathbf{S} = 3dxdy$. The flux is $3(2)(1) = 6$.
Interpretation. Only the component normal to the surface contributes.

Example 3.19

Flux through a sphere.
Problem. Compute the outward flux of $\mathbf{F} = \mathbf{r}$ through a sphere of radius R.

Solution

On the sphere, $\mathbf{F} \cdot \hat{\mathbf{n}} = R$ and the area is $4\pi R^2$. Therefore $\Phi = 4\pi R^3$.
Interpretation. Gauss' theorem gives the same result because $\nabla \cdot \mathbf{F} = 3$ and the enclosed volume is $4\pi R^3/3$.

Example 3.20

Mass from a nonuniform density.
Problem. A ball of radius R has density $\rho(r) = \rho_0(1 - r/R)$. Find its mass.

Solution

Using spherical shells,

$$M = 4\pi\rho_0 \int_0^R \left(1 - \frac{r}{R}\right) r^2 dr = \frac{\pi}{3}\rho_0 R^3.$$

Interpretation. Volume integration accumulates a local material property over the body.

Example 3.21

Cylindrical Jacobian.
Problem. Evaluate the volume of a cylinder $0 \le r \le a$, $0 \le z \le h$.

Solution

The volume element is $dV = r\,dr\,d\theta\,dz$, hence

$$V = \int_0^h \int_0^{2\pi} \int_0^a r\,dr\,d\theta\,dz = \pi a^2 h.$$

Interpretation. The factor r is the Jacobian that accounts for the widening of angular coordinate cells.

Example 3.22

Moment of inertia of a uniform disk.
Problem. Find I_z for a disk of radius R, surface density σ.

Solution

$$I_z = \int r^2\,dm = \int_0^{2\pi} \int_0^R r^2 \sigma r\,dr\,d\theta = \frac{1}{2}(\pi R^2 \sigma)R^2 = \frac{1}{2}MR^2.$$

Interpretation. The extra r^2 weights matter farther from the axis more strongly.

3.12.4 Integral theorems

Example 3.23

Green's theorem for circulation.
Problem. Evaluate $\oint_C (-y^3\,dx + x^3\,dy)$ around the unit circle counterclockwise.

Solution

Green's theorem gives

$$\iint_D \left(3x^2 + 3y^2\right) dA = 3\int_0^{2\pi} \int_0^1 r^2 r\,dr\,d\theta = \frac{3\pi}{2}.$$

Interpretation. Boundary circulation equals the accumulated local rotation over the enclosed region.

Example 3.24

Stokes' theorem on a disk.
Problem. For $\mathbf{F} = (-y, x, z)$, calculate circulation around $x^2 + y^2 = R^2$, $z = 0$.

Solution

$\nabla \times \mathbf{F} = (0, 0, 2)$. With upward normal,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S 2\,dA = 2\pi R^2.$$

Interpretation. Any surface spanning the same oriented boundary gives the same result, provided the field is regular throughout it.

Example 3.25

Gauss' theorem for a cube.
Problem. Find the total outward flux of $\mathbf{F} = (x^3, y^3, z^3)$ through the cube $[-a, a]^3$.

Solution

$\nabla \cdot \mathbf{F} = 3(x^2 + y^2 + z^2)$. Symmetry and separation give

$$\Phi = 3\left[3\left(\int_{-a}^{a} x^2 dx\right)(2a)^2\right] = 24a^5.$$

Interpretation. A six-face surface calculation is replaced by one volume integral.

Example 3.26

Electrostatic Gauss law.
Problem. A spherically symmetric charge density is constant, ρ, inside radius R. Find the electric field for $r < R$.

Solution

Symmetry gives $\mathbf{E} = E(r)\hat{\mathbf{r}}$. The enclosed charge is $Q_r = 4\pi\rho r^3/3$. Gauss' law yields

$$E(r)4\pi r^2 = \frac{Q_r}{\varepsilon_0}, \qquad \mathbf{E} = \frac{\rho r}{3\varepsilon_0}\hat{\mathbf{r}}.$$

Interpretation. Inside a uniform sphere the field grows linearly because the enclosed charge grows as r^3 while area grows as r^2.

Example 3.27

Continuity equation check.
Problem. Let $\rho(x,t) = \rho_0 e^{-t}e^{-x}$ and $j(x,t) = \rho_0 e^{-t}e^{-x}$. Verify one-dimensional conservation.

Solution

$\partial_t \rho = -\rho_0 e^{-t-x}$ and $\partial_x j = -\rho_0 e^{-t-x}$, so $\partial_t \rho + \partial_x j = -2\rho_0 e^{-t-x} \neq 0$. Therefore the proposed pair is not conserved. Choosing $j = -\rho_0 e^{-t-x}$ instead gives $\partial_t \rho + \partial_x j = 0$.
Interpretation. Conservation constrains density and current together; neither may be chosen independently.

Example 3.28

Diffusion of a Fourier mode.
Problem. Verify that $u(x,t) = Ae^{-Dk^2t}\cos(kx)$ solves $\partial_t u = D\partial_x^2 u$.

Solution

$\partial_t u = -Dk^2 u$, while $\partial_x^2 u = -k^2 u$. Hence both sides equal $-Dk^2 u$.
Interpretation. Fine spatial structure (large k) decays faster, which is why diffusion smooths profiles.

Example 3.29

Harmonic potential.
Problem. Show that $\phi = x^2 - y^2$ is harmonic and determine its field $\mathbf{E} = -\nabla\phi$.

Solution

$\nabla^2\phi = 2 - 2 = 0$, and $\mathbf{E} = (-2x, 2y)$.
Interpretation. The potential has saddle curvature but zero average curvature. In a charge-free electrostatic region, such harmonic potentials are admissible.

Example 3.30

Vector identity check.
Problem. Verify $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for $\mathbf{A} = (yz, zx, xy)$.

Solution

Directly, $\nabla \times \mathbf{A} = (0, 0, 0)$, so its divergence vanishes. More generally, equality of mixed partial derivatives makes the divergence of every smooth curl zero.
Interpretation. This identity is the mathematical reason magnetic fields written as curls automatically satisfy $\nabla \cdot \mathbf{B} = 0$.

Example 3.31

Helmholtz-style reconstruction insight.
Problem. Suppose a smooth field in a simply connected region has both zero divergence and zero curl and decays at infinity. What can be concluded?

Solution

Zero curl implies $\mathbf{F} = \nabla\phi$. Zero divergence then gives $\nabla^2\phi = 0$. With decay at infinity and no singularities, the relevant uniqueness result gives ϕ constant, so $\mathbf{F} = 0$.
Interpretation. Divergence, curl, boundary behavior, and topology jointly determine a field; local differential data alone may not suffice without global conditions.

3.13 Computational Companion

Numerical and symbolic tools are most valuable when they expose geometry rather than replace reasoning. The companion programs reproduce the field plots and verify representative identities using finite differences.

fig:ch3-computational-gallery

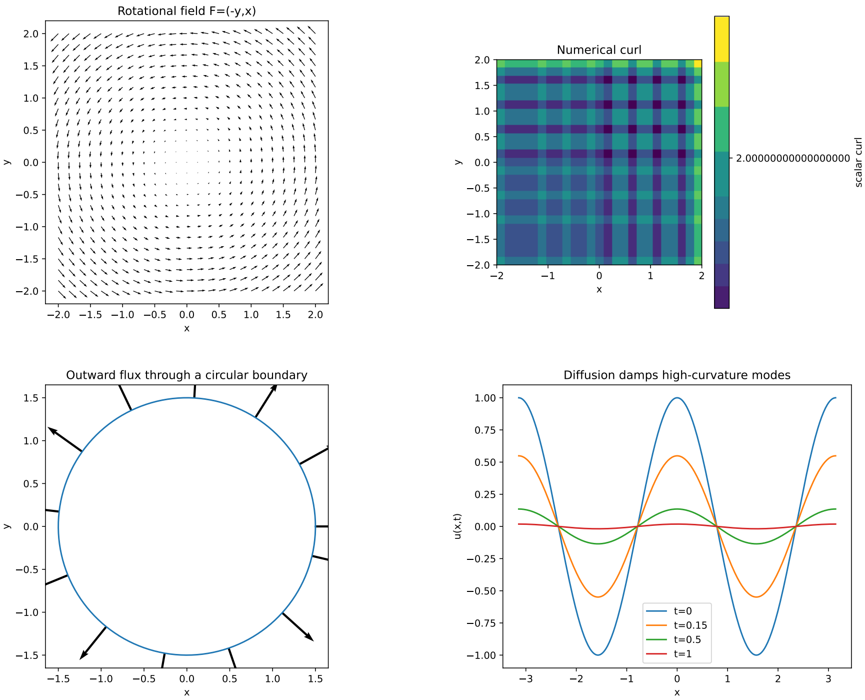


Figure 3.15: Computational field gallery: a rotational field, its curl, flux through a circle, and diffusion smoothing.

3.13.1 Python workflow

The file `examples/python/ch03/vector_calculus_companion.py` generates four publication-ready figures and reports numerical checks for divergence, curl, flux, and circulation.

Listing 3.1: Core numerical field operations used in the Chapter 3 companion.

```python
def divergence(Fx, Fy, dx, dy):
    dFx_dx = np.gradient(Fx, dx, axis=1)
    dFy_dy = np.gradient(Fy, dy, axis=0)
    return dFx_dx + dFy_dy

def scalar_curl(Fx, Fy, dx, dy):
    dFy_dx = np.gradient(Fy, dx, axis=1)
    dFx_dy = np.gradient(Fx, dy, axis=0)
    return dFy_dx - dFx_dy
```

3.13.2 Wolfram Language workflow

The matching file `examples/wolfram/ch03/vector_calculus_companion.wl` demonstrates symbolic evaluation of $\nabla \cdot \mathbf{F}$, $\nabla \times \mathbf{F}$, and the three integral theorems. Keeping both implementations makes the mathematical intent independent of one software ecosystem.

Common Mistake

A visually smooth field plot is not evidence that a derivative or integral has been computed correctly. Always test convergence under grid refinement and compare against an analytic case whose answer is known.

3.14 Deeper Physical Applications

The same four ideas recur throughout theoretical physics: a density, a current or flux, a local differential law, and a global integral statement. This section develops that architecture before later chapters attach the full physical dynamics.

3.14.1 Fluid dynamics: local expansion and rotation

For a velocity field $\mathbf{v}$, the divergence measures local volume expansion and the vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{v}$ measures local rotational tendency. Conservation of mass is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0.$$

For constant density this reduces to $\nabla \cdot \mathbf{v} = 0$, the incompressibility condition. It does not imply zero curl: shear and vortex flows can be incompressible.

Physical Interpretation

The differential equation describes what happens inside an infinitesimal control volume. Gauss' theorem converts it into a statement that the rate of mass loss inside a finite region equals the outward mass flux across its boundary.

3.14.2 Electrostatics and gravitation: sources and flux

Electrostatic and Newtonian gravitational fields share inverse-square geometry. In differential form,

$$\nabla \cdot \mathbf{E} = \frac{\rho_q}{\varepsilon_0}, \qquad \nabla \cdot \mathbf{g} = -4\pi G \rho_m.$$

The signs encode the physical distinction between positive electric sources and universally attractive gravitational mass. In source-free regions, potentials satisfy Laplace's equation. Boundary values then control the interior field through uniqueness principles.

3.14.3 Magnetism: circulation without monopole sources

Writing $\mathbf{B} = \nabla \times \mathbf{A}$ automatically gives

$$\nabla \cdot \mathbf{B} = 0.$$

Thus magnetic flux lines do not begin or end at ordinary isolated sources. Stokes' theorem relates the circulation of the vector potential around a boundary to magnetic flux through a spanning surface:

$$\oint_{\partial S} \mathbf{A} \cdot d\mathbf{l} = \iint_S \mathbf{B} \cdot d\mathbf{S}.$$

This relation later becomes central in electromagnetic induction, gauge theory, Berry phase, and the quantum Hall effect.

3.14.4 Diffusion, waves, and spectral structure

The Laplacian distinguishes three important dynamical patterns:

$$\partial_t u = D\nabla^2 u, \qquad \partial_t^2 u = c^2 \nabla^2 u, \qquad -\nabla^2 \psi = \lambda \psi.$$

They represent diffusion, waves, and spatial eigenmodes. The operator is the same, but the time law and boundary conditions produce fundamentally different behavior: irreversible smoothing, oscillatory propagation, or discrete mode spectra.

3.14.5 Quantum probability conservation

A wavefunction carries probability density $\rho = |\psi|^2$. Schrödinger evolution produces a probability current $\mathbf{j}$ satisfying

$$\frac{\partial |\psi|^2}{\partial t} + \nabla \cdot \mathbf{j} = 0.$$

The equation has exactly the same local-conservation structure as fluid mass and electric charge. Quantum mechanics changes what the density and current mean, not the underlying geometry of conservation.

One mathematical architecture

$$\text{local source} \longleftrightarrow \text{divergence}, \qquad \text{local rotation} \longleftrightarrow \text{curl},$$

$$\text{boundary flux} \longleftrightarrow \text{Gauss}, \qquad \text{boundary circulation} \longleftrightarrow \text{Stokes}.$$

This architecture is reused in continuum mechanics, Maxwell theory, gauge theory, and quantum transport.

3.14.6 From Chapter 3 to the rest of the book

Chapter 4 uses vector-valued curves to describe trajectories. Newtonian mechanics then interprets gradients as forces derived from potentials. Electromagnetism combines divergence and curl into a closed field theory. Quantum mechanics promotes the Laplacian to a kinetic-energy operator and conservation laws to probability flow. The reader should therefore regard vector calculus not as a preliminary language, but as part of the conceptual structure of physics itself.

3.15 Graded Exercise Bank

The following one hundred problems are cumulative. The four tiers distinguish technical fluency, theorem selection, synthesis, and open-ended mathematical modelling. Unless a problem states otherwise, assume sufficient smoothness for the indicated derivatives and integrals to exist.

Tier I: Foundations (30 problems)

1. Sketch the vector field $\mathbf{F} = (x, y)$ and identify its source-like behavior.
2. Sketch $\mathbf{F} = (-y, x)$ and indicate the direction of circulation.
3. Compute ∇f for $f = x^2 + 3y^2 - z$.
4. Compute $\nabla \cdot \mathbf{F}$ for $\mathbf{F} = (x^2, xy, z^2)$.
5. Compute $\nabla \times \mathbf{F}$ for $\mathbf{F} = (-y, x, 0)$.
6. Compute $\nabla^2 f$ for $f = x^2 + y^2 + z^2$.
7. Evaluate $\int_C x\, ds$ on the line segment from $(0, 0)$ to $(1, 1)$.
8. Evaluate $\int_C (2x, 2y) \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 2)$.
9. Parametrize the circle $x^2 + y^2 = a^2$ counterclockwise.
10. Find the unit tangent to $\mathbf{r}(t) = (t, t^2, 0)$.
11. Find an oriented normal to the plane $x + 2y + 3z = 6$.
12. Compute the flux of $\mathbf{F} = (0, 0, 1)$ through a horizontal disk of radius R.
13. Compute $\iiint_V 1\, dV$ for a rectangular box with sides a, b, c.
14. Write the cylindrical-coordinate volume element and explain the factor r.
15. Write the spherical-coordinate volume element and identify the polar-angle factor.

16. Determine whether $\mathbf{F} = (2x, 2y)$ is conservative.

17. Find a potential for $\mathbf{F} = (y, x)$.

18. Evaluate the circulation of a gradient field around a closed curve.

19. State Green's theorem with positive boundary orientation.

20. State Stokes' theorem and its orientation rule.

21. State Gauss' theorem for a smooth vector field.

22. Compute the outward flux of $\mathbf{F} = (x, y, z)$ through the unit sphere using Gauss' theorem.

23. Compute the divergence of a constant vector field.

24. Compute the curl of a constant vector field.

25. Show directly that $\nabla \times (\nabla f) = 0$ for $f = x^2y + z^3$.

26. Show directly that $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for $\mathbf{A} = (yz, zx, xy)$.

27. Interpret the sign of $\nabla \cdot \mathbf{v}$ in a compressible fluid.

28. Interpret $\nabla \times \mathbf{v} = 0$ in a simply connected fluid region.

29. Write the local conservation law for density ρ and current $\mathbf{J}$.

30. Explain why the boundary of a boundary has no boundary in the integral theorems.

Tier II: Intermediate synthesis (30 problems)

1. Find the flux of $\mathbf{F} = (x, y, 0)$ across the outward boundary of the unit disk using Green's theorem.

2. Evaluate $\oint_C(-y\,dx + x\,dy)$ for the circle of radius a.

3. Determine whether $\mathbf{F} = (yz, xz, xy)$ is conservative and find a potential when possible.

4. Evaluate the work of $\mathbf{F} = (y, x + 2y)$ along two different paths from $(0, 0)$ to $(1, 1)$ and compare.

5. Compute the surface area of the paraboloid $z = x^2 + y^2$ above the disk $x^2 + y^2 \leq 1$.

6. Compute the flux of $\mathbf{F} = (x, y, z)$ through the upper hemisphere and its base separately.

7. Use Gauss' theorem to find the flux of $\mathbf{F} = r^2\mathbf{r}$ through a sphere of radius R.

8. Use Stokes' theorem to compute the circulation of $\mathbf{F} = (-y, x, z)$ around a circle in the plane $z = c$.

9. Use Green's theorem to compute the area enclosed by an ellipse.

10. Derive the polar-coordinate area element from a Jacobian determinant.

11. Derive the cylindrical-coordinate scale factors.

12. Compute $\nabla^2(1/r)$ for $r \neq 0$.

13. Verify that $u = e^{kx}\cos ky$ is harmonic in two dimensions.

14. Solve the steady one-dimensional diffusion equation with fixed endpoint temperatures.

15. For $\mathbf{v} = (ax, by, cz)$, determine the condition for incompressibility.

16. For rigid rotation $\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r}$, compute the vorticity.

17. Derive the integral continuity equation from the local one using Gauss' theorem.

18. Derive the local continuity equation from the integral form for arbitrary fixed volumes.

19. Show that a radial inverse-square field has constant flux through concentric spheres.

20. Find the electric field from $\phi = k/r$ away from the origin.

21. Compute the gravitational field produced by a spherically symmetric mass distribution outside its support.

22. Use integration by parts to derive Green's first identity.

23. Use Green's first identity to prove uniqueness for a Dirichlet problem under suitable assumptions.

24. Show that $\int_V |\nabla u|^2 dV = -\int_V u\nabla^2 u\, dV$ when $u = 0$ on the boundary.

25. Explain why a nonzero circulation around a puncture does not contradict zero curl away from the puncture.

26. Compute the circulation of $\mathbf{F} = (-y/(x^2+y^2), x/(x^2+y^2))$ around a circle centered at the origin.

27. Find the divergence and curl of $\mathbf{F} = f(r)\mathbf{r}$.

28. Derive the radial Laplacian of a spherically symmetric scalar field.

29. Compute the mean value of a harmonic function over a circle for a simple polynomial example.

30. Verify numerically or symbolically one vector identity from the chapter.

Tier III: Advanced analysis (20 problems)

1. Prove the path-independence theorem for conservative fields on a connected domain.

2. Give a counterexample showing that zero curl need not imply a global potential on a nonsimply connected domain.

3. Derive the divergence formula in orthogonal curvilinear coordinates from scale factors.

4. Derive the Laplacian in cylindrical coordinates.

5. Derive the Laplacian in spherical coordinates for a radial function.

6. Establish Green's second identity and state one consequence.

7. Use the divergence theorem to derive the electrostatic Poisson equation from Gauss' law.

8. Derive the wave equation from a first-order field system in a source-free region.

9. Prove that the flux of a curl through any closed smooth surface is zero.

10. Prove that the circulation of a gradient around any closed piecewise-smooth loop is zero.

11. Analyze the distributional divergence of $\mathbf{r}/r^3$ and determine its total source strength.

12. For a time-dependent control volume, state and derive the Reynolds transport theorem in a simple setting.

13. Derive the probability-current continuity equation from the Schrödinger equation.

14. Show how gauge transformation $\mathbf{A} \mapsto \mathbf{A} + \nabla\chi$ leaves $\mathbf{B} = \nabla \times \mathbf{A}$ unchanged.

15. Explain the differential-form pattern behind gradient, curl, divergence, and the identity $d^2 = 0$.

16. Prove a Helmholtz-type uniqueness statement under clearly stated decay and regularity assumptions.

17. Construct a divergence-free field from a vector potential and discuss gauge freedom.

18. Construct a curl-free field from a scalar potential and discuss topology.

19. Analyze boundary terms in an eigenvalue problem for the Laplacian.

20. Show that Laplacian eigenfunctions with distinct eigenvalues are orthogonal under suitable boundary conditions.

Tier IV: Challenge and computational investigations (20 problems)

1. Derive the fundamental solution of the three-dimensional Laplacian and fix its normalization by flux.

2. Derive the two-dimensional logarithmic fundamental solution of the Laplacian.

3. Use separation of variables to solve Laplace's equation inside a disk with one nontrivial boundary Fourier mode.

4. Use separation of variables to solve the heat equation on an interval with homogeneous Dirichlet data.

5. Prove an energy-decay identity for the heat equation.

6. Derive Poynting's theorem from Maxwell's equations using a vector identity.

7. Relate circulation quantization around a puncture to phase winding of a complex wavefunction.

8. Analyze a magnetic vector potential for a uniform magnetic field in two different gauges.

9. Use Stokes' theorem to explain why gauge-related potentials yield the same magnetic flux.

10. Formulate Green, Stokes, and Gauss as instances of the generalized Stokes theorem.

11. Derive the weak form of Poisson's equation and identify the natural boundary term.

12. Prove a maximum principle for harmonic functions in a bounded domain at an appropriate level of rigor.

13. Derive the multipole expansion of a localized source through the first nontrivial terms.

14. Analyze the field and flux of a line singularity using cylindrical symmetry.

15. Explain how local differential laws can fail to determine global holonomy on a multiply connected domain.

16. Develop a computational experiment comparing direct surface integration with Gauss' theorem on a deformed closed surface.

17. Develop a computational experiment comparing direct line integration with Stokes' theorem on two spanning surfaces.

18. Prove conservation of total probability from the continuity equation under decaying or no-flux boundary conditions.

19. Show how the covariant divergence generalizes flux conservation in curvilinear coordinates.

20. Write a concise synthesis connecting vector calculus, gauge potentials, Berry phases, and later Quantum Hall theory.

Selected hints and solution outlines

Foundation 22.

Since $\nabla \cdot \mathbf{r} = 3$, the flux through the unit sphere is $3(4\pi/3) = 4\pi$.

Intermediate 8.

Choose the flat disk as the spanning surface; only the normal component of $\nabla \times \mathbf{F}$ contributes.

Intermediate 17.

Integrate $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ over V and use Gauss' theorem.

Intermediate 25.

The domain excludes the origin. Topology, rather than a local derivative, controls the loop integral.

Advanced 11. Integrate over a ball with a small central ball removed, then take the inner radius to zero.

Advanced 13. Multiply the Schrödinger equation by ψ^*, subtract its conjugate, and collect the result as a divergence.

Challenge 1. Begin with spherical symmetry, solve away from the origin, and determine the coefficient from unit total flux.

Challenge 6. Dot Faraday's law with $\mathbf{H}$, dot Ampere–Maxwell with $\mathbf{E}$, subtract, and use $\nabla \cdot (\mathbf{E} \times \mathbf{H})$.

Challenge 8. Compare symmetric gauge $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$ with a Landau gauge and exhibit the gauge function.

Challenge 20.

Separate local curvature, encoded by field strength, from global phase information encoded by holonomy.

3.16 Notation, Glossary, and Repository Integration

This chapter fixes the repository-wide language for local field operators and global integral laws. Later chapters may therefore use the sequence

gradient $\longrightarrow$ local field operators $\longrightarrow$ circulation and flux $\longrightarrow$ boundary theorems

without redefining its mathematical conventions.

Canonical vector-calculus notation

For a scalar field f and vector field $\mathbf{F}$ in Euclidean three-space,

$$\nabla f, \qquad \nabla \cdot \mathbf{F}, \qquad \nabla \times \mathbf{F}, \qquad \nabla^2 f$$

denote gradient, divergence, curl, and scalar Laplacian. Directed line elements are written $d\mathbf{r}$, oriented surface elements $d\mathbf{S} = \mathbf{n}\, dS$, and volume elements dV. Closed line integrals use $\oint$. Closed surface integrals use the ordinary surface-integral symbol together with an explicit statement of closure and orientation.

Local object	Global quantity	Governing bridge
∇f	endpoint potential difference	fundamental theorem for line integrals
$\nabla \times \mathbf{F}$	boundary circulation	Green / Stokes
$\nabla \cdot \mathbf{F}$	outward boundary flux	Gauss
$\partial_t \rho + \nabla \cdot \mathbf{J}$	conserved total quantity	continuity equation

Dependency map into physics

Mechanics uses line integrals for work and potentials. Fluid dynamics uses divergence, curl, and material transport. Electromagnetism unifies all three boundary theorems in Maxwell's equations. Quantum mechanics uses the Laplacian as kinetic energy and the continuity equation for probability. Gauge theory and Quantum Hall physics later add global phase and topology to this same local-to-global framework.

3.17 Summary and Looking Ahead

Vector calculus supplies a unified language for fields. The gradient converts a scalar field into its steepest-change vector field. Divergence measures local source strength, curl measures local circulation, and the Laplacian measures accumulated spatial curvature. Line, surface, and volume integrals convert local quantities into work, circulation, flux, mass, charge, energy, and probability.

Green's, Stokes', and Gauss' theorems reveal one organizing principle:

$$\boxed{\text{integral of a derivative over an interior} = \text{integral of the field over its boundary.}}$$

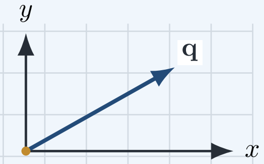
This principle underlies conservation laws and the integral formulations of electromagnetism. In Chapter 4, these tools will be applied to motion and changing physical systems; later they will become the natural language of Maxwell's equations, fluid dynamics, and quantum probability current.

Chapter 4

Ordinary Differential Equations and Dynamical Systems

Laws of change

A differential equation does not merely describe a state; it specifies how one state produces the next.



Chapter Roadmap

Physical laws as evolution equations $\longrightarrow$ first-order initial-value problems $\longrightarrow$ second-order motion $\longrightarrow$ coupled systems $\longrightarrow$ phase space, equilibrium, and stability $\longrightarrow$ nonlinear dynamics and numerical evolution.

Learning Objectives

After completing this chapter, the reader should be able to:

- translate a verbal law of change into an ordinary differential equation;
- distinguish general solutions, particular solutions, and initial-value problems;
- solve separable and first-order linear equations and interpret their characteristic time scales;
- derive and classify free, damped, and driven oscillator equations;
- rewrite higher-order equations as first-order systems and analyze them using matrices;
- read direction fields and phase portraits as geometric representations of dynamics;
- classify equilibria by linearization and connect stability to physical behavior.

4.1 Differential Equations as Laws of Change

A function records how one quantity depends on another. A differential equation goes further: it constrains the rate at which that function may change. In physics this distinction is fundamental. A position $x(t)$ is a description of motion; Newton's law is a rule that selects the physically allowed motions from all imaginable curves.

Ordinary differential equation

An *ordinary differential equation* (ODE) is a relation involving an unknown function of one independent variable and one or more of its derivatives. Its order is the order of the highest derivative that appears.

Examples include

$$\frac{dN}{dt} = -\lambda N, \qquad \text{radioactive decay,} \tag{4.1}$$

$$C\frac{dV}{dt} + \frac{1}{R}V = \frac{1}{R}V_{\text{in}}, \qquad \text{RC circuit,} \tag{4.2}$$

$$m\frac{d^2x}{dt^2} + kx = 0, \qquad \text{harmonic oscillator.} \tag{4.3}$$

The symbols differ, but the logic is the same: the present state and the law determine how the system evolves.

State plus law plus initial data

A deterministic initial-value problem has three ingredients:

$$\text{state variables} + \text{evolution law} + \text{initial data.}$$

The differential equation supplies local change; the initial condition selects one trajectory from the family of mathematically possible solutions.

4.1.1 General and particular solutions

For

$$\frac{dy}{dt} = ay,$$

separation gives the family

$$y(t) = Ce^{at}.$$

The arbitrary constant C represents missing information. If $y(0) = y_0$, then $C = y_0$ and the initial-value problem has the particular solution

$$\boxed{y(t) = y_0e^{at}}.$$

Thus integration constants are not algebraic leftovers; they encode the freedom fixed by physical preparation.

4.1.2 Existence, uniqueness, and predictability

An equation may have no solution, one solution, or many solutions through the same initial point. For a first-order equation

$$\dot{y} = f(t, y),$$

continuity of f near (t_0, y_0) supports local existence, while a local Lipschitz condition in y supports uniqueness. The precise theorem will be developed later, but its physical meaning is immediate: uniqueness is the mathematical form of deterministic prediction.

A formula is not automatically a law

Writing $y(t) = t^2$ describes one curve. Writing $\dot{y} = 2t$ specifies a family of curves, and adding $y(0) = 0$ selects the particular curve t^2. Description, law, and initial condition must not be confused.

4.1.3 Autonomous and nonautonomous equations

An ODE is *autonomous* when the independent variable does not appear explicitly:

$$\dot{y} = f(y).$$

It is *nonautonomous* when external time dependence appears:

$$\dot{y} = f(t, y).$$

Autonomous equations are naturally represented in phase space because the velocity at a state depends only on that state. Nonautonomous equations often describe externally driven systems.

From geometric motion to differential law

Newton formulated motion through fluxions and forces; Leibniz supplied the differential notation that made rates of change algebraically portable. Euler systematized differential equations as a distinct subject, while Cauchy placed initial-value problems and local existence on a more rigorous foundation. Their combined legacy is the modern idea that a physical law is an evolution rule posed together with data.

Connection to physics

The same architecture recurs throughout the book. Newton's equation evolves position and momentum; Maxwell's equations evolve electromagnetic fields; the Schrödinger equation evolves a quantum state; diffusion equations evolve density; Einstein's equations constrain the geometry of spacetime. The mathematical details change, but the central question remains: given a state now, what states are allowed next?

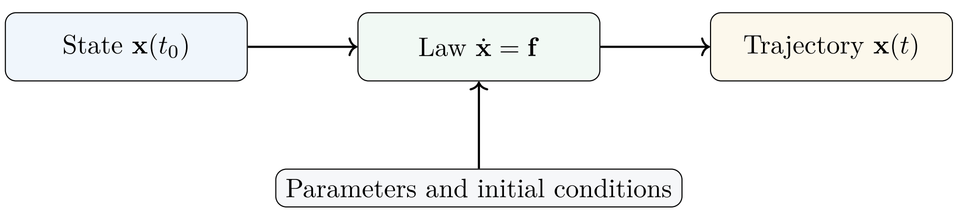


Figure 4.1: The architecture of deterministic evolution: a state, a law, and initial data define a trajectory.

4.2 First-Order Differential Equations

A first-order ODE relates a function to its first derivative. Although simple in form, such equations already model decay, charging, cooling, relaxation, transport, and population growth.

4.2.1 Separable equations

An equation is separable when it can be written

$$\frac{dy}{dt} = g(t)h(y).$$

Where $h(y) \neq 0$, rearrange and integrate:

$$\int \frac{dy}{h(y)} = \int g(t)\,dt + C.$$

The method is justified by the chain rule, not by treating dy and dt as independent numbers.

Example 4.1

A sample contains N_0 unstable nuclei and obeys $\dot{N} = -\lambda N$ with $\lambda > 0$. Find $N(t)$ and the half-life.

Solution

Separate variables:

$$\frac{dN}{N} = -\lambda\,dt.$$

Integrating gives $\ln N = -\lambda t + C$, hence

$$N(t) = N_0 e^{-\lambda t}.$$

The half-life $t_{1/2}$ satisfies $N(t_{1/2}) = N_0/2$, so

$$\boxed{t_{1/2} = \frac{\ln 2}{\lambda}}.$$

The inverse rate $\tau = 1/\lambda$ is the characteristic decay time.

4.2.2 First-order linear equations

The standard form is

$$\dot{y} + p(t)y = q(t).$$

Multiplication by the integrating factor

$$\mu(t) = \exp\left(\int p(t)\,dt\right)$$

turns the left side into a product derivative:

$$\frac{d}{dt}(\mu y) = \mu q.$$

Therefore

$$\boxed{y(t) = \frac{1}{\mu(t)}\left[C + \int^t \mu(s)q(s)\,ds\right]}.$$

Example 4.2

A body at temperature $T(t)$ is placed in an environment at constant temperature T_a. Newton's cooling law is

$$\dot{T} = -k(T - T_a), \qquad k > 0.$$

For $T(0) = T_0$, determine $T(t)$.

Solution

Set $u = T - T_a$. Then $\dot{u} = -ku$, so

$$u(t) = (T_0 - T_a)e^{-kt}.$$

Hence

$$\boxed{T(t) = T_a + (T_0 - T_a)e^{-kt}}.$$

The difference from equilibrium decays exponentially; the ambient temperature is a stable fixed point.

4.2.3 Equilibria and qualitative behavior

For an autonomous equation $\dot{y} = f(y)$, an equilibrium y_* satisfies $f(y_*) = 0$. The sign of f on either side determines whether neighboring solutions move toward or away from y_*. This one-dimensional phase-line analysis anticipates the stability theory of systems.

For the logistic equation

$$\dot{P} = rP\left(1 - \frac{P}{K}\right),$$

the equilibria are $P = 0$ and $P = K$. For $r > 0$, small positive populations grow away from zero and toward the carrying capacity K.

Example 4.3

Solve the logistic initial-value problem $\dot{P} = rP(1 - P/K)$, $P(0) = P_0 > 0$.

Solution

Partial fractions give

$$\frac{1}{P(1 - P/K)} = \frac{1}{P} + \frac{1}{K - P}.$$

Integration yields

$$\ln \frac{P}{K - P} = rt + C.$$

Using the initial condition,

$$\boxed{P(t) = \frac{K}{1 + \left(\frac{K - P_0}{P_0}\right) e^{-rt}}}.$$

The early-time behavior is approximately exponential, while nonlinear feedback produces saturation.

4.2.4 Direction fields

The differential equation $\dot{y} = f(t, y)$ assigns a slope to every point in the (t, y) plane. Drawing short line segments of slope $f(t, y)$ produces a direction field. A solution curve is tangent to this field everywhere.

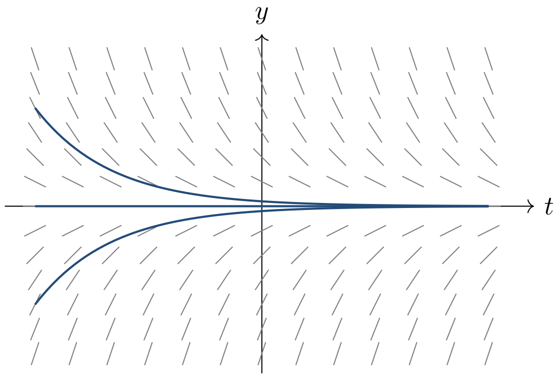


Figure 4.2: A direction field for $\dot{y} = -y$ and representative solution curves. Local slopes organize the global family.

Relaxation as universal behavior

Many systems near stable equilibrium obey $\dot{u} = -u/\tau$: a capacitor voltage, a temperature difference, a chemical imbalance, or a small perturbation of a damped device. The microscopic mechanisms differ, but the same linear relaxation equation produces the same exponential law.

4.3 Second-Order Equations and Oscillation

Second-order equations arise naturally because acceleration is the second derivative of position. The prototype is the oscillator, whose mathematics later reappears in normal modes, waves, quantum mechanics, and field theory.

4.3.1 Constant-coefficient equations

Consider

$$a\ddot{x} + b\dot{x} + cx = 0.$$

Trying $x = e^{rt}$ gives the characteristic equation

$$ar^2 + br + c = 0.$$

Its roots determine the qualitative motion: distinct real roots give sums of exponentials, a repeated root introduces a factor of t, and complex-conjugate roots produce oscillation.

4.3.2 The ideal harmonic oscillator

Hooke's law $F = -kx$ and Newton's second law produce

$$m\ddot{x} + kx = 0.$$

Defining $\omega_0 = \sqrt{k/m}$ gives

$$\ddot{x} + \omega_0^2 x = 0,$$

with solution

$$\boxed{x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)}.$$

Equivalently, $x(t) = C\cos(\omega_0 t - \phi)$. The constants encode amplitude and phase.

The velocity is $v = \dot{x}$, and the energy

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$$

is conserved because

$$\frac{dE}{dt} = \dot{x}(m\ddot{x} + kx) = 0.$$

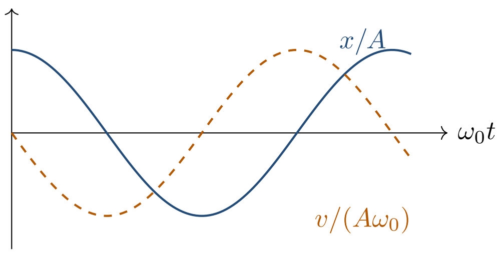


Figure 4.3: Position and velocity of an ideal oscillator. Their quarter-cycle phase offset encodes continual exchange between potential and kinetic energy.

4.3.3 Damped motion

With linear resistance $-b\dot{x}$,

$$m\ddot{x} + b\dot{x} + kx = 0.$$

Define

$$\gamma = \frac{b}{2m}, \qquad \omega_0 = \sqrt{\frac{k}{m}}.$$

The characteristic roots are

$$r = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}.$$

This produces three regimes:

- underdamped: $\gamma < \omega_0$;
- critically damped: $\gamma = \omega_0$;
- overdamped: $\gamma > \omega_0$.

For underdamping,

$$x(t) = e^{-\gamma t}\left[A\cos(\omega_d t) + B\sin(\omega_d t)\right], \qquad \omega_d = \sqrt{\omega_0^2 - \gamma^2}.$$

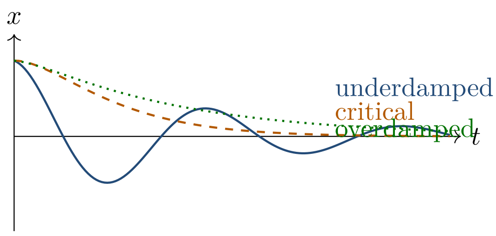


Figure 4.4: Underdamped, critically damped, and overdamped return toward equilibrium.

4.3.4 Driven oscillation and resonance

An external sinusoidal force gives

$$m\ddot{x} + b\dot{x} + kx = F_0 \cos(\Omega t).$$

After transients decay, the steady-state response has the form

$$x_{\text{ss}}(t) = A(\Omega)\cos(\Omega t - \delta),$$

where

$$A(\Omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2}}.$$

The response is largest near resonance. Damping limits the peak and produces a frequency-dependent phase lag.

Homogeneous plus particular

For a linear driven equation, the complete solution is

$$x = x_{\text{transient}} + x_{\text{steady}}.$$

The homogeneous part remembers the initial condition; damping erases that memory. The particular part follows the persistent drive.

Why oscillators dominate physics

Near a stable equilibrium, a smooth potential is approximately quadratic:

$$V(x) \approx V(x_*) + \frac{1}{2}V''(x_*)(x - x_*)^2.$$

Therefore small deviations obey an oscillator equation. Molecular vibrations, lattice modes, electrical resonances, optical cavities, and quantum field modes all inherit this local quadratic structure.

4.4 Systems of Differential Equations

Many physical states require several variables. A system of first-order equations is written compactly as

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}).$$

A higher-order equation can always be converted into such a system by introducing additional state variables.

4.4.1 From one second-order equation to two first-order equations

For

$$\ddot{x} + \omega_0^2 x = 0,$$

define $v = \dot{x}$. Then

$$\dot{x} = v, \qquad \dot{v} = -\omega_0^2 x.$$

The state is the point (x, v) in phase space. Time evolution traces a curve rather than a graph of x alone.

4.4.2 Linear systems and matrices

A linear autonomous system has the form

$$\dot{\mathbf{x}} = A\mathbf{x}.$$

If $A\mathbf{v} = \lambda\mathbf{v}$, then

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$$

is a modal solution. A basis of eigenvectors decomposes general motion into independent modes.

Example 4.4

Consider

$$\dot{x} = 3x + y, \qquad \dot{y} = x + 3y.$$

Classify and solve the system.

Solution

The matrix

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

has eigenvalues 4 and 2 with eigenvectors $(1,1)^T$ and $(1,-1)^T$. Therefore

$$\boxed{\begin{pmatrix} x \\ y \end{pmatrix} = c_1 e^{4t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{2t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}}.$$

Both modes grow, so the origin is unstable.

4.4.3 Coupled oscillators and normal modes

For two identical masses coupled by springs, the displacements satisfy a matrix equation

$$M\ddot{\mathbf{x}} + K\mathbf{x} = 0.$$

Seeking $\mathbf{x} = \mathbf{a}e^{i\omega t}$ gives

$$(K - \omega^2 M)\mathbf{a} = 0.$$

Nontrivial amplitudes require

$$\det(K - \omega^2 M) = 0.$$

The allowed frequencies are normal-mode frequencies, and the corresponding eigenvectors specify collective patterns of motion.

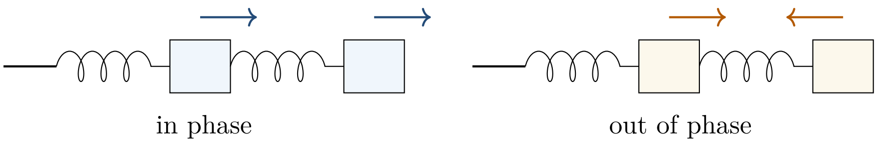


Figure 4.5: Two elementary normal modes of a symmetric coupled system: in-phase and out-of-phase motion.

4.4.4 Matrix exponential

The formal solution of $\dot{\mathbf{x}} = A\mathbf{x}$ is

$$\boxed{\mathbf{x}(t) = e^{At}\mathbf{x}(0)},$$

where

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \cdots .$$

This is the finite-dimensional prototype of evolution operators used later in quantum mechanics.

From modes to particles and fields

Normal-mode analysis is not confined to springs. Vibrations of molecules, acoustic modes, electromagnetic cavity modes, phonons in crystals, and free quantum fields all reduce to eigenvalue problems. Diagonalizing the coupling matrix reveals the independent dynamical degrees of freedom.

4.5 Phase Space, Equilibria, and Stability

A phase space contains all variables needed to specify the instantaneous state. A deterministic autonomous law assigns a velocity vector to every point, producing a vector field of possible evolution.

4.5.1 Trajectories in state space

For $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, a trajectory is a curve whose tangent vector equals $\mathbf{f}$. Distinct trajectories cannot cross where uniqueness holds: crossing would assign two futures to the same state.

For the harmonic oscillator,

$$\dot{x} = v, \qquad \dot{v} = -\omega_0^2 x,$$

energy conservation gives

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = E.$$

Thus phase trajectories are ellipses. Every closed orbit represents periodic motion.

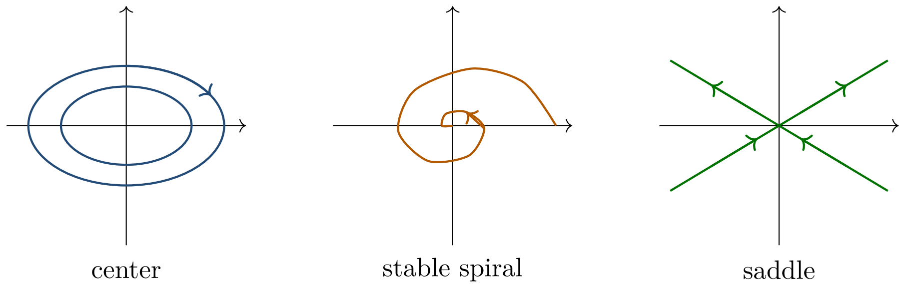


Figure 4.6: Representative phase portraits: a center, a stable spiral, and a saddle.

4.5.2 Linearization near an equilibrium

An equilibrium $\mathbf{x}_*$ satisfies $\mathbf{f}(\mathbf{x}_*) = 0$. Write $\mathbf{x} = \mathbf{x}_* + \boldsymbol{\eta}$. To first order,

$$\dot{\boldsymbol{\eta}} = J(\mathbf{x}_*)\boldsymbol{\eta},$$

where J is the Jacobian matrix. The eigenvalues of J determine local behavior.

For a two-dimensional system:

- negative real parts imply local attraction;

- positive real parts imply instability;
- opposite signs produce a saddle;
- purely imaginary eigenvalues require further analysis but often indicate a center in conservative linear systems.

4.5.3 Lyapunov viewpoint

A scalar function $V(\mathbf{x})$ can establish stability without solving the equations explicitly. If V has a strict minimum at equilibrium and $\dot{V} \leq 0$ along trajectories, then V acts like a generalized energy that cannot increase.

For the damped oscillator,

$$V(x, v) = \frac{1}{2}mv^2 + \frac{1}{2}kx^2,$$

and

$$\dot{V} = -bv^2 \leq 0.$$

Energy decreases monotonically, so trajectories spiral toward the origin.

Geometry can replace explicit solution

A phase portrait may reveal fixed points, periodic orbits, attraction, repulsion, and invariant regions even when no closed-form solution exists. Modern dynamical systems theory therefore studies the geometry of trajectories, not only formulas for $x(t)$.

4.5.4 A first glimpse of nonlinear dynamics

Nonlinear systems can exhibit multiple equilibria, limit cycles, bifurcations, and sensitive dependence. The equation

$$\dot{x} = \mu x - x^3$$

changes stability as the parameter μ passes through zero. For $\mu < 0$, only $x = 0$ is stable. For $\mu > 0$, the origin becomes unstable and two stable equilibria $x = \pm\sqrt{\mu}$ appear.

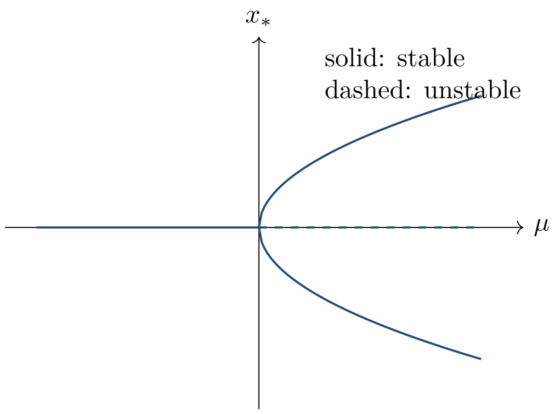


Figure 4.7: A supercritical pitchfork bifurcation: varying a parameter changes both the number and stability of equilibria.

Stability is often more important than exact solvability

Engineers ask whether a bridge oscillation decays, physicists ask whether an equilibrium persists under perturbation, and control theorists ask whether feedback drives a state toward a target. These are stability questions. Exact formulas may be unavailable, but local eigenvalues and energy-like functions can still answer them.

4.6 Visual Synthesis: From Equation to Dynamics

The same dynamical law can be represented in several complementary ways. No single representation is sufficient in every problem.

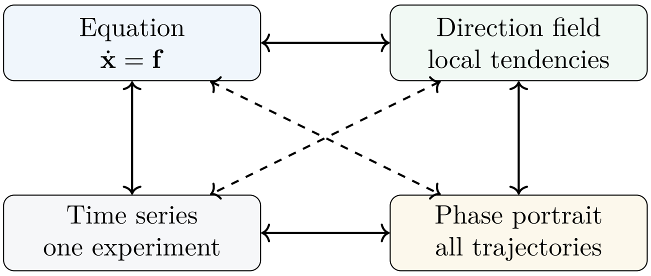


Figure 4.8: Four complementary representations of an ODE: symbolic equation, direction field, time series, and phase portrait.

Equation. Best for derivation, parameter dependence, and exact manipulation.

Direction field. Displays local tendencies without solving globally.

Time series. Shows what an observer records as time passes.

Phase portrait. Reveals the global organization of all initial conditions.

Core results of Commits 36–38

- An ODE combines a local law of change with initial data to determine a trajectory.
- Separable and linear first-order equations model decay, relaxation, and saturation.
- Second-order linear equations classify free, damped, and driven oscillations through characteristic roots.
- Higher-order equations become first-order systems, making matrix and eigenvalue methods available.
- Phase space turns dynamics into geometry; equilibria and their stability are read from vector fields and linearization.
- Nonlinear parameters can reorganize the set of equilibria through bifurcations.

Looking ahead

The next production stage will expand this core with a larger worked-example sequence, numerical integration in Python and Wolfram Language, driven and nonlinear applications, and a systematic bridge into mechanics and quantum evolution.

4.7 Thirty Integrated Worked Examples

These examples move from scalar initial-value problems through oscillators, coupled systems, stability, numerical evolution, and later physics.

Example 4.5

Exponential decay.
Problem. Solve $\dot{N} = -\lambda N$ with $N(0) = N_0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$N = N_0 e^{-\lambda t}$, with $\tau = 1/\lambda$ and half-life $\ln 2/\lambda$.
Interpretation. The decay rate fixes a natural time scale.

Example 4.6

Newton cooling.
Problem. Solve $\dot{T} = -k(T - T_a)$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$T = T_a + (T_0 - T_a)e^{-kt}$.
Interpretation. The temperature difference from the environment decays.

Example 4.7

Logistic growth.
Problem. Solve $\dot{P} = rP(1 - P/K)$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$P = K/[1 + Ae^{-rt}]$, where $A = (K - P_0)/P_0$.
Interpretation. Growth changes from exponential to saturation.

Example 4.8

RC charging.
Problem. Solve $RC\dot{V}_C + V_C = V_0$, $V_C(0) = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$V_C = V_0(1 - e^{-t/(RC)})$.
Interpretation. The time constant is RC.

Example 4.9

Radioactive dating.
Problem. A sample retains fraction f of its isotope. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$t = -\ln f/\lambda$.
Interpretation. A measured fraction becomes an age through the decay law.

Example 4.10

Mixing tank.
Problem. A well-mixed tank of volume V receives flow q at concentration c_{in}. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$\dot{S} = qc_{in} - (q/V)S$, so $S = Vc_{in} + (S_0 - Vc_{in})e^{-qt/V}$.
Interpretation. Conservation appears as input minus output.

Example 4.11

Linear drag.
Problem. Solve $m\dot{v} = mg - bv$, $v(0) = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$v = v_T(1 - e^{-t/\tau})$, $v_T = mg/b$, $\tau = m/b$.
Interpretation. Terminal speed is a stable equilibrium.

Example 4.12

Unstable equilibrium.
Problem. Solve $\dot{x} = \alpha x$ near $x = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$x = x_0 e^{\alpha t}$; for $\alpha > 0$ perturbations grow.
Interpretation. Linear growth signals instability.

Example 4.13

Exact equation.
Problem. Solve $(2xy + 1)dx + (x^2 + 2y)dy = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$M_y = N_x = 2x$, so $x^2y + x + y^2 = C$.
Interpretation. Exactness reveals a conserved level set.

Example 4.14

Bernoulli equation.
Problem. Solve $\dot{y} + y = xy^2$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

With $u = y^{-1}$, $\dot{u} - u = -x$; solve the linear equation and invert.
Interpretation. A nonlinear equation can hide a linear structure.

Example 4.15

Undamped oscillator.
Problem. Solve $\ddot{x} + \omega_0^2 x = 0$, $x(0) = A$, $\dot{x}(0) = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$x = A \cos \omega_0 t$.
Interpretation. Energy alternates between kinetic and potential forms.

Example 4.16

Energy portrait.
Problem. Find oscillator speed at position x. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$v = \pm\omega_0\sqrt{A^2 - x^2}$ from energy conservation.
Interpretation. The phase curve is an ellipse.

Example 4.17

Underdamped motion.
Problem. Solve $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$ for $\gamma < \omega_0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$x = e^{-\gamma t}(C \cos \omega_d t + D \sin \omega_d t)$, $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$.
Interpretation. Oscillation survives inside a shrinking envelope.

Example 4.18

Critical damping.
Problem. Classify $\ddot{x} + 2\omega_0\dot{x} + \omega_0^2 x = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$x = (C + Dt)e^{-\omega_0 t}$.
Interpretation. This is the fastest nonoscillatory return.

Example 4.19

Driven amplitude.
Problem. Find the steady amplitude of a damped driven oscillator. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$A(\Omega) = (F_0/m)/\sqrt{(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2}$.
Interpretation. Damping limits and broadens resonance.

Example 4.20

Driven phase.
Problem. Find the steady-state phase lag. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$\tan\delta = 2\gamma\Omega/(\omega_0^2 - \Omega^2)$.
Interpretation. The response shifts phase across resonance.

Example 4.21

Small-angle pendulum.
Problem. Linearize $\ddot{\theta} + (g/L)\sin\theta = 0$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$\sin\theta \approx \theta$ gives $\ddot{\theta} + (g/L)\theta = 0$.
Interpretation. The approximation is local in amplitude.

Example 4.22

Coupled oscillators.
Problem. Find the normal frequencies of two equal coupled masses. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

Modes $(1, 1)$ and $(1, -1)$ give $\omega_+^2 = k/m$ and $\omega_-^2 = (k + 2\kappa)/m$.
Interpretation. Normal coordinates decouple collective motion.

Example 4.23

Matrix exponential.
Problem. Solve $\dot{\mathbf{x}} = A\mathbf{x}$ for diagonal A. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$\mathbf{x}(t) = e^{At}\mathbf{x}_0 = (e^{at}x_0, e^{bt}y_0)^T$.
Interpretation. Eigenvalues set independent rates.

Example 4.24

Stable spiral.
Problem. Classify $A = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

Eigenvalues $-1 \pm 2i$ imply inward spirals.
Interpretation. Real parts control growth; imaginary parts rotation.

Example 4.25

Saddle point.
Problem. Classify $\dot{x} = x$, $\dot{y} = -2y$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

The eigenvalues have opposite signs.
Interpretation. Stable and unstable manifolds meet at the equilibrium.

Example 4.26

Predator-prey equilibrium.
Problem. Find the nonzero equilibrium of the Lotka–Volterra equations. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$(x_*, y_*) = (c/d, a/b)$.
Interpretation. Each population makes the other population growth rate vanish.

Example 4.27

SIR threshold.
Problem. Determine when infections initially grow. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$\dot{I} = I(\beta S - \gamma)$, so growth requires $S > \gamma/\beta$.
Interpretation. Transmission must exceed removal.

Example 4.28

Conservative phase curves.
Problem. Find invariant curves for $\dot{x} = v$, $\dot{v} = -\omega_0^2 x$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$E = \frac{1}{2}v^2 + \frac{1}{2}\omega_0^2 x^2$ is constant.
Interpretation. Closed curves encode periodic motion.

Example 4.29

Pitchfork bifurcation.
Problem. Analyze $\dot{x} = \mu x - x^3$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

Equilibria are 0 and, for $\mu > 0$, $\pm\sqrt{\mu}$; use $f'(x)$ for stability.
Interpretation. A parameter creates two symmetry-broken states.

Example 4.30

Euler step.
Problem. Apply one Euler step to $\dot{y} = -2y$, $h = 0.1$, $y_0 = 1$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$y_1 = 1 + 0.1(-2) = 0.8$.
Interpretation. Euler advances along the local tangent.

Example 4.31

RK4 update.
Problem. State the fourth-order Runge–Kutta update. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$y_{n+1} = y_n + h(k_1 + 2k_2 + 2k_3 + k_4)/6$.
Interpretation. Four weighted slopes produce fourth-order global accuracy.

Example 4.32

Stiff decay.
Problem. Explain Euler instability for $\dot{y} = -100y$ and $h = 0.05$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

The amplification factor is $1 - 100h = -4$, with magnitude greater than one.
Interpretation. A stable exact solution can have an unstable discretization.

Example 4.33

RLC circuit.
Problem. Write the series RLC charge equation. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$L\ddot{Q} + R\dot{Q} + Q/C = V(t)$.
Interpretation. The circuit and the driven oscillator share one equation.

Example 4.34

Quantum evolution preview.
Problem. Interpret $i\hbar\dot{\psi} = H\psi$. **Theory.** Identify the equation type, characteristic scales, and initial data before calculating.

Solution

$\psi(t) = e^{-iHt/\hbar}\psi(0)$ for constant H.
Interpretation. The generator produces norm-preserving unitary evolution.

4.8 Computational Companion

Numerical integration is indispensable for nonlinear, coupled, and driven systems. The companion compares Euler and fourth-order Runge–Kutta methods, tests convergence, draws a nonlinear phase portrait, and computes resonance.

4.8.1 Python workflow

The executable file `examples/python/ch04/dynamical_systems_companion.py` generates the four figures below and prints numerical error checks.

Listing 4.1: Core explicit time-stepping algorithms.

```
def euler(f, t, y0):
    y = np.empty_like(t); y[0] = y0
    for n in range(len(t)-1):
        h = t[n+1]-t[n]
        y[n+1] = y[n] + h*f(t[n], y[n])
    return y
```

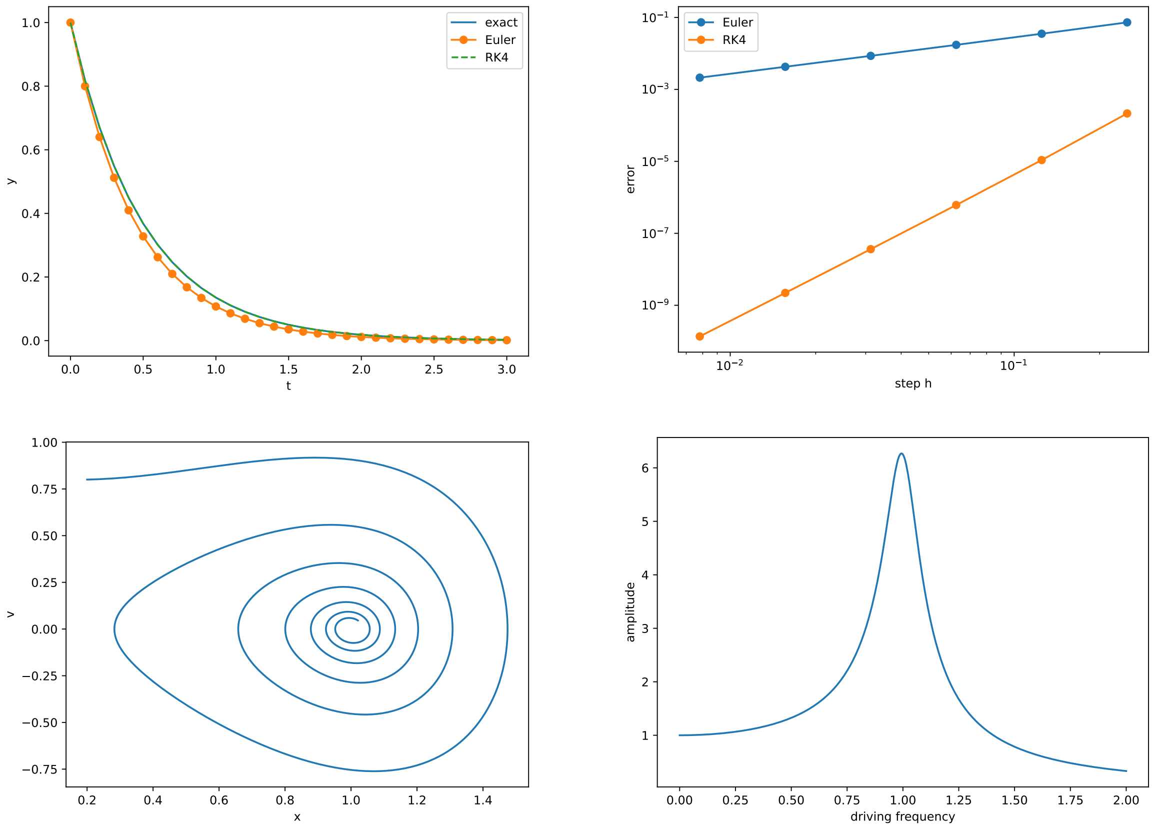


Figure 4.9: Computational dynamics gallery: numerical decay, convergence, nonlinear phase flow, and resonance.

4.8.2 Wolfram Language workflow

The matching file `examples/wolfram/ch04/dynamical_systems_companion.wl` uses `DSolve`, `NDSolve`, and `ParametricPlot`.

Common Mistake

Always test convergence under step refinement and compare against an analytic case. A smooth plot is not proof of numerical correctness.

4.9 Differential Equations Across Physics

Differential equations are the form in which local laws become predictions. The same patterns recur because inertia, storage, transport, feedback, and symmetry recur.

4.9.1 Mechanics

Newton's law $m\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, t)$ is a second-order vector equation. Introducing velocity or momentum converts it into a first-order phase-space system. Conservative forces create energy surfaces; damping contracts trajectories; forcing creates resonance and can generate chaos.

Physical Interpretation

The state-space language developed here leads directly to Lagrange's equations, Hamilton's canonical equations, normal modes, and perturbation theory.

4.9.2 Circuits

Capacitance stores electric energy, inductance stores magnetic energy, and resistance dissipates it. RC circuits obey first-order relaxation equations, while RLC circuits obey the same second-order equation as a damped driven oscillator.

4.9.3 Fields

Separation of variables converts many field equations into families of ODEs. Wave equations produce oscillator equations for modes; diffusion produces exponential decay; radial symmetry reduces electrostatic, gravitational, and quantum problems to radial ODEs.

4.9.4 Quantum evolution

The time-dependent Schrödinger equation

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$$

is a first-order linear equation in a complex vector space. For constant $\hat{H}$, an operator exponential generates unitary, norm-preserving evolution.

4.9.5 Relativity and field theory

Symmetry reduction and mode decomposition turn Maxwell, Klein–Gordon, Dirac, and Einstein equations into ODE systems. Cosmological expansion, geodesics, atomic radial functions, and renormalization-group flows all use the dynamical ideas introduced here.

Key Idea

Identify the state, write its local rate of change, locate equilibria or invariants, and determine how initial data propagate. The symbols change across physics, but the dynamical logic remains.

4.9.6 Looking ahead

State space, generators, normal modes, and stability now form a bridge to classical mechanics, electromagnetism, quantum mechanics, and nonlinear field dynamics.

4.10 Graded Exercise Bank

The following one hundred exercises form a cumulative assessment of Chapter 4. They progress from solution fluency to modelling, qualitative dynamics, numerical analysis, and research-style computation. Unless otherwise stated, constants are real, physical parameters are positive, and all functions possess the regularity required by the indicated operations.

Tier I: Foundations (30 problems)

1. Classify the equation $y' + 3y = 0$ by order, linearity, and autonomy.

2. Verify that $y = Ce^{-2t}$ solves $y' + 2y = 0$.

3. Solve $y' = 4y$ with $y(0) = 3$.

4. Solve $y' = -ky$ with $y(0) = N_0$ and identify the characteristic time.

5. Find all equilibria of $y' = y(1 - y)$.

6. Determine the sign of y' for $y' = y(2 - y)$ on the intervals separated by its equilibria.

7. Solve the separable equation $y' = t(1 + y^2)$ with $y(0) = 0$.

8. Solve $y' + y = t$ using an integrating factor.

9. Solve $y' - 2y = e^{3t}$ with $y(0) = 1$.

10. Find the half-life of a sample satisfying $N' = -\lambda N$.

11. Write Newton's cooling law as an initial-value problem for an object placed in a bath of constant temperature.

12. Solve the charging equation $RC\, q' + q = CV_0$ with $q(0) = 0$.

13. Convert $x'' + 4x = 0$ into a first-order system.

14. Find the characteristic roots of $x'' + 5x' + 6x = 0$.

15. Solve $x'' + 9x = 0$ with $x(0) = A$ and $x'(0) = 0$.

16. Compute the conserved energy of $x = A\cos(\omega t)$ for $mx'' + kx = 0$.

17. Classify the damping regime of $x'' + 4x' + 4x = 0$.

18. Find the general solution of $x'' + 2x' + 5x = 0$.

19. Find a particular solution of $x'' + x = \cos 2t$.

20. State the resonance condition for an undamped driven oscillator.

21. Find the equilibrium points of $x' = x - y$, $y' = x + y$.

22. Write $\mathbf{x}' = A\mathbf{x}$ for the system $x' = 2x + y$, $y' = x + 2y$.

23. Find the eigenvalues of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

24. Classify the origin for $\mathbf{x}' = \mathrm{diag}(-1, -3)\mathbf{x}$.

25. Classify the origin for $\mathbf{x}' = \mathrm{diag}(1, -1)\mathbf{x}$.

26. Sketch the phase portrait of $x' = y$, $y' = -x$.

27. Evaluate one Euler step for $y' = -y$, $y(0) = 1$, with step $h = 0.1$.

28. Evaluate two Euler steps for the same problem and compare with $e^{-0.2}$.

29. Explain the distinction between a direction field and a phase portrait.

30. Explain why an initial condition selects one trajectory from a solution family.

Tier II: Intermediate synthesis (30 problems)

1. Solve the logistic initial-value problem $P' = rP(1 - P/K)$, $P(0) = P_0$, and identify the carrying capacity.

2. Determine the stability of every equilibrium of $y' = y(y-1)(2-y)$.

3. Solve $y' + (2/t)y = t^2$ for $t > 0$.

4. Solve the Bernoulli equation $y' + y = y^2$ by the substitution $u = 1/y$.

5. Determine whether $(2xy + y^2)\,dx + (x^2 + 2xy)\,dy = 0$ is exact and solve it.

6. Model a tank containing a dissolved substance when well-mixed fluid enters and leaves at equal rates.

7. Derive the terminal velocity for a falling body with linear drag $mv' = mg - bv$.

8. Solve $mv' = F_0 - bv$ with $v(0) = 0$ and interpret the relaxation time.

9. Find the current in an RL circuit satisfying $LI' + RI = V_0$, $I(0) = 0$.

10. Solve the discharging RC equation and find the time at which the voltage falls to ten percent of its initial value.

11. Solve $x'' - x' - 6x = 0$ with $x(0) = 0$, $x'(0) = 1$.

12. For $mx'' + bx' + kx = 0$, derive the critical-damping condition.

13. Find the amplitude and phase lag of the steady response to $mx'' + bx' + kx = F_0 \cos \Omega t$.

14. Show that the average power absorbed by a driven oscillator is maximized near resonance.

15. Derive the small-angle pendulum equation from $\theta'' + (g/\ell)\sin\theta = 0$.

16. Estimate the relative period error caused by the small-angle approximation at a moderate initial angle.

17. Diagonalize $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$ and solve $\mathbf{x}' = A\mathbf{x}$.

18. Find the matrix exponential of a planar rotation generator $A = \begin{pmatrix} 0 & -\omega \\ omega & 0 \end{pmatrix}$.

19. Derive the normal-mode frequencies of two identical masses coupled by three identical springs.

20. Show that the symmetric and antisymmetric normal modes evolve independently.

21. Classify a linear planar system from its trace and determinant.

22. Linearize $x' = x - x^3 - y$, $y' = x - y$ at each equilibrium.

23. Use a quadratic Lyapunov function to establish stability of $x' = -ax$, $y' = -by$ for $a, b > 0$.

24. Analyze the bifurcation of $x' = \mu x - x^3$ as μ passes through zero.

25. Compare Euler and midpoint methods on $y' = -2y$ over a fixed interval.

26. Derive the local truncation error of Euler's method by Taylor expansion.

27. Apply one RK4 step to $y' = t - y$, $y(0) = 1$.

28. Explain absolute stability for the test equation $y' = \lambda y$.

29. Determine the Euler stability restriction when $\lambda < 0$ is real.

30. Construct a numerical phase portrait for the damped pendulum and identify its attractor.

Tier III: Advanced analysis (20 problems)

1. Prove uniqueness for $y' = f(t, y)$ under a local Lipschitz condition by estimating the difference of two solutions.
2. Give an example in which an initial-value problem has more than one solution because the Lipschitz condition fails.
3. Derive the variation-of-constants formula for $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$.
4. Prove that e^{At} satisfies the semigroup property for a constant matrix A.
5. Derive Abel's identity for the Wronskian of a second-order linear equation.
6. Use the Wronskian to test linear independence of two oscillator solutions.
7. Construct the Green function for a forced oscillator with specified homogeneous initial conditions.
8. Show that damping makes the mechanical energy nonincreasing for $mx'' + bx' + V'(x) = 0$.
9. Analyze the nonlinear pendulum in phase space and identify separatrices.
10. Derive the energy-dependent period integral for the nonlinear pendulum.
11. Apply the Hartman–Grobman idea informally to a hyperbolic equilibrium and state its limitation.
12. Classify all phase portraits of a real $2imes2$ linear system, including repeated and complex eigenvalues.
13. Show that a nontrivial periodic orbit cannot occur in a one-dimensional autonomous ODE.
14. Use the Bendixson–Dulac criterion to exclude periodic orbits in a specified planar model.
15. Nondimensionalize the driven damped oscillator and identify its independent dimensionless parameters.
16. Derive the modified equation associated with Euler's method through second order.
17. Prove the fourth-order global accuracy of RK4 under suitable smoothness assumptions at an outline level.
18. Explain stiffness using a two-timescale linear system and compare explicit and implicit Euler stability.
19. Derive Crank–Nicolson evolution for a linear system and discuss norm preservation for skew-Hermitian generators.
20. Show how separation of variables converts a linear PDE mode into an ODE eigenvalue problem.

Tier IV: Challenge and computational investigations (20 problems)

1. Implement Euler, midpoint, and RK4 methods without using a library ODE solver; compare error versus step size on a problem with known solution.
2. Implement adaptive step-size control using step doubling and test it near a rapidly varying transient.
3. Compute and plot the resonance curve of an RLC circuit for several damping values.
4. Numerically locate the separatrix of the nonlinear pendulum and compare it with the analytic energy condition.
5. Construct a bifurcation diagram for $x' = \mu x - x^3$ and annotate stable and unstable branches.
6. Explore a saddle-node bifurcation numerically for $x' = \mu - x^2$.

7. Simulate the Lotka–Volterra model, derive its nontrivial equilibrium, and compare linearized and nonlinear trajectories.

8. Add logistic limitation to a predator–prey model and investigate whether a stable cycle persists.

9. Integrate the Duffing oscillator for several forcing amplitudes and compare phase portraits and Poincar'e sections.

10. Simulate the Lorenz equations and estimate sensitivity to initial conditions from two nearby trajectories.

11. Estimate the largest Lyapunov exponent of a chosen nonlinear system.

12. Construct a symplectic Euler integrator for the harmonic oscillator and compare long-time energy drift with standard Euler.

13. Implement velocity Verlet for a one-dimensional potential and test time reversibility.

14. Solve a two-mass coupled oscillator both by direct integration and by normal-mode decomposition.

15. Build a numerical propagator for a two-level quantum system and test norm conservation.

16. Compare explicit Euler, implicit Euler, and trapezoidal evolution on a stiff decay system.

17. Infer parameters of an exponential-decay model from synthetic noisy data and quantify uncertainty.

18. Fit a damped sinusoid to simulated data and discuss parameter correlations.

19. Create an interactive direction-field and solution-curve viewer for a user-entered first-order ODE.

20. Write a short modelling report that begins from a physical assumption, derives an ODE, nondimensionalizes it, solves or simulates it, and evaluates the model's limitations.

Selected guidance

Foundation 7.
Separate variables and integrate: $\arctan y = t^2/2 + C$; then impose the initial condition.

Foundation 17.
The repeated characteristic root is $r = -2$; remember the second independent solution te^{-2t}.

Intermediate 13.
Substitute $x_p = a\cos\Omega t + b\sin\Omega t$, or use complex amplitudes, and separate magnitude from phase.

Intermediate 21.
For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, use $\tau = a + d$ and $\Delta = ad - bc$, then examine $\tau^2 - 4\Delta$.

Intermediate 29.
Euler is stable for $|1 + h\lambda| < 1$; solve this inequality for a negative real λ.

Advanced 8. Differentiate $E = \frac{1}{2}m\dot{x}^2 + V(x)$ and use the equation of motion to obtain $\dot{E} = -b\dot{x}^2$.

Advanced 13.
Uniqueness prevents a trajectory from returning to the same state with a different derivative in one dimension.

Challenge 12.

Track both phase-space geometry and the numerical energy error over many periods; short-time agreement alone is not sufficient.

Challenge 15.

Use a unitary exponential or a norm-preserving time integrator as the reference method.

Challenge 20.

State assumptions, dimensions, parameters, initial data, validation criteria, and failure modes explicitly.

4.11 Repository Integration and Chapter Dependencies

The vocabulary of this chapter is now part of the book-wide glossary and notation system. In particular, ordinary differential equation, initial-value problem, direction field, phase space, equilibrium point, stability, bifurcation, and stiffness are used consistently in later chapters. The canonical state-space convention is

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}), \qquad \mathbf{x}(t_0) = \mathbf{x}_0,$$

while autonomous systems omit the explicit time argument.

Dependency map: from laws of change to physics

$$\begin{array}{ccc} \text{initial-value problem} & \longrightarrow \text{trajectory} & \longrightarrow \text{phase flow} \\ \downarrow & \downarrow & \downarrow \\ \text{Newtonian motion} & \text{circuit dynamics} & \text{quantum evolution} \end{array}$$

Chapter 1 supplies vectors and eigenmodes; Chapter 2 supplies linearization and Jacobians; Chapter 3 supplies spatial field operators. This chapter combines them into evolution laws. Later mechanics chapters use second-order equations and phase space, electromagnetism uses coupled first-order field evolution, and quantum mechanics uses linear state evolution generated by a Hamiltonian.

4.12 Summary and Looking Ahead

An ordinary differential equation converts a local law of change into a trajectory. First-order equations describe relaxation, transport between compartments, growth, and circuit response. Second-order equations describe inertia, oscillation, damping, and resonance. Systems of equations expose eigenmodes and coupled motion, while phase space replaces a single graph by a geometric flow of possible states. Equilibria, linearization, Lyapunov functions, and bifurcations distinguish persistence from instability and reveal how qualitative behavior changes with parameters.

Analytical formulas remain indispensable, but numerical integration is part of the mathematical theory rather than an afterthought. Step size, consistency, stability, stiffness, invariants, and error control determine whether a computed trajectory represents the differential equation or merely the algorithm.

The central pattern is

$$\boxed{\begin{array}{c}\text{physical assumption} \longrightarrow \text{differential equation} \longrightarrow \text{initial data} \\ \longrightarrow \text{evolution} \longrightarrow \text{interpretation}\end{array}}$$

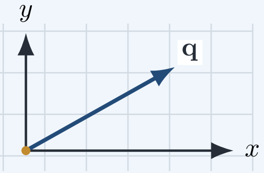
This pattern will recur throughout the book. Newton's laws become dynamical systems; Maxwell's equations propagate fields; the Schrödinger equation evolves quantum states; and relativistic field equations determine the geometry and matter content of spacetime. The next chapter extends the mathematical foundation needed to analyze these theories with greater structural precision.

Chapter 5

Complex Analysis: Geometry, Analyticity, and Physical Phase

The geometry of phase

Complex numbers turn oscillation into rotation and unite algebra, geometry, and analysis in a single plane.



Chapter Roadmap

Why real numbers are not enough $\longrightarrow$ the complex plane and polar form $\longrightarrow$ multiplication as rotation and scaling $\longrightarrow$ Euler's formula and roots $\longrightarrow$ complex functions as plane mappings $\longrightarrow$ limits, derivatives, analyticity, and the Cauchy–Riemann equations.

Learning Objectives

After completing this chapter, the reader should be able to:

- perform algebraic operations with complex numbers and interpret them geometrically;
- move fluently between Cartesian, polar, and exponential forms;
- explain why multiplication combines scaling with rotation;
- derive and use Euler's formula and de Moivre's theorem;
- interpret a complex function as a mapping between planes;
- define complex differentiability and derive the Cauchy–Riemann equations;
- connect complex phase to oscillations, waves, circuits, and quantum states.

5.1 Why Real Numbers Are Not Enough

The real numbers describe position along a line. They are sufficient for ordering, distance, and one-dimensional change, but they are not closed under every algebraic operation. The equation

$$x^2 + 1 = 0$$

has no real solution because $x^2 \geq 0$ for every real x. Mathematics enlarges the number system by introducing a number i satisfying

$$\boxed{i^2 = -1}.$$

A complex number is then

$$z = x + iy, \qquad x, y \in \mathbb{R}.$$

Real and imaginary parts

For $z = x + iy$,

$$\Re z = x, \qquad \Im z = y.$$

Two complex numbers are equal precisely when their real and imaginary parts agree separately.

The extension from $\mathbb{R}$ to $\mathbb{C}$ is not an artificial trick. It is the smallest natural setting in which polynomial equations factor completely and in which rotation and oscillation become algebraic operations.

From "impossible" numbers to a geometric plane

Cardano encountered square roots of negative numbers while solving cubic equations. Bombelli showed that these expressions could be manipulated consistently. Euler connected them to exponentials and trigonometry, while Argand and Gauss made their geometry explicit by representing $x + iy$ as a point in a plane. Cauchy and Riemann then revealed that differentiability in this plane is far more restrictive and powerful than differentiability on the real line.

Why physics keeps finding complex numbers

A sinusoidal motion may be written as the real part of $Ae^{i\omega t}$; an AC voltage uses the same phase language; a plane wave contains $e^{i(kx-\omega t)}$; and a quantum state carries a complex amplitude whose phase influences interference. Complex numbers are therefore a compact language for magnitude and phase together.

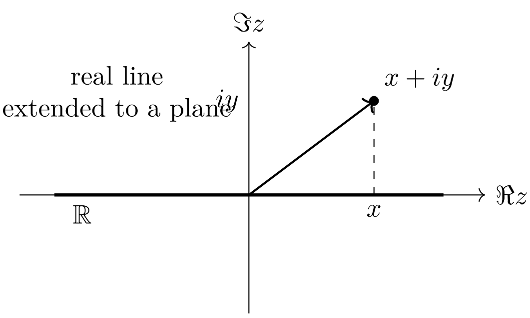


Figure 5.1: Extending the real line to the complex plane adds a perpendicular direction associated with the imaginary unit.

5.2 The Complex Plane

The number $z = x + iy$ is represented by the point (x, y) in the Argand plane. The horizontal axis records $\Re z$ and the vertical axis records $\Im z$. This identification turns complex arithmetic into geometry.

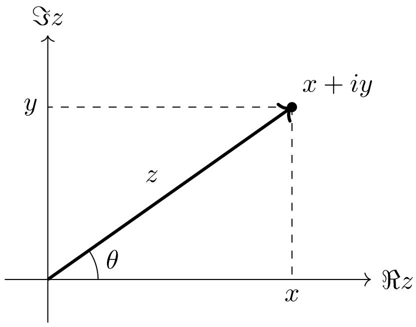


Figure 5.2: A complex number as both a point and a directed displacement in the Argand plane.

The complex conjugate of $z = x + iy$ is $\bar{z} = x - iy$. Geometrically, conjugation reflects a point across the real axis. The modulus is the Euclidean distance from the origin,

$$|z| = \sqrt{x^2 + y^2}, \qquad z\bar{z} = |z|^2.$$

When $z \neq 0$,

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

Example 5.1

For $z = 3 + 4i$,

$$|z| = 5, \qquad \bar{z} = 3 - 4i, \qquad \frac{1}{z} = \frac{3 - 4i}{25}.$$

The familiar 3–4–5 triangle is now the algebraic identity $z\bar{z} = 25$.

Addition is vector addition:

$$(x_1 + iy_1) + (x_2 + iy_2) = (x_1 + x_2) + i(y_1 + y_2).$$

The triangle inequality $|z + w| \leq |z| + |w|$ is therefore exactly the geometric triangle inequality.

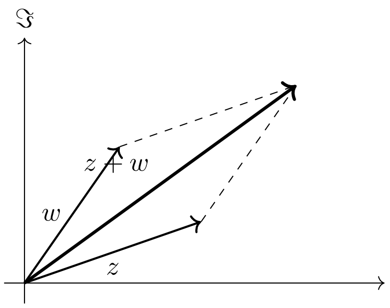


Figure 5.3: Complex addition is the parallelogram law for planar vectors.

5.3 Polar Form and Euler's Formula

For $z \neq 0$, define $r = |z|$ and an argument θ by

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

Then

$$z = r(\cos\theta + i\sin\theta).$$

The argument is multivalued: θ and $\theta + 2\pi n$ describe the same point. A selected representative is called a principal argument.

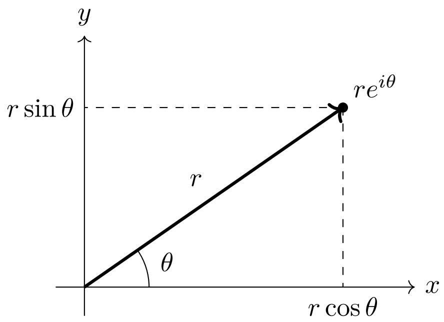


Figure 5.4: Cartesian and polar descriptions of the same complex number.

Euler's formula states

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}.$$

Comparing power series gives

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \cdots \tag{5.1}$$

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right). \tag{5.2}$$

Thus

$$\boxed{z = re^{i\theta}}.$$

Oscillation is rotation viewed from one axis

The point $e^{i\omega t}$ moves uniformly around the unit circle. Its real projection is $\cos\omega t$ and its imaginary projection is $\sin\omega t$. Harmonic motion is a shadow of uniform rotation in the complex plane.

The identity $e^{i\pi} + 1 = 0$ has a direct geometric meaning: rotation by π carries 1 to -1.

5.4 Multiplication, Division, Powers, and Roots

Let $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$. Then

$$z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}, \qquad \frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1-\theta_2)}.$$

Multiplication multiplies lengths and adds angles; division divides lengths and subtracts angles.

Multiplication by $i = e^{i\pi/2}$ rotates every vector counterclockwise through 90°:

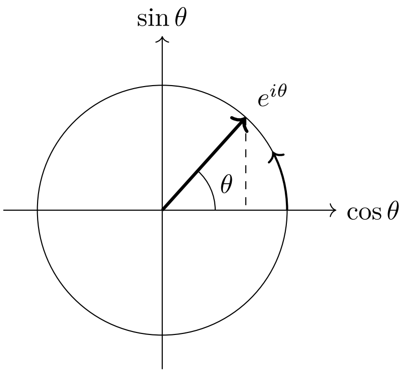


Figure 5.5: Uniform complex rotation generates sine and cosine as coordinate projections.

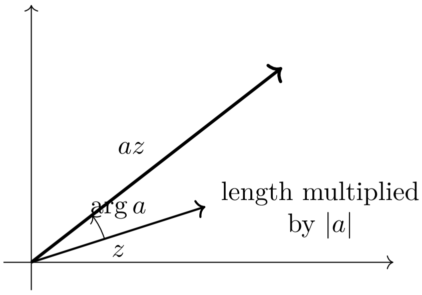


Figure 5.6: Multiplication by a fixed complex number produces a dilation followed by a rotation.

$$i(x + iy) = -y + ix.$$

Applying the operation twice gives a 180° rotation, the geometric content of $i^2 = -1$.

De Moivre's theorem follows immediately:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta), \qquad z^n = r^n e^{in\theta}.$$

The nth roots of $z = re^{i\theta}$ are

$$w_k = r^{1/n}\exp\left[i\frac{\theta + 2\pi k}{n}\right], \qquad k = 0, 1, \ldots, n-1.$$

They lie at equally spaced angles on a circle.

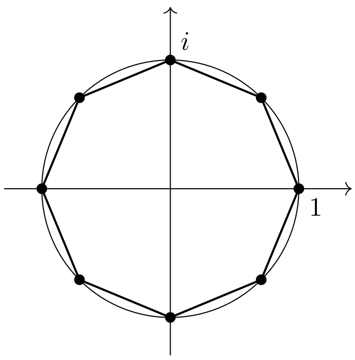


Figure 5.7: The eighth roots of unity form a regular polygon on the unit circle.

Example 5.2

Solving $z^3 = 8$ gives

$$z_k = 2e^{i2\pi k/3}, \qquad k = 0, 1, 2.$$

Thus the roots are 2, $-1+i\sqrt{3}$, and $-1-i\sqrt{3}$. Their sum vanishes by rotational symmetry.

5.5 Complex Functions as Mappings

A complex function maps one plane to another:

$$w = f(z), \qquad z = x + iy, \qquad w = u + iv.$$

Equivalently, $u = u(x, y)$ and $v = v(x, y)$. The function therefore transforms points, curves, angles, and regions.

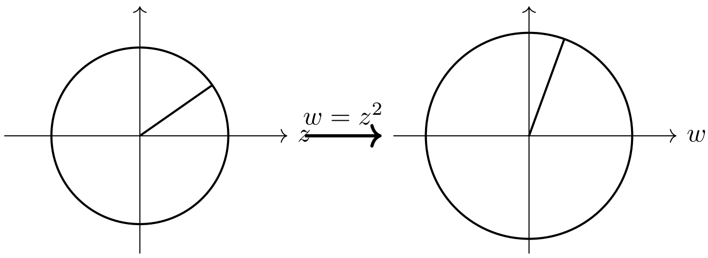
For $f(z) = z^2$,

$$(x + iy)^2 = (x^2 - y^2) + i(2xy),$$

so

$$u = x^2 - y^2, \qquad v = 2xy.$$

In polar form, $z = re^{i\theta}$ gives $w = r^2e^{i2\theta}$: radii are squared and angles are doubled.



domain: angle θ image: angle 2θ

Figure 5.8: The map $w = z^2$ doubles arguments and squares moduli.

Basic geometric maps include

$$f(z) = z + a \qquad \text{translation,} \tag{5.3}$$

$$f(z) = az \qquad \text{rotation and scaling,} \tag{5.4}$$

$$f(z) = \bar{z} \qquad \text{reflection,} \tag{5.5}$$

$$f(z) = \frac{1}{z} \qquad \text{inversion with angular reversal.} \tag{5.6}$$

The linear fractional or Möbius transformation

$$f(z) = \frac{az + b}{cz + d}, \qquad ad - bc \neq 0,$$

combines translations, rotations, dilations, and inversion. Such maps send generalized circles—circles or straight lines—to generalized circles.

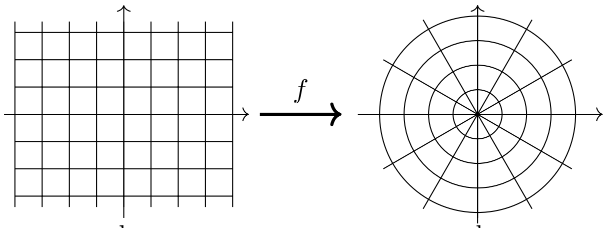


Figure 5.9: A preview of a Möbius transformation: grid geometry is reorganized while local angle structure is preserved away from singular points.

A function is more than a formula

On the real line, a graph often captures a function. For $f : \mathbb{C} \to \mathbb{C}$, a graph would require four real dimensions. The useful replacement is to study how the function maps families of curves, domains, and local grids.

5.6 Complex Differentiability and Analyticity

The derivative of a complex function is defined by the same-looking limit used in real calculus:

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}.$$

The crucial difference is that Δz may approach zero from infinitely many directions in the plane. A complex derivative exists only if every approach gives the same value.

Write

$$f(z) = u(x, y) + iv(x, y).$$

Approaching along the real direction, $\Delta z = \Delta x$, gives

$$f'(z) = u_x + iv_x.$$

Approaching along the imaginary direction, $\Delta z = i\Delta y$, gives

$$f'(z) = v_y - iu_y.$$

Equality requires the Cauchy–Riemann equations

$$\boxed{u_x = v_y, \qquad u_y = -v_x}.$$

Under suitable continuity assumptions on the first partial derivatives, these equations are also sufficient for complex differentiability.

Holomorphic and analytic

A function is *holomorphic* on an open domain if it is complex differentiable at every point of that domain. Holomorphic functions are analytic: locally they possess convergent power-series expansions. This remarkable equivalence is one of the central results developed later in the chapter.

Example 5.3

For $f(z) = z^2$,

$$u = x^2 - y^2, \qquad v = 2xy.$$

Then

$$u_x = 2x = v_y, \qquad u_y = -2y = -v_x.$$

The Cauchy–Riemann equations hold everywhere and $f'(z) = 2z$.

For $f(z) = \bar{z} = x - iy$, one has $u = x$ and $v = -y$. Thus $u_x = 1$ while $v_y = -1$, so the Cauchy–Riemann equations fail everywhere. Conjugation is geometrically simple but nowhere holomorphic.

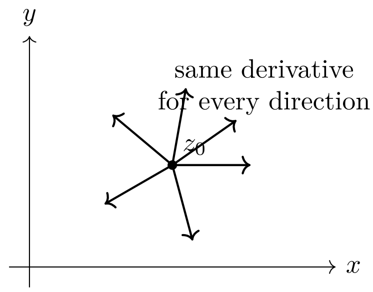


Figure 5.10: Complex differentiability demands one linear action for every approach direction; the Cauchy–Riemann equations encode this compatibility.

Differentiating the Cauchy–Riemann equations shows that both components are harmonic:

$$\nabla^2 u = u_{xx} + u_{yy} = 0, \qquad \nabla^2 v = v_{xx} + v_{yy} = 0.$$

Thus analytic functions naturally generate pairs of harmonic potentials.

Electrostatic potential and stream function

In two-dimensional electrostatics, a charge-free potential satisfies Laplace's equation. Its harmonic conjugate supplies orthogonal field lines. The same mathematics describes incompressible, irrotational fluid flow: equipotential curves and streamlines cross at right angles. Complex analyticity packages both scalar fields into one function.

5.7 Visual Synthesis and Looking Ahead

The first stage of complex analysis may be summarized as a chain of increasingly powerful identifications:

$$\begin{aligned} \text{number} \longleftrightarrow \text{point and vector} &\longleftrightarrow \text{modulus and phase,} \\ &\longleftrightarrow \text{rotation and dilation} \longleftrightarrow \text{plane mapping.} \end{aligned}$$

What makes complex analysis special

Real differentiability controls change along one axis. Complex differentiability requires consistency across every direction in a plane. This stronger requirement forces rigidity: harmonic components, angle-preserving local maps, power-series representations, and powerful integral identities.

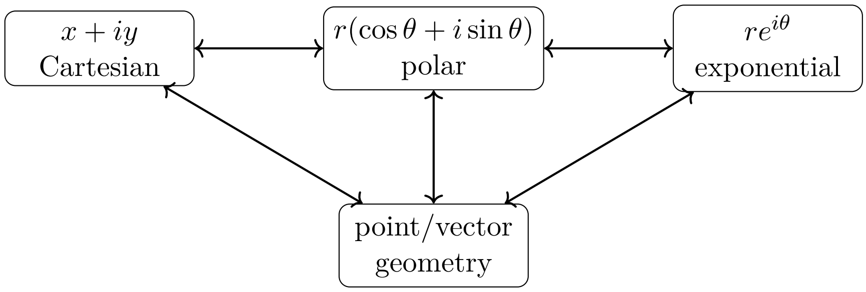


Figure 5.11: Equivalent representations of a complex number and the operations each representation makes transparent.

The next production stage develops this rigidity in full. Contours replace intervals, Cauchy's theorem makes path deformation possible, Cauchy's integral formula recovers interior values from boundary data, and Laurent series classify singularities. Residues then convert local information near poles into global integral results.

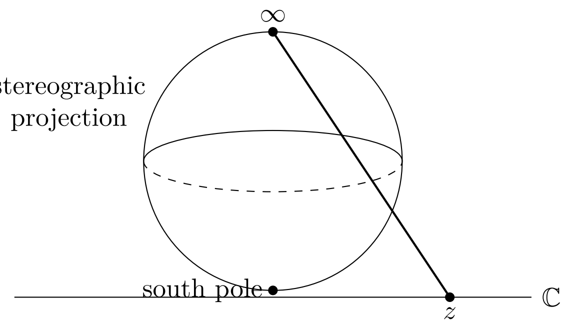


Figure 5.12: Preview of the Riemann sphere, where the extended complex plane includes a single point at infinity.

Chapter Roadmap

Next: contour integration $\longrightarrow$ Cauchy's theorem $\longrightarrow$ Cauchy's integral formula $\longrightarrow$ Taylor and Laurent series $\longrightarrow$ singularities and residues $\longrightarrow$ conformal mapping and physical applications.

5.8 Thirty Integrated Worked Examples

The examples below develop a working language for complex arithmetic, plane mappings, analyticity, and elementary contour integration. Each solution is paired with an interpretation so that symbolic manipulation remains connected to geometry and physics.

5.8.1 Arithmetic, polar form, and roots

Example 5.4

Cartesian arithmetic.
Problem. Compute $(3 - 2i) + (1 + 5i)$ and $(3 - 2i)(1 + 5i)$.

Solution

Addition gives $4+3i$. Multiplication gives $3+15i-2i-10i^2 = 13+13i$.
Interpretation. Addition combines displacement vectors; multiplication mixes scaling and rotation.

Example 5.5

Division by conjugation.
Problem. Simplify $(2+3i)/(1-i)$.

Solution

Multiply numerator and denominator by $1+i$:

$$\frac{2+3i}{1-i} = \frac{(2+3i)(1+i)}{2} = \frac{-1+5i}{2}.$$

Interpretation. Conjugation converts a complex denominator into the real number $|1-i|^2$.

Example 5.6

Modulus and argument.
Problem. Write $z = -1+\sqrt{3}i$ in polar form.

Solution

$r = 2$ and the point lies in quadrant II, so $\theta = 2\pi/3$. Hence $z = 2e^{2\pi i/3}$.
Interpretation. Cartesian coordinates locate the point; polar data separate distance from direction.

Example 5.7

A product in polar form.
Problem. Multiply $z_1 = 2e^{i\pi/6}$ and $z_2 = 3e^{-i\pi/4}$.

Solution

$$z_1 z_2 = 6e^{i(\pi/6-\pi/4)} = 6e^{-i\pi/12}.$$

Interpretation. Moduli multiply while arguments add.

Example 5.8

De Moivre's theorem.
Problem. Evaluate $(\cos\theta + i\sin\theta)^5$.

Solution

By de Moivre's theorem,

$$(\cos\theta + i\sin\theta)^5 = \cos 5\theta + i\sin 5\theta.$$

Interpretation. Repeated multiplication performs repeated rotation.

Example 5.9

Cube roots of unity.
Problem. Solve $z^3 = 1$.

Solution

$$z_k = e^{2\pi ik/3}, \qquad k = 0, 1, 2.$$

Thus 1, $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$, and $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$ are the roots.
Interpretation. The solutions are equally spaced vertices of a regular triangle on the unit circle.

Example 5.10

Square roots of a complex number.
Problem. Find the square roots of $3 + 4i$.

Solution

Let $(a + ib)^2 = 3 + 4i$. Then $a^2 - b^2 = 3$ and $2ab = 4$. Since $a^2 + b^2 = 5$, one obtains $a = 2$, $b = 1$. The roots are $\pm(2 + i)$.
Interpretation. Squaring doubles argument and squares modulus; taking square roots reverses both operations with two possible angles.

Example 5.11

Euler's identity.
Problem. Evaluate $e^{i\pi}$.

Solution

Euler's formula gives $e^{i\pi} = \cos\pi + i\sin\pi = -1$.
Interpretation. A rotation by π on the unit circle reaches the point -1.

5.8.2 Mappings and complex functions

Example 5.12

Affine mapping.
Problem. Describe $w = (1 + i)z + 2$.

Solution

The factor $1 + i = \sqrt{2}e^{i\pi/4}$ rotates by $\pi/4$ and scales by $\sqrt{2}$; adding 2 translates two units along the real axis.
Interpretation. Every nonconstant affine complex map is a similarity transformation.

Example 5.13

The square map.
Problem. Map the ray $z = re^{i\theta_0}$ under $w = z^2$.

Solution

$$w = r^2 e^{2i\theta_0}.$$

The ray angle doubles and radial distances square.
Interpretation. The map stretches nonuniformly and doubles angular separation.

Example 5.14

Inversion.
Problem. Find the image of $z = re^{i\theta}$ under $w = 1/z$.

Solution

$$w = r^{-1}e^{-i\theta}.$$

Interpretation. Inversion reciprocates distance from the origin and reverses orientation angle.

Example 5.15

A line under inversion.
Problem. Determine the image of the vertical line $x = a \neq 0$ under $w = 1/z$.

Solution

Writing $w = u + iv$ and $z = 1/w = (u - iv)/(u^2 + v^2)$ gives

$$\frac{u}{u^2 + v^2} = a,$$

or $(u - 1/(2a))^2 + v^2 = 1/(4a^2)$.
Interpretation. A line not through the origin maps to a circle through the origin.

Example 5.16

Exponential periodicity.
Problem. Solve $e^z = 1$.

Solution

With $z = x + iy$, $e^z = e^x(\cos y + i \sin y)$. Therefore $x = 0$ and $y = 2\pi n$, so $z = 2\pi i n$. **Interpretation.** The complex exponential wraps horizontal strips around the punctured plane.

Example 5.17

Complex sine on the imaginary axis.
Problem. Evaluate $\sin(iy)$.

Solution

Using $\sin z = (e^{iz} - e^{-iz})/(2i)$,

$$\sin(iy) = i \sinh y.$$

Interpretation. Trigonometric and hyperbolic functions are different slices of the same complex exponential structure.

5.8.3 Differentiability and analyticity

Example 5.18

Derivative of z^2.
Problem. Compute the complex derivative of $f(z) = z^2$.

Solution

$$\frac{f(z+h) - f(z)}{h} = 2z + h \longrightarrow 2z.$$

Thus $f'(z) = 2z$ for all z.
Interpretation. The local scale factor is $|2z|$ and the local rotation is $\arg(2z)$ away from the origin.

Example 5.19

Failure of differentiability for conjugation.
Problem. Test $f(z) = \bar{z}$ at the origin.

Solution

Along real increments, $\bar{h}/h = 1$; along imaginary increments, $\bar{h}/h = -1$. The limit depends on direction, so no complex derivative exists.
Interpretation. Complex differentiability is much stricter than differentiability in two real variables.

Example 5.20

Cauchy–Riemann test.
Problem. Determine whether $f(z) = x^2 - y^2 + i2xy$ is analytic.

Solution

Here $u = x^2 - y^2$, $v = 2xy$. Since $u_x = 2x = v_y$ and $u_y = -2y = -v_x$, the Cauchy–Riemann equations hold everywhere. In fact $f(z) = z^2$.
Interpretation. The real and imaginary parts are locked together as harmonic conjugates.

Example 5.21

A nonanalytic real-valued function.
Problem. Test $f(z) = |z|^2$.

Solution

$u = x^2 + y^2$, $v = 0$. The equations require $2x = 0$ and $2y = 0$, so they hold only at the origin, not on a neighbourhood. The function is nowhere analytic.
Interpretation. Satisfying the equations at one isolated point is not enough for analyticity.

Example 5.22

Finding a harmonic conjugate.
Problem. Given $u = x^2 - y^2$, find v so that $u + iv$ is analytic.

Solution

From $v_y = u_x = 2x$, $v = 2xy + g(x)$. Then $v_x = 2y + g'(x) = -u_y = 2y$, so g is constant. Thus $v = 2xy + C$.
Interpretation. The analytic function is $z^2 + iC$.

Example 5.23

Harmonicity.
Problem. Verify that the real part of e^z is harmonic.

Solution

$u = e^x \cos y$. Then $u_{xx} = e^x \cos y$ and $u_{yy} = -e^x \cos y$, so $\nabla^2 u = 0$.
Interpretation. Real and imaginary parts of analytic functions solve Laplace's equation.

Example 5.24

Constructing an analytic function from a potential.
Problem. Let $u = e^x \cos y$. Find the analytic function.

Solution

The harmonic conjugate is $v = e^x \sin y + C$, so $f(z) = e^z + iC$.
Interpretation. Equipotential and stream-function curves meet orthogonally where $f'(z) \neq 0$.

5.8.4 Elementary contour integrals

Example 5.25

Integral along a line segment.
Problem. Evaluate $\int_C z\,dz$ from 0 to $1+i$ along the straight segment.

Solution

Set $z(t) = (1+i)t$, $0 \le t \le 1$. Then

$$\int_0^1 (1+i)t(1+i)\,dt = \frac{(1+i)^2}{2} = i.$$

Interpretation. Because z has the antiderivative $z^2/2$, the value depends only on endpoints.

Example 5.26

Integral around a circle.
Problem. Evaluate $\oint_{|z|=R} z^n\,dz$ for integer n.

Solution

With $z = Re^{it}$,

$$\oint z^n\,dz = iR^{n+1} \int_0^{2\pi} e^{i(n+1)t}\,dt.$$

The result is 0 unless $n = -1$, in which case it is $2\pi i$.
Interpretation. The exceptional integrand $1/z$ detects one winding around the origin.

Example 5.27

Orientation reversal.
Problem. What happens to $\int_C f(z)\,dz$ if the path orientation is reversed?

Solution

The reversed parametrization changes dz by a minus sign, hence

$$\int_{-C} f(z)\,dz = -\int_C f(z)\,dz.$$

Interpretation. Complex line integrals are oriented geometric quantities.

Example 5.28

Path independence for an entire integrand.
Problem. Evaluate $\int_C e^z \, dz$ from z_0 to z_1.

Solution

Since e^z is its own antiderivative,

$$\int_C e^z \, dz = e^{z_1} - e^{z_0}$$

for every piecewise smooth path C.
Interpretation. Analytic antiderivatives convert a path problem into endpoint data.

Example 5.29

A path-dependent integral.
Problem. Compare $\int_C \bar{z} \, dz$ from 0 to $1+i$ along two paths.

Solution

Along the straight line $z = (1+i)t$, the integral is $\int_0^1 (1-i)t(1+i)dt = 1$. Along the broken path $0 \to 1 \to 1+i$, the two pieces give $1/2$ and $1/2+i$, totaling $1+i$.
Interpretation. The nonanalytic integrand has no global antiderivative, so the route matters.

Example 5.30

Winding number preview.
Problem. Evaluate $\oint_C dz/z$ when C winds twice counterclockwise around the origin.

Solution

Each positive winding contributes $2\pi i$, so the result is $4\pi i$.
Interpretation. The integral records topological winding rather than metric details of the curve.

5.8.5 Integrated applications

Example 5.31

Phasor differentiation.
Problem. Differentiate $Ae^{i\omega t}$ repeatedly.

Solution

Each derivative multiplies the phasor by $i\omega$:

$$\frac{d^n}{dt^n} A e^{i\omega t} = (i\omega)^n A e^{i\omega t}.$$

Interpretation. Differentiation becomes algebraic multiplication, one reason complex notation simplifies oscillations.

Example 5.32

Series impedance.
Problem. Find the impedance of a resistor and inductor in series.

Solution

With time dependence $e^{i\omega t}$, $V_R = RI$ and $V_L = i\omega LI$. Therefore

$$Z = \frac{V}{I} = R + i\omega L.$$

Interpretation. Resistance and reactance are the real and imaginary components of one response coefficient.

Example 5.33

Interference of two amplitudes.
Problem. Let $\psi_1 = A$ and $\psi_2 = Ae^{i\phi}$. Find $|\psi_1 + \psi_2|^2$.

Solution

$$|A(1 + e^{i\phi})|^2 = 2A^2(1 + \cos\phi) = 4A^2\cos^2(\phi/2).$$

Interpretation. Relative phase changes measurable intensity even though a common global phase does not.

Example 5.34

Lowest-Landau-level polynomial preview.
Problem. Describe the zeros of $\psi(z) = z^m e^{-|z|^2/(4\ell_B^2)}$.

Solution

The Gaussian never vanishes, while z^m has a zero of multiplicity m at the origin. Its phase winds by $2\pi m$ around that point.
Interpretation. Analytic zeros encode vortices and angular momentum in two-dimensional quantum states.

Example 5.35

Complex potential for uniform flow.
Problem. Interpret $F(z) = Uz$ in two-dimensional ideal flow.

Solution

$F = \phi + i\psi = Ux + iUy$. Thus $\phi = Ux$ and $\psi = Uy$; the velocity is uniform and directed along x.
Interpretation. Analytic functions package velocity potential and stream function into one object.

5.9 Computational Companion: Seeing Complex Structure

Complex analysis is unusually visual. A computation should therefore expose both algebraic values and the deformation of geometric objects. The companion programs in `examples/python/ch05` and `examples/wolfram/ch05` generate the figures in this section and provide reproducible numerical checks.

5.9.1 Domain colouring and phase

A complex value $f(z) = \rho e^{i\theta}$ can be displayed by assigning hue to θ and brightness or saturation to ρ. Zeros appear as phase vortices, while poles reverse the radial behaviour.

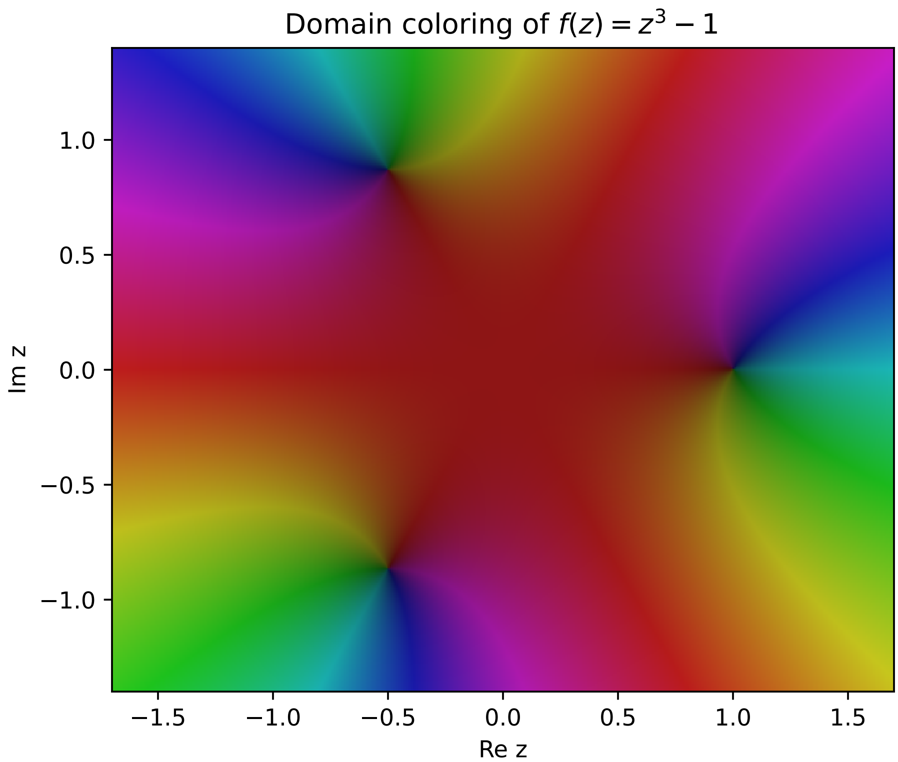


Figure 5.13: Domain colouring of $f(z) = z^3 - 1$. The three phase vortices locate the roots of unity.

5.9.2 Roots of unity

The script verifies $z_k^n = 1$ numerically and plots the regular polygon formed by the roots.

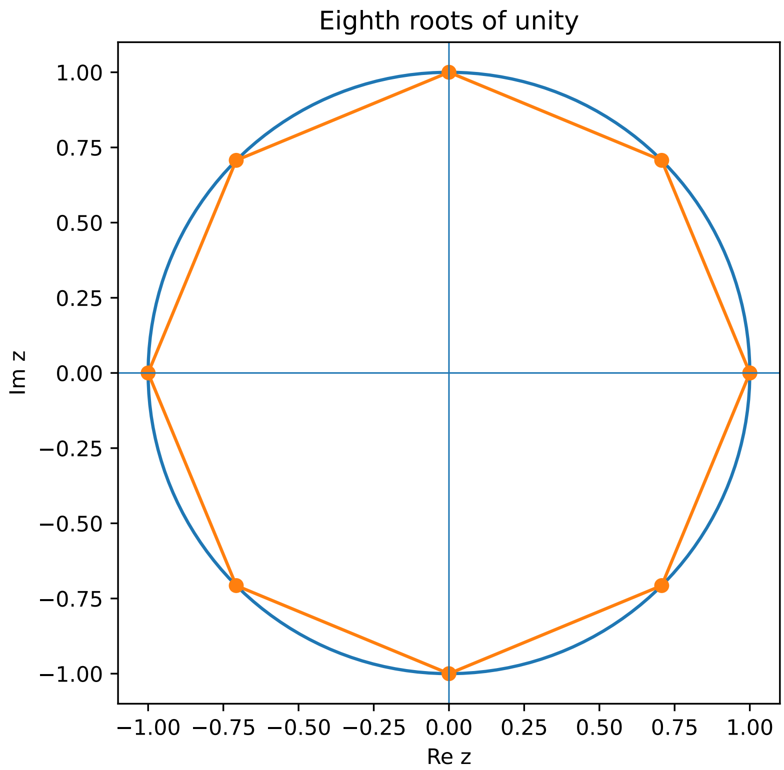


Figure 5.14: The eighth roots of unity, equally spaced on the unit circle.

5.9.3 Plane mappings

A rectangular grid is sampled in the z-plane and mapped pointwise under $w = z^2$. The resulting curves make angular doubling and radial squaring visible.

5.9.4 Numerical contour integration

For a parametrized contour $z(t)$, the integral is approximated by a complex trapezoidal rule. Refinement tests confirm

$$\oint_{|z|=1} \frac{dz}{z} = 2\pi i, \qquad \oint_{|z|=1} z^3\, dz = 0.$$

Reproducible computation

Run `python examples/python/ch05/complex_analysis_companion.py`. The program regenerates all four figures, prints root residuals, and reports contour-integral errors. The Wolfram Language file mirrors the symbolic and graphical tasks.

5.10 Complex Analysis as a Language of Physics

Complex numbers are not decorative notation added after the physics is solved. They often reveal the natural geometry of the law itself: oscillation becomes rotation, coupled real equations become one complex equation, and two-dimensional potential fields become analytic functions.

5.10.1 Oscillations, waves, and phase

A real sinusoid may be represented as the real part of a rotating complex amplitude,

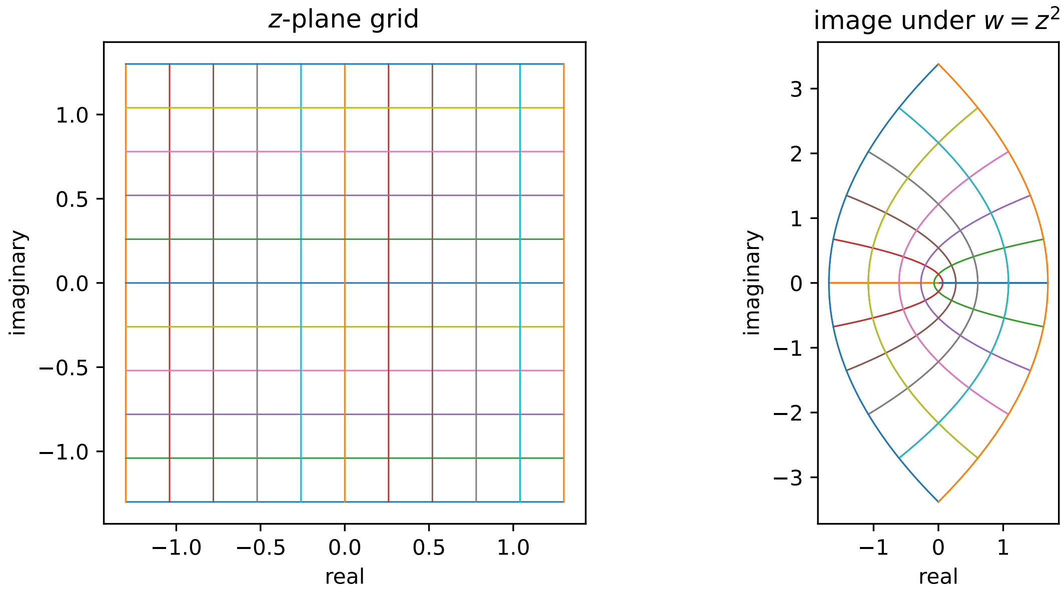


Figure 5.15: A Cartesian grid before and after the map $w = z^2$.

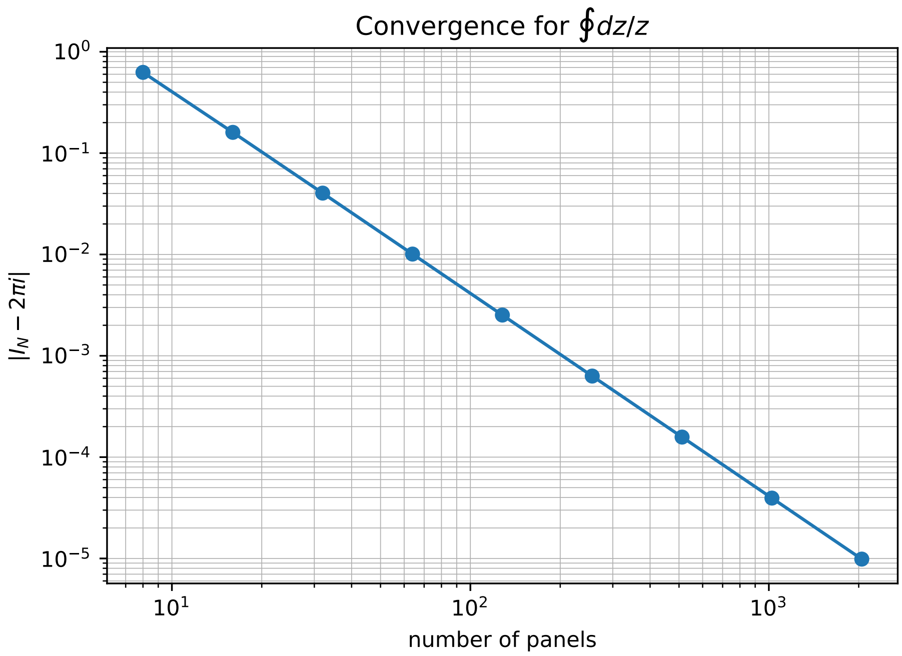


Figure 5.16: Convergence of the numerical contour integral of $1/z$ around the unit circle.

$$x(t) = \Re\Big(Ae^{i(\omega t+\phi)}\Big).$$

Differentiation becomes multiplication by $i\omega$, while phase shifts become multiplication by $e^{i\phi}$. A traveling wave is written

$$\Psi(x,t) = Ae^{i(kx-\omega t)},$$

with the measurable real field obtained by taking a real part or an appropriate quadratic observable. This notation separates amplitude, phase, wavelength, and frequency without repeatedly expanding trigonometric identities.

5.10.2 Alternating-current circuits

For sinusoidal steady state, resistor, inductor, and capacitor laws combine into the complex impedance

$$Z(\omega) = R + i\omega L + \frac{1}{i\omega C}.$$

The magnitude $|Z|$ controls current amplitude and $\arg Z$ controls phase lag. Resonance is then read from the cancellation of inductive and capacitive reactances. The same algebraic architecture reappears in mechanical response functions and optical susceptibilities.

5.10.3 Electrostatics and ideal fluid flow

Let $F(z) = \phi(x,y) + i\psi(x,y)$ be analytic. The Cauchy–Riemann equations imply that both ϕ and ψ are harmonic and that their level curves meet orthogonally. In electrostatics, ϕ may be a potential; in incompressible irrotational flow, ϕ is a velocity potential and ψ a stream function. Conformal maps can transform a simple solution into one adapted to a more complicated boundary.

5.10.4 Quantum amplitudes and interference

A quantum state is represented by a complex amplitude ψ. The probability density is

$$\rho = |\psi|^2 = \psi^*\psi,$$

so an overall phase $e^{i\alpha}$ leaves a single-state probability unchanged. Relative phases, however, enter cross terms and control interference. The time-dependent Schrödinger equation

$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi$$

uses multiplication by i to generate norm-preserving phase rotation in Hilbert space.

5.10.5 Two-dimensional quantum motion and the Hall effect

For motion in the plane, it is natural to combine coordinates as

$$z = x + iy, \qquad \bar{z} = x - iy.$$

In the lowest Landau level, a broad class of wavefunctions has the schematic form

$$\psi(z,\bar{z}) = f(z)\exp\left(-\frac{|z|^2}{4\ell_B^2}\right),$$

where $f(z)$ is analytic. The polynomial degree tracks angular momentum, while zeros of f carry phase winding. This language prepares the later discussion of guiding centres, edge states, Laughlin factors, Berry phases, and fractional quantum Hall correlations.

5.10.6 Optics, Fourier analysis, and Green functions

Complex amplitudes encode optical interference and diffraction. Fourier modes use e^{ikx} because differentiation acts diagonally on them. Green functions and response functions are often studied as analytic functions of a complex frequency or energy. Their poles identify natural modes and decay rates, while contour deformation converts oscillatory integrals into sums of residue contributions.

5.10.7 Analytic continuation in statistical and quantum field theory

Complex time and complex energy allow one to relate oscillatory real-time evolution to exponentially weighted Euclidean formulations. Analytic continuation must be handled with care because poles, branch points, and branch cuts encode physical thresholds and causal prescriptions. The contour methods developed later in this chapter become tools for propagators, dispersion relations, thermal sums, and scattering amplitudes.

Looking ahead

Later chapters will use complex analysis repeatedly: Fourier transforms in Chapter 6, quantum amplitudes and unitary evolution, Green functions and propagators, Berry phases, Landau-level wavefunctions, and the analytic many-body structure of quantum Hall states.

Chapter 6

Fourier Analysis

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

Chapter 7

Probability Theory

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

PART II

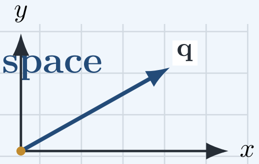
Classical Physics Foundations

Chapter 8

Motion

A trajectory is a curve through space

Position becomes a function of time. Its first and second derivatives produce velocity and acceleration, the fundamental kinematic vectors.



"Nothing in physics is truly static. Every physical system evolves with time, and understanding motion is the first step toward understanding nature."

Chapter Roadmap

Position $\longrightarrow$ displacement $\longrightarrow$ velocity $\longrightarrow$ acceleration $\longrightarrow$ trajectory.

Learning Objectives

After completing this chapter, the reader should be able to:

- understand the concept of motion;
- describe position as a function of time;
- distinguish between average and instantaneous quantities;
- calculate velocity and acceleration;
- understand derivative notation;
- describe motion using vectors;
- interpret graphs of position, velocity and acceleration;
- understand parametric curves and trajectories.

8.1 What is Motion?

One of the most fundamental questions in physics is:

What does it mean for an object to move?

Suppose we observe an electron.
At time

$$t_1$$

its position is

$$\mathbf{r}_1 = (2, 1, 0) \text{ nm}.$$

A short time later,

$$t_2,$$

its position becomes

$$\mathbf{r}_2 = (2.3, 1.1, 0) \text{ nm}.$$

The position has changed.
Therefore,

Key Idea

Motion is the change of position with time.

Motion is therefore not a property of space alone.
It requires two ingredients:

- position,
- time.

8.2 Time

Time is represented by

$$t.$$

Unlike spatial coordinates, time is an independent variable.
Instead of writing

$$\mathbf{r},$$

we now write

$$\boxed{\mathbf{r}(t).}$$

This notation means

"The position depends on time."

8.3 Functions

A function assigns exactly one output to every input.

Examples include

$$x(t),$$

$$y(t),$$

and

$$z(t).$$

Together they define the complete position vector

$$\boxed{\mathbf{r}(t) = x(t)\hat{\mathbf{x}} + y(t)\hat{\mathbf{y}} + z(t)\hat{\mathbf{z}}.}$$

Every moving particle therefore traces a curve through space.

8.4 Trajectory

The collection of all positions occupied by a moving particle is called its trajectory.

Examples include

- a straight line,
- a circle,
- an ellipse,
- a spiral,
- a completely irregular path.

Later, quantum mechanics will replace these classical trajectories by wavefunctions, but the geometric picture remains an important intuition.

8.5 Average Velocity

Suppose a particle moves

$$\Delta x = 10 \text{ m}$$

during

$$\Delta t = 2 \text{ s}.$$

Its average velocity is

$$\boxed{v_{\text{avg}} = \frac{\Delta x}{\Delta t}.}$$

For the above example,

$$v_{\text{avg}} = 5 \text{ m/s}.$$

Average velocity tells us only what happened over an interval.

It does not describe the motion at every instant.

8.6 Instantaneous Velocity

Imagine measuring the motion over smaller and smaller time intervals.

As

$$\Delta t \to 0,$$

the average velocity becomes the instantaneous velocity.

Therefore,

$$\boxed{v = \frac{dx}{dt}.}$$

In three dimensions,

$$\boxed{\mathbf{v} = \frac{d\mathbf{r}}{dt}.}$$

Expanding into components,

$$\boxed{\mathbf{v} = \frac{dx}{dt}\hat{\mathbf{x}} + \frac{dy}{dt}\hat{\mathbf{y}} + \frac{dz}{dt}\hat{\mathbf{z}}.}$$

Velocity therefore points in the direction of motion.

8.7 Speed

Velocity is a vector.

Speed is a scalar.

The speed is simply the magnitude of the velocity vector,

$$\boxed{v = |\mathbf{v}|.}$$

Thus,

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

8.8 Acceleration

Velocity itself may change.

The rate of change of velocity is called acceleration.

Average acceleration is

$$\boxed{a_{\text{avg}} = \frac{\Delta v}{\Delta t}.}$$

Instantaneous acceleration is

$$\boxed{\mathbf{a} = \frac{d\mathbf{v}}{dt}.}$$

Since

$$\mathbf{v} = \frac{d\mathbf{r}}{dt},$$

we obtain

$$\boxed{\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2}.}$$

Acceleration measures how rapidly the velocity changes.

8.9 Dot Notation

In analytical mechanics it is common to use Newton's dot notation.

Instead of writing

$$\frac{dx}{dt},$$

we write

$$\boxed{\dot{x}.}$$

Similarly,

$$\boxed{\ddot{x} = \frac{d^2x}{dt^2}.}$$

For vectors,

$$\boxed{\dot{\mathbf{r}} = \mathbf{v},}$$

and

$$\boxed{\ddot{\mathbf{r}} = \mathbf{a}.}$$

This notation will be used extensively in the Lagrangian and Hamiltonian formulations developed later in this book.

8.10 Uniform Circular Motion

One of the most important examples of motion is uniform circular motion.

A particle moving around a circle of radius

$$R$$

with angular velocity

$$\omega$$

has position

$$x(t) = R\cos(\omega t),$$

$$y(t) = R\sin(\omega t).$$

Differentiating,

$$v_x = -R\omega\sin(\omega t),$$

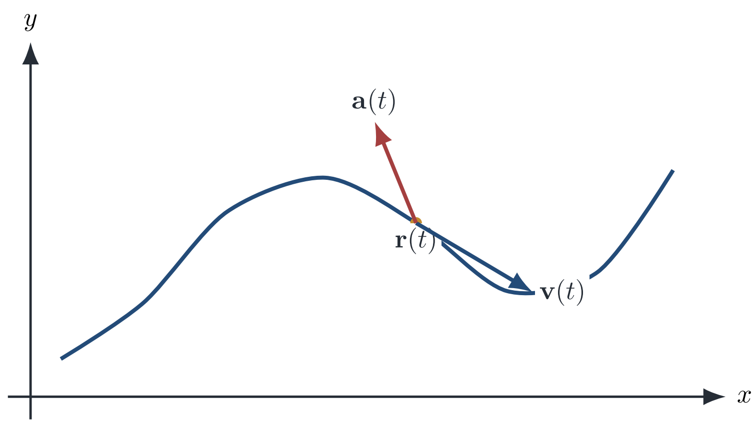


Figure 8.1: A parametric trajectory with instantaneous velocity tangent to the curve and acceleration changing the velocity vector.

$$v_y = R\omega \cos(\omega t).$$

The speed is constant,

$$v = R\omega,$$

while the direction changes continuously.

The acceleration always points toward the centre of the circle,

$$\boxed{a = \frac{v^2}{R}.}$$

Later, we shall discover that electrons in a magnetic field naturally execute cyclotron motion, which is closely related to this simple example.

How to read the figure

The position vector identifies the point on the curve. Velocity is tangent to the trajectory. Acceleration need not point along the direction of motion.

8.11 Motion as Parametric Curves

The position vector

$$\mathbf{r}(t)$$

defines a parametric curve.

The velocity vector is tangent to this curve,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt},$$

while the acceleration describes how the tangent vector changes.

These geometric ideas later become central in differential geometry and general relativity.

8.12 Connection to Future Chapters

The concepts introduced here appear everywhere in physics.

Newton's Second Law uses

$$\mathbf{a}.$$

The Lagrangian depends upon

$$\mathbf{r} \quad \text{and} \quad \dot{\mathbf{r}}.$$

The Hamiltonian depends upon

$$\mathbf{r} \quad \text{and} \quad \mathbf{p}.$$

Quantum mechanics introduces

$$\psi(\mathbf{r}, t),$$

which also depends on position and time.

Thus, understanding motion is an essential prerequisite for all later chapters.

8.13 Summary

In this chapter we introduced

- motion,
- time,
- functions,
- trajectories,
- average velocity,
- instantaneous velocity,
- speed,
- acceleration,
- dot notation,
- circular motion,
- parametric curves.

The next chapter develops the mathematical tools needed to describe changing quantities rigorously. We shall introduce limits, derivatives, partial derivatives and the differential operators that form the language of electromagnetism and quantum mechanics.

Quick Formula Sheet

Quick Formula Sheet

Quantity or operation	Formula
Displacement	$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1)$
Average velocity	$\overline{\mathbf{v}} = \Delta\mathbf{r}/\Delta t$
Instantaneous velocity	$\mathbf{v} = d\mathbf{r}/dt$
Speed	$v = \|\mathbf{v}\|$
Acceleration	$\mathbf{a} = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$
Uniform circular motion	$\|\mathbf{a}\| = v^2/R = \omega^2 R$

Notation and Units

Notation and Units

Symbol	Meaning	SI unit
t	Time	s
$\mathbf{r}(t)$	Position as a function of time	m
$\Delta\mathbf{r}$	Displacement	m
$\mathbf{v}$	Velocity	$\mathrm{m\,s^{-1}}$
v	Speed	$\mathrm{m\,s^{-1}}$
$\mathbf{a}$	Acceleration	$\mathrm{m\,s^{-2}}$
ω	Angular velocity	$\mathrm{rad\,s^{-1}}$

Physical Interpretation

Velocity is tangent to the trajectory, whereas acceleration describes how that tangent vector changes. Consequently, an object may accelerate even when its speed remains constant, as in uniform circular motion.

Common Mistake

Average velocity and instantaneous velocity are different concepts. The average uses a finite time interval; the instantaneous value is obtained from the limiting derivative at one time.

Historical Note 8.1: Galileo and mathematical motion

Galileo's studies of falling bodies and projectiles helped replace qualitative Aristotelian descriptions with mathematical laws involving distance, time, and acceleration. This transition prepared the way for Newtonian mechanics.

Learning Outcomes Check

Learning Outcomes Check

The reader should now be able to:

- interpret a trajectory $\mathbf{r}(t)$;
- distinguish distance, displacement, speed, and velocity;
- obtain velocity and acceleration by differentiation;
- describe circular and parametric motion;
- connect kinematics to later Lagrangian and Hamiltonian descriptions.

Exercises

Basic

Exercise 8.1: One-dimensional motion

Difficulty: ⋆ ○ ○ ○ ○ Let $x(t) = 4t^2 - 2t + 1$. Determine the velocity and acceleration.

Exercise 8.2: Average and instantaneous velocity

Difficulty: ⋆ ⋆ ○ ○ ○ For $\mathbf{r}(t) = (t^2, 2t, t^3)$, calculate the average velocity from $t = 1$ to $t = 2$ and the instantaneous velocity at $t = 2$.

Intermediate

Exercise 8.3: Circular trajectory

Difficulty: ⋆ ⋆ ○ ○ ○ For $\mathbf{r}(t) = R(\cos\omega t, \sin\omega t, 0)$, derive $\mathbf{v}(t)$ and $\mathbf{a}(t)$ and show that $\mathbf{a} = -\omega^2\mathbf{r}$.

Programming

Exercise 8.4: Numerical trajectory

Difficulty: ⋆ ⋆ ⋆ ○ ○ Sample a three-dimensional parametric trajectory at discrete times, estimate its velocity using finite differences, and compare the estimate with the analytic derivative.

Looking Ahead

Looking Ahead

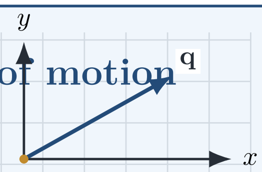
Kinematics uses derivatives, but a rigorous understanding of derivatives requires limits and multivariable calculus. The next chapter develops these tools and introduces ∇, divergence, curl, and the Laplacian.

Chapter 9

Newtonian Mechanics

Forces determine the evolution of motion

Newton's laws connect interactions to momentum and acceleration, turning kinematics into predictive dynamics.



"To every action there is always opposed an equal reaction."

— Isaac Newton, *Philosophiae Naturalis Principia Mathematica* (1687)

Chapter Roadmap

Inertia ⟶ force ⟶ momentum ⟶ equations of motion ⟶ free-body analysis.

Learning Objectives

After completing this chapter, the reader should be able to:

- understand the concept of force;
- distinguish between mass and weight;
- explain Newton's three laws of motion;
- calculate the motion of particles under applied forces;
- understand inertia and momentum;
- appreciate why Newtonian mechanics is an approximation to relativity and quantum mechanics.

9.1 Classical Mechanics

Classical mechanics describes the motion of macroscopic objects.

Examples include

- planets,
- automobiles,
- projectiles,
- pendulums,
- satellites.

For objects moving much slower than the speed of light and much larger than atoms, Newton's laws provide an extremely accurate description of nature.

9.2 The Concept of Force

In everyday language we often speak of pushing or pulling an object.

Physics introduces the more general concept of a force.

A force is any interaction capable of changing the motion of an object.
Examples include

- gravity,
- electric forces,
- magnetic forces,
- friction,
- tension,
- spring forces.

The SI unit of force is the newton (N),

$$1\ \mathrm{N} = 1\ \mathrm{kg\,m/s^2}.$$

9.3 Newton's First Law

Newton's first law states

> Every body remains at rest or continues moving with constant velocity unless acted upon by a net external force.

This law introduces the concept of inertia.
Objects naturally resist changes in their state of motion.

9.4 Inertial Reference Frames

Newton's laws are valid only in inertial reference frames.
An inertial frame is one in which an isolated object moves with constant velocity.
Accelerating reference frames require additional fictitious forces such as

- centrifugal force,
- Coriolis force.

9.5 Mass

Mass measures inertia.
Objects with larger mass require larger forces to produce the same acceleration.
Mass should not be confused with weight. Mass measures inertia, whereas weight is the gravitational force acting on that mass.

Common Mistake

Mass is measured in kilograms; weight is a force measured in newtons.

9.6 Newton's Second Law

How to read the figure

The normal force and weight cancel vertically when there is no vertical acceleration. The horizontal net force determines the horizontal acceleration.

The most famous equation in classical mechanics is

$$\boxed{\mathbf{F} = m\mathbf{a}.}$$

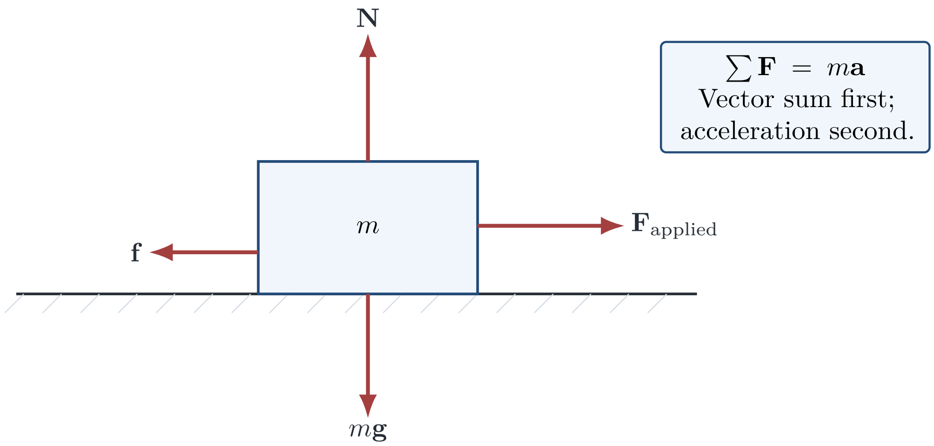


Figure 9.1: Free-body diagram for a block on a horizontal surface. Only forces acting on the selected body are shown.

Since

$$\mathbf{a} = \frac{d^2\mathbf{r}}{dt^2},$$

we may also write

$$\boxed{m\frac{d^2\mathbf{r}}{dt^2} = \mathbf{F}.}$$

This is a second-order differential equation.
Given the force, solving this equation determines the particle's motion.

9.7 Momentum

Momentum is defined by

$$\boxed{\mathbf{p} = m\mathbf{v}.}$$

Using momentum,
Newton's second law becomes

$$\boxed{\mathbf{F} = \frac{d\mathbf{p}}{dt}.}$$

This form remains valid even when the mass changes.

9.8 Newton's Third Law

Newton's third law states

> For every action there exists an equal and opposite reaction.

If object A exerts a force upon object B,
then object B simultaneously exerts an equal and opposite force upon object A.
Mathematically,

$$\boxed{\mathbf{F}_{AB} = -\mathbf{F}_{BA}.}$$

9.9 Free-Body Diagrams

To analyse motion, all forces acting on an object should first be identified.

Typical forces include

- gravity,
- normal force,
- friction,
- tension,
- external applied forces.

The vector sum of these forces determines the acceleration.

9.10 Worked Example

Worked Example

A force $\mathbf{F} = (10, 0, 0)\,\mathrm{N}$ acts on a $2\,\mathrm{kg}$ mass. Find the acceleration.

Solution

Newton's second law gives

$$\mathbf{a} = \frac{\mathbf{F}}{m} = \frac{(10, 0, 0)}{2} = (5, 0, 0)\,\mathrm{m/s^2}.$$

9.11 Limitations of Newtonian Mechanics

Although remarkably successful, Newton's theory is not universal.

For

- velocities approaching the speed of light,
- atomic dimensions,
- strong gravitational fields,

more advanced theories become necessary.

These include

- Special Relativity,
- General Relativity,
- Quantum Mechanics.

Nevertheless, Newtonian mechanics remains the starting point of nearly every modern physical theory.

9.12 Connection to the Next Chapter

Newton's equation

$$m\mathbf{a} = \mathbf{F}$$

describes motion once the force is known.

In the next chapter we shall study the most important forces in nature, beginning with gravitational, electric and magnetic interactions.

Eventually we shall derive the Lorentz force,

Key Equation 9.1: Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

which forms the classical foundation of the Quantum Hall Effect.

9.13 Summary

This chapter introduced

- force;
- inertia;
- inertial frames;
- mass;
- Newton's three laws;
- momentum;
- free-body diagrams;
- limitations of classical mechanics.

These ideas provide the bridge between elementary kinematics and the electromagnetic forces that govern charged particles.

Quick Formula Sheet

Quick Formula Sheet

Quantity or operation	Formula
Momentum	$\mathbf{p} = m\mathbf{v}$
Newton's second law	$\mathbf{F}_{\text{net}} = d\mathbf{p}/dt$
Constant-mass form	$\mathbf{F}_{\text{net}} = m\mathbf{a}$
Weight near Earth's surface	$\mathbf{W} = m\mathbf{g}$
Action–reaction pair	$\mathbf{F}_{A\to B} = -\mathbf{F}_{B\to A}$
Impulse	$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F}\,dt = \Delta\mathbf{p}$

Notation and Units

Notation and Units

Symbol	Meaning	SI unit
m	Inertial mass	kg
$\mathbf{F}$	Force	N
$\mathbf{F}_{\text{net}}$	Vector sum of forces	N
$\mathbf{p}$	Momentum	$\mathrm{kg\,m\,s^{-1}}$
$\mathbf{g}$	Gravitational acceleration	$\mathrm{m\,s^{-2}}$
$\mathbf{J}$	Impulse	$\mathrm{N\,s}$

Physical Interpretation

Newton's second law does not merely state that force causes motion. It states that net force controls the rate of change of momentum. The familiar equation $\mathbf{F} = m\mathbf{a}$ is the constant-mass special case.

Common Mistake

Mass and weight are not interchangeable. Mass measures inertia and is expressed in kilograms. Weight is a gravitational force and is expressed in newtons. Also, the two forces in an action–reaction pair act on different bodies, so they do not cancel in a free-body diagram of one body.

Historical Note 9.1: The *Principia*

Newton published the *Philosophiae Naturalis Principia Mathematica* in 1687. It unified terrestrial and celestial motion using a small set of mathematical laws and became the foundation of classical mechanics.

Learning Outcomes Check

Learning Outcomes Check

The reader should now be able to:

- identify inertial frames;
- distinguish mass, weight, momentum, and force;
- apply Newton's three laws;
- construct and interpret free-body diagrams;
- derive equations of motion from a specified net force;
- state the limits of Newtonian mechanics.

Exercises

Basic

Exercise 9.1: Net force

Difficulty: ⋆ ◦ ◦ ◦ ◦ A 4 kg body experiences forces 12 N east and 5 N west. Determine its acceleration.

Exercise 9.2: Mass and weight

Difficulty: ⋆ ◦ ◦ ◦ ◦ Calculate the weight of a 70 kg person on Earth using $g = 9.81\,\mathrm{m\,s^{-2}}$.

Intermediate

Exercise 9.3: Inclined plane

Difficulty: ⋆ ⋆ ◦ ◦ ◦ A block slides without friction down an incline of angle θ. Draw the free-body diagram and derive its acceleration along the plane.

Advanced

Exercise 9.4: Momentum form

Difficulty: ⋆ ⋆ ⋆ ◦ ◦ Starting from $\mathbf{F} = d\mathbf{p}/dt$, explain why $\mathbf{F} = m\mathbf{a}$ requires constant mass.

Challenge

Exercise 9.5: Magnetic force preview

Difficulty: ⋆ ⋆ ⋆ ⋆ ◦ The magnetic force is $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$. Explain why it can change the direction of velocity without changing the particle's speed.

Looking Ahead

Looking Ahead

The next stage identifies the forces that enter Newton's equation. Of special importance is the Lorentz force,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}),$$

which later leads to cyclotron motion, Landau quantization, and the Quantum Hall Effect.

Visual Language Preview for Quantum Mechanics

This unnumbered design page fixes the visual conventions that later chapters will use. It is a template, not a substitute for the full physics discussion.

Chapter 10

Lagrangian Mechanics

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

Chapter 11

Hamiltonian Mechanics

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

Chapter 12

Classical Fields

Chapter Roadmap

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Chapter Development Status

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PART III

Foundations of Quantum Mechanics

Chapter 13

The Crisis of Classical Physics

Chapter Roadmap

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Chapter Development Status

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Chapter 14

Blackbody Radiation

Chapter Roadmap

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Chapter Development Status

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Chapter 15

Photoelectric Effect

Chapter Roadmap

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Chapter Development Status

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Chapter 16

Atomic Spectra

Chapter Roadmap

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Chapter Development Status

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Chapter 17

Wave–Particle Duality

Chapter Roadmap

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Chapter Development Status

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Chapter 18

Schrödinger Equation

Chapter Roadmap

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Chapter Development Status

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Chapter 19

Operators and Observables

Chapter Roadmap

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Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

Chapter 20

Hilbert Spaces

Chapter Roadmap

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Chapter Development Status

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Chapter 21

Angular Momentum and Spin

Chapter Roadmap

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Chapter Development Status

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Chapter 22

Hydrogen Atom

Chapter Roadmap

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Chapter Development Status

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PART IV

Advanced Quantum Mechanics

Chapter 23

Many-Particle Systems

Chapter Roadmap

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Chapter Development Status

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Chapter 24

Approximation Methods

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

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Chapter 25

Scattering Theory

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

Chapter 26

Quantum Statistics

Chapter Roadmap

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Chapter Development Status

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Chapter 27

Relativistic Quantum Mechanics

Chapter Roadmap

Planned chapter. The directory and chapter position are fixed by the approved master architecture.

Chapter Development Status

This chapter is reserved in the canonical structure. Existing manuscript material will be migrated here without changing the approved chapter order.

Chapter 28

Introduction to Quantum Field Theory

Chapter Roadmap

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Chapter Development Status

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PART V

Appendices

Chapter A

Mathematical Tables

This appendix is reserved by the approved master architecture.

Chapter B

Physical Constants

This appendix is reserved by the approved master architecture.

Chapter C

Notation

This appendix is reserved by the approved master architecture.

Solutions

This appendix is reserved by the approved master architecture.

Chapter E

Bibliography

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Glossary

bifurcation a qualitative change in the structure of equilibria or trajectories as a parameter varies.

chain rule The composition law stating that the derivative of a composite map is the composition of the derivatives.

conservative field a vector field representable as the gradient of a scalar potential.

continuity The property that sufficiently small changes of the input produce arbitrarily small changes of the output.

continuity equation a local conservation law relating density change to current divergence.

derivative The unique best first-order linear approximation to a differentiable function at a point.

differential The linear map that gives the first-order change of a differentiable function.

direction field a geometric display of the slope prescribed by a first-order differential equation at each point of the independent-dependent-variable plane.

directional derivative The instantaneous rate of change of a function in a specified unit direction.

divergence theorem the theorem equating outward flux through a closed surface with integrated divergence inside.

equilibrium point a state at which the dynamical vector field vanishes and the system can remain indefinitely.

Fréchet derivative A coordinate-independent derivative defined by a linear approximation whose remainder is negligible compared with the input displacement.

function A rule assigning each element of a domain exactly one element of a codomain.

gradient The vector representing the differential of a scalar field through the inner product and pointing toward steepest increase.

Green's theorem the planar boundary theorem relating circulation or flux to an area integral.

initial-value problem a differential equation together with the state specified at an initial time.

Jacobian matrix The matrix representing the derivative of a vector-valued function in chosen coordinates.

level set The set of points at which a scalar function has one fixed value.

limit The value approached by a function as its argument approaches a specified point or regime.

line integral an integral accumulating a scalar or vector field along a parametrized curve.

ordinary differential equation an equation relating an unknown function of one independent variable to one or more of its derivatives.

partial derivative The rate of change of a multivariable function with respect to one coordinate while the other coordinates are held fixed.

phase space the state space in which each point specifies a complete instantaneous state of a dynamical system.

stability the property that trajectories beginning sufficiently near a reference state remain near it under evolution.

stiffness the coexistence of widely separated dynamical time scales that severely restricts explicit numerical step sizes.

Stokes' theorem the theorem relating circulation around a boundary curve to curl flux through a spanning surface.

surface integral an integral accumulating density or flux over a parametrized surface.

tangent plane The affine plane determined by the first-order linearization of a differentiable surface at a point.

vector field a function assigning a vector to every point of a domain.

volume integral an integral accumulating a density throughout a region.

Subject Index